From Conjecture to the Lean Kernel

Optimal lattices, quantum energies and population genetics —
human proofs, Lean 4, rusty-SUNDIALS and LeanFlow

A textbook for university mathematics, physics and engineering students

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written with AI agents (Claude) under the author's direction

Version 1 (English university edition), 2026

doi:10.5281/zenodo.23665873 (not peer reviewed)

To Ilene, Alexis and Aurelia.

The four letters of ALIX and the three of KAL are hidden in a few theorems of this book; the attentive reader will find them in the Lean code of Chapter 5.

Preface

This book tells a recent mathematical story as it actually unfolded: a sequence of conjectures, proofs, computer verifications and, at times, refutations. It is written for undergraduate and graduate students in mathematics, physics and engineering. The starting point is a theorem of openai/math, whose proof, produced by an AI and not peer reviewed, is checked by the Lean 4 kernel: the triangular lattice minimizes the energy of every point configuration in the plane, for a wide class of repulsive interactions. The end point is a series of open questions in quantum physics and population genetics.

Every result is presented three times.

- First as a *human proof*, step by step, at the level of an undergraduate course in analysis, linear algebra and probability.
- Then as a *Lean 4* statement checked by the kernel.
- Finally as a numerical computation with *rusty-SUNDIALS* (CVODE solvers in Rust) and *LeanFlow* (data certificates checked by Lean), illustrated on public datasets.

A single rule guides the reading: every claim carries a *confidence tier*, following the Elenchus method.

- **[Tier A]**: checked by the Lean kernel. This requires three checks: the axiom list is contained in `propext`, `Classical.choice` and `Quot.sound`; the kernel replays the proof (`leanchecker`); and a search for constructions that bypass the kernel finds nothing. The axiom list alone can be fooled; Chapter 1 explains how.
- **[Tier A without replay]**: a variant of **[Tier A]** when the kernel replay has not (yet) been run for the module concerned; the axiom list, the planted witnesses and the independent recompilation are still required.
- **[Tier L]**: a published result, read and cited precisely.
- **[Tier C]**: a human argument, not machine-checked.
- **[Tier X]**: a floating-point computation. It guides research but proves nothing.

Read the tier before the sentence.

Honesty.

- The results in this book were produced by AI agents (Claude) under the author's direction.
- Each paper was audited by a separate AI agent; no human mathematician has reviewed them yet.
- Several results depend on theorems of the openai/math release, which were also produced by an AI and have not yet been peer reviewed.

- Several chapters report *negative results*: a refuted conjecture, a method that does not extend, a transition that does not take place. They are as instructive as the others.
- No conjecture is declared proved beyond its tier. In particular, the Sandier–Serfaty conjecture is *not* proved here: Chapter 2 explains exactly what was checked and what remains to be done.

What is new in this edition. This is the English university edition of the French textbook *De la conjecture au noyau Lean*. The proofs, numbers, tiers, DOIs and caveats are identical. For undergraduate and graduate readers, short *University remarks*, each marked **[Tier L]**, state the general textbook form of a tool used in the text (dominated convergence on a measure space, Bernstein's and Bochner's theorems, the Poisson summation formula in $\mathbb{R}^d$, the theory of absorbing Markov chains), with a reference where the bibliography has one and the mark "(standard)" otherwise; they add no new claim. Each of Chapters 1 to 5 ends with two additional harder exercises, titled "(†) University exercise"; the dagger is a label, not a difficulty rating, and these exercises are distinct from the difficulty stars that some chapters already attach to their exercises. The figures are those of the French edition, reused unchanged, so the labels inside some figures are in French; every caption is in English. Likewise, the code listings and logs are reproduced exactly as they were run, so their comments are in French.

Outline.

- Chapter 1 lays the foundations: lattices, completely monotone functions and formal verification.
- Chapter 2 follows the thread of the formalizations, from universal optimality to the potentials of physics and to the Sandier–Serfaty conjecture.
- Chapter 3 presents a quantum surprise: for some interactions, the triangular lattice is no longer optimal once the zero-point energy is taken into account.
- Chapter 4 checks classical theorems of population genetics in Lean, up to the isothermal theorem of evolutionary graph theory.
- Chapter 5 gathers five use cases with data, code and certificates.

Each chapter ends with exercises.

Further reading. Every cited paper has a Zenodo DOI. The code is public: AutoevolveAI, rusty-SUNDIALS, LeanFlow, SocrateAI-Scientific-Genetic and SocrateAI-Scientific-Agora-LeanMaster.

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Chapter 1

Foundations: lattices, completely monotone functions and formal verification

This chapter sets the stage. It assumes only first-year undergraduate analysis and linear algebra, plus the basic theory of Fourier series. The proofs are written by hand, as one would write them in an exam; the machine (Lean 4) enters only in the last part, to say what "computer-verified" means and, above all, what it does not mean.

How to read this book

Every claim carries a *tier* label that says *who checked what*. These tiers, taken from the Elenchus method [10] and detailed in §1.4.6, are the following.

- **[Tier A]**: a statement checked by the Lean 4 kernel, with three controls: axioms on the whitelist, kernel replay (`leanchecker`) and a search for backdoors (§1.4).
- **[Tier A without replay]**: as **[Tier A]**, but without the kernel replay (§1.4).
- **[Tier L]**: a result from the literature, read and precisely referenced.
- **[Tier C]**: a human argument, written out in detail in this book, but not machine-checked.
- **[Tier X]**: a floating-point numerical exploration; it illustrates, it never proves.

Warning : honesty about where the results come from

All results in this book were produced by AI agents (Claude) under the author's direction, and then audited by a *separate* AI agent. **No human has audited these results.** Moreover, everything that relies on the `openai/math` repository is *conditional* on that repository's theorems, which are themselves AI-generated and have not been peer reviewed [1]. A **[Tier A]** result guarantees that the *Lean* statement follows from the allowed axioms; it does not guarantee that this statement faithfully expresses the mathematical or physical statement one has in mind.

1.1 Lattices in the plane

1.1.1 Lattice, covolume, density

Definition 1.1 (Lattice in the plane)**.** Let (u, v) be a basis of $\mathbb{R}^2$. The *lattice* it generates is $\Lambda = \mathbb{Z}u \oplus \mathbb{Z}v = \{mu + nv : (m, n) \in \mathbb{Z}^2\}$. Its *covolume* is $\mathrm{covol}(\Lambda) = |\det(u, v)|$, the area of the parallelogram $P = \{su + tv : s, t \in [0, 1)\}$, called the *fundamental cell*. The covolume does not depend on the chosen basis (two bases differ by a matrix in $\mathrm{GL}_2(\mathbb{Z})$, of determinant ± 1).

We write $N_\Lambda(R) = \#\{x \in \Lambda : |x| \le R\}$ for the number of points of Λ in the closed disk of radius R. A lattice is an example of a *locally finite configuration*: every disk contains finitely many points.

Proposition 1.2 (Density of a lattice)**.** *If $V = \mathrm{covol}(\Lambda)$, then $N_\Lambda(R)/(\pi R^2) \to 1/V$ as $R \to \infty$. More precisely, with $\delta = |u| + |v|$ and for $R > \delta$,*

$$\pi(R - \delta)^2 \;\le\; V\, N_\Lambda(R) \;\le\; \pi(R + \delta)^2.$$

[Tier C]

Human proof. The translates $x + P$, $x \in \Lambda$, *tile* the plane: every point $y = su' + tv'$ can be written uniquely as $y = x + p$ with $x \in \Lambda$ (integer part of the coordinates in the basis (u, v)) and $p \in P$. Each cell $x + P$ has area V and diameter at most δ.

- If $|x| \le R$, then $x + P \subset \overline{D}(0, R + \delta)$. The cells are disjoint, hence $V\, N_\Lambda(R) \le \pi(R + \delta)^2$.
- If $|y| \le R - \delta$, the point y lies in a cell $x + P$ with $|y - x| \le \delta$, hence $|x| \le R$. Thus $\overline{D}(0, R - \delta) \subset \bigcup_{|x| \le R}(x + P)$ and $\pi(R - \delta)^2 \le V\, N_\Lambda(R)$.

Divide by πR^2 and let R tend to infinity. □

We say that Λ has *density* $1/V$. A lattice of covolume 1 has density 1: this is the normalisation used throughout the book, which allows different lattices to be compared at an equal number of points per unit area.

1.1.2 The triangular lattice

Definition 1.3 (Triangular lattice A)**.** Let $a > 0$ with $a^2 = 2/\sqrt{3}$. The *triangular* (or hexagonal) *lattice* of covolume 1 is

$$A = \mathbb{Z}\,u \oplus \mathbb{Z}\,v, \qquad u = a(1, 0), \quad v = a\big(\tfrac{1}{2}, \tfrac{\sqrt{3}}{2}\big).$$

The choice of a comes from the covolume: $\det(u, v) = a^2\sqrt{3}/2 = 1$. Numerically $a \approx 1.0746$.

Proposition 1.4 (Norms in A)**.** *For $(m, n) \in \mathbb{Z}^2$ we have $|mu + nv|^2 = a^2\,(m^2 + mn + n^2)$. In particular the minimal distance between two points of A is a, attained by exactly 6 vectors (so the central point has 6 nearest neighbours, at the vertices of a regular hexagon).* ***[Tier C]***

Human proof. Since $|u|^2 = |v|^2 = a^2$ and $u \cdot v = a^2/2$, we have $|mu + nv|^2 = a^2(m^2 + n^2 + mn)$. Now $m^2 + mn + n^2 = (m + \frac{n}{2})^2 + \frac{3}{4}n^2 \ge 0$, with equality only for $m = n = 0$. The integer solutions of $m^2 + mn + n^2 = 1$ satisfy $\frac{3}{4}n^2 \le 1$, hence $n \in \{-1, 0, 1\}$; solving each case gives $(\pm 1, 0)$, $(0, \pm 1)$, $\pm(1, -1)$. There are six of them, and the value 1 is the smallest nonzero value because $m^2 + mn + n^2$ is a nonnegative integer. □

The first values of $k = m^2 + mn + n^2$ and their multiplicities, compared with those of the square lattice $\mathbb{Z}^2$ (also of covolume 1), are given in Table 1.1. These numbers were counted by machine computation **[Tier X]**; they are classical.

The triangular lattice has neighbours that are farther away ($|x|^2 \approx 1.155$ against 1) and more numerous (6 against 4): it "balances" the repulsion better, and this is what the universal optimality theorem will quantify.

Lattice	k (value of the form)	multiplicity	$\lvert x\rvert^2$
A	1, 3, 4, 7, 9	6, 6, 6, 12, 6	$a^2k \approx 1.155,\ 3.464,\ 4.619,\ 8.083,\ 10.392$
$\mathbb{Z}^2$	1, 2, 4, 5, 8	4, 4, 4, 8, 4	1, 2, 4, 5, 8

Table 1.1: First shells of the lattices A and $\mathbb{Z}^2$ (covolume 1).

1.1.3 Dual lattice

Definition 1.5 (Dual lattice)**.** If $\Lambda = M\mathbb{Z}^2$ with M the matrix with columns u, v, the *dual lattice* is $\Lambda^* = \{y \in \mathbb{R}^2 : x \cdot y \in \mathbb{Z} \ \ \forall x \in \Lambda\} = M^{-T}\mathbb{Z}^2$. Its covolume is $1/\operatorname{covol}(\Lambda)$.

Lemma 1.6 (A is similar to its dual)**.** *The lattice A^* is the image of A under a rotation; in particular A and A^* have the same norms, with the same multiplicities.* ***[Tier C]***

Human proof. Let $M = a\begin{pmatrix} 1 & 1/2 \\ 0 & \sqrt{3}/2 \end{pmatrix}$. A direct computation gives $M^{-T} = \frac{1}{a}\begin{pmatrix} 1 & 0 \\ -1/\sqrt{3} & 2/\sqrt{3} \end{pmatrix}$. Its two columns b_1, b_2 satisfy $|b_1|^2 = |b_2|^2 = \frac{1}{a^2}\cdot\frac{4}{3} = \frac{\sqrt{3}}{2}\cdot\frac{4}{3} = a^2$ and $b_1 \cdot b_2 = -\frac{1}{a^2}\cdot\frac{2}{3} = -\frac{a^2}{2}$: two vectors of norm a at an angle of 120°. Then $(b_1, b_1 + b_2)$ is a basis of A^* consisting of two vectors of norm a at an angle of 60°, that is, the basis of A rotated. Hence A^* is the image of A under a rotation. □

1.1.4 Energy of a lattice

We study identical particles that repel each other through a *pair potential* g, a function of the *square* $t = |x - y|^2$ of the distance (the convention of [2] and [1]: Gaussians $e^{-\alpha t}$ are then the basic building blocks). The Riesz potential $|x - y|^{-s}$ reads $g(t) = t^{-s/2}$.

Definition 1.7 (Energy of a lattice)**.** For $g : (0, \infty) \to [0, \infty)$, the energy per particle of a lattice Λ is $E_g(\Lambda) = \sum_{x \in \Lambda \setminus \{0\}} g(|x|^2) \in [0, +\infty]$.

For a locally finite configuration C of density 1, we define the *lower energy* as the average, over the points of the disk of radius R, of the interactions with the other points of the disk, followed by a limit inferior:

$$E_g(C; R) = \frac{1}{\#C_R} \sum_{x \in C_R} \sum_{y \in C_R \setminus \{x\}} g(|x - y|^2), \qquad E_g(C) = \liminf_{R \to \infty} E_g(C; R) \in [0, \infty],$$

with $C_R = C \cap \overline{D}(0, R)$. Each pair is counted twice, which indeed gives $\sum_{a \neq 0} g(|a|^2)$ for a lattice. **[Tier C]** The exact definition used by the `openai/math` repository, in $[0, \infty]$ arithmetic, will be quoted in §1.3.

Proposition 1.8 (Riesz energy of a lattice)**.** *For $g(t) = t^{-s/2}$ with $s > 0$, the energy $E_g(\Lambda) = \sum_{x \neq 0} |x|^{-s}$ is finite if and only if $s > 2$.* ***[Tier C]***

Human proof. Let $d > 0$ be the smallest nonzero norm of Λ and V its covolume. There are finitely many nonzero points of norm $\le d$; for $k \ge 0$ let $S_k = \{x \in \Lambda : 2^k d < |x| \le 2^{k+1} d\}$. By Proposition 1.2, $\#S_k = N_\Lambda(2^{k+1}d) - N_\Lambda(2^k d) = \frac{3\pi d^2}{V} 4^k + O(2^k)$, so there exist $0 < c < C$ with $c\,4^k \le \#S_k \le C\,4^k$ for $k \ge k_0$. On S_k we have $(2^{k+1}d)^{-s} \le |x|^{-s} < (2^k d)^{-s}$. The sum over S_k therefore lies between positive multiples of $4^k 2^{-ks} = 2^{k(2-s)}$. The series $\sum_k 2^{k(2-s)}$ converges if and only if $s > 2$. □

Remark 1.9. For $0 < s \le 2$ the energy of a lattice is infinite: the optimality inequality $\infty \le \infty$ is then true but void, and a meaningful comparison would require a *regularisation*. **[Tier C]**

1.1.5 Theta series and the Poisson summation formula

For $\alpha > 0$, the *Gaussian theta series* of the lattice Λ is

$$\Theta_\Lambda(\alpha) = \sum_{x \in \Lambda} e^{-\pi \alpha |x|^2} \quad (> 1).$$

(Not to be confused with the arithmetic Gauss sums $\sum e^{2\pi i n^2/p}$: here we are dealing with sums of *real* exponentials of quadratic forms.) The series converges, by comparison with a geometric series via the lower bound $|mu + nv|^2 \geq c(m^2 + n^2)$. The energy of a Gaussian interaction $g(t) = e^{-\pi \alpha t}$ is therefore $E_g(\Lambda) = \Theta_\Lambda(\alpha) - 1$.

The key result is that Θ_Λ transforms simply when one passes from α to $1/\alpha$ and from Λ to Λ^*.

Theorem 1.10 (Poisson summation for Gaussians)**.** *For every lattice Λ in the plane of covolume V and every $\alpha > 0$,*

$$\Theta_\Lambda(\alpha) = \frac{1}{V\,\alpha}\ \Theta_{\Lambda^*}\Big(\frac{1}{\alpha}\Big).$$

For the triangular lattice ($V = 1$, $A^* \simeq A$): $\Theta_A(\alpha) = \alpha^{-1}\Theta_A(1/\alpha)$. ***[Tier C]*** *(The pointwise convergence of the Fourier series used in Step 3 is Fourier's theorem on the torus:* ***[Tier L]****.)*

Human proof in three steps. Convention: $\hat f(\xi) = \int f(x) e^{-2\pi i\, \xi \cdot x}\, dx$.

Step 1: Fourier transform of a Gaussian in dimension 1. Let $\varphi(x) = e^{-\pi \alpha x^2}$ and $h = \hat\varphi$. Differentiating under the integral sign (dominated by $|x| e^{-\pi \alpha x^2}$, which is integrable), $h'(\xi) = \int (-2\pi i x)\varphi(x) e^{-2\pi i \xi x} dx$. Now $\varphi'(x) = -2\pi \alpha x \varphi(x)$, so $-2\pi i x \varphi(x) = \frac{i}{\alpha}\varphi'(x)$ and

$$h'(\xi) = \frac{i}{\alpha}\int \varphi'(x) e^{-2\pi i \xi x} dx = \frac{i}{\alpha}\Big([\varphi e^{-2\pi i \xi x}]_{-\infty}^{+\infty} + 2\pi i \xi \int \varphi(x) e^{-2\pi i \xi x} dx \Big) = -\frac{2\pi \xi}{\alpha}\, h(\xi),$$

(the bracket vanishes because $\varphi \to 0$ at $\pm\infty$). This linear differential equation gives $h(\xi) = h(0)\, e^{-\pi \xi^2/\alpha}$, and $h(0) = \int e^{-\pi \alpha x^2} dx = \alpha^{-1/2}$ (Gaussian integral $\int e^{-\pi x^2} dx = 1$, computed in polar coordinates). Hence $\widehat{e^{-\pi \alpha x^2}}(\xi) = \alpha^{-1/2} e^{-\pi \xi^2/\alpha}$. In dimension 2, by Fubini, $\widehat{e^{-\pi \alpha |x|^2}}(\xi) = \alpha^{-1} e^{-\pi |\xi|^2/\alpha}$.

Step 2: periodisation. Write $\Lambda = M\mathbb{Z}^2$ and set $f(x) = e^{-\pi \alpha |Mx|^2}$ for $x \in \mathbb{R}^2$. The function $F(x) = \sum_{n \in \mathbb{Z}^2} f(x + n)$ is well defined (locally normal convergence, together with that of its derivatives), $\mathcal{C}^\infty$, and $\mathbb{Z}^2$-periodic. Its Fourier coefficients are, for $k \in \mathbb{Z}^2$,

$$c_k = \int_{[0,1]^2} F(x) e^{-2\pi i k \cdot x} dx = \int_{\mathbb{R}^2} f(x) e^{-2\pi i k \cdot x} dx = \hat f(k),$$

(the translates of $[0,1]^2$ are put together to form $\mathbb{R}^2$). The change of variables $u = Mx$ ($dx = du/V$) gives $\hat f(k) = \frac{1}{V}\int e^{-\pi \alpha |u|^2} e^{-2\pi i (M^{-T}k)\cdot u} du = \frac{1}{V\alpha} e^{-\pi |M^{-T}k|^2/\alpha}$ by Step 1.

Step 3: evaluation at 0. The c_k are summable over $\mathbb{Z}^2$ and F is continuous; the Fourier series of F therefore converges normally to a continuous function with the same coefficients, which is F (uniqueness of the coefficients, Fejér's theorem). At $x = 0$:

$$\sum_{n \in \mathbb{Z}^2} e^{-\pi \alpha |Mn|^2} = F(0) = \sum_{k \in \mathbb{Z}^2} c_k = \frac{1}{V\alpha} \sum_{k \in \mathbb{Z}^2} e^{-\pi |M^{-T}k|^2/\alpha}.$$

The left-hand side is $\Theta_\Lambda(\alpha)$, and $M^{-T}\mathbb{Z}^2 = \Lambda^*$: the right-hand side is $\frac{1}{V\alpha}\Theta_{\Lambda^*}(1/\alpha)$. For A, Lemma 1.6 gives $\Theta_{A^*} = \Theta_A$. □

University remark. **[Tier L]** The same three steps prove the Poisson summation formula in $\mathbb{R}^d$: for a lattice $\Lambda \subset \mathbb{R}^d$ and a Schwartz function f, $\sum_{x \in \Lambda} f(x) = \mathrm{covol}(\Lambda)^{-1} \sum_{y \in \Lambda^*} \hat f(y)$. With $f(x) = e^{-\pi \alpha |x|^2}$, whose Fourier transform is $\alpha^{-d/2} e^{-\pi |\xi|^2/\alpha}$, it yields $\Theta_\Lambda(\alpha) = \mathrm{covol}(\Lambda)^{-1} \alpha^{-d/2}\, \Theta_{\Lambda^*}(1/\alpha)$, the identity behind the Jacobi theta transformation and the functional equation of Epstein zeta functions (standard; see also [5] for the planar case).

Example 1.11 (Jacobi theta function)**.** In dimension 1, the same proof gives, for $\vartheta(\alpha) = \sum_{n\in\mathbb{Z}} e^{-\pi\alpha n^2}$, the relation $\vartheta(1/\alpha) = \sqrt{\alpha}\,\vartheta(\alpha)$. It shows that $\vartheta(\alpha) \sim \alpha^{-1/2}$ as $\alpha \to 0$: for small α the sum "sees" the Gaussian integral. **[Tier C]**

Numerics : the Poisson formula with its controls (code `code/ch1_theta.py`)

For A, we compare $\Theta_A(\alpha)$ with $\alpha^{-1}\Theta_A(1/\alpha)$ (positive control), and then with $\Theta_A(1/\alpha)$ *without* the factor α^{-1} (negative control). Actual output of the script (in double precision):

α	$\Theta_A(\alpha)$	gap to $\alpha^{-1}\Theta_A(1/\alpha)$	gap without the factor
0.5	2.008477918598	$4.4 \cdot 10^{-16}$	1.0
2	1.004238959299	$2.2 \cdot 10^{-16}$	1.0
3	1.000112671200	$8.9 \cdot 10^{-16}$	2.0

The negative control is discriminating ($\alpha \neq 1$); at $\alpha = 1$ both gaps vanish, so that point tests nothing. The script also compares $\Theta_A(\alpha) \leq \Theta_{\mathbb{Z}^2}(\alpha)$ at eight values of $\alpha \in [0.1, 5]$ (always true), for example $\Theta_A(1) = 1.159595267$ against $\Theta_{\mathbb{Z}^2}(1) = 1.180340599$. **[Tier X]**: this illustrates Montgomery's theorem below, without proving it.

Theorem 1.12 (Montgomery, 1988)**.** *For every lattice Λ in the plane of covolume 1 and every $\alpha > 0$, $\Theta_A(\alpha) \leq \Theta_\Lambda(\alpha)$, with equality if and only if Λ is the image of A under a rotation.* ***[Tier L]*** *(cited second-hand) [5]*

Montgomery thus solves the optimality problem *among lattices*, for every Gaussian interaction. We return to this in §1.3 to pass from Gaussians to completely monotone potentials. The difficulty lies elsewhere: comparing A not with other lattices, but with *all* point configurations of density 1.

1.2 Completely monotone functions

1.2.1 Definition and first examples

Definition 1.13 (Completely monotone function)**.** A function $g : (0,\infty) \to \mathbb{R}$ is *completely monotone* (CM) if it is of class $\mathcal{C}^\infty$ and $(-1)^n g^{(n)}(t) \geq 0$ for every $n \in \mathbb{N}$ and every $t > 0$.

The case $n = 0$ says $g \geq 0$; the case $n = 1$ says that g is nonincreasing; the case $n = 2$ says that g is convex; and so on. The inequalities are *non-strict*: constants are CM. In the `openai/math` repository, a function that is $\mathcal{C}^\infty$ on $(0,\infty)$, nonnegative and CM is called *admissible* [1]; since $n = 0$ gives nonnegativity, the two notions coincide. This coincidence is itself a Lean theorem: `ProgCM.isCompletelyMonotoneOn_iff_admissible` **[Tier A]** (library of programme P3 [12]).

Example 1.14 (Exponential, powers)**.**

1. $g(t) = e^{-at}$, $a \geq 0$: $(-1)^n g^{(n)}(t) = a^n e^{-at} \geq 0$. **[Tier C]**
2. $g(t) = t^{-s}$, $s \geq 0$: $(-1)^n g^{(n)}(t) = s(s+1)\cdots(s+n-1)\,t^{-s-n} \geq 0$ (immediate induction). **[Tier C]** For $s = \frac{1}{2}$ this is the Coulomb repulsion $1/r$ in the variable $t = r^2$. The CM property of $t \mapsto t^{-s/2}$ (Riesz) is also checked by the kernel: `TriangularRiesz.riesz_admissible` **[Tier A]**.

Example 1.15 (Counterexamples)**.** $g(t) = t$ is not CM ($-g' = -1 < 0$), nor is $\cos t$ (it takes negative values), nor e^t ($-g' < 0$). A subtler counterexample, of physical origin (a charge–quadrupole combination), is $q(t) = t^{-1/2} - 2(t+1)^{-1/2} + (t+4)^{-1/2}$: we have $q(3) = 3^{-1/2} - 2\cdot 4^{-1/2} + 7^{-1/2} \approx 0.5774 - 1 + 0.3780 < 0$, so q is not nonnegative, let alone CM. The rigorous verification of $q(3) < 0$ and the non-CM property of q are checked by the kernel (`ProgCM.quadrupole_not_cm`) **[Tier A]**.

1.2.2 Closure properties: human proofs

Proposition 1.16 (Sum, product)**.** *Let f, g be CM and $c \geq 0$. Then $f+g$, cf and fg are CM.* ***[Tier C]*** *(Lean version: programme P3 [12],* ***[Tier A]****)*

Human proof. The sum and multiplication by a nonnegative constant are immediate by linearity of differentiation. For the product, the Leibniz rule gives

$$(-1)^n (fg)^{(n)} = \sum_{k=0}^{n} \binom{n}{k} \left[(-1)^k f^{(k)}\right] \left[(-1)^{n-k} g^{(n-k)}\right],$$

since $(-1)^n = (-1)^k (-1)^{n-k}$. Each bracket is nonnegative, hence so is the sum. □

A *Bernstein function* is the right notion for composing with a CM function.

Definition 1.17 (Bernstein function)**.** A $\mathcal{C}^\infty$ function $h : (0,\infty) \to [0,\infty)$ is a *Bernstein function* if its derivative h' is completely monotone (Lean definition `IsBernsteinOn`; Theorem 1.19 additionally requires $h > 0$). In particular h is nondecreasing.

Example 1.18. The following are Bernstein functions: $h(t) = at + b$ $(a, b \geq 0)$; $h(t) = t^\alpha$ for $0 < \alpha \leq 1$, since $h'(t) = \alpha\, t^{-(1-\alpha)}$ is CM by Example 1.14; $h(t) = 1 - e^{-at}$ $(a > 0)$, since $h' = a e^{-at}$; $h(t) = \ln(1+t)$, since $h'(t) = 1/(1+t)$. In particular $h(t) = \sqrt{t}$ (the case $\alpha = \frac{1}{2}$) is one. **[Tier C]**

University remark. **[Tier L]** Bernstein functions admit a Lévy–Khintchine representation: $h : (0,\infty) \to [0,\infty)$ is a Bernstein function if and only if $h(t) = a + bt + \int_{(0,\infty)} (1 - e^{-st})\, \nu(ds)$ with $a, b \geq 0$ and ν a positive measure satisfying $\int_{(0,\infty)} \min(1, s)\, \nu(ds) < \infty$. This follows by integrating the Bernstein–Widder representation (Theorem 1.22) of the CM function h'; for instance $\sqrt{t} = \frac{1}{2\sqrt{\pi}} \int_0^\infty (1 - e^{-st})\, s^{-3/2}\, ds$ (standard; see [28]).

Theorem 1.19 (Subordination: CM $\circ$ Bernstein)**.** *If f is CM and h is a* strictly positive *Bernstein function, then $f \circ h$ is CM.* ***[Tier C]*** *Lean version:* `ProgCM.IsCompletelyMonotoneOn.comp_bernstein` ***[Tier A]****, whose proof follows exactly the induction below [12].*

Human proof. The composition $f \circ h$ is well defined (since $h > 0$) and of class $\mathcal{C}^\infty$. We prove by strong induction on n the property

$$\mathcal{P}(n): \quad \text{for every CM } f \text{ and every Bernstein } h, \quad (-1)^n (f \circ h)^{(n)} \geq 0 \text{ on } (0,\infty).$$

It is essential to quantify over *all* functions f and h: the induction hypothesis will be applied to a function other than f.

Base case. $\mathcal{P}(0)$: $f \circ h \geq 0$ since $f \geq 0$.

Inductive step. Assume $\mathcal{P}(k)$ for every $k \leq n$. Let f be CM and h Bernstein. We have $(f \circ h)' = (f' \circ h)\, h'$, hence

$$-(f \circ h)' = G \cdot h', \qquad G = (-f') \circ h.$$

Now $-f'$ is CM (it is f differentiated once, with the sign: $(-1)^m (-f')^{(m)} = (-1)^{m+1} f^{(m+1)} \geq 0$). By $\mathcal{P}(k)$ applied to the CM function $-f'$, $(-1)^k G^{(k)} \geq 0$ for every $k \leq n$. On the other hand h' is CM by definition. Differentiating n times and using the Leibniz rule,

$$(-1)^{n+1} (f \circ h)^{(n+1)} = (-1)^n \big(G\, h'\big)^{(n)} = \sum_{k=0}^{n} \binom{n}{k} \left[(-1)^k G^{(k)}\right] \left[(-1)^{n-k} (h')^{(n-k)}\right] \geq 0.$$

Hence $\mathcal{P}(n+1)$. □

Corollary 1.20 (The square-root case)**.** *If f is CM, then $t \mapsto f(\sqrt{t})$ is CM. In particular, for $\kappa \geq 0$, $t \mapsto e^{-\kappa\sqrt{t}}$ is CM, and so is the planar Yukawa potential $e^{-\kappa r}/r$ written in the variable $t = r^2$: $g_\kappa(t) = e^{-\kappa\sqrt{t}}\, t^{-1/2}$.* ***[Tier C]*** *(Lean:* `ProgCM.IsCompletelyMonotoneOn.comp_sqrt`*,* `ProgCM.phi_cm` ***[Tier A]****.)*

Human proof. Theorem 1.19 with $h = \sqrt{\cdot}$ (Example 1.18) gives the first sentence. With $f(u) = e^{-\kappa u}$ we obtain that $e^{-\kappa\sqrt{t}}$ is CM. The Yukawa potential is the *product* of this function with $t^{-1/2}$, which is CM by Example 1.14; Proposition 1.16 concludes. □

Remark 1.21 (Direct check on one case). For $\psi(t) = e^{-\sqrt{t}}$ we have $\psi'(t) = -\frac{1}{2\sqrt{t}} e^{-\sqrt{t}}$ and $\psi''(t) = \left(\frac{1}{4t} + \frac{1}{4t^{3/2}}\right) e^{-\sqrt{t}}$. The signs $-\psi' \geq 0$ and $\psi'' \geq 0$ are as announced. A machine checks the signs up to order 8 at seven points (script `code/ch1_theta.py`): $e^{-\sqrt{t}}$ passes, $\cos t$ (negative control) fails at every order **[Tier X]**. This is not a proof: the proof is the induction above.

Physics and biology : why physicists meet CM functions

A potential that decays with distance while flattening out "more and more slowly" (Coulomb, screened Yukawa, dipole) is typically CM. The five physical potentials of Table 1.2 include dipolar excitons $2\big(1/r - 1/\sqrt{r^2 + d^2}\big)$ (doi: 10.5281/zenodo.23646103) and the Yukawa potential (doi: 10.5281/zenodo.23646105). In both cases admissibility is a **[Tier A]** theorem; the physical consequence (the triangular lattice has minimal energy) remains conditional on `openai/math`.

1.2.3 Bernstein's theorem and Gaussian mixtures

Theorem 1.22 (Bernstein–Widder)**.** *A function $g : (0, \infty) \to \mathbb{R}$ is completely monotone if and only if there exists a positive measure μ on $[0, \infty)$ such that*

$$g(t) = \int_{[0,\infty)} e^{-\alpha t}\, d\mu(\alpha) \qquad (t > 0).$$

The measure μ is then unique. ***[Tier L]*** *(classical statement, cited second-hand) [6, 7]*

The geometric reading is essential: with $t = |x|^2$, we have $g(|x|^2) = \int e^{-\alpha|x|^2} d\mu(\alpha)$, that is, *every completely monotone potential is a positive mixture of Gaussians.* The "converse" direction is easy and we prove it; the forward direction (CM ⇒ mixture) is a deep result, which we cite, and which is *not* formalised in the work described here.

Proposition 1.23 (Mixtures are CM (easy direction))**.** *Let μ be a positive measure on $[0, \infty)$ such that $\int e^{-\alpha t} d\mu(\alpha) < \infty$ for every $t > 0$. Then $g(t) = \int e^{-\alpha t} d\mu(\alpha)$ is CM.* ***[Tier C]*** *(Lean:* `ProgCM.laplace_measure` ***[Tier A]****, a statement of this same direction.)*

Human proof. Fix $t_0 > 0$. For $t \geq t_0$ and $n \in \mathbb{N}$, we have $0 \leq \alpha^n e^{-\alpha t} \leq C_n\, e^{-\alpha t_0/2}$ with $C_n = \sup_{\alpha \geq 0} \alpha^n e^{-\alpha t_0/2} < \infty$, and $\alpha \mapsto e^{-\alpha t_0/2}$ is μ-integrable by hypothesis. The theorem on differentiation under the integral sign (uniform domination on $[t_0, \infty)$, a consequence of Lebesgue's dominated convergence theorem) shows, by induction on n, that g is of class $\mathcal{C}^\infty$ and $(-1)^n g^{(n)}(t) = \int \alpha^n e^{-\alpha t} d\mu(\alpha) \geq 0$. □

Example 1.24 (A CM potential that the closure rules do not see)**.** For $c > 0$, $\varphi(t) = t^{-1/2} - (t + c)^{-1/2}$ is a *difference* of two CM functions, and nothing in Proposition 1.16 allows us to conclude. But Euler's formula $u^{-1/2} = \frac{1}{\sqrt{\pi}} \int_0^\infty \alpha^{-1/2} e^{-\alpha u} d\alpha$ $(u > 0)$ gives

$$\varphi(t) = \frac{1}{\sqrt{\pi}} \int_0^\infty \alpha^{-1/2} \big(1 - e^{-c\alpha}\big)\, e^{-\alpha t}\, d\alpha,$$

a mixture with *positive* weight ($1 - e^{-c\alpha} > 0$): φ is CM by Proposition 1.23. This is the dipolar exciton potential, up to the factor 2. **[Tier C]**

From the energy of Gaussians to that of mixtures. Let $g(t) = \int e^{-\pi\alpha t} d\nu(\alpha)$ be a mixture, with ν positive. For a finite configuration C_R, by positivity (Tonelli's theorem) the double sum can be moved under the integral:

$$E_g(C; R) = \int E_{e^{-\pi\alpha\cdot}}(C; R)\, d\nu(\alpha).$$

Fatou's lemma (valid in $[0, \infty]$) then gives $E_g(C) \geq \int \liminf_R E_{e^{-\pi\alpha\cdot}}(C; R)\, d\nu(\alpha) = \int E_{e^{-\pi\alpha\cdot}}(C)\, d\nu(\alpha)$. If we know that, for each α, no configuration has a Gaussian energy lower than that of A, we deduce $E_g(C) \geq \int (\Theta_A(\alpha) - 1)\, d\nu(\alpha) = E_g(A)$, by Tonelli on the lattice sum. **[Tier C]** *The Gaussian case is therefore the heart of the problem*; it is this case, for every value of α, that the theorem of the next part asserts. The same idea, applied to *lattices* with Montgomery's Theorem 1.12, gives: $E_g(A) \leq E_g(\Lambda)$ for every lattice Λ of covolume 1 and every CM g (a mixture ν of Gaussians), since $E_g(\Lambda) = \int (\Theta_\Lambda(\alpha) - 1)\, d\nu(\alpha)$. **[Tier C]** (conditional on 1.12 and 1.22, **[Tier L]**)

1.3 Universal optimality

1.3.1 Statement

Cohn and Kumar [3] defined a notion of *universal optimality*: a configuration is universally optimal if it minimises the energy for *every* completely monotone potential of the squared distance, simultaneously. They conjectured that this happens for many remarkable configurations, including the triangular lattice in the plane. Cohn, Kumar, Miller, Radchenko and Viazovska [2] proved the conjecture for the E_8 lattice in dimension 8 and the Leech lattice in dimension 24 **[Tier L]**, building on Viazovska's work [4]. Dimension 2 remained open in the literature we have read.

The `openai/math` repository contains a statement and a proof in Lean 4 of the planar case [1]. Here is the statement as it appears in the challenge file, first in English and then in Lean.

Theorem 1.25 (Universal optimality of the triangular lattice (`openai/math`))**.** *Let g be admissible (Definition 1.13) and $C \subset \mathbb{R}^2$ a locally finite configuration of centred density* 1*. Then*

$$\mathcal{E}_g(A) \;\leq\; E_g(C) \qquad \text{and} \qquad \mathcal{E}_g(A) = E_g(A), \qquad \text{where} \quad \mathcal{E}_g(A) = \sum_{x \in A \setminus 0} g(|x|^2),$$

with values in $[0, \infty]$, and $E_g(\cdot)$ the lower energy over closed centred disks (Lean: `latticeEnergy g` *for $\mathcal{E}_g(A)$ and* `energy g` *for E_g). **[Tier A]** for the Lean statement* `OAI.AtomicTriangular.universal_energy_minimum`*, **conditional**: the proof is AI-generated and has not been peer reviewed.*

Lean 4 : the main statement (extract from the challenge file, where the proof is `sorry` by construction)

```
theorem universal_energy_minimum (g : ℝ → ℝ) (C : Set Plane)
    (hg : AdmissiblePotential g) (hC : LocallyFinite C) (hd : DensityOne C) :
    latticeEnergy g ≤ energy g C ∧ latticeEnergy g = energy g A
```

Three reading remarks, essential for what follows.

1. The energy is a *limit inferior* over *closed disks centred at* 0. The CKMRV definition [2] additionally requires the limit to exist for the optimal configuration. The notion used here is therefore weaker **[Tier C]**.

2. The theorem does not say that A is the only minimiser, and uniqueness is false in general: corollary D3 of textbook v1 [11] establishes non-uniqueness by removing one point of A (a "vacancy"), which changes neither the density nor the lower energy. **[Tier A]** (conditional)

3. For $s \leq 2$ (Riesz), the energy of A is infinite (Proposition 1.8): the inequality is then true but void. Content appears for $s > 2$.

1.3.2 The thread of the formalisations

The rest of the book follows a natural sequence: each step takes the previous result as a black box. Table 1.2 summarises it. The DOI numbers and tiers are those of the record documents.

#	Step	Tier	Reference
1	Upstream theorem (`openai/math`), planar case	**[Tier L]** and conditional **[Tier A]**	[1]
2	Corollaries D1 to D4: Riesz, every density, CKMRV sense and non-uniqueness, open or closed disks	**[Tier A]**, conditional on 1	[11] (textbook v1)
3	Five physical potentials (dipolar excitons, Yukawa, Rytova–Keldysh, $\sigma = 3$ enstrophy, Bogoliubov structure factor)	**[Tier A]** for the replayed modules (dipolar, Yukawa, Keldysh), **[Tier A without replay]** for $\sigma = 3$ and Bogoliubov; Keldysh↔Struve: **[Tier C]**	5 Zenodo deposits
4	CM/Bernstein library (P3): subordination, equivalence with `AdmissiblePotential`	**[Tier A]**	[12]
5	Quantum surprise: the zero-point energy, QU-1 false for 8 CM potentials out of 12; certificate at $a = 1$; no triangular→oblique transition	**[Tier X]**, then **[Tier A without replay]** (logic) and **[Tier C]** (ball-arithmetic integrals); negative result	Zenodo
6	Sandier–Serfaty (Abrikosov lattices): audit, partial port to Lean, failure of the extension to the 3D case	**[Tier C]**, partial **[Tier A without replay]**; negative results	Zenodo
7	Navier–Stokes (corpus `openai/NavierStokesAndEuler`): rebuild and kernel replay	**[Tier A]** for the kernel replay of the corpus statements; faithfulness to Fefferman's alternative (C): **[Tier C]**; no "Clay problem solved"	doi: 10.5281/ zenodo.23661381
8	Population genetics: Moran fixation, Price equation, isothermal theorem	**[Tier A]**	Zenodo

Table 1.2: The sequence of formalisations, from the foundations (this chapter) to the applications.

Warning : what the table does not claim

No row means "conjecture proved" beyond its tier. Rows 5 and 6 contain *negative* results: they say that certain plausible extensions fail (or are not established), and this is information of the same value as the positive results. Step 7 is an audit of a third-party corpus: the credit belongs to its authors. We do not claim that a Clay problem is solved.

1.4 Lean 4 for the student

1.4.1 Propositions and proofs

Lean 4 [8] is both a programming language and a *proof assistant*. For an undergraduate reader, the following dictionary is enough to read the statements of this book (this is the Curry–Howard correspondence).

- A *proposition* is a mathematical statement; in Lean, it is an object of type `Prop`.
- A *proof* of the proposition P is an object `h : P`. The Lean kernel checks that a term really has the announced type.
- $P \wedge Q$ is written `P âĹğ Q`: a proof is a *pair* $\langle$`hP, hQ`$\rangle$.
- $P \Rightarrow Q$ is written `P âĘŠ Q`: a proof is a *function* that turns every proof of P into a proof of Q.
- $\forall n,\ P(n)$ is written `âĹĂ n, P n`: a proof is a function that assigns to each n a proof of $P(n)$.

The Mathlib library [9] provides many already proved theorems (analysis, algebra).

1.4.2 A minimal example, with its actual output

The following file was compiled as is with Lean 4.34.0-rc2, without any library (output reproduced identically by the independent auditor). It contains three true theorems and two false "theorems" $1 = 2$ that can be proved only by cheating: one with `sorry`, the other with an axiom added by the user.

Lean 4 : `T.lean` (compiled on 2026-10-26)

```
theorem and_swap' (p q : Prop) (h : p ∧ q) : q ∧ p := ⟨h.2, h.1⟩

theorem add_comm' (a b : Nat) : a + b = b + a := Nat.add_comm a b

theorem two_mul' (n : Nat) : 2 * n = n + n := by
  omega

theorem bad : 1 = 2 := by sorry

axiom cheat : False
theorem bad2 : 1 = 2 := cheat.elim

#print axioms and_swap'
#print axioms two_mul'
#print axioms bad
#print axioms bad2
```

The first two theorems are *proof terms*: the object is written directly. The third uses a *tactic* (`by omega`): a small program that *constructs* the proof term; the kernel then checks the term produced. Here is the actual output:

```
T.lean:8:8: warning: declaration uses `sorry`
'and_swap'' does not depend on any axioms
'two_mul'' depends on axioms: [propext, Quot.sound]
'bad' depends on axioms: [sorryAx]
'bad2' depends on axioms: [cheat]
```

The compiler's exit code was 0: **Lean accepts a file that contains `sorry`**, emitting only a warning. *A zero exit code does not mean "everything is proved".*

1.4.3 Axioms, and the gate

Every proof in mathematics rests on axioms. The classical mathematics of Mathlib uses three: `propext` (two equivalent propositions are equal), `Classical.choice` (the axiom of choice) and `Quot.sound` (compatibility of quotients). We call them the *whitelist*:

$$\{\texttt{propext},\ \texttt{Classical.choice},\ \texttt{Quot.sound}\}.$$

Any other axiom in the footprint of a theorem disqualifies it. In the output above, the footprint of `bad` contains `sorryAx` (the internal axiom hidden behind `sorry`) and that of `bad2` contains the user axiom `cheat`: both false theorems are therefore *detected*. This is the *negative control* of the axiom gate; the *positive control* is that `and_swap'` and `two_mul'` pass. A measurement without these two controls proves nothing.

Warning : `#print axioms` is not enough

The A2 audit [13] of the `openai/NavierStokesAndEuler` corpus (2 440 modules rebuilt, kernel replay of 53 752 + 34 648 declarations) showed that a declaration *added directly to the kernel with its type checking disabled* (option `debug.skipKernelTC`) is displayed by `#print axioms` as "does not depend on any axioms", even though it can establish `False`. The audit's positive control (such a false declaration, deliberately fabricated) was *invisible* to `#print axioms` and *caught* by the full kernel replay (`leanchecker`) (a **[Tier X]**-tier observation in the A2 paper: it concerns a deliberately invalid declaration). The lesson: the axiom audit is necessary, not sufficient.

The **[Tier A]** gate of this book therefore consists of *four* controls (compilation, then the three checks of the preface):

1. compilation of the project without errors;
2. the axiom list of each retained theorem is contained in the whitelist;
3. full *kernel replay* of the declarations by the tool `leanchecker`, which rechecks every term without trusting the compiler;
4. a textual search for the words `skipKernelTC`, `addDecl`, `native_decide` and `implemented_by` in the sources, which are the known backdoors.

To this is added an organisational rule: whoever *produces* a proof is not the one who *verifies* it (here, two separate AI agents, and never a human).

When the `leanchecker` replay has not been done, we write **[Tier A without replay]**: axiom list, planted witnesses and independent recompilation, without kernel replay.

1.4.4 Reading the upstream statement; a short corollary

Once the main theorem is available, a corollary often reduces to checking a hypothesis. Here is the Riesz case, quoted from the file `formal_cert/Triangular/Riesz.lean` of the repository (the ellipsis marks an omitted part):

Lean 4 : Riesz corollary (extract, layout modified)

```
noncomputable def riesz (s : ℝ) : ℝ → ℝ := fun t => t ^ (-(s / 2))

theorem riesz_admissible {s : ℝ} (hs : 0 < s) : AdmissiblePotential (riesz s) := by
  ...

theorem triangular_riesz_optimal {s : ℝ} (hs : 0 < s) (C : Set Plane)
    (hC : LocallyFinite C) (hd : DensityOne C) :
    latticeEnergy (riesz s) ≤ energy (riesz s) C :=
  (universal_energy_minimum (riesz s) C (riesz_admissible hs) hC hd).1
```

The last line "applies" the main theorem to $g = \text{riesz } s$ by supplying the three proofs of the hypotheses, and then takes the first component `.1` of the conjunction. This is exactly a *composition of functions* in the sense of the Curry–Howard dictionary. The hypothesis `riesz_admissible` is the complete monotonicity property of Example 1.14 (2). The "every density" corollary (D2) is re-exported in Chapter 5 under the name `ALIX`.

1.4.5 The induction of §1.2, on the Lean side

The subordination Theorem 1.19 has been formalised [12]. The general statement and its square-root special case are the following (extracts from the file `Bernstein.lean` in the folder `formal_cert/Programmes/CompletelyMonotone/`, proofs omitted):

Lean 4 : subordination (statements)

```
theorem IsCompletelyMonotoneOn.comp_bernstein {f g : ℝ → ℝ} (hf :
    IsCompletelyMonotoneOn f)
    (hg : IsBernsteinOn g) (hpos : ∀ t, 0 < t → 0 < g t) :
    IsCompletelyMonotoneOn (fun t => f (g t))

theorem IsCompletelyMonotoneOn.comp_sqrt {f : ℝ → ℝ} (hf : IsCompletelyMonotoneOn f) :
    IsCompletelyMonotoneOn (fun t => f (Real.sqrt t))
```

One recognises the structure of the human theorem. The hypotheses `hf`, `hg` and `hpos` are the three hypotheses of the theorem (f CM, h Bernstein, $h > 0$), `fun t => f (g t)` is the function $f \circ h$, and `comp_sqrt` is obtained from the general case by supplying the proof that $\sqrt{\cdot}$ is a Bernstein function and strictly positive on $(0, \infty)$. The repository states that the proof proceeds by strong induction on the order, quantified over all CM functions f, with the formula $(f \circ h)' = -((-f') \circ h) \cdot h'$ and the Leibniz lemma: this is the same reasoning as the human proof. **[Tier A]** for the statements above, **[Tier C]** for our human write-up, whose correspondence with the Lean proof is a comment by the author, not a theorem.

Remark 1.26 (Why `hpos` appears in Lean). In paper mathematics, one tacitly assumes that $f \circ h$ makes sense. Lean does not: f is a function defined everywhere on $\mathbb{R}$, but constrained only on $(0, \infty)$. If $h \equiv 0$ (which is a Bernstein function), $f \circ h$ would be the spurious constant $f(0)$, about which the hypothesis says nothing. The hypothesis $h > 0$ is not nitpicking: it is what makes the statement *true*.

Warning : summary of the tiers of this chapter

- **[Tier L]**: Poisson summation on the torus (Fourier's theorem), Montgomery's theorem and the Bernstein–Widder theorem (forward direction), both cited second-hand, the CKMRV result in dimensions 8 and 24.
- **[Tier C]**: all human proofs of this chapter (counting, Riesz energy, Poisson summation, closure of CM functions, subordination, mixtures, Fatou reduction).
- **[Tier A]** (with kernel replay for the Triangular modules, the CM/Bernstein library and P7; see the following chapters): the Lean statements cited (equivalence with `AdmissiblePotential`, subordination, `laplace_measure`, `quadrupole_not_cm`, the upstream statement), the last one being *conditional*.
- **[Tier X]**: the script `code/ch1_theta.py`, which illustrates without proving.

1.4.6 The Elenchus tiers

The Elenchus method [10] attaches to each statement of the book one of four tiers, according to the *nature of the available proof*.

Tier	What is guaranteed	What is not
[Tier A]	The Lean statement is accepted by the kernel, with the four controls of the gate above	That the Lean statement faithfully expresses the intended mathematical or physical statement
[Tier L]	The result appears in a source that was read and precisely referenced (for example a theorem of a published paper)	That we have rechecked it ourselves
[Tier C]	A complete human argument, written in this book, reviewed by an independent AI agent	Any machine verification, and any human review
[Tier X]	A floating-point computation, with positive and negative controls	Nothing: a numerical exploration never supports a claim

Two composition rules, used throughout what follows.

1. *The tier of a consequence is the weakest of the tiers of its hypotheses.* A corollary of `openai/math` is *conditional* **[Tier A]**: the kernel has checked the implication, not the upstream hypothesis, whose proof has not been peer reviewed.

2. *A negative result is a result.* When a plausible extension fails, we write it down with its tier.

Warning : adequacy of the statement

The kernel certifies that the statement `universal_energy_minimum`, *with the repository's definitions*, follows from the whitelisted axioms. It does not certify that `DensityOne` is the density the physicist has in mind, nor that `energy` (a limit inferior over closed centred disks) is the CKMRV energy. No human has audited this adequacy. This is why every chapter repeats the tier, and it is the reason for the word "conditional".

Exercises

Hints follow the list of exercises. The exercises are of increasing difficulty (from * to ***). The two final "(†) University exercises" are harder than all of these; their "(†)" is a label, not a difficulty rating, and they have no hints.

Exercise 1.1 (*)**.** Show that the change-of-basis matrix between two bases of the same lattice lies in $\mathrm{GL}_2(\mathbb{Z})$ and deduce that the covolume does not depend on the basis. Compute the covolume and the minimal distance of the lattice $\mathbb{Z}u \oplus \mathbb{Z}v$ with $u = (1, 0)$, $v = (1/2, \sqrt{3}/2)$, then normalise it to density 1. Recover $a^2 = 2/\sqrt{3}$.

Exercise 1.2 (*)**.** In A, determine all vectors of norm $a\sqrt{3}$ and their number (hint: $m^2 + mn + n^2 = 3$). Compare with the case of $\mathbb{Z}^2$ (shell $|x|^2 = 2$). Justify that $a > 1$: at equal covolume, the minimal distance of A exceeds that of $\mathbb{Z}^2$.

Exercise 1.3 (**)**.** Write out the proof of Proposition 1.2 for $\Lambda = \mathbb{Z}^2$ and deduce that $\#\{(m, n) \in \mathbb{Z}^2 : m^2 + n^2 \le R^2\} = \pi R^2 + O(R)$. Is this error $O(R)$ optimal? (Look up the literature on the *Gauss circle problem*.)

Exercise 1.4 (**)**.** Prove that $\sum_{x \in A\setminus 0} |x|^{-s}$ diverges for $s \le 2$ and converges for $s > 2$. Using $t^{-s/2} = \frac{\pi^{s/2}}{\Gamma(s/2)} \int_0^\infty \alpha^{s/2-1} e^{-\pi\alpha t} d\alpha$, express this sum as an integral of $\Theta_A(\alpha) - 1$ against $\alpha^{s/2-1} d\alpha$ and recover the condition $s > 2$ from the behaviour of Θ_A at 0 and at $+\infty$.

Exercise 1.5 (**)**.** (a) Prove the relation $\vartheta(1/\alpha) = \sqrt{\alpha}\,\vartheta(\alpha)$ of Example 1.11 by redoing the one-dimensional proof. (b) Deduce an algorithm that computes $\vartheta(\alpha)$ to within 10^{-12} by summing only over $|n| \le 6$, whatever $\alpha > 0$. (c) Compute $\Theta_{\mathbb{Z}^2}(1) = \vartheta(1)^2$ and compare with the value $1.1803\ldots$ in the text.

Exercise 1.6 (*)**.** Show that if f is CM and $a > 0$, $b \ge 0$, then $t \mapsto f(at + b)$ is CM. Deduce that $t \mapsto (t + 1)^{-s} e^{-at}$ is CM for $s, a \ge 0$. Show by a computation that $t \mapsto t^{-1} e^{-t}$ is CM as well, without using the composition theorem.

Exercise 1.7 (**)**.** (a) Check that $h(t) = \ln(1 + t)$ is a Bernstein function. (b) Deduce that $t \mapsto (\ln(1 + t))^{-s}$ is CM for $s \ge 0$ (pay attention to the positivity of h on $(0, \infty)$). (c) Show that $t \mapsto 1/(1 + \sqrt{t})$ is CM by applying Corollary 1.20. (d) Why can the hypothesis $h > 0$ not be omitted in Theorem 1.19? (Consider $h \equiv 0$.)

Exercise 1.8 (***)**.** Let $g(t) = t^{-1/2} - 2(t + 1)^{-1/2} + (t + 4)^{-1/2}$. Noting that g is a mixture of Gaussians with a *signed weight*, explain why positivity of the weight fails, and recover in this way that g is not CM (compare with $g(3) < 0$).

Exercise 1.9 (*)**.** Among the following four outputs of `#print axioms`, which satisfy the axiom gate? Which satisfy the *full* four-control gate? (i) `[propext, Classical.choice, Quot.sound]`; (ii) `[propext, sorryAx]`; (iii) `does not depend on any axioms`; (iv) `[cheat, propext]`. Explain why (iii) alone is not enough to conclude.

Exercise 1.10 (**)**.** Let Λ be a lattice and $g \ge 0$ an arbitrary potential. Show that the disk energy $E_g(\Lambda; R)$ defined above converges, as $R \to \infty$, to $\sum_{a \in \Lambda\setminus 0} g(|a|^2)$ (finite or not). Thus, for a lattice, the lower energy and the lattice energy coincide.

Exercise 1.11 ((†) University exercise: Poisson summation in $\mathbb{R}^d$)**.** Let $\Lambda = M\mathbb{Z}^d \subset \mathbb{R}^d$ be a lattice, $M \in \mathrm{GL}_d(\mathbb{R})$, with dual $\Lambda^* = M^{-T}\mathbb{Z}^d$ and $\Theta_\Lambda(\alpha) = \sum_{x \in \Lambda} e^{-\pi\alpha|x|^2}$. (a) Adapting the three steps of the proof of Theorem 1.10, prove that for every $\alpha > 0$

$$\Theta_\Lambda(\alpha) = \frac{1}{\mathrm{covol}(\Lambda)}\, \alpha^{-d/2}\, \Theta_{\Lambda^*}\Big(\frac{1}{\alpha}\Big), \qquad \mathrm{covol}(\Lambda) = |\det M|.$$

(b) Deduce that $\Theta_\Lambda(\alpha) \sim \mathrm{covol}(\Lambda)^{-1}\alpha^{-d/2}$ as $\alpha \to 0^+$, and recover the case $d = 1$ of Example 1.11. (c) Show that if Λ^* is the image of Λ under an orthogonal map (so that $\mathrm{covol}(\Lambda) = 1$), then $\alpha \mapsto \alpha^{d/4}\Theta_\Lambda(\alpha)$ is invariant under $\alpha \mapsto 1/\alpha$.

Exercise 1.12 ((†) University exercise: Exponentials of Bernstein functions)**.** Let h be a Bernstein function with $h > 0$ on $(0, \infty)$. (a) Show that $t \mapsto e^{-u\,h(t)}$ is CM for every $u \geq 0$, and that $t \mapsto 1/h(t)$ is CM. (b) Show that if h_1, h_2 are Bernstein functions with $h_2 > 0$, then $h_1 \circ h_2$ is a Bernstein function. (Hint: $(h_1 \circ h_2)' = (h_1' \circ h_2)\, h_2'$; use Theorem 1.19 and Proposition 1.16.) (c) Deduce that $t \mapsto \left(\ln(1 + \sqrt{t})\right)^{-1}$ and $t \mapsto e^{-\sqrt{t+t^{1/2}}}$ are CM.

Hints

1.1. If M, M' are two basis matrices, $M' = MP$ with P integral and $M = M'P'$ with P' integral: $PP' = I$, so $\det P \det P' = 1$ with two integers, hence $\det P = \pm 1$. For the last part: $\det(u, v) = \sqrt{3}/2$; rescale by the factor a with $a^2\sqrt{3}/2 = 1$.

1.2. $m^2 + mn + n^2 = 3$: $(1, 1)$, $(-1, -1)$, $(2, -1)$, $(-2, 1)$, $(1, -2)$, $(-1, 2)$, i.e. six vectors. Since $a^2 = 2/\sqrt{3} > 1$ ($4 > 3$), $a > 1$; the minimal distance of $\mathbb{Z}^2$ is 1.

1.3. Take $\delta = \sqrt{2}$ for $\mathbb{Z}^2$: $\pi(R - \sqrt{2})^2 \leq N(R) \leq \pi(R + \sqrt{2})^2$, hence $N(R) = \pi R^2 + O(R)$; optimality of the exponent 1 is not asked for (the best known exponent is below 1, see the literature).

1.4. For divergence, reuse the lower bound on $\#S_k$. For the integral: $\Theta_A(\alpha) - 1 \sim 6e^{-\pi a^2 \alpha}$ as $\alpha \to \infty$ (convergence) and $\Theta_A(\alpha) \sim \alpha^{-1}$ as $\alpha \to 0$ (by Poisson); the integral $\int_0 \alpha^{s/2-1}\alpha^{-1} d\alpha$ converges iff $s/2 - 2 > -1$, i.e. $s > 2$.

1.5. (b) For $\alpha \geq 1$ sum directly; for $\alpha < 1$ apply $\vartheta(\alpha) = \alpha^{-1/2}\vartheta(1/\alpha)$. (c) $\vartheta(1) = 1.0864348\ldots$, hence $\Theta_{\mathbb{Z}^2}(1) = 1.18034\ldots$

1.6. $f(at + b)$: differentiate n times and factor out a^n. For the last one: product of two CM functions (Proposition 1.16); or a direct computation via Leibniz.

1.7. (b) Set $f(u) = u^{-s}$, which is CM. (d) For $h \equiv 0$, $f \circ h$ is not defined on $(0, \infty)$ (or takes a spurious value $f(0)$ if one extends).

1.8. $t^{-1/2} = \pi^{-1/2} \int \alpha^{-1/2} e^{-\alpha t} d\alpha$; the weight of g is $\pi^{-1/2}\alpha^{-1/2}(1 - 2e^{-\alpha} + e^{-4\alpha})$, which is negative for some α (study $1 - 2x + x^4$ with $x = e^{-\alpha}$). Injectivity of the Laplace transform (including for signed measures) shows that g has no representation with a positive weight; more directly, $g(3) < 0$ concludes.

1.9. (i) passes. (ii) fails (`sorryAx`). (iii) passes the axiom gate alone, but the declaration might have been inserted by bypassing the kernel's type checker. (iv) fails.

1.10. The term of $x \in \Lambda_R$ is $\sum_{a \in (\Lambda_R - x)\setminus 0} g(|a|^2)$: it is bounded above by $\sum_{a \neq 0} g(|a|^2)$, whence the upper bound on $E_g(\Lambda; R)$. For the lower bound, fix $r > 0$: if $|x| \leq R - r$, the term of x contains all $a \neq 0$ with $|a| \leq r$. Hence $E_g(\Lambda; R) \geq \frac{N_\Lambda(R-r)}{N_\Lambda(R)} \sum_{0<|a|\leq r} g(|a|^2)$; the ratio tends to 1 (Proposition 1.2). Let R and then r tend to infinity.

Chapter 2

The flow of formalisations: from universal optimality to physical potentials

In Chapter 1, a theorem from elsewhere — the triangular lattice minimises the energy of every potential that is a completely monotone function of the squared distance — was stated, read and checked by the Lean kernel. This chapter tells what happened next: how, following a method fixed in advance, that theorem was applied to three interactions from materials physics; how a small library turned these applications into a few lines each; and how the same method, turned towards a neighbouring and much harder problem (the Sandier–Serfaty conjecture), served above all to state precisely what is not *proved.*

Warning : read before the whole chapter

- Everything that rests on openai/math is *conditional* on the upstream theorems, which are AI-generated and *not peer reviewed.*
- Every result in this chapter was produced by AI agents (Claude) under the author's direction, then audited by a *separate* AI agent. **No human has audited these results**, neither the faithfulness of the Lean statements nor the "human" proofs written here.
- A **[Tier A]** concerns the Lean statement, never its physical reading.
- The Sandier–Serfaty conjecture is **not proved** in this book (Section 2.5).

2.1 Overview: a flow, not a catalogue

The starting point (Chapter 1) is the upstream theorem `universal_energy_minimum` of openai/-math: for every *admissible* potential g, the triangular lattice A of covolume one has energy less than or equal to that of every locally finite configuration of density one **[Tier A]**, conditionally on the upstream [1]. In the human literature, universal optimality is proved in dimensions 8 and 24 [2] **[Tier L]**, and Petrache and Serfaty wrote in 2019 that the planar case remained open [15] **[Tier L]**: the upstream theorem is therefore an AI claim on an open problem, accepted here only as a Lean statement checked by the kernel from the upstream definitions. We recall the two statements, quoted verbatim.

Lean 4 — openai/math, `OAI/Analysis/Triangular/Energy/Basic.lean:26-29`

```
def AdmissiblePotential (g : ℝ → ℝ) : Prop :=
  ContDiffOn ℝ (⊤ : ℕ∞) g (Set.Ioi 0) ∧
  (∀ t : ℝ, 0 < t → 0 ≤ g t) ∧
  ∀ (r : ℕ) (t : ℝ), 0 < t → 0 ≤ (-1 : ℝ) ^ r * iteratedDeriv r g t
```

Lean 4 — openai/math, `OAI/Analysis/Triangular/Energy/Universal.lean:9-11` (proof omitted)

```
theorem universal_energy_minimum (g : ℝ → ℝ) (C : Set Plane)
    (hg : AdmissiblePotential g) (_hC : LocallyFinite C) (hd : DensityOne C) :
    latticeEnergy g ≤ energy g C ∧ latticeEnergy g = energy g A := by
```

(These two excerpts are quoted from the pinned copy of openai/math; the working copy `results/openai_math/hypotheses/night_2026-10-08/TRI_D3/src/` in the repository is identical to it for these two files.) The variable of g is the *square* $t = r^2$ of the distance; "admissible" means: C^∞ on $(0, +\infty)$, nonnegative, and $(-1)^r g^{(r)}(t) \geq 0$ for every r, that is, *completely monotone* (CM) on $(0, +\infty)$. The energies take values in $[0, +\infty]$.

The flow of this chapter is as follows (Figure 2.1).

1. **Upstream** (Chapter 1): universal optimality in dimension two, **[Tier A]** conditional, and our corollaries D1–D4 (Riesz, every density, CKMRV sense, open or closed disks), published in textbook v1 (doi:10.5281/zenodo.23295142) [11].
2. **Three physical potentials** (Section 2.3): dipolar excitons, planar Yukawa, Rytova–Keldysh. For each: a proof of complete monotonicity in r^2, the Lean statement, the consequence.
3. **The generalisation** (Section 2.4): a CM/Bernstein library whose subordination theorem turns these three results into corollaries of two or three lines.
4. **A neighbouring problem** (Section 2.5): the Sandier–Serfaty conjecture, where the same method produces a reproduction, an audit, a Lean statement, a *partial* formalisation and a negative result in dimension three — but no proof.

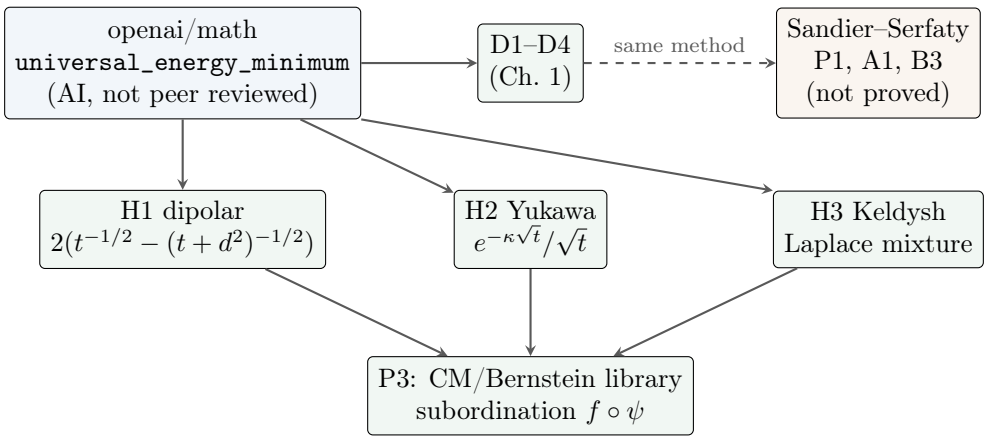


Figure 2.1: The flow of formalisations in this chapter. Green: **[Tier A]** Lean statements (conditional on the upstream); blue: cited upstream; orange: work whose mathematical status remains **[Tier C]** or lower.

2.2 The method: preregister, falsify, let the kernel judge

Before the first computation, one must say how one will know that one was wrong. The method followed in all the work of this chapter comes down to four rules. It is tooled by Elenchus [10], a claims register ("ledger") equipped with an automatic gate.

2.2.1 Rule 1: preregister

One writes down, *before* seeing the results, what one is looking for, what one will report, and what would make the project fail. Two real examples.

- For the dipolar excitons, the preregistration `DIP-TRI-1` was committed (commit `4feb027`) *before* the solver code (commit `17ea6be`) in rusty-SUNDIALS [16]. It fixed the quantity to be reported (a truncated image sum plus the uniform tail). The paper reports exact Ewald energies instead, and says so explicitly as a *deviation*: the truncated sum left a non-monotone residual.
- For the formalisation of the Sandier–Serfaty certificate (A1), the prior probabilities were written in `preregistration.json`: 0.6 that a few pilot stages succeed, 0.2 that all of them succeed, 0.35 for one complete lemma of the reduction, 0.02 for the whole; and a *stopping criterion*: no announcement of a formal proof as long as all stages and the whole reduction are not checked by the kernel [20]. The criterion applies (Section 2.5.3).

2.2.2 Rule 2: a falsification gate, with positive and negative controls

Every experiment, every theorem, every program comes with a *positive control* (a case where the answer must be "yes") and a *negative control* (a case where it must be "no"), run before any number is reported. A gate that accepts everything checks nothing. In this chapter, every admissibility statement has its counterexample checked by the kernel: $t^{-1/2} - 2(t + d^2)^{-1/2}$ for the dipoles, $e^{-\kappa\sqrt{t}} \cos t$ and $g_1 - 2g_2$ for Yukawa, $e^{-t} - 2e^{-2t}$ for the Laplace mixtures, cos and a "junk" composition for the library. For the Sandier–Serfaty certificate, three deliberately weakened versions of the program were run; Section 2.5.2 says honestly that one of them *passes*.

2.2.3 Rule 3: the kernel is the judge — but not its exit code

A Lean file that contains `sorry`, or a planted axiom, *compiles with exit code 0*. The exit code is therefore not the gate. Here is the planted-witness file of the CM library, quoted verbatim.

Lean 4 — `formal_cert/Programmes/CompletelyMonotone/Controls.lean:11-18`

```
theorem ProgCM.ctl_sorry : ProgCM.IsCompletelyMonotoneOn Real.cos := sorry

axiom ProgCM.ctl_cheat : False

theorem ProgCM.ctl_axiom : ProgCM.IsCompletelyMonotoneOn Real.cos := ProgCM.ctl_cheat.
    elim

#print axioms ProgCM.ctl_sorry
#print axioms ProgCM.ctl_axiom
```

Both "proofs" of the false statement "cos is CM" compile with exit code 0; only the command `#print axioms` unmasks them (`sorryAx`, resp. `ProgCM.ctl_cheat`) [12]. Hence the definition of **[Tier A]** (fact sheet `FACTS.md`): the list of axioms is contained in {`propext`, `Classical.choice`, `Quot.sound`}, plus a kernel replay (`leanchecker`), plus a textual search for `skipKernelTC`, `addDecl`, `native_decide`, `implemented_by`.

Remark 2.1 (Why `#print axioms` is not enough). The A2 audit of openai's Navier–Stokes corpus showed that a declaration forged with the option `debug.skipKernelTC` can display an impeccable axiom list. A full kernel replay closes this backdoor. In the interest of honesty: the source papers of these sections did not report a `leanchecker` replay. This replay was carried out for this book on 2026-10-26 (record `results/kernel_replay_2026-10-26/`), on the modules cited here: exit code 0 for each of the dipolar modules (`DipolarExciton`, `DipolarExcitonMultipolar`, `DipolarExcitonStep1`), the Yukawa modules (`Yukawa`, `YukawaMix`), the Keldysh module (`KeldyshCM`) and the CM/Bernstein library (`Basic`, `Bernstein`, `Laplace`, `Corollaries`, `Applications`, `Examples`, `Limits`), and the textual search finds no forbidden token outside the planted-control files. The replay covers the declarations of these modules; Mathlib and the Lean core are loaded, not replayed. The modules of Section 2.5 (Sandier–Serfaty) are in the replay plan but had not been replayed when this book was assembled: their statements carry the label **[Tier A without replay]**.

2.2.4 Rule 4: the producer is not the verifier

The agent that writes a theorem is never the one that declares it verified. A second AI agent recompiles the files *from source*, in its own directory, plants its own controls, reads the statements adversarially, and returns a verdict (for instance `ACCEPT_WITH_CORRECTIONS`). The Elenchus register assigns each claim a tier — A (kernel), L (literature read and pinned), C (human argument not machine-checked), X (floating-point exploration, which never supports a claim) — and the gate checks that no row is ranked above what it rests on, transitively. It raises a flag `LEDGER_UNAUDITED_TIER_A` for every Tier A row for which no human has certified that the statement means what its gloss says. These flags remain open in all the work of this chapter: an AI audit does not close them.

Warning : what a Tier A certifies

The kernel certifies that *a Lean statement* is proved from the Lean definitions. It certifies neither that these definitions faithfully translate the physics, nor that the upstream (openai/math) is correct in the human sense: it certifies that, *if* one accepts the upstream definitions and theorems, the conclusion follows.

2.3 Three physical potentials, three proofs

The three subsections follow the same plan: (i) the physical model **[Tier C]**; (ii) a step-by-step proof that the potential, viewed as a function of $t = r^2$, is completely monotone **[Tier C]**, alongside the verified Lean statement **[Tier A]**; (iii) the consequence: optimality of the triangular lattice (conditional), and finiteness of the energy when it is proved; (iv) a negative control.

Definition 2.2 (complete monotonicity)**.** A function $f\colon (0,+\infty) \to \mathbb{R}$ is *completely monotone* (CM) if it is of class C^∞ and, for every $n \in \mathbb{N}$ and every $t > 0$, $(-1)^n f^{(n)}(t) \geq 0$. In other words: $f \geq 0$, $f' \leq 0$, $f'' \geq 0$, $f''' \leq 0$, and so on.

Example 2.3. $t \mapsto e^{-at}$ $(a \geq 0)$ is CM since $(-1)^n \frac{d^n}{dt^n} e^{-at} = a^n e^{-at} \geq 0$. The function cos is not: $\cos \pi = -1 < 0$. Nor is $\sqrt{t}$: it is increasing.

Remark 2.4 (why the variable $t = r^2$?). The upstream theorem is about $g(\|x-y\|^2)$. Being CM in r^2 is more demanding than being CM in r: e^{-r} is CM in r, and we shall see that $e^{-\sqrt{t}}$ is also CM in t, but this requires an argument (Section 2.3.2).

2.3.1 Dipolar excitons in a bilayer

Physics and biology : the model [Tier C]

In a heterobilayer of transition-metal dichalcogenides (for instance $MoSe_2/WSe_2$), an "indirect" exciton is an electron in one layer, at height d, bound to a hole in the other layer, at height 0, at the same in-plane position. Two excitons at in-plane distance r form four Coulomb pairs: electron–electron and hole–hole at distance r (repulsion $+1/r$ each), and two cross pairs at distance $\sqrt{r^2+d^2}$ (attraction $-1/\sqrt{r^2+d^2}$ each). The internal binding energy, a constant, is neglected. In units of $e^2/(4\pi\varepsilon)$:

$$V(r) = \frac{2}{r} - \frac{2}{\sqrt{r^2+d^2}}, \qquad g_d(t) = 2\big(t^{-1/2} - (t+d^2)^{-1/2}\big), \quad V(r) = g_d(r^2).$$

At large distance $V(r) = d^2/r^3 + O(r^{-5})$: the dipole–dipole law. This pair counting is **[Tier C]**; the algebraic identity $g_d(r^2) = 2(1/r - 1/\sqrt{r^2+d^2})$ is checked by the kernel (`g_inplane`, **[Tier A]**: the module `DipolarExcitonStep1.lean` was verified only by its producer in the source paper [16]; it has since been replayed by `leanchecker`, exit code 0).

Lean 4 — `formal_cert/Autoresearch/DipolarExciton.lean:22`

```
noncomputable def g (d : ℝ) : ℝ → ℝ := fun t => 2 * (t ^ (-(1 / 2 : ℝ)) - (t + d ^ 2)
    ^ (-(1 / 2 : ℝ)))
```

Theorem 2.5 (H1, admissibility of the dipolar potential)**.** *For every real d, g_d is completely monotone on $(0,+\infty)$. More generally, for $s>0$ and $c\geq 0$, $\mathrm{shiftDiff}_{s,c}(t) = t^{-s/2} - (t+c)^{-s/2}$ is CM.* ***[Tier A]*** *(Lean statement); proof below* ***[Tier C]****.*

The proof rests on *Pochhammer symbols*. For $a\in\mathbb{R}$ and $r\in\mathbb{N}$, write

$$\mathrm{desc}_r(a) = a(a-1)\cdots(a-r+1), \qquad \mathrm{asc}_r(a) = a(a+1)\cdots(a+r-1),$$

with $\mathrm{desc}_0 = \mathrm{asc}_0 = 1$ (falling and rising factorials; Mathlib: `descPochhammer`, `ascPochhammer`).

Lemma 2.6 (derivatives of a shifted power)**.** *For $a\in\mathbb{R}$, $c\geq 0$, $t>0$ and $r\in\mathbb{N}$:* $\dfrac{d^r}{dt^r}(t+c)^a = \mathrm{desc}_r(a)\,(t+c)^{a-r}$.

Human proof. Induction on r. For $r=0$ it is the identity. If the formula holds at order r, differentiate $(t+c)^{a-r}$, which is legitimate since $t+c>0$: one obtains $(a-r)(t+c)^{a-r-1}$, and $\mathrm{desc}_r(a)(a-r) = \mathrm{desc}_{r+1}(a)$. □

Lean 4 — `DipolarExciton.lean:32-33` and `:36-37` (proofs omitted)

```
lemma iteratedDeriv_rpow (a : ℝ) (r : ℕ) (t : ℝ) :
    iteratedDeriv r (fun x : ℝ => x ^ a) t = (descPochhammer ℝ r).eval a * t ^ (a - r)
    := by
```

```
lemma iteratedDeriv_shift (a c : ℝ) (r : ℕ) (t : ℝ) :
    iteratedDeriv r (fun x : ℝ => (x + c) ^ a) t = (descPochhammer ℝ r).eval a * (t + c
    ) ^ (a - r) := by
```

By linearity of differentiation, for $t>0$:

$$g_d^{(r)}(t) = 2\,\mathrm{desc}_r\!\left(-\tfrac{1}{2}\right)\Big(t^{-1/2-r} - (t+d^2)^{-1/2-r}\Big). \tag{2.1}$$

Lean 4 — `DipolarExciton.lean:41-44` (proof omitted)

```
lemma iteratedDeriv_g (d : ℝ) (r : ℕ) {t : ℝ} (ht : 0 < t) :
    iteratedDeriv r (g d) t =
      2 * ((descPochhammer ℝ r).eval (-(1 / 2 : ℝ)) *
        (t ^ (-(1 / 2 : ℝ) - r) - (t + d ^ 2) ^ (-(1 / 2 : ℝ) - r))) := by
```

Lemma 2.7 (the sign)**.** $(-1)^r \operatorname{desc}_r(-\frac{1}{2}) = \operatorname{asc}_r(\frac{1}{2}) = \frac{1}{2}\cdot\frac{3}{2}\cdots\left(r-\frac{1}{2}\right) > 0.$

Human proof. $(-1)^r \operatorname{desc}_r(-\frac{1}{2}) = \prod_{k=0}^{r-1}(-1)\left(-\frac{1}{2}-k\right) = \prod_{k=0}^{r-1}\left(\frac{1}{2}+k\right)$, a product of strictly positive numbers. □

Lean 4 — `DipolarExciton.lean:56-57` (proof omitted)

```
lemma sign_desc (r : ℕ) :
    (-1 : ℝ) ^ r * (descPochhammer ℝ r).eval (-(1 / 2 : ℝ)) = (ascPochhammer ℝ r).eval
    (1 / 2 : ℝ) := by
```

Lemma 2.8 (the bracket)**.** *For $c \geq 0$, $e \leq 0$ and $t > 0$: $t^e - (t+c)^e \geq 0$.*

Human proof. For $e \leq 0$, $x \mapsto x^e$ is nonincreasing on $(0, +\infty)$; and $t \leq t + c$. □

Lean 4 — `DipolarExciton.lean:62-63` (proof omitted)

```
lemma diff_nonneg (c e : ℝ) (hc : 0 ≤ c) (he : e ≤ 0) {t : ℝ} (ht : 0 < t) :
    0 ≤ t ^ e - (t + c) ^ e := by
```

Proof of Theorem 2.5. Regularity. $t \mapsto t^{-1/2}$ and $t \mapsto (t+d^2)^{-1/2}$ are C^∞ wherever their base is strictly positive, hence on $(0, +\infty)$. *Signs.* By (2.1) and Lemma 2.7,

$$(-1)^r g_d^{(r)}(t) = 2\operatorname{asc}_r\left(\tfrac{1}{2}\right)\left(t^{-1/2-r} - (t+d^2)^{-1/2-r}\right),$$

the product of a positive number and a nonnegative bracket (Lemma 2.8 with $e = -\frac{1}{2} - r \leq 0$ and $c = d^2 \geq 0$). The case $r = 0$ gives $g_d \geq 0$. Replacing $\frac{1}{2}$ by $\frac{s}{2}$ ($s > 0$) and d^2 by $c \geq 0$ changes nothing: this is the statement for $\operatorname{shiftDiff}_{s,c}$. Finally $g_d = 2\operatorname{shiftDiff}_{1,d^2}$ and a positive multiple of a CM function is CM, which gives the result for *every* real d. □

Lean 4 — `DipolarExciton.lean:68` and `DipolarExcitonMultipolar.lean:37, 59, 102` (proofs omitted)

```
theorem dipolar_admissible {d : ℝ} (_hd : 0 < d) : AdmissiblePotential (g d) := by
```

```
noncomputable def shiftDiff (s c : ℝ) : ℝ → ℝ := fun t => t ^ (-(s / 2)) - (t + c) ^
    (-(s / 2))
```

```
theorem shiftDiff_admissible {s c : ℝ} (hs : 0 < s) (hc : 0 ≤ c) : AdmissiblePotential
    (shiftDiff s c) := by
```

```
theorem dipolar_admissible_all (d : ℝ) : AdmissiblePotential (AutoresearchDipolar.g d)
    := by
```

Remark 2.9 (Bernstein reading **[Tier C]**). Equivalently, $t^{-1/2}-(t+c)^{-1/2}=\pi^{-1/2}\int_0^\infty e^{-ut}\,(1-e^{-uc})\,u^{-1/2}\,du$ is a positive mixture of exponentials, which makes complete monotonicity evident [16]. This identity is not checked in Lean; the Pochhammer proof above avoids it.

Consequence 1: optimality at density one [Tier A], conditional. We apply the upstream theorem to g_d. The Lean proof is one line: the first component of `universal_energy_minimum`.

Lean 4 — `DipolarExciton.lean:94-96` (statement and complete proof)

```
theorem triangular_dipolar_optimal {d : ℝ} (hd : 0 < d) (C : Set Plane) (hC :
    LocallyFinite C)
    (hdens : DensityOne C) : latticeEnergy (g d) ≤ energy (g d) C :=
  (universal_energy_minimum (g d) C (dipolar_admissible hd) hC hdens).1
```

Consequence 2: the inequality is not $\top\le\top$ [Tier A]. An inequality in $[0,+\infty]$ whose left-hand side equals $+\infty$ says nothing. One must therefore show that $\mathrm{latticeEnergy}(g_d)<+\infty$.

Lemma 2.10. *For $c\ge 0$ and $t>0$: $t^{-1/2}-(t+c)^{-1/2}\le\frac{c}{2}\,t^{-3/2}$. In particular $g_d(t)\le d^2t^{-3/2}$.*

Human proof. Set $a=t^{-1/2}$ and $b=(t+c)^{-1/2}$, so that $0<b\le a$, $a^2t=1$ and $b^2(t+c)=1$. Then $a^2-b^2=a^2b^2\big((t+c)-t\big)=c\,a^2b^2$, i.e. $(a-b)(a+b)=c\,a^2b^2$. Since $b\le a$, $2b^2\le a^2+ab=a(a+b)$, so $2(a-b)(a+b)\le c\,a^3(a+b)$; dividing by $a+b>0$: $a-b\le\frac{c}{2}a^3=\frac{c}{2}t^{-3/2}$. □

The sum $\sum_{a\in A\setminus\{0\}}\|a\|^{-3}$ converges (Riesz exponent $3>2$, Chapter 1), hence $\mathrm{latticeEnergy}(g_d)\le d^2\,\mathrm{latticeEnergy}(t^{-3/2})<+\infty$.

Lean 4 — `DipolarExciton.lean:119-120, 159, 167` (proofs omitted)

```
lemma diff_le (c : ℝ) (hc : 0 ≤ c) {t : ℝ} (ht : 0 < t) :
    t ^ (-(1 / 2 : ℝ)) - (t + c) ^ (-(1 / 2 : ℝ)) ≤ c / 2 * t ^ (-(3 / 2 : ℝ)) := by
```

```
lemma g_le (d : ℝ) {t : ℝ} (ht : 0 < t) : g d t ≤ d ^ 2 * riesz 3 t := by
```

```
theorem latticeEnergy_dipolar_lt_top (d : ℝ) : latticeEnergy (g d) < ⊤ := by
```

Consequence 3: every density $\rho>0$ [Tier A], conditional. If C has density ρ, then $\sqrt{\rho}\,C$ has density one and $\|\sqrt{\rho}x-\sqrt{\rho}y\|^2=\rho\|x-y\|^2$: the energy of C for g is that of $\sqrt{\rho}\,C$ for $t\mapsto g(t/\rho)$, which remains CM (each derivative gains the factor $\rho^{-r}>0$). Corollary D2 of Chapter 1 then gives the optimality of $\rho^{-1/2}A$, and $g_d(t/\rho)\le d^2\rho^{3/2}t^{-3/2}$ gives finiteness.

Lean 4 — `DipolarExcitonMultipolar.lean:125-129` (statement and complete proof) and `:175-176`

```
theorem triangular_dipolar_optimal_any_density {d : ℝ} (hd : 0 < d) {ρ : ℝ} (hρ : 0 < ρ
    ) :
    LocallyFinite ((Real.sqrt ρ)⁻¹ • A) ∧ TriangularDensity.DensityRho ρ ((Real.sqrt ρ)
    ⁻¹ • A) ∧
      ∀ C : Set Plane, LocallyFinite C → TriangularDensity.DensityRho ρ C →
        energy (AutoresearchDipolar.g d) ((Real.sqrt ρ)⁻¹ • A) ≤ energy (
    AutoresearchDipolar.g d) C :=
  TriangularDensity.triangular_optimal_any_density _ (AutoresearchDipolar.
    dipolar_admissible hd) hρ
```

```
theorem energy_dipolar_any_density_lt_top {d : ℝ} (hd : 0 < d) {ρ : ℝ} (hρ : 0 < ρ) :
    energy (AutoresearchDipolar.g d) ((Real.sqrt ρ)⁻¹ • A) < ⊤ := by
```

Negative controls [Tier A]. (i) If the weight of the cross attraction is doubled, $h_d(t) = t^{-1/2} - 2(t+d^2)^{-1/2}$, then at $t = d^2$: $h_d(d^2) = \frac{1}{d} - \frac{2}{\sqrt{2}\,d} < \frac{1}{d} - \frac{2}{2d} = 0$, so h_d is not admissible (`h_not_admissible`). (ii) For $c < 0$ and $t = -2c > 0$, we have $t + c = -c \in (0, t)$, whence $(t+c)^{-s/2} > t^{-s/2}$ and $\mathrm{shiftDiff}_{s,c}(t) < 0$ (`shiftDiff_not_admissible_of_neg`). The sign of the shift matters.

Warning : scope

Finiteness is proved for g_d only; for the family $\mathrm{shiftDiff}_{s,c}$ no finiteness is proved, and for $c = 0$ one has $\mathrm{shiftDiff}_{s,0} \equiv 0$: the conclusion is then the trivial $0 \le$ energy. The model is classical: rigid dipoles, no kinetic energy, no dielectric screening, no moiré potential. In the experimental density window, real excitons form a quantum gas, not a classical crystal **[Tier C]** [16]. Paper: doi:10.5281/zenodo.23646103.

2.3.2 The planar Yukawa potential

Physics and biology : the model [Tier C]

Charged particles confined to a plane but immersed in a three-dimensional screening medium (dusty plasmas, charged colloids) interact through the screened Coulomb potential $V(r) = e^{-\kappa r}/r$, $\kappa > 0$. In the variable $t = r^2$: $g_\kappa(t) = e^{-\kappa\sqrt{t}}/\sqrt{t}$. **This is not** the Green's function $K_0(\kappa r)$ of the two-dimensional Debye–Hückel equation, which is not treated here [17].

Lean 4 — `formal_cert/Autoresearch/Yukawa.lean:28, 31`

```
noncomputable def phi (κ : ℝ) : ℝ → ℝ := fun t => Real.exp (-κ * Real.sqrt t)
```

```
noncomputable def g (κ : ℝ) : ℝ → ℝ := fun t => Real.exp (-κ * Real.sqrt t) / Real.
    sqrt t
```

The plan is: $g_\kappa = t^{-1/2} \cdot \varphi_\kappa$ with $\varphi_\kappa(t) = e^{-\kappa\sqrt{t}}$; we show that the product of two CM functions is CM, then that φ_κ is CM by strong induction. We say that f has the *sign pattern up to order n* at t if $(-1)^i f^{(i)}(t) \ge 0$ for $i = 0, \dots, n$.

Lemma 2.11 (Leibniz signs)**.** *If f and h are n times differentiable at t and both have the sign pattern up to order n at t, then $(-1)^n (fh)^{(n)}(t) \ge 0$. Consequently, the product of two CM functions is CM.*

Human proof. Leibniz's rule: $(fh)^{(n)} = \sum_{i=0}^{n} \binom{n}{i} f^{(i)} h^{(n-i)}$. Write $(-1)^n = (-1)^i (-1)^{n-i}$:

$$(-1)^n (fh)^{(n)} = \sum_{i=0}^{n} \binom{n}{i} \left[(-1)^i f^{(i)}\right] \left[(-1)^{n-i} h^{(n-i)}\right],$$

a sum of products of nonnegative numbers. Regularity and nonnegativity of a product are immediate. □

Lean 4 — `Yukawa.lean:38-41` (proof omitted; it takes about a dozen lines)

```
lemma mul_sign {f h : ℝ → ℝ} {t : ℝ} (r : ℕ) (hf : ContDiffAt ℝ r f t) (hh :
    ContDiffAt ℝ r h t)
    (Hf : ∀ i ≤ r, 0 ≤ (-1 : ℝ) ^ i * iteratedDeriv i f t)
    (Hh : ∀ i ≤ r, 0 ≤ (-1 : ℝ) ^ i * iteratedDeriv i h t) :
    0 ≤ (-1 : ℝ) ^ r * iteratedDeriv r (f * h) t := by
```

Lemma 2.12 ($t^{-s/2}$ is CM)**.** *For $s > 0$, $p(t) = t^{-s/2}$ satisfies $(-1)^n p^{(n)}(t) = \operatorname{asc}_n(\frac{s}{2})\, t^{-s/2-n} > 0$.*

Human proof. This is Lemma 2.6 with $c = 0$ and $a = -\frac{s}{2}$, followed by the sign computation of Lemma 2.7 with $\frac{s}{2}$ in place of $\frac{1}{2}$. (Lean: `TriangularRiesz.riesz_admissible`, Chapter 1.) □

Proposition 2.13 ($e^{-\kappa\sqrt{t}}$ is CM)**.** *For $\kappa > 0$, $\varphi_\kappa(t) = e^{-\kappa\sqrt{t}}$ is completely monotone on $(0, +\infty)$.*

Human proof. By the chain rule, on $(0, +\infty)$:

$$\varphi_\kappa'(t) = -\frac{\kappa}{2\sqrt{t}}\, e^{-\kappa\sqrt{t}} = -\frac{\kappa}{2}\, t^{-1/2}\, \varphi_\kappa(t). \tag{$*$}$$

We prove by *strong induction* on n the property $\mathcal{P}(n)$: "for every $m \le n$ and every $t > 0$, $(-1)^m \varphi_\kappa^{(m)}(t) \ge 0$".

$\mathcal{P}(0)$: $\varphi_\kappa > 0$.

$\mathcal{P}(n) \Rightarrow \mathcal{P}(n+1)$: only the order $m = n+1$ is new. The two sides of $(*)$ are equal on the open set $(0, +\infty)$, hence so are their derivatives of all orders: $\varphi_\kappa^{(n+1)} = (\varphi_\kappa')^{(n)} = -\frac{\kappa}{2}\,(t^{-1/2}\varphi_\kappa)^{(n)}$. By Lemma 2.12, $t^{-1/2}$ has the sign pattern at every order; by $\mathcal{P}(n)$, φ_κ has it up to order n. Lemma 2.11 gives $(-1)^n (t^{-1/2}\varphi_\kappa)^{(n)} \ge 0$, whence

$$(-1)^{n+1}\varphi_\kappa^{(n+1)} = \frac{\kappa}{2}\,(-1)^n (t^{-1/2}\varphi_\kappa)^{(n)} \ge 0,$$

since $\kappa > 0$. The step from n to $n+1$ uses the signs of φ_κ only up to order n: this is why strong induction suffices and why no Faà di Bruno formula is needed. Regularity follows from the fact that exp and $\sqrt{\cdot}$ are C^∞ on $(0, +\infty)$. □

Lean 4 — `Yukawa.lean:82-83, 94-95, 121` (proofs omitted)

```
lemma phi_hasDerivAt (κ : ℝ) {t : ℝ} (ht : 0 < t) :
    HasDerivAt (phi κ) (-(κ / 2) * (riesz 1 t * phi κ t)) t := by
```

```
lemma phi_sign {κ : ℝ} (hκ : 0 < κ) :
    ∀ n : ℕ, ∀ m ≤ n, ∀ t : ℝ, 0 < t → 0 ≤ (-1 : ℝ) ^ m * iteratedDeriv m (phi κ) t
    := by
```

```
theorem phi_admissible {κ : ℝ} (hκ : 0 < κ) : AdmissiblePotential (phi κ) :=
```

Note the shape of `phi_sign`: the quantification "$\forall\, m \le n$" is exactly the strong induction of the human proof.

Theorem 2.14 (H2, admissibility of Yukawa)**.** *For $\kappa > 0$, g_κ is admissible.* ***[Tier A]***

Human proof. For $t > 0$, $g_\kappa(t) = t^{-1/2}\varphi_\kappa(t)$: a product of two CM functions (Lemmas 2.12, 2.11 and Proposition 2.13). In Lean, $g_\kappa(0) = 0$ by the convention $x/0 = 0$; but admissibility only looks at $(0, +\infty)$, and two functions that agree on an open set have the same derivatives there (`admissible_congr`). □

Lean 4 — `Yukawa.lean:133-134` (statement and complete proof)

```
theorem yukawa_admissible {κ : ℝ} (hκ : 0 < κ) : AdmissiblePotential (g κ) :=
  admissible_congr (admissible_mul (riesz_admissible (s := 1) one_pos) (phi_admissible h
    κ)) (g_eqOn κ)
```

Consequence: optimality and finite energy [Tier A], conditional. Optimality at density one (`triangular_yukawa_optimal`) is obtained as for the dipoles. For finiteness: for $x \geq 0$, $e^x \geq x^2/2$; with $x = \kappa\sqrt{t}$,

$$g_\kappa(t) = \frac{1}{e^{\kappa\sqrt{t}}\sqrt{t}} \leq \frac{2}{\kappa^2\, t\sqrt{t}} = \frac{2}{\kappa^2}\, t^{-3/2},$$

and the Riesz sum with exponent 3 converges. The hypothesis $\kappa > 0$ is used here: for $\kappa = 0$, $g_0 = t^{-1/2}$ is the Riesz potential with exponent 1, whose lattice sum diverges.

Lean 4 — `Yukawa.lean:160, 175` (proofs omitted)

```
lemma g_le {κ : ℝ} (hκ : 0 < κ) {t : ℝ} (ht : 0 < t) : g κ t ≤ 2 / κ ^ 2 * riesz 3 t
    := by

theorem latticeEnergy_yukawa_lt_top {κ : ℝ} (hκ : 0 < κ) : latticeEnergy (g κ) < ⊤ :=
    by
```

Finite mixtures [Tier A]. The admissible functions form a convex cone (derivatives are linear), so every mixture $\sum_i w_i g_{\kappa_i}$ with $w_i \geq 0$, $\kappa_i > 0$ is admissible, with finite energy since it is bounded by $\left(\sum_i w_i\, 2/\kappa_i^2\right) t^{-3/2}$; the result also holds at every density (`YukawaMix.lean`).

Lean 4 — `formal_cert/Autoresearch/YukawaMix.lean:129-130, 138-139` (proof omitted)

```
noncomputable def yukawaMix {ι : Type*} (s : Finset ι) (w κ : ι → ℝ) : ℝ → ℝ :=
  fun t => ∑ i ∈ s, w i * g (κ i) t

theorem yukawaMix_admissible {ι : Type*} (s : Finset ι) {w κ : ι → ℝ} (hw : ∀ i ∈ s, 0
    ≤ w i)
    (hκ : ∀ i ∈ s, 0 < κ i) : AdmissiblePotential (yukawaMix s w κ) := by
```

Negative controls [Tier A]. (a) $c_\kappa(t) = e^{-\kappa\sqrt{t}}\cos t$ takes the value $-e^{-\kappa\sqrt{\pi}} < 0$ at $t = \pi$ (`c_not_admissible`). (b) The signed mixture $g_1 - 2g_2$: at $t = \frac{1}{4}$, $\sqrt{t} = \frac{1}{2}$ and $g_1(\frac{1}{4}) - 2g_2(\frac{1}{4}) = 2e^{-1/2} - 4e^{-1} = 2e^{-1}(e^{1/2} - 2) < 0$ since $e < 4$ (`signedMix_not_admissible`). These controls show that the hypotheses of the lemmas are not vacuous; they do *not* show that the triangular lattice ceases to be optimal for these functions: nothing here decides that.

Warning : scope

Classical model of point particles; no kinetic energy; not the K_0 kernel. The companion paper (doi:10.5281/zenodo.23646105) also contains a **[Tier X]** numerical study with CVODE that corrects a published truncation erratum; it supports no statement of this chapter [17].

2.3.3 The Rytova–Keldysh potential

Physics and biology : the model [Tier L] / [Tier C]

In a monolayer of a two-dimensional semiconductor (MoS_2, WSe_2, ...) suspended in vacuum, two charges interact through the Rytova–Keldysh potential, in the form of Berkelbach, Hybertsen and Reichman [26]:

$$V(r) = \frac{\pi}{2r_0}\big[\mathbf{H}_0 - Y_0\big](r/r_0), \qquad \widehat{V}(k) = \frac{2\pi}{k(1+r_0 k)},$$

where $\mathbf{H}_0$ is the Struve function, Y_0 the Neumann function, and $r_0 > 0$ a screening length (in units $e^2/(4\pi\varepsilon_0) = 1$). For $r_0 \to 0$ one recovers Coulomb, $2\pi/k$. The form is cited **[Tier L]** ([26], after [27] for $\mathbf{K}_0 = \mathbf{H}_0 - Y_0$); the Fourier transform is **[Tier C]**.

Mathlib, at the pinned version, knows neither Struve nor Neumann functions. The idea is therefore to *define* the potential in Lean by a manifestly completely monotone integral representation — a mixture of Gaussians in r, that is, a mixture of exponentials in $t = r^2$ — and then to prove *by hand*, at Tier C, that this integral is indeed the Struve form. The gap between the two is precisely the "fidelity gap" of this formalisation.

Step 0: the weight. For $x \geq 0$ set

$$\operatorname{erfcx}(x) = \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-(y^2+2xy)}\, dy.$$

The change of variable $s = y + x$ gives $\operatorname{erfcx}(x) = e^{x^2}\frac{2}{\sqrt{\pi}}\int_x^\infty e^{-s^2} ds = e^{x^2}\operatorname{erfc}(x)$, the "scaled" complementary error function. For $r_0 > 0$, the weight is $w(a) = r_0^{-1}\operatorname{erfcx}(\sqrt{a}/r_0)$.

Lean 4 — `formal_cert/Autoresearch/KeldyshCM.lean:190-191, 244, 248-249`

```
noncomputable def erfcx (x : ℝ) : ℝ :=
  2 / Real.sqrt Real.pi * ∫ y in Ioi (0 : ℝ), Real.exp (-(y ^ 2 + 2 * x * y))

noncomputable def w (r0 a : ℝ) : ℝ := 1 / r0 * erfcx (Real.sqrt a / r0)

noncomputable def keldysh (r0 t : ℝ) : ℝ :=
  ∫ a in Ioi (0 : ℝ), w r0 a / (2 * a) * Real.exp (-t / (4 * a))
```

In other words, the Lean object is

$$\mathcal{K}_{r_0}(t) = \int_0^\infty \frac{w(a)}{2a}\, e^{-t/(4a)}\, da. \tag{2.2}$$

Lemma 2.15 (bounds on the weight). *For $x \geq 0$, $0 < \operatorname{erfcx}(x) \leq 1$; for $x > 0$, $\operatorname{erfcx}(x) \leq \frac{1}{\sqrt{\pi}\,x}$. Hence, for $a > 0$, $0 < w(a) \leq \frac{1}{r_0}$ and $w(a) \leq \frac{1}{\sqrt{\pi}\sqrt{a}}$.*

Human proof. The integrand is continuous and strictly positive, so $\mathrm{erfcx}(x) > 0$. For $y > 0$ and $x \geq 0$, $y^2 + 2xy \geq y^2$, so the integral is at most $\int_0^\infty e^{-y^2} dy = \frac{\sqrt{\pi}}{2}$, whence $\mathrm{erfcx} \leq 1$. If $x > 0$, bound from below by $2xy$ instead: $\int_0^\infty e^{-2xy} dy = \frac{1}{2x}$, whence $\mathrm{erfcx}(x) \leq \frac{2}{\sqrt{\pi}} \cdot \frac{1}{2x}$. For w, substitute $x = \sqrt{a}/r_0$. □

Step 1: a generic Laplace-mixture lemma. Let (α, μ) be a measure space, k, m measurable, $k \geq 0$ and $m \geq 0$ almost everywhere, and assume $\int k^n e^{-ck} m \, d\mu < \infty$ for every n and every $c > 0$. Set $G_n(t) = \int (-k)^n e^{-tk} m \, d\mu$.

Theorem 2.16 (Laplace mixture)**.** *$g = G_0$ is admissible, and $g^{(n)}(t) = G_n(t)$ for $t > 0$.* ***[Tier A]***

Human proof. Fix $t > 0$ and let s range over $(t/2, +\infty)$, a neighbourhood of t. The derivative in s of $(-k)^n e^{-sk} m$ is $(-k)^{n+1} e^{-sk} m$, whose absolute value satisfies $k^{n+1} e^{-sk} m \leq k^{n+1} e^{-(t/2)k} m$ since $s \geq t/2$ and $k \geq 0$. The bound does not depend on s and is integrable (hypothesis with $c = t/2$). The theorem on differentiation under the integral sign (dominated convergence) gives $G_n'(t) = G_{n+1}(t)$. By induction, G_0 is of class C^N for every N and $g^{(n)} = G_n$. Finally $(-1)^n G_n(t) = \int k^n e^{-tk} m \, d\mu \geq 0$, since the integrand is nonnegative almost everywhere. □

University remark. **[Tier L]** The step used above is the standard theorem on differentiation under the integral sign: let $U \subset \mathbb{R}$ be open, (X, μ) a measure space, $F \colon U \times X \to \mathbb{R}$ with $F(s, \cdot)$ integrable for every $s \in U$, $s \mapsto F(s, x)$ differentiable for almost every x, and $|\partial_s F(s, x)| \leq H(x)$ for all $s \in U$ and almost every x, with H integrable and independent of s. Then $s \mapsto \int F(s, x) \, d\mu(x)$ is differentiable on U with derivative $\int \partial_s F(s, x) \, d\mu(x)$. It follows from the mean value theorem and Lebesgue's dominated convergence theorem (standard).

Lean 4 — `KeldyshCM.lean:31-36, 134-135` (proof omitted)

```
structure MixtureData (μ : Measure α) (k m : α → ℝ) : Prop where
  k_meas : Measurable k
  m_meas : Measurable m
  k_nonneg : ∀ᵐ x ∂μ, 0 ≤ k x
  m_nonneg : ∀ᵐ x ∂μ, 0 ≤ m x
  integrable : ∀ (n : ℕ) (c : ℝ), 0 < c → Integrable (fun x => k x ^ n * Real.exp (-(c
      * k x)) * m x) μ
```

```
theorem laplace_admissible (h : MixtureData μ k m) :
    AdmissiblePotential (fun t => ∫ x, Real.exp (-(t * k x)) * m x ∂μ) := by
```

This theorem is re-exported under the name `ALEXIS` (Chapter 5).

Step 2: the Keldysh instance. Take Lebesgue measure on $(0, +\infty)$, $k(a) = \frac{1}{4a}$ and $m(a) = \frac{w(a)}{2a}$. The moments must be checked: $F_n(a) = (4a)^{-n} e^{-c/(4a)} \frac{w(a)}{2a}$ is integrable. Split at $a = 1$. On $(0, 1]$, with $u = \frac{1}{4a}$: $F_n = 2w(a)\, u^{n+1} e^{-cu}$, and $e^{cu} \geq (cu)^N / N!$ gives $u^N e^{-cu} \leq N!/c^N$; F_n is bounded by $2r_0^{-1}(n+1)!/c^{n+1}$ on an interval of finite length. On $(1, +\infty)$, $u \leq \frac{1}{4}$, so $F_n \leq \frac{w(a)}{2a} \leq \frac{1}{2\sqrt{\pi}} a^{-3/2}$ (Lemma 2.15), and $\int_1^\infty a^{-3/2} da = 2$. Theorem 2.16 applies.

Lean 4 — `KeldyshCM.lean:354` (proof omitted)

```
theorem keldysh_admissible {r0 : ℝ} (hr0 : 0 < r0) : AdmissiblePotential (keldysh r0) :=
    by
```

Step 3: the statement is not that of the zero function [Tier A]. In Lean, the Bochner integral of a non-integrable function is 0, and 0 is admissible: admissibility alone could be vacuous. But for $t > 0$ the integrand of (2.2) is strictly positive and integrable, so $\mathcal{K}_{r_0}(t) > 0$.

Lean 4 — `KeldyshCM.lean:379` (proof omitted)

```
theorem keldysh_pos {r0 t : ℝ} (hr0 : 0 < r0) (ht : 0 < t) : 0 < keldysh r0 t := by
```

Step 4: corollaries, and a caveat [Tier A] / [Tier C]. The product with $e^{-\varepsilon t}$ ($\varepsilon \geq 0$) remains admissible (Lemma 2.11), and the upstream theorem gives the two inequalities `triangular_keldysh_optimal` and `triangular_keldyshReg_optimal`.

Lean 4 — `KeldyshCM.lean:390-392` (statement and complete proof)

```
theorem triangular_keldysh_optimal {r0 : ℝ} (hr0 : 0 < r0) (C : Set Plane) (hC :
    LocallyFinite C)
    (hdens : DensityOne C) : latticeEnergy (keldysh r0) ≤ energy (keldysh r0) C :=
  (universal_energy_minimum (keldysh r0) C (keldysh_admissible hr0) hC hdens).1
```

Warning : $\top \leq \top$

$V(r) \sim 1/r$ at large distance, and $\sum_{0<|x|\leq R} 1/|x|$ grows like R on a planar lattice. One therefore *expects* latticeEnergy$(\mathcal{K}_{r_0}) = +\infty$: the first inequality would then read $\top \leq \top$ [Tier C]. This is neither proved nor refuted in Lean. For $\varepsilon > 0$, the regularised sum should be finite [Tier C], but **this finiteness is not proved**. The physically relevant energy of a Wigner crystal, neutralised by a uniform background ("jellium"), is covered by neither of the two statements [18].

Step 5: the link with the Struve form [Tier C].

Theorem 2.17 (not machine-checked)**.** *For $r_0 > 0$ and $r > 0$: $V(r) = \mathcal{K}_{r_0}(r^2)$.*

Human proof. All integrands are nonnegative: every exchange of integrals is justified by Tonelli's theorem.

(a) A Laplace form in r. Formula DLMF 11.5.2 with $\nu = 0$ gives $\mathbf{K}_0(z) = \frac{2}{\pi}\int_0^\infty e^{-zu}(1+u^2)^{-1/2}du$ for $\operatorname{Re} z > 0$ [Tier L] [27]. With $z = r/r_0$: $V(r) = \frac{1}{r_0}\int_0^\infty e^{-ru/r_0}(1+u^2)^{-1/2}du$. This already suffices to see that V is CM *in* r.

(b) Subordination. For $\rho > 0$ and $t > 0$, $e^{-\rho\sqrt{t}} = \int_0^\infty e^{-ta}\,\frac{\rho}{2\sqrt{\pi}}\,a^{-3/2}e^{-\rho^2/(4a)}\,da$ (Laplace transform of the Lévy density, a classical formula used without citation in the source paper). Take $\rho = u/r_0$, $t = r^2$, insert into (a) and exchange the integrals:

$$V(r) = \int_0^\infty e^{-ta}\,\frac{a^{-3/2}}{2\sqrt{\pi}\,r_0^2}\left[\int_0^\infty \frac{u}{\sqrt{1+u^2}}e^{-\lambda u^2}du\right]da, \qquad \lambda = \frac{1}{4r_0^2 a}.$$

(c) The inner integral is elementary. With $v = \sqrt{1+u^2}$, $dv = u(1+u^2)^{-1/2}du$:

$$\int_0^\infty \frac{u}{\sqrt{1+u^2}}e^{-\lambda u^2}du = e^{\lambda}\int_1^\infty e^{-\lambda v^2}dv = \frac{1}{2}\sqrt{\frac{\pi}{\lambda}}\operatorname{erfcx}(\sqrt{\lambda}).$$

With $\sqrt{\pi/\lambda} = 2r_0\sqrt{\pi a}$, one obtains $V(r) = \int_0^\infty e^{-ta}\rho(a)\,da$ where $\rho(a) = \dfrac{\operatorname{erfcx}\big(1/(2r_0\sqrt{a})\big)}{2r_0\,a}$.

(d) Change of variable. Set $a = \frac{1}{4b}$, whence $|da| = \frac{db}{4b^2}$ and $e^{-ta} = e^{-t/(4b)}$. Then $\rho(a)\,da = \frac{4b\,\mathrm{erfcx}(\sqrt{b}/r_0)}{2r_0} \cdot \frac{db}{4b^2} = \frac{w(b)}{2b}\,db$, so $V(r) = \int_0^\infty \frac{w(b)}{2b} e^{-r^2/(4b)} db = \mathcal{K}_{r_0}(r^2)$. □

Numerical checks at 30 digits (mpmath) confirm each step, with two planted errors detected **[Tier X]** [18]; they do not replace the proof. The reader must accept Theorem 2.17 to transfer the Lean statement, which concerns (2.2), to the Struve form. Paper: doi:10.5281/zenodo.23646107.

2.4 The generalisation: a CM/Bernstein library

The three preceding proofs share a family resemblance: a difference of translates (dipoles), a product and a composition with $\sqrt{t}$ (Yukawa), a Laplace mixture (Keldysh). The P3 library [12] (doi:10.5281/zenodo.23653842) isolates these mechanisms once and for all. Gate summary: **130 theorems** declared, each with its `#print axioms` line, all within the allowed list, no warnings; an audit by a separate AI agent rebuilt the 11 files and 5 prerequisite modules from source and found 0 off-list constants among 167 **[Tier A]**. No human audit; the 20 `LEDGER_UNAUDITED_TIER_A` flags remain open.

2.4.1 Two definitions and a bridge

Definition 2.18 (Bernstein function)**.** $\psi\colon (0,+\infty) \to \mathbb{R}$ is a *Bernstein function* if it is C^∞, nonnegative ($\psi \geq 0$), and ψ' is CM. Examples: $at + b$ ($a, b \geq 0$), t^α for $0 \leq \alpha \leq 1$, in particular $\sqrt{t}$; $1 - e^{-at}$; $\ln(1+t)$.

University remark. **[Tier L]** Bernstein functions admit a Lévy–Khintchine representation: ψ is a Bernstein function if and only if $\psi(t) = a + bt + \int_{(0,\infty)} (1 - e^{-tu})\,\nu(du)$ for constants $a, b \geq 0$ and a positive measure ν on $(0,\infty)$ with $\int \min(1,u)\,\nu(du) < \infty$ [28]. For example, $\sqrt{t} = \frac{1}{2\sqrt{\pi}} \int_0^\infty (1 - e^{-tu})\,u^{-3/2}\,du$. This representation is not formalised here; the library works only with the derivative-sign definition above.

Lean 4 — `formal_cert/Programmes/CompletelyMonotone/Basic.lean:28-30, 33-34` and `Bernstein.lean:24-25`

```
def IsCompletelyMonotoneOn (f : ℝ → ℝ) : Prop :=
  ContDiffOn ℝ (⊤ : ℕ∞) f (Set.Ioi 0) ∧
  ∀ (n : ℕ) (t : ℝ), 0 < t → 0 ≤ (-1 : ℝ) ^ n * iteratedDeriv n f t
```

```
theorem isCompletelyMonotoneOn_iff_admissible {f : ℝ → ℝ} :
    IsCompletelyMonotoneOn f ↔ OAI.AtomicTriangular.AdmissiblePotential f := by
```

```
def IsBernsteinOn (g : ℝ → ℝ) : Prop :=
  ContDiffOn ℝ (⊤ : ℕ∞) g (Set.Ioi 0) ∧ (∀ t : ℝ, 0 < t → 0 ≤ g t) ∧
    IsCompletelyMonotoneOn (deriv g)
```

The bridge `isCompletelyMonotoneOn_iff_admissible` is an *equivalence* **[Tier A]**: the clause $g \geq 0$ of the upstream is the case $n = 0$ of the third clause. Every CM function of the library can therefore be used directly in `universal_energy_minimum`.

2.4.2 The closure lemmas

Write $h_n(t) = (-1)^n f^{(n)}(t)$ for f CM.

(i) *Cone.* Derivatives are linear: cf ($c \geq 0$), $f + g$, and every finite nonnegative combination of CM functions is CM. Nonnegative constants are CM.

(ii) *Product.* By the Leibniz lemma 2.11, fg is CM.

(iii) *Derivative.* $h_n' = (-1)^n f^{(n+1)} = -h_{n+1} \le 0$, so each h_n is nonincreasing (mean value theorem). In particular $-f'$ is CM, since $(-1)^n(-f')^{(n)} = h_{n+1} \ge 0$.

(iv) *Translate.* For $c \ge 0$, $t \mapsto f(t+c)$ is CM: its derivatives are $f^{(n)}(t+c)$ and $t + c > 0$.

(v) *Difference of translates.* For $c \ge 0$, $f(t) - f(t+c)$ is CM, since $(-1)^n\big[f(t) - f(t+c)\big]^{(n)} = h_n(t) - h_n(t+c) \ge 0$ because h_n is nonincreasing. No integral representation is used.

Lean 4 — `Basic.lean:160-161` (proof omitted; it takes ten lines)

```
theorem sub_comp_add_const (hf : IsCompletelyMonotoneOn f) {c : ℝ} (hc : 0 ≤ c) :
    IsCompletelyMonotoneOn (fun t => f t - f (t + c)) := by
```

Item (v) is the abstract version of the dipolar proof: the Pochhammer computation is replaced there by the mere monotonicity of h_n.

2.4.3 The subordination theorem

Theorem 2.19 (subordination)**.** *If f is CM and ψ is a Bernstein function that is strictly positive on $(0, +\infty)$, then $f \circ \psi$ is CM.* ***[Tier A]****; proof below* ***[Tier C]****.*

Human proof. Regularity comes from the stability of C^∞ under composition (ψ maps $(0,+\infty)$ into $(0,+\infty)$). For the signs, we prove by induction on n the property

$$\mathcal{Q}(n) \ : \text{“for } \textit{every} \text{ CM function } f \text{ and every } m \le n,\ (-1)^m (f \circ \psi)^{(m)} \ge 0 \text{ on } (0,+\infty)\text{”}.$$

$\mathcal{Q}(0)$: $f(\psi(t)) \ge 0$ since $\psi(t) > 0$ and $f \ge 0$.

$\mathcal{Q}(n) \Rightarrow \mathcal{Q}(n+1)$: let f be CM; the orders $m \le n$ are covered by $\mathcal{Q}(n)$. For $m = n+1$, set $F = -f'$, which is CM by (iii). The chain rule gives $(f \circ \psi)' = (f' \circ \psi)\,\psi' = -(F \circ \psi)\,\psi'$, so

$$(-1)^{n+1}(f \circ \psi)^{(n+1)} = (-1)^n\big[(F \circ \psi)\,\psi'\big]^{(n)} \ge 0$$

by the Leibniz lemma, applied to $u = F \circ \psi$ (whose signs up to order n are given by $\mathcal{Q}(n)$ *applied to* F) and to $v = \psi'$ (CM by the definition of a Bernstein function). It is because $\mathcal{Q}(n)$ quantifies over *all* CM functions that the induction closes without Faà di Bruno's formula. □

Lean 4 — `Bernstein.lean:49-51` (proof omitted) and `:85-87` (statement and complete proof)

```
theorem comp_sign (hg : IsBernsteinOn g) (hpos : ∀ t, 0 < t → 0 < g t) :
    ∀ n : ℕ, ∀ f : ℝ → ℝ, IsCompletelyMonotoneOn f → ∀ m ≤ n, ∀ t : ℝ, 0 < t →
      0 ≤ (-1 : ℝ) ^ m * iteratedDeriv m (fun t => f (g t)) t := by
```

```
theorem IsCompletelyMonotoneOn.comp_bernstein {f g : ℝ → ℝ} (hf :
    IsCompletelyMonotoneOn f)
    (hg : IsBernsteinOn g) (hpos : ∀ t, 0 < t → 0 < g t) : IsCompletelyMonotoneOn (fun
    t => f (g t)) :=
  ⟨hf.1.comp hg.1 (fun s hs => hpos s hs), fun n t ht => hg.comp_sign hpos n f hf n
    le_rfl t ht⟩
```

The Lean statement `comp_sign` carries “$\forall f$” *inside* the induction on n: it is literally the property $\mathcal{Q}(n)$.

Remark 2.20 (the hypothesis $\psi > 0$ is necessary **[Tier A]**). f is constrained only on $(0, +\infty)$, and $\psi \equiv 0$ is a Bernstein function; then $f \circ \psi$ is the "junk" constant $f(0)$, which may be negative. The control `junk_comp_zero_not_cm` realises this case. The theorem is classical in spirit ("Bochner subordination"); the source paper did not locate its number in the reference monograph [28]. The contribution is the machine-checked proof on $(0, +\infty)$.

Instances. $\psi(t) = t^\alpha$ $(0 \le \alpha \le 1)$ is a Bernstein function: $\psi' = \alpha t^{-(1-\alpha)}$ is CM by Lemma 2.12. Hence, with $f(u) = e^{-au}$: e^{-at^α} is CM (stretched exponential), and with $\alpha = \frac{1}{2}$: $e^{-\kappa\sqrt{t}}$ is CM for every $\kappa \ge 0$, in one line.

Lean 4 — `Bernstein.lean:160-161` (statement and complete proof)

```
theorem phi_cm {κ : ℝ} (hκ : 0 ≤ κ) : IsCompletelyMonotoneOn (AutoresearchYukawa.phi κ)
    :=
  ((exp_neg_mul hκ).comp_sqrt).congr' fun t _ => by simp only [AutoresearchYukawa.phi,
    neg_mul]
```

Compare with the strong induction of Proposition 2.13: that one required $\kappa > 0$ and an ad hoc induction; the library covers $\kappa \ge 0$.

Laplace mixtures, weakened hypothesis. The theorem `laplace` of `Laplace.lean:49` no longer assumes that all moments are finite, but only that $e^{-ck}m$ is integrable for every $c > 0$: the inequality $u^n \le n!\,(c/2)^{-n} e^{(c/2)u}$ $(u \ge 0)$ gives $k^n e^{-ck} m \le \frac{n!}{(c/2)^n} e^{-(c/2)k} m$, whence all the moments. This is the *easy* direction of the Bernstein–Widder theorem (a Laplace transform of a positive measure is CM); the hard direction (every CM function is such a transform [7]) is *not* proved.

University remark. **[Tier L]** The full statement (Bernstein's theorem, also called the Bernstein–Widder theorem): a function $f\colon (0,\infty) \to \mathbb{R}$ is completely monotone if and only if $f(t) = \int_{[0,\infty)} e^{-tu}\,\mu(du)$ for a positive Borel measure μ on $[0,\infty)$ such that the integral is finite for every $t > 0$; the measure μ is then unique [6, 7]. The measure need not be finite: $1/t = \int_0^\infty e^{-tu}\,du$. The preceding paragraph concerns only the "if" direction; the "only if" direction is not proved in this book.

2.4.4 The three potentials become corollaries

Lean 4 — `Corollaries.lean:24-25, 28-30, 38`

```
theorem yukawa_cm {κ : ℝ} (hκ : 0 ≤ κ) : IsCompletelyMonotoneOn (AutoresearchYukawa.g κ
    ) :=
  ((riesz_cm zero_le_one).mul (phi_cm hκ)).congr (AutoresearchYukawa.g_eqOn κ)
```

```
theorem dipolar_cm (d : ℝ) : IsCompletelyMonotoneOn (AutoresearchDipolar.g d) :=
  (((rpow_neg (by norm_num : (0 : ℝ) ≤ 1 / 2)).sub_comp_add_const (sq_nonneg d)).
    const_mul
    zero_le_two).congr' fun _ _ => rfl
```

```
theorem keldysh_cm {r0 : ℝ} (hr0 : 0 < r0) : IsCompletelyMonotoneOn (AutoresearchKeldysh
    .keldysh r0) := by
```

Reading: Yukawa = Riesz × $(\exp \circ \sqrt{\ })$; dipole = 2× (difference of translates of $t^{-1/2}$); Keldysh = Laplace mixture (eight-line proof, omitted).

Warning : what the library does not shorten

The phrase "one-line corollaries" is only half true. The P3 paper *measured* the dependency closure of the three results H1, H2, H3 taken together: it *increases* when one goes through the library. Only the *marginal* cost of a new potential becomes small. For Keldysh, the short proof still relies on the heavy bounds of `keldysh_data` (erfcx, integrability) from the original module: the weakened hypothesis is *available* but not *exercised*. Mathlib already contains a neighbouring notion, `AbsolutelyMonotoneOn` (without alternating signs); the library is not ready for Mathlib [12].

New consequences [Tier A], conditional for optimality. Soft-core Yukawa $g_\kappa(t+b^2)$ (translate), stretched exponential e^{-at^α}, screened dipolar exciton $2(g_\kappa(t)-g_\kappa(t+d^2))$, continuous mixtures of Yukawa potentials. An instructive example, the quadrupoles $+1,-2,+1$ in the layers $0,d,2d$ ($c=d^2$):

- the *field* of a quadrupole on a test charge, $f(t)-2f(t+c)+f(t+4c)$ with $f=t^{-1/2}$, is *not* CM: at $t=3$, $c=1$, it equals $\frac{1}{\sqrt{3}}-1+\frac{1}{\sqrt{7}}<0$ (`quadrupole_not_cm`);
- the *interaction* of two aligned quadrupoles counts the pairs ($1+4+1=6$ at $\Delta z=0$, $4\times(-2)=-8$ at $|\Delta z|=d$, $2\times 1=2$ at $|\Delta z|=2d$) and equals $2\big(3f(t)-4f(t+c)+f(t+4c)\big)$ [Tier C] for the counting; it *is* CM for every CM f (Exercise 2.7).

Lean 4 — `Applications.lean:172, 175-176` (proof omitted)

```
noncomputable def quadrupolarPair (f : ℝ → ℝ) (c t : ℝ) : ℝ := 2 * (3 * f t - 4 * f (t
    + c) + f (t + 4 * c))
```

```
theorem quadrupolarPair_cm {f : ℝ → ℝ} (hf : IsCompletelyMonotoneOn f) {c : ℝ} (hc : 0
    ≤ c) :
  IsCompletelyMonotoneOn (quadrupolarPair f c) := by
```

For these new potentials, no finiteness of the optimal energy is proved: for slowly decaying tails, the optimality inequality may read $\top \le \top$.

2.5 A neighbouring problem: the Sandier–Serfaty conjecture

Warning : status

The Sandier–Serfaty conjecture is not proved in this book. A manuscript in the openai/math release claims a proof of it; this manuscript is AI-generated and not peer reviewed. What we have done — reproduce a certificate, audit a reduction, write a Lean statement, formalise a small part — constitutes neither a proof, nor a verification, nor a peer review.

2.5.1 What the conjecture says

Physics and biology : from vortices to lattices [Tier C]

In a type-II superconductor in a magnetic field, the field penetrates in the form of quantised flux tubes, the *vortices*, which arrange themselves in a lattice (the Abrikosov lattice, observed to be triangular). Sandier and Serfaty derived from the Ginzburg–Landau model a *renormalised energy* W that governs the position of the vortices at next order [22] **[Tier L]**. The link with vortex physics is a **[Tier C]** gloss here.

Here is the formulation of the manuscript, which follows Sandier–Serfaty **[Tier L]**. Let $\Lambda \subset \mathbb{R}^2$ be locally finite, of bounded density, and $\nu_\Lambda = \sum_{p\in\Lambda} \delta_p$. A locally integrable field $E : \mathbb{R}^2 \to \mathbb{R}^2$ is *admissible* ($E \in \mathcal{A}_1$) if, in the sense of distributions,

$$\operatorname{div} E = 2\pi(\nu_\Lambda - 1), \qquad \operatorname{curl} E = 0.$$

Each point of Λ is a unit charge; the "-1" is a uniform neutralising background. Near a charge, $|E|^2 \sim 1/|x-p|^2$ is not integrable: one removes disks of radius η and compensates,

$$W(E,\chi_R) = \lim_{\eta\to 0}\Big[\tfrac{1}{2}\int_{\mathbb{R}^2\setminus\bigcup_p B(p,\eta)} \chi_R|E|^2 + \pi\ln\eta \sum_{p\in\Lambda}\chi_R(p)\Big], \qquad W(E) = \limsup_{R\to\infty}\frac{W(E,\chi_R)}{4R^2},$$

where χ_R is a smooth cutoff supported in the square $[-R,R]^2$. The sign $+\pi\ln\eta$ is the right one: $\frac{1}{2}\int_{\eta<|x-p|<\rho}|x-p|^{-2}dx = \pi\ln\rho - \pi\ln\eta$.

- **Conjecture** (Sandier–Serfaty 2012, Conjecture 1): $W(E) \geq W(E_\triangle)$ for every $E \in \mathcal{A}_1$, where $E_\triangle$ is the periodic field of the triangular lattice $\Lambda_\triangle$ of covolume one **[Tier L]**.
- Sandier–Serfaty prove it *among lattices* (their Theorem 2) and write that minimisation in general remains open **[Tier L]** [22].
- Bétermin–Sandier show that it is equivalent to a value conjectured by Brauchart–Hardin–Saff for the next-order term of the optimal logarithmic energy of N points on the sphere [23, 24] **[Tier L]**, and give $W(E_\triangle) = \pi\ln\frac{2\sqrt{2}\,\pi}{\sqrt{3}\,\Gamma(1/3)^3} \approx -4.1504128$ **[Tier L]**.

What the openai/math manuscript claims. A proof [14], via (1) a reduction from the plane to square tori $\mathbb{R}^2/(\sqrt{n}\,\mathbb{Z}^2)$, (2) a trial potential glued on the Voronoi sectors, *exactly calibrated* to the triangular lattice, and (3) a finite certificate in exact interval arithmetic (144-bit integers), organised in twelve stages. This manuscript is AI-generated and not peer reviewed.

2.5.2 P1: reproduce and audit, without proving

- **Reproduction of the certificate [Tier X].** The unchanged C++ program was recompiled and run on a different toolchain and a different machine (g++ 11.4); running the certificate took about 478 s; the final line is `OK` and the 22 printed integer pairs are identical, 22 out of 22, to the original record. *Negative controls*: three one-line weakened versions; two are rejected by the program's assertions, **the third one passes** (the outcomes of the last two had been predicted in writing before the computations finished). The reproduction shows that the program behaves as recorded; it tests none of the analytic bounds it is supposed to certify.
- **Audit of the reduction [Tier C].** An AI agent checked Sections 2 to 5 of the manuscript step by step (31 numbered steps and 14 lines of the certificate step): *no step was classified*

as a gap, with declared limits (the spline pieces outside $[p, R_0]$, the values of the grid step and trust in the execution are not independently checked). This is neither a proof nor a peer review.

- **Locked Lean statement.** The conjecture is written as a Lean proposition (`Prop`), **not proved**, whose statement region is locked by fingerprint (`statement_lock.py`).

Lean 4 — `formal_cert/Programmes/SandierSerfatyStatement.lean:153-160` (a definition, not a theorem)

```
def SandierSerfatyStatement : Prop :=
  (∃ χ, IsSquareCutoffFamily χ) ∧
    ∃ E₀ : Plane → Plane, Admissible triLattice E₀ ∧ IsPeriodic triLattice E₀ ∧
      ∀ χ : ℝ → Plane → ℝ, IsSquareCutoffFamily χ →
        ∃ w₀ : ℝ → ℝ, (∀ R : ℝ, 1 < R → HasLocalEnergy triLattice E₀ (χ R) (w₀ R)) ∧
          WSq w₀ ≠ ⊤ ∧ WSq w₀ ≠ ⊥ ∧
          ∀ (Λ : Set Plane) (E : Plane → Plane) (w : ℝ → ℝ), Admissible Λ E →
            (∀ R : ℝ, 1 < R → HasLocalEnergy Λ E (χ R) (w R)) → WSq w₀ ≤ WSq w
```

Only ten non-degeneracy lemmas are **[Tier A without replay]**: the admissible class is nonempty (the field $E = -\pi x$, with no charge), the background constant 2π matters (with 0 or -2π in its place, this field is no longer admissible), the cutoff classes are nonempty. Warning: the witness of `competitor_nonvacuous` is this field $-\pi x$, whose energy is $+\infty$ **[Tier C]**; it shows that the hypotheses on a competitor are *satisfiable*, not that the inequality constrains a competitor. The existence of the limit in η is *assumed* (`HasLocalEnergy`) and the triangular minimiser is an existential witness: these are declared deviations from the manuscript's formulation [19]. P1 paper: doi:10.5281/zenodo.23653827.

2.5.3 A1: formalise a part, and measure the rest

Interval arithmetic in Lean. An interval is a pair of integers (u, v) read as $[u/Q, v/Q]$ with $Q = 2^{144}$. Products are rounded *outwards*: lower bound by $\lfloor \cdot/Q \rfloor$, upper bound by $\lceil \cdot/Q \rceil$, so that the true real result is never lost. Toy example with $Q = 100$: $[1.20, 1.30] \times [0.50, 0.60]$, i.e. $(120, 130) \times (50, 60)$, has endpoint products $6000, 7200, 6500, 7800$, whence $(\lfloor 6000/100 \rfloor, \lceil 7800/100 \rceil) = (60, 78) = [0.60, 0.78]$ [20].

Two verification levels (an axis distinct from Tiers A/L/C/X): **L1** (kernel re-execution): the Lean transcription of a stage, evaluated by the kernel with `decide +kernel` (and not `native_decide`, which adds an axiom and trusts compiled code), reproduces bit for bit the integers printed by the C++. **L2** (real statement): a Lean theorem asserts that a *real number* belongs to the interval.

Result (saved logs)	Value	Tier
Stages complete at L1	4 of 12 (`init_constants`, `bcons/eps`, `calcLam`, `calcK`)	**[Tier A without replay]** for `init_constants`, `bcons/eps`; **[Tier C]** for `calcLam`, `calcK` (no independent recompilation)
of which conditional	`calcLam`, `calcK` (tables as premises), `bcons` (coefficients c_m)	
Stages complete at L2	1 of 12 (`init_constants`: $\ln 2, \pi, p, R_0, a$ real)	**[Tier A without replay]**
Share of interval operations re-executed	1 685 129 of 1 238 829 689, i.e. 0.136 %	**[Tier X]**
Planted false statements rejected	11	**[Tier X]**
Projected cost of the rest at L1	854 to 2 240 core-hours	**[Tier X]**

The 22 printed pairs and the final assertions of the C++ are in addition checked as arithmetic on the exact endpoints (level "E", weaker), except the bound test of the `core_test` stage; the table-separation test is checked only window by window. Independent verification: the audit by a second AI agent rebuilt 13 Lean modules from source, but *not* `CalcK.lean`, `CalcLam.lean` nor 28 chunk files; the L1 results of `calcLam` and `calcK` therefore have not had a producer $\neq$ verifier pass **[Tier C]** [20]. The heaviest remaining work is level L2 of the analytic content (radial functions, lattice coefficients, quadrature remainders), then 30 further steps of the reduction: a project of several months **[Tier C]** [20]. A1 paper: doi:10.5281/zenodo.23656804.

The lemma `signed_endpoints` [Tier A without replay]. It is the only lemma of the reduction proved completely. It handles a Voronoi sector whose foot of the perpendicular falls outside its edge. Setting: a kernel $G(d, s)$, even in s and locally integrable for $d \geq d_0$, and its primitive $g_d(x) = \int_0^x G(d, s)\, ds$.

Manuscript	Lean (`SignedEndpoints.lean:79-85`, verbatim)
$K < 0$, $d_0 > 0$; $G(d, \cdot)$ even	`(hK : K < 0) (hd₀ : 0 < d₀), heven : ∀ d, d₀ ≤ d → ∀ s, G d (-s) = G d s`
locally integrable	`hint : ∀ d, d₀ ≤ d → ∀ a b : ℝ, IntervalIntegrable (G d ) volume a b`
$2g_d(x) \geq K$ for $x \geq 0$	`hpos : ∀ d, d₀ ≤ d → ∀ x, 0 ≤ x → K ≤ 2 * gPrim G d x`
$\int_0^\infty \max\{0, -G(d, s)\}\, ds \leq -K$	`hneg : ∀ d, d₀ ≤ d → ∫⁻ s in Ioi 0, ENNReal.ofReal (-G d s) ≤ ENNReal.ofReal (-K)`
conclusion for $\lvert\alpha\rvert, \lvert\beta\rvert < \frac{\pi}{2}$, $\alpha+\beta > 0$	`K ≤ gPrim G d (d * Real.tan α) + gPrim G d (d * Real.tan β)`

Lean 4 — `formal_cert/Programmes/SandierSerfaty/SignedEndpoints.lean:33, 79-85` (proof omitted)

```
noncomputable def gPrim (G : ℝ → ℝ → ℝ) (d x : ℝ) : ℝ := ∫ s in (0 : ℝ)..x, G d s
```

```
theorem signed_endpoints (G : ℝ → ℝ → ℝ) (K d₀ : ℝ) (hK : K < 0) (hd₀ : 0 < d₀)
    (heven : ∀ d, d₀ ≤ d → ∀ s, G d (-s) = G d s)
    (hint : ∀ d, d₀ ≤ d → ∀ a b : ℝ, IntervalIntegrable (G d) volume a b)
    (hpos : ∀ d, d₀ ≤ d → ∀ x, 0 ≤ x → K ≤ 2 * gPrim G d x)
    (hneg : ∀ d, d₀ ≤ d → ∫⁻ s in Ioi 0, ENNReal.ofReal (-G d s) ≤ ENNReal.ofReal (-K)
    ) :
```

```
∀ d, d₀ ≤ d → ∀ α β : ℝ, |α| < π / 2 → |β| < π / 2 → 0 < α + β →
  K ≤ gPrim G d (d * Real.tan α) + gPrim G d (d * Real.tan β) := by
```

The integral of the negative part is a `lintegral` ($\int^-$, with values in $[0,+\infty]$): an infinite integral *violates* the hypothesis, instead of producing the junk value 0 of the Bochner integral.

Theorem 2.21 (signed endpoints)**.** *Under the hypotheses of the table, for $d \geq d_0$ and $|\alpha|, |\beta| < \frac{\pi}{2}$ with $\alpha + \beta > 0$: $g_d(d\tan\alpha) + g_d(d\tan\beta) \geq K$.* ***[Tier A without replay]***

Human proof. (1) Oddness. By evenness of $G(d,\cdot)$ and the change of variable $s \mapsto -s$, $g_d(-x) = -g_d(x)$.

(2) Two nonnegative angles. Then $x = d\tan\alpha \geq 0$ and $y = d\tan\beta \geq 0$; the hypothesis $2g_d \geq K$ gives $g_d(x) \geq K/2$ and $g_d(y) \geq K/2$; add them.

(3) Opposite signs, $\alpha < 0$. Then $\alpha + \beta > 0$ forces $\beta > -\alpha > 0$. Set $u = -d\tan\alpha > 0$ and $v = d\tan\beta$; since tan is strictly increasing on $(-\frac{\pi}{2}, \frac{\pi}{2})$, we have $u < v$. By oddness, $g_d(d\tan\alpha) + g_d(d\tan\beta) = g_d(v) - g_d(u) = \int_u^v G(d,s)\,ds$.

(4) Central estimate. $\int_u^v G \geq -\int_u^v G^- \geq -\int_0^\infty G^- \geq K$, where $G^- = \max(0,-G)$, by the hypothesis on the negative part and $0 < u$.

(5) The case $\beta < 0$ is symmetric; both negative is excluded by $\alpha + \beta > 0$. □

Without $\alpha + \beta > 0$, the statement is false (Exercise 2.9); the refutation is checked by the kernel (`ctl_signed_without_sum_false`), and the same integral kernel $G \equiv 1$ satisfies all the other hypotheses (`ctl_signed_hyps`), so that the theorem is not vacuous.

Warning : what `signed_endpoints` does not do

The lemma is stated for an *abstract* G. The concrete G requires the radial data f_0, e of Section 4 of the manuscript, which are not defined in Lean, and nothing derives its two hypotheses from the certificate: the lemma **is not connected to the certificate**. The preregistered stopping criterion applies: **no formal proof of the conjecture is claimed**. The faithfulness of the statement to the manuscript was compared by an agent, not by a human.

2.5.4 B3: why the method does not carry over as is to dimension three

In dimension three, the analogue is the Wigner conjecture: the body-centred cubic (bcc) lattice would minimise the jellium energy among *all* configurations; it is known to be optimal only among Bravais lattices of covolume one, for the Coulomb Born energy (Muratov–Novaga, Theorem 1, p. 4 [25]) **[Tier L]**. The B3 study (doi:10.5281/zenodo.23655002) [21] splits the planar proof into 13 steps and classifies each one: dimension-independent, transferable, or specifically planar. The incidence count (Euler's identity per cell) is *not* the obstacle.

The obstacle: exact calibration. The whole planar proof is a lower bound via a trial potential H: $W(h) = F(H) + \frac{1}{8\pi}\int |\nabla(h-H)|^2 \geq F(H)$ (dual identity, **[Tier C]**), and the bound is optimal only if H *equals* the true Green's function h at the optimal lattice. In dimension two this is possible: in each sector, $\cos 6m\varphi = T_{6m}(d/t)$ (Chebyshev polynomial, $t = d/\cos\varphi$), so $h_\triangle$ depends on only two variables and belongs to the trial class. In dimension three, the restriction of the bcc Green's function to a circle of fixed radius on a face carries a $\cos 6\psi$ mode (hexagonal faces) or a $\cos 4\psi$ mode (square faces) **[Tier X]**, whereas an axial H is constant there **[Tier C]**.

Result, with its exact scope. For the radial, axial and "intrinsic hierarchical" trial classes, a projection argument **[Tier C]**, evaluated in floating point, forces a deficit: the certifiable constant is at most $W_{\mathrm{bcc}} - \delta$ with $\delta \geq 8.751 \times 10^{-5}$ (hierarchical) or $\delta \geq 1.058 \times 10^{-4}$ (axial, 1.18 times the fcc–bcc gap 8.969×10^{-5}); these values are **[Tier X]**. For richer classes (which read data from neighbouring faces), the negative result rests only on a counting heuristic (about 12 conditions for 3 multipliers, against 3 for 3 in dimension two) **[Tier C]** and **is not a theorem**. Nothing is proved about bcc.

2.6 Exercises

Exercises marked **(Lean)** point to a kernel-checked reference solution in the repository **[Tier A]**; the solutions sketched here are **[Tier C]**.

Exercise 2.1 (dipolar control)**.** Let $d > 0$ and $h_d(t) = t^{-1/2} - 2(t + d^2)^{-1/2}$. Compute $h_d(d^2)$ and show that it is strictly negative. Deduce that h_d is not completely monotone. Interpret physically: which pair-counting error would produce h_d?

Exercise 2.2 (a signed mixture)**.** Let $\sigma(t) = e^{-t} - 2e^{-2t}$. Show that $\sigma(\ln \frac{3}{2}) = -\frac{2}{9}$. Why does Theorem 2.16 require $m \geq 0$? Exhibit a signed measure whose Laplace transform is σ.

Exercise 2.3 (the tail bound)**.** Prove Lemma 2.10 again using the mean value theorem applied to $x \mapsto x^{-1/2}$ on $[t, t + c]$. Which constant does one obtain? Compare with $\frac{c}{2}$.

Exercise 2.4 (the heart of the Yukawa tail bound)**.** (a) Show that $e^x \geq x^2/2$ for $x \geq 0$. (b) Deduce $g_\kappa(t) \leq \frac{2}{\kappa^2} t^{-3/2}$. (c) Show that, in the lattice A, the number of points in the annulus $k \leq |a| < k+1$ is $O(k)$ (hint: Voronoi cells of area 1, disjoint, contained in an enlarged annulus), and conclude that $\sum_{a \neq 0} |a|^{-3} < \infty$. (d) Why does the argument fail for $\kappa = 0$?

Exercise 2.5 (moments of the Keldysh mixture)**.** Show that for $u \geq 0$, $c > 0$ and $N \in \mathbb{N}$, $u^N e^{-cu} \leq N!/c^N$. Deduce that $a \mapsto (4a)^{-n} e^{-c/(4a)} \frac{w(a)}{2a}$ is bounded on $(0, 1]$. (Lean: `pow_mul_exp_neg_le`.)

Exercise 2.6 (the $\top \leq \top$ trap)**.** Show that for a planar lattice, $\sum_{0<|a|\leq R} |a|^{-1}$ grows like R. Deduce why one *expects* $\mathrm{latticeEnergy}(\mathcal{K}_{r_0}) = +\infty$. Why does this not make the theorem `triangular_keldysh_optimal` false, but only vacuous?

Exercise 2.7 (quadrupoles)**.** Let f be CM and $c \geq 0$, $g(t) = f(t) - f(t + c)$ and $\Delta_a h(t) = h(t) - h(t + a)$. (a) Check that $3f(t) - 4f(t + c) + f(t + 4c) = \Delta_c g(t) + \Delta_{2c} g(t) + \Delta_{3c} g(t)$. (b) Deduce, using item (v) of Section 2.4, that the interaction of two quadrupoles is CM. (c) Show that the field of a single quadrupole, $t^{-1/2} - 2(t + 1)^{-1/2} + (t + 4)^{-1/2}$, is negative at $t = 3$, using $\sqrt{3} > 1.7$ and $\sqrt{7} > 2.6$.

Exercise 2.8 (CM or Bernstein?)**.** Show that: (a) $1/t$ is CM and positive but is not a Bernstein function; (b) $e^{-1/t}$ is not CM (hint: the sign of its derivative); (c) t^α is a Bernstein function for $0 \leq \alpha \leq 1$ but not for $\alpha = 2$. Explain why (b) shows that, in Theorem 2.19, the inner function cannot be just any positive function.

Exercise 2.9 (the hypothesis $\alpha + \beta > 0$)**.** With $G \equiv 1$, $K = -1$, $d_0 = d = 1$: check the four hypotheses of `signed_endpoints`, then show that for $\alpha = \beta = -\frac{\pi}{4}$ one has $g_1(\tan \alpha) + g_1(\tan \beta) = -2 < K$. Which step of the proof of Theorem 2.21 uses $\alpha + \beta > 0$?

Exercise 2.10 (interval arithmetic)**.** With $Q = 100$, encode $a = [-1.25, 0.33]$ as $(-125, 33)$ and $b = [0.50, 0.67]$ as $(50, 67)$. (a) Compute the four endpoint products and the outward-rounded product interval. (b) With Euclidean integer division $\lfloor \cdot \rfloor$, show that $x \mapsto -\lfloor -x/Q \rfloor$ computes $\lceil x/Q \rceil$. (c) Why would rounding to nearest be incorrect for a certificate?

Exercise 2.11 ((**Lean**) a negative control)**.** Starting from `h_neg_at : h d (d ^ 2) < 0` (for $d > 0$), prove in Lean $\neg$ `AdmissiblePotential (h d)`. Hint: `intro H`, then use the second component `H.2.1` of the definition. Reference solution: `formal_cert/Autoresearch/DipolarExciton.lean:113-116` **[Tier A]**.

Exercise 2.12 ((**Lean**) corollaries of the library)**.** (a) State in Lean that $t \mapsto \mathrm{riesz}\, s\, t - \mathrm{riesz}\, s\, (t+c)$ is CM for $s, c \geq 0$ and prove it in one line using `riesz_cm` and `sub_comp_add_const`. Reference solution: `Corollaries.lean:33-35` **[Tier A]**. (b) Same question for $e^{-at^{\alpha}}$, $a \geq 0$, $0 \leq \alpha \leq 1$, using `exp_neg_comp` and `IsBernsteinOn.rpow`. Reference solution: `Corollaries.lean:59-61` **[Tier A]**. (c) Why is the condition `(fun _ ht => Real.rpow_pos_of_pos ht α)` required?

Exercise 2.13 ((†) University exercise: Laplace representations of Riesz and dipolar potentials)**.** (a) For $s > 0$ and $t > 0$, show that $t^{-s} = \frac{1}{\Gamma(s)} \int_0^{\infty} u^{s-1} e^{-tu}\, du$ (substitute $v = tu$ in the integral defining $\Gamma(s)$). Deduce directly, by differentiation under the integral sign, that t^{-s} is completely monotone, and recover the formula $(-1)^n \frac{d^n}{dt^n} t^{-s} = s(s+1)\cdots(s+n-1)\, t^{-s-n}$. (b) Using (a) with $s = \frac{1}{2}$ and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, prove the identity of the remark "Bernstein reading": for $c \geq 0$, $t^{-1/2} - (t+c)^{-1/2} = \pi^{-1/2} \int_0^{\infty} e^{-ut}\, (1 - e^{-uc})\, u^{-1/2}\, du$, and deduce once more that $\mathrm{shiftDiff}_{1,c}$ is CM. (c) More generally, show that if $f(t) = \int_{[0,\infty)} e^{-tu}\, \mu(du)$ for a positive measure μ, then for $c \geq 0$ the function $f(t) - f(t+c)$ is the Laplace transform of the positive measure $(1 - e^{-cu})\, \mu(du)$.

Exercise 2.14 ((†) University exercise: Jacobi's theta identity and a square-lattice Gaussian sum)**.** Take for granted the Poisson summation formula on $\mathbb{R}$: for a Schwartz function ϕ, $\sum_{n \in \mathbb{Z}} \phi(n) = \sum_{k \in \mathbb{Z}} \widehat{\phi}(k)$ with $\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi}\, dx$. (a) Show that the Fourier transform of $x \mapsto e^{-\pi a x^2}$ $(a > 0)$ is $\xi \mapsto a^{-1/2} e^{-\pi \xi^2 / a}$. (b) Deduce Jacobi's identity $\theta(a) = a^{-1/2}\, \theta(1/a)$, where $\theta(a) = \sum_{n \in \mathbb{Z}} e^{-\pi a n^2}$. (c) Show that $\sum_{x \in \mathbb{Z}^2} e^{-\pi a \|x\|^2} = \theta(a)^2$, and hence $\sum_{x \in \mathbb{Z}^2} e^{-\pi a \|x\|^2} = a^{-1} \sum_{x \in \mathbb{Z}^2} e^{-\pi \|x\|^2 / a}$. (d) Deduce that the Gaussian lattice energy of the square lattice, $E(a) = \sum_{x \in \mathbb{Z}^2 \setminus \{0\}} e^{-\pi a \|x\|^2}$, satisfies $E(a) = a^{-1} - 1 + O(a^{-1} e^{-\pi/a})$ as $a \to 0^+$, and $E(a) \sim 4 e^{-\pi a}$ as $a \to +\infty$. *Answer.* d) $E(a) = a^{-1} \theta(1/a)^2 - 1$ and $\theta(1/a)^2 = 1 + 4e^{-\pi/a} + O(e^{-2\pi/a})$.

Chapter 3

The quantum surprise: zero-point energy and lattices

Chapter 1 told a success story: for every *completely monotone* repulsive interaction $V(r) = g(r^2)$, the triangular lattice minimizes the *static* energy $\frac{1}{2}\sum_{R\neq 0} V(|R|)$ among all density-one configurations (the openai/math upstream statement, **[Tier A]** conditional on upstream theorems that were produced by an AI and are not peer reviewed; see also the textbook version 1 [11]). This chapter asks the following question, a very natural one for a physicist: *what becomes of this result once quantum mechanics is taken into account?* A real crystal is never at rest. Even at absolute zero, each atom vibrates about its site, and this vibration costs energy, the *zero-point* energy.

We shall see that the answer is a surprise. For a large family of Gaussians, the triangular lattice is not a local minimum of the zero-point energy: it is a local *maximum* (**[Tier X]**). We then give a *certified* version, in which the logic and the arithmetic are checked by the Lean kernel (**[Tier A]**) and the integrals are enclosed by ball arithmetic (**[Tier C]**). Next we ask whether this produces a genuine phase transition (the answer is negative, within the self-consistent harmonic approximation). We close with a cousin of the same mathematics: quantum fluids, with the Feynman–Bijl bound and the Bogoliubov structure factor.

Warning : what this chapter claims, and what it does not

All results were produced by AI agents (Claude) under the author's direction, and audited by a separate AI agent. **No human has audited them.** Nothing in this chapter depends on a theorem of the openai/math upstream. Montgomery's theorem on Gaussians, used for the static energy, is cited second hand. The statements about the zero-point energy are Tier X or Tier C. Only the logic, the algebra and the rational arithmetic are Tier A. The **[Tier A]** label always attaches to the Lean statement, never to its physical reading.

3.1 The harmonic crystal from first principles

3.1.1 The quantum harmonic oscillator and its minimal energy

A particle of mass m in the potential $\frac{1}{2}m\omega^2 x^2$ is described by a wave function $\psi : \mathbb{R} \to \mathbb{C}$ of norm $\int |\psi|^2 = 1$. Its mean energy is

$$\langle H\rangle_\psi = \frac{\hbar^2}{2m}\int_{\mathbb{R}} |\psi'(x)|^2\,dx + \frac{m\omega^2}{2}\int_{\mathbb{R}} x^2|\psi(x)|^2\,dx.$$

In classical mechanics the minimal energy is 0 (the particle at rest at the bottom of the well). In quantum mechanics this is no longer possible.

Proposition 3.1 (Zero-point energy of the oscillator)**.** *For every normalized C^1 function ψ such that $x\psi$ and ψ' are square integrable and $x|\psi(x)|^2 \to 0$ as $x \to \pm\infty$, one has $\langle H\rangle_\psi \geq \frac{1}{2}\hbar\omega$, with equality if and only if $\psi(x) = c\,e^{-m\omega x^2/2\hbar}$ with $|c|$ fixed by the normalization.* ***[Tier C]***

Human proof. Set $\phi = m\omega\, x\psi + \hbar\psi'$. Expanding $\int |\phi|^2 \geq 0$:

$$0 \leq \int |\phi|^2 = m^2\omega^2 \int x^2|\psi|^2 + \hbar^2 \int |\psi'|^2 + 2m\omega\hbar\,\mathrm{Re}\int x\,\overline{\psi}\,\psi'.$$

Now $2\,\mathrm{Re}(\overline{\psi}\psi') = (|\psi|^2)'$, so integrating by parts (the boundary term $[x|\psi|^2]$ vanishes by hypothesis) $2\,\mathrm{Re}\int x\overline{\psi}\psi' = \int x(|\psi|^2)' = -\int|\psi|^2 = -1$. What remains is

$$0 \leq m^2\omega^2 \int x^2|\psi|^2 + \hbar^2 \int |\psi'|^2 - m\omega\hbar = 2m\langle H\rangle_\psi - m\omega\hbar,$$

that is, $\langle H\rangle_\psi \geq \hbar\omega/2$. Equality forces $\phi = 0$, i.e. the linear differential equation $\psi' = -(m\omega/\hbar)x\psi$, whose solutions are exactly $c\,e^{-m\omega x^2/2\hbar}$. □

The energy $\frac{1}{2}\hbar\omega$ is the *zero-point energy*. Physically, confining the particle (small $\int x^2|\psi|^2$) forces ψ to vary rapidly (large $\int|\psi'|^2$): this is the uncertainty principle.

Several independent modes [Tier C]. If a system has n coordinates $q_1, \dots, q_n$ and a Hamiltonian $H = \sum_j \left(\frac{p_j^2}{2m} + \frac{1}{2}m\omega_j^2 q_j^2\right)$ (*decoupled* modes), the product wave function $\prod_j \psi_j(q_j)$ shows that the ground-state energy is $\sum_j \frac{1}{2}\hbar\omega_j$. For a crystal, the first step is therefore to find the coordinates that decouple the vibrations: these are the *phonons*.

3.1.2 Warm-up: the linear chain

Take N masses m on a circle, with equilibrium positions na, joined to their neighbours by springs of stiffness K, and let u_n be the displacement of mass n (with $u_{n+N} = u_n$). The potential energy is $\frac{K}{2}\sum_n (u_{n+1} - u_n)^2$ and Newton's equation reads $m\ddot{u}_n = K(u_{n+1} + u_{n-1} - 2u_n)$.

We look for solutions $u_n = e^{i(kna - \omega t)}$ with $k \in \frac{2\pi}{Na}\mathbb{Z}$ (periodicity condition). Substituting: $-m\omega^2 = K(e^{ika} + e^{-ika} - 2) = -2K(1 - \cos ka) = -4K\sin^2(ka/2)$, hence

$$\omega(k) = 2\sqrt{K/m}\,\big|\sin(ka/2)\big|.$$

The discrete Fourier transform decouples the N modes exactly (it diagonalizes the circulant matrix of the potential energy), so the zero-point energy per particle is

$$\frac{1}{N}\sum_k \frac{\hbar\omega(k)}{2} \xrightarrow[N\to\infty]{} \frac{\hbar}{2}\cdot\frac{1}{2\pi}\int_{-\pi}^{\pi} 2\sqrt{K/m}\,|\sin(\theta/2)|\,d\theta = \frac{2}{\pi}\,\hbar\sqrt{K/m}$$

(a Riemann sum, $\theta = ka$). Three ideas are worth keeping. **(i)** The frequencies are the square roots of the eigenvalues of a k-dependent matrix. **(ii)** As $k \to 0$, $\omega \approx a\sqrt{K/m}\,|k|$: this is sound, and the frequency vanishes like $|k|$ (a conical singularity). **(iii)** The zero-point energy is an *average over the Brillouin zone* $\theta \in [-\pi, \pi]$. All of this is **[Tier C]**.

3.1.3 The planar lattice: dynamical matrix and zero-point energy

A planar Bravais lattice of covolume one is determined, up to rotation, by its *shape* $\tau = x + iy$ ($y > 0$), with basis $a_1 = (1/\sqrt{y}, 0)$, $a_2 = (x/\sqrt{y}, \sqrt{y})$. The squared length of the vector $ma_1 + na_2$ is

$$Q_{mn}(\tau) = \frac{(m + nx)^2 + n^2y^2}{y}.$$

The triangular lattice corresponds to $\rho = e^{i\pi/3} = \frac{1}{2} + i\frac{\sqrt{3}}{2}$, the square lattice to $\tau = i$.

The Hessian of a radial potential. For $V(R) = g(|R|^2)$ with $R \in \mathbb{R}^2$, one has $\partial_i V = 2g'(|R|^2)R_i$ and then

$$\operatorname{Hess} V(R) = 2g'(|R|^2)\, I + 4g''(|R|^2)\, RR^{\mathsf{T}}. \tag{3.1}$$

For the Gaussian $g(u) = e^{-\pi a u}$: $g' = -\pi a\, g$, $g'' = \pi^2 a^2 g$, hence $\operatorname{Hess} V(R) = (-2\pi a\, I + 4\pi^2 a^2 RR^{\mathsf{T}})e^{-\pi a|R|^2}$ **[Tier C]**.

The equations of motion. Place a particle of mass $m = 1$ at each point R of the lattice L, and let u_R be its small displacement. To second order, the potential energy is $\frac{1}{4}\sum_{R\neq R'}(u_R - u_{R'})^{\mathsf{T}}\operatorname{Hess} V(R - R')(u_R - u_{R'})$ and Newton gives $\ddot{u}_0 = -\sum_{R\neq 0}\operatorname{Hess} V(R)\,(u_0 - u_R)$. With the plane wave $u_R = e\, e^{i(\mathbf{k}\cdot R - \omega t)}$ ($e \in \mathbb{C}^2$), one obtains $\omega^2 e = \sum_{R\neq 0}(1 - e^{i\mathbf{k}\cdot R})\operatorname{Hess} V(R)\, e$. The terms R and $-R$ have the same Hessian, so one may symmetrize: $1 - e^{i\mathbf{k}\cdot R}$ becomes $1 - \cos \mathbf{k}\cdot R$.

Definition 3.2 (Dynamical matrix, stability, zero-point energy)**.** For $\mathbf{k} \in \mathbb{R}^2$, the *dynamical matrix* is the real symmetric 2×2 matrix

$$D(\mathbf{k}) = \sum_{R\in L\setminus\{0\}} \big(1 - \cos \mathbf{k}\cdot R\big)\operatorname{Hess} V(R).$$

The lattice is *dynamically stable* if $D(\mathbf{k})$ is positive semidefinite for every $\mathbf{k}$. Its two frequencies are then $\omega_j(\mathbf{k}) = \sqrt{\lambda_j(D(\mathbf{k}))}$, and the *harmonic zero-point energy* per particle (units $\hbar = m = 1$) is

$$E_{\mathrm{zp}}(\tau) = \frac{1}{2}\Big\langle \omega_1 + \omega_2 \Big\rangle_{\mathrm{ZB}} = \frac{1}{2}\Big\langle \operatorname{tr} D(\mathbf{k})^{1/2} \Big\rangle_{\mathrm{ZB}},$$

where $\langle\cdot\rangle_{\mathrm{ZB}}$ is the average over a cell of the dual lattice (the Brillouin zone). **[Tier C]**

The formula $\operatorname{tr} D^{1/2}$ is the heart of this chapter: the zero-point energy is the trace of the *square root* of a lattice sum. Unlike the static energy, it is not a sum over pairs. For a positive semidefinite 2×2 matrix, $\sqrt{\lambda_1} + \sqrt{\lambda_2} = \sqrt{\operatorname{tr} D + 2\sqrt{\det D}}$ (square both sides), a formula used by the numerical codes.

A sum rule [Tier C]. The average of $\cos \mathbf{k}\cdot R$ over the zone vanishes for $R \neq 0$. Hence $\langle \operatorname{tr} D\rangle_{\mathrm{ZB}} = \sum_{R\neq 0}\Delta V(R)$, and by Cauchy–Schwarz and then Jensen (Exercise 3.4),

$$E_{\mathrm{zp}} \le \sqrt{\langle \operatorname{tr} D\rangle_{\mathrm{ZB}}/2} \qquad \text{(Einstein bound)}. \tag{3.2}$$

For the Gaussian, Poisson summation gives $\langle \operatorname{tr} D\rangle = 4\pi a - (4\pi^2/a)\, S(1/a)$ with $S(\alpha) = \sum_{P\in L}|P|^2 e^{-\pi\alpha|P|^2}$ **[Tier C]**. Luo and Wei prove that, for $\alpha \ge 1$, the minimum of $S(\alpha;\cdot)$ over density-one lattices is attained at the hexagonal lattice [35] **[Tier L]**; hence, for $a \le 1$, the triangular lattice *maximizes* $\langle \operatorname{tr} D\rangle$ **[Tier C]**. An upper bound that is maximal at ρ, however, says nothing about the extrema of E_{zp} itself.

University remark. **[Tier L]** The Poisson summation formula in $\mathbb{R}^d$ states that for a lattice $L \subset \mathbb{R}^d$ of covolume $|L|$, with dual lattice $L^* = \{P : P\cdot R \in \mathbb{Z}\ \forall R \in L\}$, and a Schwartz function f, $\sum_{R\in L} f(R + x) = |L|^{-1}\sum_{P\in L^*}\hat{f}(P)\, e^{2i\pi P\cdot x}$, with $\hat{f}(\xi) = \int f(x)e^{-2i\pi\xi\cdot x}dx$ (standard). Applied to $f = \Delta V$ with $V = e^{-\pi a|x|^2}$, it rewrites the direct sum for $\langle \operatorname{tr} D\rangle$ as a sum over the reciprocal lattice, which converges faster when a is small; this is the "Poisson sum in reciprocal space" used by the numerical implementations below.

3.1.4 The role of the triangular point: symmetry and isotropy

The energy of a lattice depends only on the lattice up to rotation; it is therefore invariant under $\tau \mapsto \tau + 1$ and $\tau \mapsto -1/\tau$, hence under the whole group $SL_2(\mathbb{Z})$. The element $\gamma(\tau) = (\tau - 1)/\tau$

fixes ρ, since $\rho^2 = \rho - 1$, and its derivative at ρ is $1/\rho^2 = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$: this is the rotation by $-120°$. If F is C^2 and invariant, its gradient at ρ is fixed by this rotation, hence zero, and its Hessian commutes with it, hence is scalar. Thus

$$F(\tau) - F(\rho) = c\,\frac{|\tau - \rho|^2}{y_\rho^2} + O(|\tau - \rho|^3), \qquad c = \tfrac{1}{4}\Delta F(\rho), \quad \Delta = y^2(\partial_x^2 + \partial_y^2). \tag{3.3}$$

The sign of c decides everything: local minimum if $c > 0$, local maximum if $c < 0$. The two linear-algebra facts are checked by the kernel **[Tier A without replay]**; the step "invariance $\Rightarrow$ zero gradient, scalar Hessian" (chain rule, regularity of E_{zp}) is **[Tier C]**.

Lean 4 : `formal_cert/Programmes/ZeroPoint/Basic.lean`, lines 91–92, 168–171 and 201

```
theorem lap_Q (m n x y : ℝ) (hy : y ≠ 0) :
    y ^ 2 * (2 * n ^ 2 / y + 2 * (m + n * x) ^ 2 / y ^ 3) = 2 * Q m n x y := by
theorem rot120_comm_symm (p q r s : ℝ) (hs : s ^ 2 = 3 / 4)
    (h11 : -1 / 2 * p + -s * q = p * (-1 / 2) + q * s)
    (h12 : -1 / 2 * q + -s * r = p * -s + q * (-1 / 2)) :
    q = 0 ∧ r = p := by
theorem rho_sq : rho ^ 2 = rho - 1 := by
```

Reading: `lap_Q` says that $\Delta Q = 2Q$ for the length function (one recognizes $y^2(Q_{xx} + Q_{yy})$, with $Q_{xx} = 2n^2/y$ and $Q_{yy} = 2(m + nx)^2/y^3$). `rot120_comm_symm` says that a symmetric matrix $\left(\begin{smallmatrix} p & q \\ q & r \end{smallmatrix}\right)$ commuting with the rotation by 120° (with entries $-\frac{1}{2}$ and $\pm s$, $s^2 = \frac{3}{4}$) satisfies $q = 0$ and $r = p$. The module contains 19 printed declarations, all with axioms in {`propext`, `Classical.choice`, `Quot.sound`}, and trap controls (a `sorry` and a planted axiom) caught by `#print axioms` [29]. **[Tier A without replay]**

The static energy. For the Gaussian, Montgomery's theorem [5] asserts that the triangular lattice minimizes the static energy $E_{\mathrm{st}} = \frac{1}{2}\sum_{R\neq 0} e^{-\pi a|R|^2}$ among lattices of covolume one, for every $a > 0$ (**[Tier L]**, cited second hand through Deng–Luo [34]). For a potential of amplitude ε and range l, and a mass m, the total harmonic energy is

$$E(\tau) = E_{\mathrm{st}}(\tau) + \hbar_{\mathrm{eff}}\, E_{\mathrm{zp}}(\tau), \qquad \hbar_{\mathrm{eff}} = \frac{\hbar}{\sqrt{m\varepsilon l^2}}, \tag{3.4}$$

in units of ε and l **[Tier C]**. The parameter $\hbar_{\mathrm{eff}}$ measures "how quantum" the system is.

3.2 The QU-1 conjecture and its numerical refutation

3.2.1 The statement, and the rules of the game

Definition 3.3 (Conjecture QU-1)**.** For every completely monotone function g, the triangular lattice minimizes $E_{\mathrm{zp}}(\tau)$ among planar Bravais lattices of covolume one.

It was the natural "quantum companion" of static universal optimality. It was *preregistered* with a prior probability of 0.6 of surviving a numerical falsification gate (G1), an attempted proof for Gaussians (G2) to be launched *only if* G1 survived [29]. The key rule of the Elenchus method applies here: *try to refute before trying to prove.*

3.2.2 The counterexample [Tier X]

For the Gaussian $g(t) = e^{-\pi t}$ ($a = 1$), Table 3.1 compares the triangular lattice and the rhombic lattice $\tau = 0.5 + 0.966025\,i$.

Table 3.1: Gaussian $a = 1$: zero-point energy (spectral zone rule, quadrature error of about 4.4×10^{-15}), stability margin $\min \lambda/|\mathbf{k}|^2$ over the zone, and exact long-wavelength limit **[Tier X]** [29].

lattice τ	E_{zp}	ΔE_{zp}	$\min \lambda/\lvert\mathbf{k}\rvert^2$	long wavelength
triangular ρ	1.5033523522	0	0.03222	0.23826
$0.5 + 0.966025\,i$	1.4902783183	-1.3074×10^{-2}	0.03493	0.10466
i (square, negative control)	–	–	-0.1475 (unstable)	-0.1475

The rhombic lattice is dynamically stable with a clear margin, and its zero-point energy is *lower* than that of the triangular lattice. Four implementations of two formulas (direct sum, and Poisson sum in reciprocal space) agree; a short independent verification program (`human_check.py`, whose core is reproduced in the paper [29]) returns 1.490278318 and 1.503352352. Of the 625 shapes sampled for this potential, 65 are stable, and the 64 stable non-triangular shapes all have lower energy: around ρ, the triangular lattice is a *local maximum* within the stable region **[Tier X]**.

Twelve potentials. The conjecture is numerically refuted for 8 of the 12 completely monotone potentials tested (Table 3.2). The `riesz:4` case is weak: it rests on a single distant competitor, whose stability margin is 0.08 % of the triangular value, and there the triangular point is a local minimum. The four survivors are *not* proofs: a sample of 567 shapes and two dense paths are not an exhaustive search.

Table 3.2: QU-1 verdicts and nature of the triangular point (sign of c, interval over 12 directions) **[Tier X]** [29]. `gauss:`a is $e^{-\pi a r^2}$, `yukawa:`κ is $e^{-\kappa r}/r$, `riesz:`s is r^{-s}, `dipolar:`d is the dipolar exciton.

potential	QU-1 verdict	c at ρ	nature of ρ
`gauss:0.25`	survives	$+0.19 \ldots +0.19$	min
`gauss:0.5`	**refuted**	$-0.47 \ldots -0.45$	max
`gauss:1`	**refuted**	$-0.98 \ldots -0.95$	max
`gauss:2`	survives	$+0.48 \ldots +0.51$	min
`gauss:4`	survives	$+0.31 \ldots +0.32$	min
`yukawa:1`	**refuted**	$-0.40 \ldots -0.38$	max
`yukawa:2`	**refuted**	$-0.32 \ldots -0.31$	max
`dipolar:0.1`	**refuted**	$-0.33 \ldots -0.32$	max
`dipolar:0.5`	**refuted**	$-0.61 \ldots -0.58$	max
`riesz:3`	**refuted**	$-0.31 \ldots -0.29$	max
`riesz:4`	**refuted** (weak)	$+0.34 \ldots +0.36$	min
`riesz:6`	survives	$+3.38 \ldots +3.45$	min

3.2.3 The Gaussian family: an interval of local maxima [Tier X]

For $V = e^{-\pi a r^2}$, the curvature $c(a) = \frac{1}{4}\Delta E_{\mathrm{zp}}(\rho)$ was computed for 23 values of $a \in [0.1, 8]$, and its sign changes were then located by bisection (Table 3.3).

Proposition 3.4 (Numerical result, P2)**.** *For the Gaussian $e^{-\pi a r^2}$, among the values $0.15 \le a \le 4$ explored, the curvature $c(a)$ is negative exactly on the numerical interval $a \in (0.37668, 1.58148)$; on this interval, the triangular lattice is a* strict local maximum *of E_{zp}.* ***[Tier X]***

The endpoints are numerical roots obtained by bisection; the scan does not exclude further sign changes between its points, and at $c = 0$ the second-order test is inconclusive. The restricted

Table 3.3: Curvature $c(a)$ of E_{zp} at the triangular point, Gaussian $e^{-\pi a r^2}$ (excerpt) **[Tier X]** [29].

a	0.25	0.35	0.4	0.5	0.8	1	1.2	1.4	1.6	2
$c(a)$	$+0.1882$	$+0.0796$	-0.0799	-0.4605	-1.0707	-0.9620	-0.6598	-0.3035	$+0.0285$	$+0.4945$

version QU-1′ (strict local minimum outside the interval, strict local maximum inside) is a conjecture supported at Tier X, not proved.

3.2.4 The mechanism: concavity of the trace of the square root

Why does the analogy with the static case fail? The static energy is *linear* in V. A completely monotone potential is a positive mixture of Gaussians (Bernstein's theorem, Chapter 1), and a property that holds for each Gaussian passes to the mixture. The zero-point energy, by contrast, contains $\operatorname{tr}\sqrt{D}$: it is not linear in V, and the mixture argument does not apply **[Tier C]**.

More precisely, at fixed $\mathbf{k}$, write $\Phi(D) = \operatorname{tr} D^{1/2}$. Differentiating $\Phi(D(\tau))$ twice gives two terms:

$$\Delta E_{\mathrm{zp}} = \underbrace{\tfrac{1}{2}\Big\langle y^2 \textstyle\sum_i \operatorname{tr}\big(\Phi'(D)\,\partial_i^2 D\big)\Big\rangle}_{\text{Lin (linear response)}} + \underbrace{\tfrac{1}{2}\Big\langle y^2 \textstyle\sum_i \Phi''(D)[\partial_i D, \partial_i D]\Big\rangle}_{\text{Conc} \le 0}.$$

The second term is nonpositive by *concavity* of the square root on positive semidefinite matrices (the matrix analogue of the concavity of $\sqrt{\cdot}$ on $\mathbb{R}_+$) **[Tier C]**. The first has no controlled sign. For $a = 1$, the numerical decomposition gives $\Delta E_{\mathrm{zp}}(\rho) = -3.848 = -0.799 + (-3.048)$: concavity dominates **[Tier X]**. For $a = 0.25$ and $a = 2$, on the contrary, a large term Lin > 0 wins.

Physics and biology : where is the energy gained?

For the rhombic competitor at $a = 1$, the gain comes mainly from a *softening of the lower branch near the zone boundary*. The mean squared frequency decreases only slightly: the Einstein bound (3.2) explains only 24 % of the gain; the remaining 76 % come from the Jensen gap, that is, from a more spread-out spectrum. Since $\sqrt{\cdot}$ is concave, spreading the spectrum at fixed quadratic mean lowers the mean frequency. These numbers are **[Tier X]**, and their reading is **[Tier C]**.

Warning : the competitors are not free of constraints

The triangular lattice has the lowest static energy; a rhombic competitor is therefore held in its shape only by a constraint. QU-1 compares a functional over imposed shapes, and it is refuted as such. The physical question (total energy, each lattice at its equilibrium shape) is that of Section 3.4.

3.3 B2: a certified result

A floating-point computation, however robust, proves nothing. The B2 programme sought a *certified* statement at $a = 1$ [30]. The preregistered statement asked for a certified two-sided enclosure of the Laplacian $\Delta E_{\mathrm{zp}}(\rho) < 0$, verified in Lean. **This is not what was obtained.** What was obtained is a statement that is weaker in one respect (a one-sided, local bound along a single path) and different in another (it uses neither isotropy nor criticality).

The setting. Start from the triangular lattice L_ρ (one vector on the x-axis) and shear it by the symmetric matrix of determinant one

$$P_t = \begin{pmatrix} 1 & t \\ t & 1+t^2 \end{pmatrix}, \qquad L_t = P_t L_\rho, \qquad E(t) = \frac{1}{2} \int_{[-\frac{1}{2},\frac{1}{2}]^2} \operatorname{tr} D_t(\kappa)^{1/2}\, d\kappa,$$

with $V = e^{-\pi r^2}$, and with the lattice labels and the variable κ (reduced coordinate) held fixed as t varies. The shape of the lattice L_t is $\tau(t) = \rho + \sqrt{3}\,t + O(t^2)$ **[Tier C]**.

Theorem 3.5 (B2). *For every t with $0 < |t| \le 10^{-5}$, one has $E(t) < E(0)$. Consequently the triangular lattice is not a local minimum of the harmonic zero-point energy for $e^{-\pi r^2}$.* ***[Tier C]*** *overall; logic and arithmetic* ***[Tier A without replay]****; integrals* ***[Tier C]*** *(ball arithmetic).*

The proof consists of five building blocks. We give them in order, each with its tier.

3.3.1 Block 1: a trace inequality for 2×2 matrices

Lemma 3.6 (Concavity of $\operatorname{tr}\sqrt{\cdot}$ in tangent form). *Let X be real symmetric positive definite, S real symmetric, and $P = X^{-1}$. Then*

$$2 \operatorname{tr} S \le \operatorname{tr} X + \operatorname{tr}(X^{-1} S^2), \qquad \text{with equality if } S = X.$$

[Tier A without replay] *for 2×2 matrices (the only case used);* ***[Tier C]*** *in arbitrary size.*

Human proof. Scalar case first. For $x > 0$ and $s \in \mathbb{R}$: $x + s^2/x - 2s = (s-x)^2/x \ge 0$. This is exactly the AM–GM inequality, and it is also the tangent line to the concave function $\sqrt{\cdot}$.

Matrix case. Set $M = S - X$, which is symmetric. Expanding and using $PX = XP = I$:

$$\operatorname{tr}(PM^2) = \operatorname{tr}(PS^2) - \operatorname{tr}(PSX) - \operatorname{tr}(PXS) + \operatorname{tr}(PX^2) = \operatorname{tr}(PS^2) - 2\operatorname{tr} S + \operatorname{tr} X,$$

since $\operatorname{tr}(PSX) = \operatorname{tr}(XPS) = \operatorname{tr} S$ (cyclic invariance of the trace), $\operatorname{tr}(PXS) = \operatorname{tr} S$ and $PX^2 = X$. It therefore suffices to show $\operatorname{tr}(PM^2) \ge 0$. If m_1, m_2 are the columns of M, then $M^2 = MM^{\mathsf{T}}$ and $\operatorname{tr}(PMM^{\mathsf{T}}) = \operatorname{tr}(M^{\mathsf{T}}PM) = m_1^{\mathsf{T}} P m_1 + m_2^{\mathsf{T}} P m_2 \ge 0$, since P is positive definite. For equality, $S = X$ gives $M = 0$.

Positivity of the quadratic form, by hand. If $P = \left(\begin{smallmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{smallmatrix}\right)$ with $p_{11} > 0$ and $\det P \ge 0$, then

$$p_{11}\,(p_{11}a^2 + 2p_{12}ab + p_{22}b^2) = (p_{11}a + p_{12}b)^2 + (p_{11}p_{22} - p_{12}^2)\,b^2 \ge 0,$$

and one divides by $p_{11} > 0$. □

Lean 4 : `formal_cert/Programmes/ZeroPointCert/Basic.lean`, lines 49–50 and 64–78

```
theorem quad_nonneg (p11 p12 p22 a b : ℝ) (hp : 0 < p11) (hdet : 0 ≤ p11 * p22 - p12 ^
    2) :
    0 ≤ p11 * a ^ 2 + 2 * p12 * a * b + p22 * b ^ 2 := by

theorem two_trace_le_of_inverse (x11 x12 x22 p11 p12 p22 s11 s12 s22 : ℝ)
    (hp : 0 < p11) (hdet : 0 ≤ p11 * p22 - p12 ^ 2)
    (h11 : x11 * p11 + x12 * p12 = 1) (h12 : x11 * p12 + x12 * p22 = 0)
    (h21 : x12 * p11 + x22 * p12 = 0) (h22 : x12 * p12 + x22 * p22 = 1) :
    2 * (s11 + s22) ≤ (x11 + x22) +
      (p11 * (s11 ^ 2 + s12 ^ 2) + 2 * p12 * (s12 * (s11 + s22)) + p22 * (s12 ^ 2 + s22
    ^ 2)) := by
  have key : (x11 + x22) +
      (p11 * (s11 ^ 2 + s12 ^ 2) + 2 * p12 * (s12 * (s11 + s22)) + p22 * (s12 ^ 2 + s22
    ^ 2)) - 2 * (s11 + s22)
      = (p11 * (s11 - x11) ^ 2 + 2 * p12 * (s11 - x11) * (s12 - x12) + p22 * (s12 - x12)
```

```
    ^ 2)
      + (p11 * (s12 - x12) ^ 2 + 2 * p12 * (s12 - x12) * (s22 - x22) + p22 * (s22 -
   x22) ^ 2) := by
   linear_combination (2 * s11 - x11) * h11 + (2 * s12 - x12) * h12 + (2 * s12 - x12) *
    h21
     + (2 * s22 - x22) * h22
have q1 := quad_nonneg p11 p12 p22 (s11 - x11) (s12 - x12) hp hdet
have q2 := quad_nonneg p11 p12 p22 (s12 - x12) (s22 - x22) hp hdet
linarith
```

Side by side. The four hypotheses `h11`–`h22` are the four entries of $XP = I$. The right-hand side is $\operatorname{tr} X + \operatorname{tr}(PS^2)$ written entry by entry for symmetric S. The identity `key` is our "$\operatorname{tr}(PM^2) = \ldots$", written as a sum of two quadratic forms evaluated on the two columns $(s_{11} - x_{11}, s_{12} - x_{12})$ and $(s_{12} - x_{12}, s_{22} - x_{22})$ of M; the tactic `linear_combination` checks that it follows from the four equations of $XP = I$ with the indicated coefficients. The two calls to `quad_nonneg` are the column-by-column positivity. The Lean statement is restricted to 2×2 matrices, which suffices: there are two phonon branches.

3.3.2 Block 2: the linear-response majorant [Tier C]

At each $\kappa \neq 0$, set $S = D_t(\kappa)^{1/2}$ (so $S^2 = D_t$) and $X = D_0(\kappa)^{1/2}$, which is positive definite because the triangular lattice is dynamically stable. Lemma 3.6 gives

$$\operatorname{tr} D_t^{1/2} \leq \tfrac{1}{2}\big(\operatorname{tr} X + \operatorname{tr}(X^{-1} D_t)\big).$$

Integrating: $E(t) \leq U(t) := \frac{1}{4}\int \big[\operatorname{tr} X + \operatorname{tr}(X^{-1} D_t)\big]\, d\kappa$, and $U(0) = E(0)$ by the equality case. **Why this majorant?** U is *affine* in D_t: its dependence on t is a single absolutely convergent lattice sum against the fixed matrix X^{-1}, with no square root of the perturbed matrix. The price is that U bounds E only from above. The neglected term, $\frac{1}{2}\operatorname{tr}(X^{-1}(S - X)^2) \geq 0$, can only help the conclusion "E is smaller". Lean sees only the pointwise inequality; the integration, the integrability and the identification $S = D_t^{1/2}$ are human "glue".

3.3.3 Block 3: the window bound

Since U is affine in D_t, its symmetric second difference is an integral of the second difference of the Hessians $H_n(t) = \operatorname{Hess} V(P_t R_n)$:

$$U(t) + U(-t) - 2U(0) = \tfrac{1}{4}\int \operatorname{tr}\Big(X^{-1} \textstyle\sum_n (1 - \cos 2\pi\kappa \cdot n)\big[H_n(t) + H_n(-t) - 2H_n(0)\big]\Big)\, d\kappa.$$

Entry by entry, Taylor's theorem with the Lagrange remainder gives $H_n(t) + H_n(-t) - 2H_n(0) = t^2 H_n''(\xi)$ with $|\xi| < t$. Splitting $H_n''(\xi) = H_n''(0) + (H_n''(\xi) - H_n''(0))$, one obtains for $0 < t \leq t_0 = 10^{-5}$:

$$\frac{U(t) + U(-t) - 2U(0)}{t^2} \leq \underbrace{U''(0)}_{\text{Upp0}} + \underbrace{\Big(\int \operatorname{tr} X^{-1}\Big)}_{\text{trW}} \cdot \underbrace{\sum_n \sum_{\text{entries}} \sup_{|\xi| \leq t_0} \big|H_n''(\xi) - H_n''(0)\big|}_{\text{esum}}. \tag{3.5}$$

The reduction is **[Tier C]**. The derivatives along the path, $q(t) = |P_t R_0|^2$, q', q'', and the product rule for one Hessian entry, are checked by the kernel (`hasDerivAt_qS`, `hasDerivAt_gaussS'`, `hasDerivAt_entry'`) **[Tier A without replay]**.

The numbers [Tier C] (ball arithmetic). The three quantities in (3.5) were rigorously enclosed in Arb ball arithmetic [36] (python-flint 0.6.0, 96 bits, outward rounding): the zone is cut into four triangles with vertex Γ (Duffy change of variables, which removes the conical singularity), then into 206 468 cells, with no failure, in 448 s on one core. The tails of the lattice sum are bounded explicitly (Table 3.4). This code is *not* checked by the kernel: an error in the bounds would not be detected by the Lean theorems.

Table 3.4: Enclosures certified by ball arithmetic, $a = 1$, shear path, $t_0 = 10^{-5}$ **[Tier C]** [30].

quantity	enclosure / bound	meaning
$U''(0) = \mathrm{Upp0}$	$[-1.947713, -1.900168]$	second derivative of the majorant
$\int \mathrm{tr}\, X^{-1}\, d\kappa$	$[1.825520, 1.833534]$	weight of the perturbation
esum	≤ 0.0319642	tails included
$B_{hi} = \mathrm{Upp0}_{hi} + \mathrm{trW}_{hi}\, \mathrm{esum}_{hi}$	-1.841560	upper bound in (3.5)
$\lambda_{\min}(B_0)$	≥ 0.918889	stability of L_ρ on each cell
$\|B_t - B_0\|$	≤ 0.007528	margin $0.911361 > 0$ for $\|t\| \le 10^{-5}$

3.3.4 Block 4: the rational certificate

The upper ends of the balls are copied into Lean as rationals, and the kernel checks their sign.

Lean 4 : `formal_cert/Programmes/ZeroPointCert/Basic.lean`, lines 127–144 (comments omitted)

```
noncomputable def t0 : ℝ := 1 / 100000
noncomputable def Upp0_hi : ℝ := -475041875131 / 250000000000
noncomputable def trW_hi : ℝ := 366706720019 / 200000000000
noncomputable def esum_hi : ℝ := 31964165579 / 1000000000000
noncomputable def lmin_lo : ℝ := 183777872659 / 200000000000
noncomputable def dB_hi : ℝ := 7528036227 / 1000000000000

theorem t0_pos : 0 < t0 := by norm_num [t0]

theorem cert_B_neg : Upp0_hi + trW_hi * esum_hi < 0 := by
  norm_num [Upp0_hi, trW_hi, esum_hi]

theorem cert_stability : 0 < lmin_lo - dB_hi := by
  norm_num [lmin_lo, dB_hi]
```

Checking by hand. $\mathrm{Upp0}_{hi} = -1.900167500524$, $\mathrm{trW}_{hi} = 1.833533600095$, $\mathrm{esum}_{hi} = 0.031964165579$; the product is about 0.0586074 and the sum about $-1.8415601 < 0$. `norm_num` performs the same computation in exact rational arithmetic. The LeanFlow tool `lean.certify` independently verified both inequalities, rewritten in integers **[Tier A without replay]**. What the kernel does *not* check: that these rationals really bound the integrals (that is the **[Tier C]** content of Block 3). That they coincide with the results file is checked by a script.

3.3.5 Block 5: the logic, then the assembly

Lemma 3.7 (A concave majorant excludes a local minimum)**.** *Let $E, U : \mathbb{R} \to \mathbb{R}$, $t_0 > 0$ and $B < 0$. Assume $E \le U$ on $[-t_0, t_0]$, $U(0) = E(0)$, and $U(t) + U(-t) - 2U(0) \le t^2 B$ for $0 < t \le t_0$. Then 0 is not a local minimum of E. If moreover E is even, then $E(t) < E(0)$ for every $0 < t \le t_0$.* ***[Tier A without replay]***

Human proof. For $0 < t \le t_0$: $E(t) + E(-t) \le U(t) + U(-t) \le 2U(0) + t^2 B < 2U(0) = 2E(0)$. Hence at least one of the numbers $E(t)$, $E(-t)$ is strictly less than $E(0)$. If 0 were a local

minimum, there would exist $\varepsilon > 0$ such that $E \geq E(0)$ on $(-\varepsilon, \varepsilon)$; the choice $t = \min(\varepsilon/2, t_0)$ gives a contradiction. If E is even, $2E(t) < 2E(0)$. □

Lean 4 : `formal_cert/Programmes/ZeroPointCert/Basic.lean`, lines 100–102 and 157–162

```
theorem not_isLocalMin_of_majorant {E U : ℝ → ℝ} {t0 B : ℝ} (ht0 : 0 < t0) (hB : B <
    0)
    (hEU : ∀ t, |t| ≤ t0 → E t ≤ U t) (h0 : U 0 = E 0)
    (hU : ∀ t, 0 < t → t ≤ t0 → U t + U (-t) - 2 * U 0 ≤ t ^ 2 * B) : ¬ IsLocalMin E
    0 := by

theorem triangular_not_localMin_of_certificate (E U : ℝ → ℝ) (Upp0 trW esum : ℝ)
    (hEU : ∀ t, |t| ≤ t0 → E t ≤ U t) (h0 : U 0 = E 0)
    (hwin : ∀ t, 0 < t → t ≤ t0 → U t + U (-t) - 2 * U 0 ≤ t ^ 2 * (Upp0 + trW * esum
    ))
    (hU : Upp0 ≤ Upp0_hi) (hW0 : 0 ≤ trW) (hW : trW ≤ trW_hi) (he0 : 0 ≤ esum) (he :
    esum ≤ esum_hi)
    (hsym : ∀ t, E (-t) = E t) :
    ¬ IsLocalMin E 0 ∧ ∀ t, 0 < t → t ≤ t0 → E t < E 0 := by
```

How to read the assembled theorem. Each hypothesis is a fact that the analytic work must supply: `hEU` and `h0` are Block 2, `hwin` is Block 3, `hU`, `hW`, `he` are the ball enclosures, `hsym` is the mirror symmetry $E(-t) = E(t)$ (the reflection $\mathrm{diag}(1, -1)$ preserves L_ρ and conjugates P_t into P_{-t}, a **[Tier C]** argument). The Lean theorem is thus the *exact* statement of what the Tier C parts must guarantee, not a proof of the physics. The heading of Theorem 3.5 is **[Tier C]** overall, because it rests on Tier C blocks; only its logic and its arithmetic are Tier A.

Why this excludes a local minimum of the shape problem [Tier C]. Since $\tau(t) = \rho + \sqrt{3}\,t + O(t^2)$, small $t \neq 0$ give lattices of covolume one, with shape different from ρ, arbitrarily close to ρ, and each has strictly lower zero-point energy. A local minimum at ρ would require $E_{\mathrm{zp}} \geq E_{\mathrm{zp}}(\rho)$ on a neighbourhood, which this path contradicts. Neither criticality nor isotropy at ρ is used.

Corollary, under additional hypotheses [Tier C]. If moreover E is C^2, E_{zp} is C^2 near ρ, ρ is critical and the Hessian is isotropic (Section 3.1.4), then $E''(0) \leq -1.84156$ and $\Delta E_{\mathrm{zp}}(\rho) \leq -0.92078 < 0$. This is compatible with the numerical value -3.848 of P2 **[Tier X]**, and much weaker: the method bounds E'' by U'' from one side only.

3.3.6 The controls

- **Negative control** $a = 2$ (where P2 finds a local minimum): one obtains $U''(0) \in [+12.69365, +12.75701]$, the certificate is *not* produced; LeanFlow rejects it (`false-statement`) and the file `RejectA2.lean` fails to compile (rc 1). This shows that the method does not always say "not a minimum"; it does not show that it is complete. **[Tier C]** and **[Tier A without replay]**.

- **Tampered certificate** (sign of $\mathrm{Upp0}_{hi}$ flipped): `RejectTampered.lean` fails (rc 1) **[Tier A without replay]**.

- **Planted `sorry` and axiom**: `Controls.lean` compiles with rc 0, but `#print axioms` displays `sorryAx` and `ProgZeroPointCert.ctl_cheat`. The return code is not the gate; the list of axioms is **[Tier A without replay]**.

- **Non-preregistered by-product,** $a = 1/2$: $B_{hi} = -0.918357 < 0$, same conclusion **[Tier C]**.

Warning : exact scope of Tier A here

The module `ZeroPointCert/Basic.lean` has 18 printed declarations, all with axioms exactly {`propext`, `Classical.choice`, `Quot.sound`}; it never imports `Mathlib` as a whole. A search for `skipKernelTC`, `addDecl`, `native_decide`, `implemented_by`, `sorry` and `axiom` in the files `ZeroPoint/Basic.lean`, `ZeroPointCert/Basic.lean` and `FeynmanBijl/Basic.lean` finds nothing (checked while this chapter was written). On the other hand, no `leanchecker` kernel replay (the third check in the definition of Tier A) is recorded in the folders of these three modules (P2, B2 and P6). They appear in the replay plan of 2026-10-26 (`results/kernel_replay_2026-10-26/`), but that replay had not completed for them when this book was assembled: their statements therefore carry the **[Tier A without replay]** label, which rests on compilation by the kernel, the list of axioms and the search above. The eight `LEDGER_UNAUDITED_TIER_A` flags of the ledger remain open, because no person has certified that the Lean statements say what their glosses say.

Work in progress. Two follow-ups preregistered on 2026-10-26 are *in progress*, and none of their results is used here: N1 (a classification theorem for Gaussians: is the triangular lattice a local minimum of E_{zp} if and only if a lies outside an interval (a_-, a_+)?) and N2 (extending the B2 certificate to physical potentials: Yukawa, dipolar exciton, Riesz).

3.4 B1: is there a genuine phase transition?

The static energy favours the triangular lattice; the zero-point energy (for a in the interval) disfavours it. By (3.4), the sum $E_{\mathrm{st}} + \hbar_{\mathrm{eff}} E_{\mathrm{zp}}$ must therefore change minimizer as $\hbar_{\mathrm{eff}}$ grows. The B1 programme [31] asked: *where, and is it physical?* Its preregistered stopping rule read: "if the transition lies beyond melting or disappears beyond the harmonic level, report it (negative result)". **The rule fired: this is a negative result.** B1 contains no Lean statement.

3.4.1 The balance of curvatures

By (3.3) applied to both energies, the total curvature at ρ is $c(\hbar_{\mathrm{eff}}) = c_{\mathrm{st}} + \hbar_{\mathrm{eff}}\, c_{\mathrm{zp}}$. Montgomery gives $c_{\mathrm{st}} \geq 0$, and $c_{\mathrm{zp}} < 0$ on the interval of Proposition 3.4. The triangular lattice therefore ceases to be a (harmonic) local minimum at

$$\hbar^* = -\frac{c_{\mathrm{st}}}{c_{\mathrm{zp}}} \qquad \textbf{[Tier C]} \text{ (argument), } \textbf{[Tier X]} \text{ (numbers).}$$

At $a = 1$: $c_{\mathrm{st}} = 0.11913$, $c_{\mathrm{zp}} = -0.96198$, hence $\hbar^* = 0.1238$. On $a \in [0.4, 1.5]$, $\hbar^*$ is minimal (0.0877) at $a = 0.5$ and diverges at both ends of the interval, where $c_{\mathrm{zp}} \to 0$.

A first-order transition [Tier C] and [Tier X]. The stabilizer of ρ is the dihedral group D_3. In the disc coordinate $w = (\tau - \rho)/(\tau - \bar{\rho}) = re^{i\theta}$, an invariant expansion contains $|w|^2$ and $\mathrm{Re}\, w^3$, but not w^2: a *cubic term* $r^3 \cos 3\theta$ is allowed. This is the Landau scenario of a first-order transition: a state of *finite* distortion catches up with the triangular lattice before the curvature vanishes. Numerically, at $a = 1$, a stable oblique lattice $\tau \approx 0.30 + 0.96\, i$ reaches the triangular energy already at $\Lambda_1 = 0.0942 < \hbar^*$; the slopes measured along two mirror rays are equal and opposite (ratio -1.02), as the cubic term predicts. The winners lie on the boundary of the harmonic stability region.

3.4.2 The Lindemann criterion

A crystal melts when the root-mean-square amplitude of the vibrations becomes a sizeable fraction of the nearest-neighbour distance: this is Lindemann's *empirical rule*, not a theorem. In the harmonic theory at $T = 0$, $\langle |u|^2 \rangle = \hbar_{\text{eff}}\, U(a)$ with $U = \frac{1}{2}\langle \operatorname{tr} D^{-1/2} \rangle_{\text{ZB}}$ (finite in dimension two, because the $1/|\mathbf{k}|$ singularity is integrable), and the Lindemann ratio is $\gamma = \sqrt{\langle |u|^2 \rangle}/a_{nn}$ with $a_{nn}^2 = 2/\sqrt{3}$. Since $\gamma^2 \propto \hbar_{\text{eff}}$, the estimated melting coupling $\Lambda_L(\gamma_c) = \gamma_c^2 a_{nn}^2/U(a)$ is proportional to γ_c^2: an error of $\pm 10\,\%$ in γ_c multiplies it by 0.81 or 1.21 **[Tier C]**.

Quantum Monte Carlo simulations of two-dimensional bosonic crystals give, verbatim, "$\gamma = 0.230(6)$" for dipoles, "0.235(15) for Yukawa bosons", "0.24(1) for Coulomb bosons" and "a hard-disks $\gamma = 0.279$" [37] (**[Tier C]**, quotation checked verbatim; the citation gate admits only theorems at Tier L). Hence the band $\gamma_c \in [0.23, 0.28]$.

Reading [Tier X]. At the spinodal point $\hbar^*$, $\gamma \geq 0.2798$ for all tested a (0.313 at $a = 1$): at the top of the band or beyond. At the first-order point Λ_1, γ ranges from 0.245 to 0.273 for $a = 0.5$ to 1: *inside* the band. The Lindemann test alone is therefore **inconclusive**.

3.4.3 The self-consistent harmonic approximation (SCHA)

One replaces the bare dynamical matrix by an effective matrix D_{eff}, determined variationally, so that the potential is averaged over the quantum fluctuations themselves. The relative displacement $w_R = u_R - u_0$ is then Gaussian, with covariance $C_R = \hbar_{\text{eff}} \langle (1 - \cos \mathbf{k} \cdot R) D_{\text{eff}}^{-1/2} \rangle$. For a Gaussian potential everything is explicit: with $b = \pi a$, completing the square in the Gaussian integral,

$$\left\langle e^{-b|R+w|^2} \right\rangle_{w \sim \mathcal{N}(0,C)} = \det(I + 2bC)^{-1/2} \exp\!\left(-b\, R^{\mathsf{T}} (I + 2bC)^{-1} R\right) \qquad \textbf{[Tier C]} \qquad (3.6)$$

(Exercise 3.9). The fluctuations *smooth* the potential: this is the main anharmonic effect.

Result [Tier X]. In the SCHA, at $a = 1$ and $\hbar_{\text{eff}} = 0.15$, the Laplacian $\Delta E(\rho)$ equals $+0.117$ (local minimum), whereas the harmonic theory gives $\Delta E(\rho) = 4(c_{\text{st}} + \hbar_{\text{eff}} c_{\text{zp}}) = -0.101$ (no longer a minimum). The triangular lattice remains a local minimum and the lowest of the sampled stable lattices up to $\hbar_{\text{eff}} \approx 0.27$–$0.33$; there, the SCHA Lindemann ratio is 0.60 to 0.87, far beyond any melting estimate. Two implementations (Python and the Rust module `scha.rs` of rusty-SUNDIALS, commit `9e85e15`) agree to 8.9×10^{-16}; a separate AI auditor reproduced the verdict with its own code and a different quadrature.

Warning : what the negative result says exactly

The transition *disappears in the SCHA*. This is not a proof of physical absence: the SCHA is a Gaussian variational theory that omits the second-order cubic term, and the stiffening it finds is itself of that order. Path-integral Monte Carlo (PIMC) was **not** run: **BLOCKED** (it would require fixed-area box-shape moves, and finite size masks the signal up to $M \gtrsim 12$). Only Bravais lattices were compared. The interpretation "ordinary quantum melting comes first" is an inference from Tier X numbers, not a statement from the literature.

Rydberg-dressed atoms are not Gaussian [Tier C] and [Tier X]. The interaction of atoms dressed by a Rydberg state has the form of a "soft shoulder" $U_0/(1 + (r/R_c)^6)$ [40, 39]. Its planar Fourier transform takes negative values (relative minimum -0.0448 at $kR_c = 4.82$, a **[Tier X]** computation), whereas that of the Gaussian $e^{-\pi a r^2}$ is $a^{-1} e^{-\pi k^2/a} > 0$. Henkel, Nath and Pohl write that, because of the inner plateau ($r < R_c$), the momentum-space potential

has negative contributions around $k_{\mathrm{rot}} \sim 2\pi/R_c$, and they find for this potential a roton-type softening and the formation of a supersolid [39]. Prestipino et al. write, summarizing the classical literature on cluster crystals, that the Gaussian repulsion is "a marginal case" [38]. Both quotations were checked verbatim. The more general idea that cluster crystals require a negative Fourier component is reported only by an unverified secondary reading and supports no claim here. Rydberg dressing therefore does not realize the Gaussian model of this chapter.

University remark. **[Tier L]** Bochner's theorem (standard): a continuous function $f : \mathbb{R}^d \to \mathbb{R}$ is positive definite, i.e. $\sum_{j,l} \bar{c}_j c_l f(x_j - x_l) \geq 0$ for all finite families, if and only if f is the Fourier transform of a finite positive measure. In particular a continuous integrable radial function whose planar Fourier transform takes negative values is not positive definite (the soft shoulder above is an example). The Gaussian $e^{-\pi a|x|^2}$, whose transform $a^{-1}e^{-\pi|k|^2/a}$ is positive, is positive definite, and so is every completely monotone $g(|x|^2)$ with finite mixing measure, being a positive mixture of Gaussians (standard). Singular potentials such as $1/r$ fall outside the scope of this statement.

3.5 Quantum fluids: superfluid helium and condensates

A superfluid is not a crystal, but the same mathematics appears in it: an excitation energy, a square root, and an inequality between a measurable quantity (the static structure factor $S(k)$, obtained by diffraction) and the spectrum.

3.5.1 The Bogoliubov structure factor and its envelope

For a dilute Bose gas of mass m and sound velocity c, set $h_c = \hbar c$, $k_* = 2mc/\hbar$ and $\varepsilon_0(k) = \hbar^2 k^2/2m = h_c k^2/k_*$. The Bogoliubov dispersion relation and the corresponding structure factor are

$$\varepsilon_{\mathrm{B}}(k) = \sqrt{(h_c k)^2 + (h_c k^2/k_*)^2}, \qquad S_{\mathrm{B}}(k) = \frac{\varepsilon_0(k)}{\varepsilon_{\mathrm{B}}(k)}.$$

In Lean, S_{B} is a *definition*; the relation $\varepsilon_0/S_{\mathrm{B}} = \varepsilon_{\mathrm{B}}$ is therefore an algebraic identity, not an independent N-body derivation.

Proposition 3.8 (Bogoliubov envelope)**.** *For $h_c, k, k_* > 0$: $S_{\mathrm{B}}(k)^2 = k^2/(k_*^2 + k^2)$ and $S_{\mathrm{B}}(k) \leq \sqrt{k/2k_*}$, with equality if and only if $k = k_*$.* ***[Tier A without replay]***

Human proof. $S_{\mathrm{B}}^2 = \varepsilon_0^2/\varepsilon_{\mathrm{B}}^2 = \dfrac{h_c^2 k^4/k_*^2}{h_c^2 k^2 + h_c^2 k^4/k_*^2} = \dfrac{k^2}{k_*^2 + k^2}$ (h_c cancels). Both sides being positive, $S_{\mathrm{B}}^2 \leq k/2k_* \iff 2k_* k^2 \leq k(k_*^2 + k^2) \iff 0 \leq k(k - k_*)^2$, which is true, with equality if and only if $k = k_*$. One concludes by taking square roots, which are increasing on $[0, +\infty)$. □

Lean 4 : `formal_cert/Programmes/FeynmanBijl/Basic.lean`, lines 262–268 (comments omitted) and 314–319

```
noncomputable def epsFree (hc ks k : ℝ) : ℝ := hc * k ^ 2 / ks
noncomputable def epsBog (hc ks k : ℝ) : ℝ := Real.sqrt ((hc * k) ^ 2 + (hc * k ^ 2 /
    ks) ^ 2)
noncomputable def sBog (hc ks k : ℝ) : ℝ := epsFree hc ks k / epsBog hc ks k

theorem sBog_le_sqrt {hc ks k : ℝ} (hhc : 0 < hc) (hk : 0 < k) (hks : 0 < ks) :
    sBog hc ks k ≤ Real.sqrt (k / (2 * ks)) := by
  rw [← Real.sqrt_sq (sBog_pos hhc hk hks).le]
  apply Real.sqrt_le_sqrt
  rw [sBog_sq hhc hk hks, div_le_div_iff₀ (by positivity) (by positivity)]
  nlinarith [mul_nonneg hk.le (sq_nonneg (k - ks))]
```

Side by side. The first proof line writes $S_{\mathrm{B}} = \sqrt{S_{\mathrm{B}}^2}$, the second reduces to comparing squares, the third replaces S_{B}^2 by its closed form and clears denominators, and the last is our "$0 \le k(k-k_*)^2$", supplied as a hint to `nlinarith`. This result was first proved on another Lean stack [32], then re-proved on the frozen stack of this book [33].

3.5.2 The Feynman–Bijl bound, in linear algebra

The argument of Feynman and Bijl [45] (these original papers were not read; we follow the formalized version of [33]) bounds the energy of the lowest density excitation by $\hbar^2 k^2/(2mS(k))$.

The finite setting. Let $E = \mathbb{C}^n$ with the Hermitian inner product $\langle x, y\rangle$ (linear in y), H a Hermitian matrix, ψ an eigenvector, $H\psi = E_0\psi$ with E_0 real, and ρ an arbitrary matrix, with adjoint $\rho^\dagger$. Write $[A, B] = AB - BA$ and define the *excess*

$$\mathcal{A}(X) = \langle X\psi, HX\psi\rangle - E_0\langle X\psi, X\psi\rangle = \langle X\psi, (H - E_0)X\psi\rangle.$$

In physics, $\rho = \hat{\rho}_{\mathbf{k}} = \sum_j e^{-i\mathbf{k}\cdot\mathbf{r}_j}$, ψ is the ground state and $NS(k) = \|\rho\psi\|^2$.

Lemma 3.9 (Double commutator)**.** *For every matrix ρ: $\langle\psi, [\rho^\dagger, [H, \rho]]\psi\rangle = \mathcal{A}(\rho) + \mathcal{A}(\rho^\dagger)$.* ***[Tier A without replay]***

Human proof. Expand $[\rho^\dagger, [H, \rho]] = \rho^\dagger H\rho - \rho^\dagger\rho H - H\rho\rho^\dagger + \rho H\rho^\dagger$ and take the expectation of each term in ψ. (a) $\langle\psi, \rho^\dagger H\rho\psi\rangle = \langle\rho\psi, H\rho\psi\rangle$ (definition of the adjoint). (b) $\langle\psi, \rho^\dagger\rho H\psi\rangle = E_0\|\rho\psi\|^2$ since $H\psi = E_0\psi$. (c) $\langle\psi, H\rho\rho^\dagger\psi\rangle = \langle H\psi, \rho\rho^\dagger\psi\rangle = E_0\|\rho^\dagger\psi\|^2$ (H Hermitian, E_0 real). (d) $\langle\psi, \rho H\rho^\dagger\psi\rangle = \langle\rho^\dagger\psi, H\rho^\dagger\psi\rangle$. Grouping (a)−(b) and (d)−(c), one finds $\mathcal{A}(\rho) + \mathcal{A}(\rho^\dagger)$. □

Lemma 3.10 (Rayleigh quotient in a sector)**.** *Let (e_i) be an orthonormal basis of eigenvectors of H, $He_i = \lambda_i e_i$. For every $x \neq 0$, there exists i with $\langle e_i, x\rangle \neq 0$ and $\lambda_i \le \langle x, Hx\rangle/\|x\|^2$.* ***[Tier A without replay]***

Human proof. With $c_i = \langle e_i, x\rangle$: $\langle x, Hx\rangle = \sum_i |c_i|^2\lambda_i$ and $\|x\|^2 = \sum_i |c_i|^2 > 0$ (Parseval). The quotient is therefore an average of the λ_i weighted by $w_i = |c_i|^2 \ge 0$. If all the λ_i with weight $w_i > 0$ were strictly above the average $\bar{\lambda}$, one would have $\sum w_i\lambda_i > \bar{\lambda}\sum w_i = \sum w_i\lambda_i$, a contradiction. □

Theorem 3.11 (Feynman–Bijl bound, abstract form)**.** *Assume $N, m > 0$, $S = \|\rho\psi\|^2/N > 0$, the f-sum rule $\langle\psi, [\rho^\dagger, [H, \rho]]\psi\rangle = N\hbar^2k^2/m$ and the symmetry $\mathcal{A}(\rho^\dagger) = \mathcal{A}(\rho)$. Then there exists an eigenvector e_i with $\langle e_i, \rho\psi\rangle \neq 0$ and*

$$\lambda_i - E_0 \le \frac{\hbar^2k^2}{2mS}. \textbf{\textit{[Tier A without replay]}}$$

Human proof. By Lemma 3.9 and the symmetry, $2\mathcal{A}(\rho) = N\hbar^2k^2/m$. The trial state $\phi = \rho\psi/\|\rho\psi\|$ therefore has energy $\langle\phi, H\phi\rangle = E_0 + \mathcal{A}(\rho)/\|\rho\psi\|^2 = E_0 + \hbar^2k^2/(2mS)$. Lemma 3.10 applied to $x = \rho\psi$ provides e_i. □

Lean 4 : `formal_cert/Programmes/FeynmanBijl/Basic.lean`, lines 45–46, 57–58 and 189–194

```
noncomputable def excess (H : E →ℓ[ℂ] E) (E0 : ℝ) (ψ : E) (X : E →ℓ[ℂ] E) : ℂ :=
  ⟪X ψ, H (X ψ)⟫_ℂ - (E0 : ℂ) * ⟪X ψ, X ψ⟫_ℂ

theorem double_comm_expect (hH : H.IsSymmetric) (hψ : H ψ = (E0 : ℂ) • ψ) (ρ : E →ℓ[ℂ]
    E) :
```

```
    ⟪ψ, comm (adjoint ρ) (comm H ρ) ψ⟫_ℂ = excess H E0 ψ ρ + excess H E0 ψ (adjoint ρ)
     := by

theorem feynman_bijl_bound (hH : H.IsSymmetric) (hψ : H ψ = (E0 : ℂ) • ψ) (ρ : E →ℓ[ℂ]
     E)
    {N m hbar k S : ℝ} (hN : 0 < N) (hm : 0 < m) (hS : 0 < S) (hSdef : S = ‖ρ ψ‖ ^ 2 /
    N)
    (hf : ⟪ψ, comm (adjoint ρ) (comm H ρ) ψ⟫_ℂ = ((N * hbar ^ 2 * k ^ 2 / m : ℝ) : ℂ))
    (hsym : excess H E0 ψ (adjoint ρ) = excess H E0 ψ ρ) :
    ∃ i, ⟪hH.eigenvectorBasis rfl i, ρ ψ⟫_ℂ ≠ 0 ∧
      hH.eigenvalues rfl i - E0 ≤ hbar ^ 2 * k ^ 2 / (2 * m * S) := by
```

Here `E` $\to_\ell$ $[\mathbb{C}]$ `E` denotes the $\mathbb{C}$-linear maps of E (a finite-dimensional space) and $\langle\!\langle \cdot, \cdot \rangle\!\rangle_{\mathbb{C}}$ the Hermitian inner product. The four lines `h1`–`h4` of the Lean proof of `double_comm_expect` are exactly (a)–(d). The Feynman–Bijl module has 21 declarations whose axioms are on the whitelist; it was independently recompiled by a separate AI auditor, who recovered the same axiom lists and caught the planted controls as well as two mutations of its own [33].

Where does the f-sum rule come from? **[Tier C].** For N bosons with Hamiltonian $H = \sum_j p_j^2/2m + V(\mathbf{r}_1, \ldots, \mathbf{r}_N)$, only the kinetic term fails to commute with $\hat{\rho}_{\mathbf{k}}$, and the computation with $[x, p] = i\hbar$ gives the operator identity $[\hat{\rho}_{\mathbf{k}}^\dagger, [H, \hat{\rho}_{\mathbf{k}}]] = N\hbar^2 k^2/m$. But the relation $[x, p] = i\hbar$ *has no finite-dimensional realization*: the trace of a commutator vanishes, whereas that of $i\hbar I$ does not (Exercise 3.12). This is why the Lean theorem, set in finite dimension, takes the f-sum rule as a *hypothesis*. The same holds for the symmetry $\mathbf{k} \leftrightarrow -\mathbf{k}$. A kernel-checked counterexample shows that the symmetry cannot be removed from the halving step **[Tier A without replay]**, and a kernel-checked witness shows that all the hypotheses are jointly satisfiable and that the bound is attained **[Tier A without replay]**.

Link with Bogoliubov. For $S > 0$: $\varepsilon_0/S \le \varepsilon_{\mathrm{B}} \iff S_{\mathrm{B}} \le S$, and $\varepsilon_0/S_{\mathrm{B}} = \varepsilon_{\mathrm{B}}$ (`feynman_le_epsBog_iff`, `feynman_sBog_eq`) **[Tier A without replay]**. Hence the *one-sided diffraction test* **[Tier C]**: if a measured $S(k)$ exceeds $\sqrt{k/2k_*}$, it is not the S_{B} of a homogeneous Bogoliubov gas with the same sound velocity. A point below the envelope confirms nothing.

3.5.3 The test on tabulated data: the roton region

Data [Tier L]. Table 14.1 of the Donnelly–Barenghi compilation [41] (web edition, a *secondary* compilation, measurements between 1.4 and 4 K, with no printed error bars) gives 99 points $(Q, S(Q))$ after removing $Q = 0$ (49 X-ray, 50 neutron). With $c = 238.3$ m/s [42], one finds $k_* = 2m_4 c/\hbar = 3.00379$ Å^{-1}, hence $2k_* = 6.008$ Å^{-1}.

Arithmetic [Tier X]. With a threshold $\Delta_{ds} = 0.082$ fixed *a posteriori* (the gap between the maxima of the two data sets; it is not in the preregistration), 63 points are refuted, 29 compatible and 7 undecided. In the roton region $1.7 \le Q \le 2.4$ Å^{-1}, **all 21 points are refuted**, with S between 1.62 and 2.49 times the bound. This result does not depend on the threshold: the smallest margin there is 0.328, about $4\Delta_{ds}$.

Checking by hand. At the peak, $Q = 2.058$ Å^{-1} and $S = 1.458$, whereas $\sqrt{Q/2k_*} = \sqrt{2.058/6.008} \approx 0.585$. The margin is about 0.87, well beyond the gap 0.082 between the data sets.

Reading [Tier C]. Liquid helium-4 is not a Bogoliubov gas with its own sound velocity. This is a textbook fact (helium is not dilute) and *not* a discovery. The reading is conditional on the secondary data, on temperatures of 1.4 to 4 K being taken as the ground state, and on the a posteriori threshold. For trapped Bose gases, only values read off figures exist ([Tier X]): the three absolute points of Stamper-Kurn et al. [43] lie below a bound averaged over the local density, with low test power, and the points of Steinhauer et al. [44] were multiplied by 2.3 by the authors. No decisive test is possible.

3.6 Chapter summary

Table 3.5: The claims of the chapter and their tier.

claim	tier
$E_{\mathrm{zp}} \geq \frac{1}{2}\hbar\omega$ for the oscillator; phonons of the chain	[Tier C]
identities for Q, isotropy (2×2 algebra), $\rho^2 = \rho - 1$	[Tier A without replay] (glue: [Tier C])
QU-1 refuted for 8 potentials out of 12; local maximum for $a \in (0.37668, 1.58148)$	[Tier X]
mechanism: concavity of $\mathrm{tr}\,\sqrt{\cdot}$	[Tier C] (numbers [Tier X])
inequality $2\,\mathrm{tr}\,S \leq \mathrm{tr}\,X + \mathrm{tr}(X^{-1}S^2)$, logic, rational certificate	[Tier A without replay]
enclosures of the integrals ($U''(0) \in [-1.948, -1.900]$)	[Tier C] (ball arithmetic)
B2: ρ is not a local minimum at $a = 1$	[Tier C] overall
B1: no transition in the SCHA; PIMC blocked	[Tier X] (negative)
$S_{\mathrm{B}} \leq \sqrt{k/2k_*}$; abstract Feynman–Bijl bound	[Tier A without replay] (f-sum rule: hypothesis)
^{4}He: 21/21 points of the roton region above the envelope	data [Tier L], computation [Tier X], reading [Tier C]

The methodological lesson is twofold. First, *test before proving*: the QU-1 conjecture, plausible as it was, could have absorbed months of proof attempts. Second, *separate the layers*: a statement such as Theorem 3.5 mixes a kernel-checked core (logic, arithmetic), rigorous but not kernel-checked computation (balls) and human arguments; the assembled Lean theorem makes this separation explicit, hypothesis by hypothesis.

3.7 Exercises

Exercise 3.1 (Oscillator)**.** (a) Check that $\psi_0(x) = e^{-m\omega x^2/2\hbar}$ satisfies $-\frac{\hbar^2}{2m}\psi_0'' + \frac{1}{2}m\omega^2 x^2\psi_0 = \frac{1}{2}\hbar\omega\,\psi_0$. (b) In the proof of Proposition 3.1, justify the integration by parts on $[-A, A]$ and then the passage to the limit. (c) For the trial state $\psi_\alpha(x) \propto e^{-\alpha x^2}$, compute $\langle H\rangle$ as a function of α and recover the minimum.

Exercise 3.2 (Linear chain)**.** (a) Show that the $N \times N$ matrix of the potential energy of the chain is circulant, and that the vectors $(e^{ikna})_n$, $k \in \frac{2\pi}{Na}\mathbb{Z}$, diagonalize it. (b) Establish $\frac{1}{N}\sum_k \frac{1}{2}\hbar\omega(k) \to \frac{2}{\pi}\hbar\sqrt{K/m}$. (c) Compute the sound velocity $\lim_{k\to 0}\omega/|k|$.

Exercise 3.3 (Radial Hessian)**.** (a) Prove formula (3.1). (b) Deduce $\Delta V(R) = 4(ug')'(u)$ at $u = |R|^2$, and then, for $g(u) = e^{-\pi a u}$, $\Delta V = (-4\pi a + 4\pi^2 a^2 u)e^{-\pi a u}$ (this is the function `m2f` of `ZeroPoint/Basic.lean`). (c) Why is $D(\mathbf{k})$ real symmetric?

Exercise 3.4 (Einstein bound)**.** (a) Show that $\frac{1}{2}(\omega_1 + \omega_2) \leq \sqrt{(\omega_1^2 + \omega_2^2)/2}$. (b) Using the Cauchy–Schwarz inequality for the integral over the zone, deduce (3.2). (c) When does equality

hold? Explain why an upper bound that is maximal at ρ does not, by itself, locate the extrema of E_{zp}.

Exercise 3.5 (Rotation by 120°)**.** Let $R = \left(\begin{smallmatrix} -1/2 & -s \\ s & -1/2 \end{smallmatrix}\right)$ with $s^2 = 3/4$. (a) Show that a vector g with $Rg = g$ is zero. (b) Show that a symmetric matrix H with $RH = HR$ is scalar (compare with the statement `rot120_comm_symm`). (c) Show that $\rho^2 = \rho - 1$ and that $(\tau - 1)/\tau$ fixes ρ.

Exercise 3.6 (The trace inequality)**.** (a) Prove Lemma 3.6 when X and S are diagonal. (b) Give an example of symmetric matrices X, S, with X positive definite, for which the inequality is strict, and check it numerically. (c) Show that, for fixed X, $S \mapsto \operatorname{tr} X + \operatorname{tr}(X^{-1}S^2) - 2\operatorname{tr} S$ attains its minimum 0 at $S = X$. Interpret this as "the tangent lies above a concave function".

Exercise 3.7 (The logic of the majorant)**.** (a) Write out the proof of Lemma 3.7 in full. (b) Construct E and U with $E \le U$, $U(0) = E(0)$ and $U''(0) > 0$, but where 0 is *not* a local minimum of E. Deduce that the B2 method can never certify a local minimum. (c) Why does the hypothesis $E \le U$ not suffice if $U(0) = E(0)$ is replaced by $U(0) \ge E(0)$?

Exercise 3.8 (The rational certificate)**.** (a) With the rationals of Section 3.3.4, compute $\mathrm{Upp0}_{hi} + \mathrm{trW}_{hi}\,\mathrm{esum}_{hi}$ to within 10^{-6}. (b) Up to what value of esum_{hi} would the certificate remain valid? (Answer: about 1.0363.) (c) Prove the monotonicity statement `cert_window`: if $\mathrm{Upp0} \le \mathrm{Upp0}_{hi}$, $0 \le \mathrm{trW} \le \mathrm{trW}_{hi}$ and $0 \le \mathrm{esum} \le \mathrm{esum}_{hi}$, then $\mathrm{Upp0} + \mathrm{trW}\,\mathrm{esum} \le \mathrm{Upp0}_{hi} + \mathrm{trW}_{hi}\,\mathrm{esum}_{hi}$. Where is the hypothesis $0 \le \mathrm{esum}$ used?

Exercise 3.9 (Gaussian integrals and the SCHA)**.** (a) In dimension one, for $w \sim \mathcal{N}(0, \sigma^2)$, show that $\langle e^{-b(R+w)^2}\rangle = (1 + 2b\sigma^2)^{-1/2} \exp\!\big(-bR^2/(1 + 2b\sigma^2)\big)$ by completing the square. (b) Deduce (3.6) for a covariance $C = \sigma^2 I_2$, and then for an arbitrary symmetric positive definite C by diagonalizing C. (c) Show that the planar Fourier transform of $e^{-\pi a r^2}$ is $a^{-1}e^{-\pi k^2/a}$ (with the convention $\hat f(k) = \int f(x) e^{-2i\pi k\cdot x} dx$), hence strictly positive.

Exercise 3.10 (Balance of curvatures and Lindemann)**.** (a) With $c_{\mathrm{st}} = 0.11913$ and $c_{\mathrm{zp}} = -0.96198$ ($a = 1$), recover $\hbar^* \approx 0.1238$. (b) Show that $\Lambda_L(\gamma_c) \propto \gamma_c^2$ and that $\Lambda_1/\Lambda_L = (\gamma(\Lambda_1)/\gamma_c)^2$. (c) With $\gamma(\Lambda_1) = 0.273$ at $a = 1$, state for which values of γ_c in $[0.23, 0.28]$ the first-order point precedes the estimated melting. Explain why the test is inconclusive.

Exercise 3.11 (Bogoliubov envelope)**.** (a) Prove Proposition 3.8. (b) Show the phonon regime: $\frac{k}{k_* + k} < S_{\mathrm{B}}(k) < \frac{k}{k_*}$. (c) Show that $S_{\mathrm{B}}(k) \to 1$ as $k \to \infty$ and that the free value $S \equiv 1$ violates the envelope for every $0 < k < 2k_*$: this is the positive control of the test. (d) Redo the check of the helium-4 peak.

Exercise 3.12 (Why the f-sum rule is a hypothesis)**.** (a) Show that for two square matrices A, B of size n, $\operatorname{tr}(AB - BA) = 0$. Deduce that there are no matrices X, P with $XP - PX = i\hbar I_n$. (b) In $\mathbb{C}^2$ with orthonormal basis (e_0, e_1), let $H = |e_1\rangle\langle e_1|$, $\psi = e_0$ and $\rho = |e_1\rangle\langle e_0|$. Compute $\mathcal{A}(\rho)$, $\mathcal{A}(\rho^\dagger)$ and the double commutator, and check that $\mathcal{A}(\rho) \ne f/2$: the symmetry is necessary. (c) With $\rho = |e_1\rangle\langle e_0| + |e_0\rangle\langle e_1|$, $N = \hbar = k = 1$ and $m = 1/2$, check that the bound of Theorem 3.11 is attained.

Exercise 3.13 ((†) University exercise: Harmonic chain with second neighbours)**.** Consider the ring of N unit masses of Section 3.1.2 with nearest-neighbour springs of stiffness $K_1 > 0$ and next-nearest-neighbour springs of stiffness $K_2 \ge 0$, so that the potential energy is $\frac{K_1}{2}\sum_n (u_{n+1} - u_n)^2 + \frac{K_2}{2}\sum_n (u_{n+2} - u_n)^2$ (lattice spacing $a = 1$). (a) Show that the (here 1×1) dynamical matrix is $D(k) = 2K_1(1 - \cos k) + 2K_2(1 - \cos 2k)$, i.e. $D(k) = \sum_{j \ne 0}(1 - \cos kj)\,\Phi_j$ with $\Phi_{\pm 1} = K_1$, $\Phi_{\pm 2} = K_2$, in analogy with Definition 3.2. (b) Show that $D(k) = 4\sin^2(k/2)\,\big(K_1 + 4K_2\cos^2(k/2)\big) \ge 0$, so the chain is dynamically stable, and that the sound velocity is $\sqrt{K_1 + 4K_2}$. (c) Show that the zero-point energy per particle in the limit $N \to \infty$ is

$E_{\mathrm{zp}} = \frac{\hbar}{2\pi}\int_0^{\pi} 2\sin(\theta/2)\sqrt{K_1 + 4K_2\cos^2(\theta/2)}\,d\theta$, that it reduces to $\frac{2}{\pi}\hbar\sqrt{K_1}$ when $K_2 = 0$, and compute it in closed form for general K_2 (substitute $v = \cos(\theta/2)$). (d) Verify the Einstein bound (3.2) in this setting, in the form $E_{\mathrm{zp}} \le \frac{\hbar}{2}\sqrt{\langle D\rangle}$ with $\langle D\rangle = 2K_1 + 2K_2$, and check it explicitly when $K_2 = 0$. *Answer.* c) With $v = \cos(\theta/2)$, $E_{\mathrm{zp}} = \frac{2\hbar}{\pi}\int_0^1 \sqrt{K_1 + 4K_2 v^2}\,dv = \frac{\hbar}{\pi}\Big[\sqrt{K_1+4K_2} + \frac{K_1}{2\sqrt{K_2}}\operatorname{arsinh}\big(2\sqrt{K_2/K_1}\big)\Big]$ for $K_2 > 0$. d) For $K_2 = 0$: $\frac{2}{\pi} \approx 0.637 \le \frac{\sqrt{2}}{2} \approx 0.707$ (in units of $\hbar\sqrt{K_1}$).

Exercise 3.14 ((†) University exercise: The Gaussian is completely monotone in r^2; its Fourier transform)**.** (a) Show that $g(u) = e^{-\pi a u}$ $(a > 0)$ is completely monotone on $(0, +\infty)$, i.e. $(-1)^n g^{(n)}(u) \ge 0$ for all $n \ge 0$, and identify its Bernstein measure (a Dirac mass at $t = \pi a$). (b) Using Fubini and the one-dimensional identity $\int_{\mathbb{R}} e^{-\pi a x^2 - 2i\pi\xi x}dx = a^{-1/2}e^{-\pi\xi^2/a}$ (proved by completing the square or by an ODE in ξ), show that the Fourier transform on $\mathbb{R}^2$ of $x \mapsto e^{-\pi a|x|^2}$ is $a^{-1}e^{-\pi|\xi|^2/a}$, consistently with Exercise 3.9(c). (c) Deduce that for every positive measure μ on $(0, +\infty)$ with $\int t^{-1}d\mu(t) < \infty$ (which makes f integrable), the radial function $f(x) = \int e^{-t|x|^2}d\mu(t)$ has a nonnegative Fourier transform on $\mathbb{R}^2$, and give its formula. *Answer.* c) $\hat{f}(\xi) = \int \frac{\pi}{t}\,e^{-\pi^2|\xi|^2/t}\,d\mu(t) \ge 0$.

Chapter 4

Verified population genetics: fixation, selection and graphs

A mutation appears in a population of N individuals. Will it disappear, or will it eventually be carried by everyone? This is the question of *fixation*. It is both very concrete (will a bacterium that has become resistant to an antibiotic take over the colony?) and entirely accessible with the tools of a first course in probability and analysis. All one needs is finite Markov chains, geometric sums, the binomial theorem, linear recurrence relations and a little topology of $\mathbb{R}$.

This chapter follows the same thread as the previous ones. Each statement receives an undergraduate-level *human proof*, and next to it the statement *as the Lean 4 kernel checked it*, with the file name and line number. The two do not always coincide: sometimes the Lean proof takes a different route from the human proof, and we point this out every time.

Warning : status of this chapter

- All the results of this chapter are **classical**: Moran (1958) [55], Fisher, Wright, Price (1970) [56], Lieberman–Hauert–Nowak (2005) [54]. No mathematical novelty is claimed. What is specific to the project is a formalisation checked by the Lean kernel. We have not carried out a systematic search for earlier formalisations.
- The Lean files, the proofs and this text were produced by AI agents (Claude) under the author's direction. A separate AI agent audited them: recompilation from source, axiom audit, planted controls. The verdicts are recorded in `AUDIT.md` and `verification.md`. **No human has audited these results.**
- Nothing here depends on the openai/math theorems of Chapters 1 to 3.

The tiers, recalled. Here is what each label means in this chapter.

- **[Tier A]** means that the cited *Lean statement* is checked by the kernel. Its axioms lie in {`propext`, `Classical.choice`, `Quot.sound`}, and the module has passed a kernel replay (`leanchecker`) as well as a search for the soundness escapes `skipKernelTC`, `addDecl`, `native_decide` and `implemented_by`. Tier A concerns the Lean statement, *never* its biological reading.
- **[Tier L]** means a published source, read and located to the page.
- **[Tier C]** means a human argument, including a proof in this book, that is not checked by a machine.
- **[Tier X]** means a computation (floating point, or exact rational arithmetic by a script) that illustrates and checks, but never establishes anything.

Kernel replay coverage. The original audits checked two things: the list of axioms, and a recompilation by a separate agent. The N4 audit added a `leanchecker` replay of the modules `Programmes.PopGen.Absorption`, `Programmes.PopGen.MoranModel` and the three modules `Programmes.EvoGraph.*`. Its negative control was a falsified theorem under `debug.skipKernelTC`, and the replay did reject it. The other modules cited here had no replay: `Autoresearch.PopGenFisher`, `Autoresearch.PopGenWrightFisher`, `Programmes.PopGen.WrightFisherFixation`, `Programmes.PopGen.MoranFixation` and `Programmes.PopGen.Price`. For these, the agent writing this chapter ran `leanchecker` at the time of writing (2026-10-26), on the existing `.olean` files, all newer than their sources, with the pinned Lean 4.34.1 toolchain; it also searched with `grep` for the forbidden tokens. The result is given in §4.6. This replay is not a human audit.

The bridge to probability [Tier C]. Throughout the chapter, P denotes a stochastic matrix on a finite state space. We define P^n by the Chapman–Kolmogorov recursion $P^0 = I$, $P^{n+1}(i,j) = \sum_k P^n(i,k)P(k,j)$ (this is `kPow` in Lean). If (X_n) is the Markov chain with transition matrix P started at i, then by definition $\mathbb{P}_i(X_n = j) = P^n(i,j)$. If moreover the state N is absorbing, $\{X_n = N\} = \{T_N \le n\}$, where T_N is the hitting time of N. The limit of $P^n(i,N)$ is therefore $\mathbb{P}_i(T_N < \infty)$, the *fixation probability*. This identification is standard, but it *is not* a Lean statement: no path space is constructed. The Lean theorems are about limits of matrix powers, and that is the formulation we adopt, for instance "the probability of absorption at N after n steps tends to …".

4.1 Hardy–Weinberg and one-locus selection

4.1.1 The model

A gene (a *locus*) has two alleles A_1 and A_2, with frequencies p and $q = 1 - p$. Individuals are diploid: each carries two copies, hence one of the three genotypes A_1A_1, A_1A_2 or A_2A_2. Generations do not overlap. We further assume random mating: a zygote is formed from two gametes drawn independently from the gamete pool.

Proposition 4.1 (Hardy–Weinberg)**.** *Start from genotype frequencies (x_{11}, x_{12}, x_{22}) summing to 1, and set $p = x_{11} + x_{12}/2$. After one generation of random mating, the frequencies are $(p^2, 2pq, q^2)$, summing to 1, and the frequency of allele A_1 is still p. These proportions are stationary from the first generation on.* ***[Tier A]*** *(`hwStep_sum`, `hwStep_allele`, `hwStep_stationary`, `PopGenFisher.lean`, ll. 129–148).*

Human proof. A gamete carries A_1 with probability $x_{11} + \frac{1}{2}x_{12} = p$. Since the two gametes are independent, we obtain A_1A_1 with probability p^2, A_1A_2 with probability $2pq$ (two possible orders) and A_2A_2 with probability q^2. The sum is $(p+q)^2 = 1$. In the new generation, the frequency of A_1 is $p^2 + \frac{1}{2}\cdot 2pq = p(p+q) = p$: the map is therefore idempotent. □

Lean 4 : Hardy–Weinberg (`formal_cert/Autoresearch/PopGenFisher.lean`, ll. 125–127 and 145)

```
noncomputable def hwStep (x : ℝ × ℝ × ℝ) : ℝ × ℝ × ℝ :=
  let p := x.1 + x.2.1 / 2
  (p ^ 2, 2 * p * (1 - p), (1 - p) ^ 2)
theorem hwStep_stationary (x : ℝ × ℝ × ℝ) : hwStep (hwStep x) = hwStep x := by
```

The statement `hwStep_stationary` holds for every triple, even an unnormalised one, because $p^2 + p(1-p) = p$ is an identity. On the other hand, `hwStep` ignores x_{22}: it implicitly assumes a normalised input (remark by the verifier, `H12/verification.md`).

4.1.2 Viability selection

The genotypes A_1A_1, A_1A_2 and A_2A_2 survive to adulthood with *fitnesses* $w_{11}, w_{12}, w_{22} > 0$. Before selection, the zygotes are in Hardy–Weinberg proportions. From this we derive the *mean fitness* and the new allele frequency:

$$\bar{w}(p) = p^2 w_{11} + 2pq\, w_{12} + q^2 w_{22}, \qquad p' = \frac{p^2 w_{11} + pq\, w_{12}}{\bar{w}(p)}.$$

Lean 4 : the model (`PopGenFisher.lean`, ll. 19–24)

```
noncomputable def wbar (w11 w12 w22 p : ℝ) : ℝ :=
  p ^ 2 * w11 + 2 * p * (1 - p) * w12 + (1 - p) ^ 2 * w22

/-- Allele frequency after one generation of viability selection. -/
noncomputable def pnext (w11 w12 w22 p : ℝ) : ℝ :=
  (p ^ 2 * w11 + p * (1 - p) * w12) / wbar w11 w12 w22 p
```

Lemma 4.2. *Let $w_{11}, w_{12}, w_{22} > 0$ and $p \in [0,1]$. Then:*

(i) $\bar{w}(p) > 0$ *and* $p' \in [0,1]$*;*

(ii) $\Delta p := p' - p = \dfrac{pq\, s}{\bar{w}(p)}$*, with* $s = p(w_{11} - w_{12}) + q(w_{12} - w_{22})$*;*

(iii) $\bar{w}'(p) = 2s$*, whence Wright's form* $\Delta p = \dfrac{pq\, \bar{w}'(p)}{2\bar{w}(p)}$.

[Tier A] *(`wbar_pos`, `pnext_mem_Icc`, `delta_p_eq`, `hasDerivAt_wbar`, `delta_p_eq_deriv`, `PopGenFisher.lean`, ll. 26–91).*

Human proof. (i) If $p > 0$, the term $p^2 w_{11}$ is > 0 and the other two are ≥ 0. If $p = 0$, $\bar{w} = w_{22} > 0$. The numerator of p' is ≥ 0, and $\bar{w} - (p^2 w_{11} + pq w_{12}) = pq w_{12} + q^2 w_{22} \geq 0$, so $p' \leq 1$. (ii) We bring everything over a common denominator, using $1 - p = q$ and $1 - 2p = q - p$:

$$p^2 w_{11} + pq w_{12} - p\,\bar{w} = p\big[p w_{11}(1-p) + q w_{12}(1-2p) - q^2 w_{22}\big] = pq\big[p w_{11} + (q-p) w_{12} - q w_{22}\big] = pq\, s,$$

then divide by $\bar{w} > 0$. (iii) Expanding, $\bar{w}(x) = w_{22} + 2(w_{12} - w_{22})x + (w_{11} - 2w_{12} + w_{22})x^2$, so $\bar{w}'(p) = 2(w_{12} - w_{22}) + 2(w_{11} - 2w_{12} + w_{22})p = 2s$. □

In Lean, the derivative in (iii) is not a hand-written polynomial. It is the real derivative `deriv` of the function $x \mapsto \bar{w}(x)$, obtained from an explicit `HasDerivAt` statement (ll. 64–82).

4.1.3 The exact increment identity

Theorem 4.3 (Monotonicity of mean fitness)**.** *Let $w_{11}, w_{12}, w_{22} > 0$ and $p \in [0,1]$. With the notation of Lemma 4.2, we have the exact identity*

$$\bar{w}(p') - \bar{w}(p) = \frac{pq\, s^2 \left(\bar{w}(p) + p\, w_{11} + q\, w_{22}\right)}{\bar{w}(p)^2}, \tag{4.1}$$

and consequently $\bar{w}(p') \geq \bar{w}(p)$: mean fitness never decreases from one generation to the next.
[Tier A] *(`wbar_increment` and `fisher_mean_fitness_nondecreasing`, `PopGenFisher.lean`, ll. 95–119).*

Human proof. Set $W = \bar{w}(p)$, $a = w_{11} - 2w_{12} + w_{22}$ and $\Delta = p' - p = pqs/W$ (Lemma 4.2).

Step 1: Taylor is exact. $\bar{w}$ is a polynomial of degree 2 in x, with derivative $2s$ at p and second derivative $2a$. Taylor's formula to order 2 is therefore exact:

$$\bar{w}(p + \Delta) = W + 2s\,\Delta + a\,\Delta^2.$$

Step 2: substitute Δ.

$$\bar{w}(p') - W = 2s \cdot \frac{pqs}{W} + a\,\frac{p^2q^2s^2}{W^2} = \frac{pq\,s^2}{W^2}\,(2W + a\,pq).$$

Step 3: the key identity. Expanding, one checks that

$$W + a\,pq = p^2w_{11} + 2pqw_{12} + q^2w_{22} + pq(w_{11} - 2w_{12} + w_{22}) = pw_{11}(p+q) + qw_{22}(p+q) = pw_{11} + qw_{22}.$$

Thus $2W + a\,pq = W + pw_{11} + qw_{22}$, which gives exactly (4.1).

Step 4: the sign. The factors p, q, s^2, W^2 are ≥ 0 (and $W > 0$), and $W + pw_{11} + qw_{22} > 0$. The quotient is therefore ≥ 0. □

Lean 4 : the increment (`PopGenFisher.lean`, ll. 95–99 and 110–112)

```
theorem wbar_increment {w11 w12 w22 p : ℝ} (h11 : 0 < w11) (h12 : 0 < w12) (h22 : 0 <
    w22)
    (hp0 : 0 ≤ p) (hp1 : p ≤ 1) :
    wbar w11 w12 w22 (pnext w11 w12 w22 p) - wbar w11 w12 w22 p
      = p * (1 - p) * (p * (w11 - w12) + (1 - p) * (w12 - w22)) ^ 2
          * (wbar w11 w12 w22 p + p * w11 + (1 - p) * w22) / (wbar w11 w12 w22 p) ^ 2 :=
     by
theorem fisher_mean_fitness_nondecreasing {w11 w12 w22 p : ℝ}
    (h11 : 0 < w11) (h12 : 0 < w12) (h22 : 0 < w22) (hp0 : 0 ≤ p) (hp1 : p ≤ 1) :
    wbar w11 w12 w22 p ≤ wbar w11 w12 w22 (pnext w11 w12 w22 p) := by
```

Side by side. The Lean proof does not go through Taylor. It clears denominators (`field_simp`), then concludes with `ring`. The docstring states that the identity was first found with `sympy`, then checked by the kernel. The human proof above explains *why* the factor $\bar{w} + pw_{11} + qw_{22}$ appears: it is the identity of Step 3.

Remark 4.4 (Hypotheses **[Tier C]**). *Strict* positivity of the fitnesses is sufficient, but not necessary: by (4.1), $w \geq 0$ with $\bar{w}(p) > 0$ suffices, since then $\bar{w} + pw_{11} + qw_{22} \geq \bar{w} > 0$. This is a correction by the independent verifier (`H12/verification.md`). The Lean theorem is correct, but narrower than it could be. *Non-negativity*, however, cannot be dropped. The control `fisher_fails_without_positivity` takes $w = (-1, 1, 0)$ and $p = \frac{1}{2}$: then $\bar{w} = \frac{1}{4} > 0$, but $p' = 0$ and $\bar{w}(p') = 0 < \frac{1}{4}$ **[Tier A]** (ll. 164–167).

Remark 4.5 (The mean goes up, but p may move away from equilibrium **[Tier A]**). Under underdominance, $w = (1, \frac{1}{2}, 1)$, there is an interior equilibrium $p = \frac{1}{2}$. Yet, starting from $p = \frac{2}{5}$, we get $p' = \frac{7}{19} < \frac{2}{5}$: the frequency moves *away* from the equilibrium, and $\bar{w}$ increases all the same ($\frac{19}{25} < \frac{277}{361}$). This is `underdominance_moves_away` (ll. 155–158), and the numerical values were recomputed in exact rational arithmetic **[Tier X]**. The increase of $\bar{w}$ therefore says nothing about convergence to a polymorphism.

Warning : this is not Fisher's "fundamental theorem"

Theorem 4.3 covers a single locus, two alleles, discrete generations, viability selection, constant fitnesses and random mating. It is the monotonicity of mean fitness *for this model*. Fisher's fundamental theorem, in the sense of Price's (1972) reading in terms of

the "partial change" [57] and additive genetic variance, is a different, finer statement, which *is not formalised* here **[Tier C]**. The original Lean docstring spoke of "Fisher's fundamental theorem". The verifier required it to be renamed, and the module header now says "one-locus, two-allele mean-fitness monotonicity".

4.2 Neutral Wright–Fisher: martingale, harmonic functions, absorption

4.2.1 The chain

We now turn to *chance* alone. Consider a population of N genes, of which i are of mutant type. The next generation is obtained by drawing N genes *with replacement* from the present generation. Without selection, the number j of mutants in the next generation follows the binomial distribution $\mathcal{B}(N, i/N)$:

$$P(i,j) = \binom{N}{j}\left(\frac{i}{N}\right)^{j}\left(1-\frac{i}{N}\right)^{N-j}, \qquad 0 \le i, j \le N.$$

Lean 4 : the transition kernel (`formal_cert/Autoresearch/PopGenWrightFisher.lean`, ll. 25–26)

```
noncomputable def wfP (N : ℕ) (i j : Fin (N + 1)) : ℝ :=
  (N.choose (j : ℕ) : ℝ) * ((i : ℕ) / N : ℝ) ^ (j : ℕ) * (1 - ((i : ℕ) / N : ℝ)) ^ (N
    - (j : ℕ))
```

The positive control `ctl_positive_wfP_value` establishes `wfP 2 1 1 = 1 / 2`: the division is indeed real division, not the integer division of $\mathbb{N}$.

Proposition 4.6. *Let $N \ge 1$.*

(i) *P is stochastic, and the states 0 and N are absorbing.*

(ii) (Martingale) *For every i, $\sum_j P(i,j)\, j = i$. In other words, $h(j) = j/N$ is* harmonic*: $Ph = h$.*

(iii) *From every state $0 \le i < N$, we have $P(i,0) = (1-i/N)^N > 0$.*

[Tier A] *(`wfP_nonneg`, `wfP_sum_one`, `wfP_zero`, `wfP_last`, `wfP_mean`, `frac_harmonic`, `wfP_interior_zero_pos`, `PopGenWrightFisher.lean`, ll. 34–118).*

Human proof. (i) By the binomial theorem, $\sum_j \binom{N}{j} x^j (1-x)^{N-j} = (x+1-x)^N = 1$. At $i = 0$ we have $x = 0$, and with the convention $0^0 = 1$ the row is δ_0; likewise at $i = N$. (ii) We use $j\binom{N}{j} = N\binom{N-1}{j-1}$ for $j \ge 1$. With $x = i/N$ and $y = 1-x$,

$$\sum_{j=0}^{N} \binom{N}{j} x^j y^{N-j}\, j = N \sum_{j=1}^{N} \binom{N-1}{j-1} x^j y^{N-j} = Nx\,(x+y)^{N-1} = Nx = i.$$

(iii) This is the term $j = 0$, which equals $(1-i/N)^N$, strictly positive as soon as $i < N$. □

Identity (ii) says that the number X_n of mutants is a *martingale*: its conditional expectation at the next step is its present value **[Tier C]** (probabilistic reading). By induction on n, we obtain $\sum_j P^n(i,j)\, j = i$ for every n. This computation requires no measure theory.

University remark. **[Tier L]** *Optional stopping.* In the path-space language, (X_n) is a martingale with values in $\{0,\dots,N\}$, hence bounded, and the absorption time $T = \min(T_0, T_N)$ is almost surely finite (this follows, for instance, from Lemma 4.10 below). The optional stopping theorem for bounded martingales then gives $\mathbb{E}_i[X_T] = X_0 = i$, that is $N\,\mathbb{P}_i(X_T = N) = i$, so the fixation probability is i/N. This is the standard martingale proof [69, 68]; it is *not* the route taken in Lean, which never constructs the path space.

4.2.2 Harmonic functions and the maximum principle

Definition 4.7. Let P be a stochastic matrix on $\{0,\dots,N\}$ whose states 0 and N are absorbing. A function $g : \{0,\dots,N\} \to \mathbb{R}$ is *harmonic in the interior* if $\sum_j P(i,j)g(j) = g(i)$ for every $0 < i < N$.

The point of this notion is the following. The fixation probability $u(i) = \mathbb{P}_i(T_N < \infty)$ satisfies *formally* the "first-step" equation $u(i) = \sum_j P(i,j)u(j)$, with $u(0) = 0$ and $u(N) = 1$ **[Tier C]**. If one knows that only one harmonic function has these boundary values, one therefore knows u.

University remark. **[Tier L]** *Hitting probabilities as harmonic functions.* For a Markov chain on a countable state space and a target set A, the vector of hitting probabilities $u(i) = \mathbb{P}_i(\text{hit } A)$ is the *minimal* non-negative solution of the system $u = 1$ on A, $u(i) = \sum_j P(i,j)u(j)$ off A; on a finite state space from which $\{0, N\}$ is reached almost surely, the bounded solution with prescribed boundary values is unique, which is exactly what makes the first-step equation determine u [68] (standard).

Theorem 4.8 (Maximum principle, uniqueness)**.** *Let $N \geq 1$, and let d be a function harmonic in the interior for the Wright–Fisher kernel, with $d(0) = d(N) = 0$. Then $d \leq 0$, and by applying the same reasoning to $-d$, $d = 0$. Consequently, $i \mapsto i/N$ is* the unique *function harmonic in the interior that equals 0 at 0 and 1 at N.* ***[Tier A]*** *(`max_principle`, `harmonic_unique`, `wright_fisher_fixation_harmonic`, `PopGenWrightFisher.lean`, ll. 122–187).*

Human proof. Let m be a point where d attains its maximum; it exists because the set is finite. If $m \in \{0, N\}$, then $\max d = 0$. Otherwise m is interior, and

$$0 = d(m) - \sum_j P(m,j)d(j) = \sum_j P(m,j)\big(d(m) - d(j)\big),$$

where we used $\sum_j P(m,j) = 1$. Each term is ≥ 0, since $P \geq 0$ and $d(m)$ is maximal; they are therefore all zero. But $P(m,0) > 0$ (Proposition 4.6(iii)), so $d(m) = d(0) = 0$. In both cases, $\max d = 0$. For uniqueness, if g is a solution, then $d = g - h$ with $h(j) = j/N$ is harmonic in the interior and vanishes on the boundary. □

Lean 4 : the maximum principle (`PopGenWrightFisher.lean`, ll. 122–126)

```
theorem max_principle {N : ℕ} (hN : 0 < N) (d : Fin (N + 1) → ℝ) (h0 : d 0 = 0)
    (hN' : d (Fin.last N) = 0)
    (hharm : ∀ i : Fin (N + 1), 0 < (i : ℕ) → (i : ℕ) < N → d i = ∑ j, wfP N i j * d
     j) :
    ∀ i, d i ≤ 0 := by
```

The Lean proof follows the human proof exactly. `Finset.exists_max_image` supplies the maximum point, then `Finset.sum_eq_zero_iff_of_nonneg` kills each term, and `wfP_interior_zero_pos` concludes.

Warning : what H11 did not prove

Module H11 [48] stops there. It establishes that i/N is *the* harmonic solution, but the step "the fixation probability is harmonic" is not formalised in it. The verifier put it bluntly: "this file does NOT kernel-check 'fixation probability $= i/N$'" (`H11/NOTE.md`). It suggested closing the gap *without* measure theory, by proving $P^n(i,N) \to i/N$. This is what programme P7 did, and it is the subject of the next subsection.

4.2.3 Absorption: $P^n(i,N) \to i/N$

We write $I_n(i) = \sum_{0<k<N} P^n(i,k)$ for the *interior mass* after n steps, that is, the probability of not yet having been absorbed (`interiorMass`).

Theorem 4.9 (Neutral fixation)**.** *For every $N \geq 1$ and every $0 \leq i \leq N$,*

$$P^n(i,N) \xrightarrow[n\to\infty]{} \frac{i}{N}, \qquad P^n(i,0) \xrightarrow[n\to\infty]{} 1 - \frac{i}{N},$$

and the sequence $\big(P^n(i,N)\big)_n$ is nondecreasing. ***[Tier A]*** *(`wright_fisher_fixation_limit`, `wright_fisher_fixation_mono`, `Programmes/PopGen/WrightFisherFixation.lean`, ll. 49–58). The general theorem `fixation_limit` from which it follows reappears under the name `KAL`.*

We first give an undergraduate-level proof **[Tier C]**, which relies on the *geometric decay* of the interior mass. The Lean proof takes a different route, presented afterwards.

Lemma 4.10 (Geometric decay **[Tier C]**)**.** *Set $\delta = \min_{0<k<N} P(k,0)$. Then $\delta \geq N^{-N} > 0$ and, for every n, $I_{n+1}(i) \leq (1-\delta)\, I_n(i)$. Consequently $I_n(i) \leq (1-\delta)^n$.*

Human proof. For $0 < k < N$, we have $P(k,0) = (1-k/N)^N \geq (1/N)^N$. Next,

$$I_{n+1}(i) = \sum_{0<j<N} \sum_{k} P^n(i,k) P(k,j) = \sum_{k} P^n(i,k) \sum_{0<j<N} P(k,j).$$

If $k \in \{0, N\}$, the inner sum is zero, because the state is absorbing. If k is interior, it is at most $1 - P(k,0) \leq 1 - \delta$. Hence $I_{n+1}(i) \leq (1-\delta) \sum_{0<k<N} P^n(i,k) = (1-\delta) I_n(i)$, and we conclude by induction, since $I_0 \leq 1$. □

Human proof of Theorem 4.9. Set $h(j) = j/N$. By Proposition 4.6, $Ph = h$ at every state, hence $P^n h = h$, which reads

$$\frac{i}{N} = \sum_{j} P^n(i,j)\, \frac{j}{N} = P^n(i,N) + \sum_{0<j<N} P^n(i,j)\, \frac{j}{N}.$$

The last term lies between 0 and $I_n(i)$, which tends to 0 by Lemma 4.10. We deduce $P^n(i,N) \to i/N$ by squeezing. Next, $P^n(i,0) + P^n(i,N) + I_n(i) = 1$ (the rows of P^n sum to 1), whence $P^n(i,0) \to 1 - i/N$. Finally, $P^{n+1}(i,N) = \sum_k P^n(i,k) P(k,N) \geq P^n(i,N) P(N,N) = P^n(i,N)$. □

University remark. **[Tier L]** *The fundamental matrix.* Order the states so that the interior states $\{1, \dots, N-1\}$ come first and write $P = \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix}$, where Q is the interior-to-interior block and R the interior-to-boundary block. Lemma 4.10 says that $Q^n \to 0$ geometrically, so the series $\sum_{n \geq 0} Q^n$ converges and equals the *fundamental matrix* $N_Q = (I - Q)^{-1}$; its entry $N_Q(i,k)$ is the expected number of visits to k starting from i. The absorption probabilities are the entries of $B = N_Q R$, and the vector of expected absorption times is $N_Q \mathbf{1}$ [68] (standard). This is literature only: as recorded in §4.6, absorption times are not formalised in this chapter.

Lean 4 : the general theorem and the Wright–Fisher case

`formal_cert/Programmes/PopGen/Absorption.lean`, ll. 32–34, 45–53 and 302–308:

```
noncomputable def kPow (P : Fin (N + 1) → Fin (N + 1) → ℝ) : ℕ → Fin (N + 1) → Fin
    (N + 1) → ℝ
  | 0 => fun i j => if i = j then 1 else 0
  | n + 1 => fun i j => ∑ k, kPow P n i k * P k j
structure AbsorbingChain (P : Fin (N + 1) → Fin (N + 1) → ℝ) : Prop where
  nonneg : ∀ i j, 0 ≤ P i j
  rowSum : ∀ i, ∑ j, P i j = 1
  absorb0 : ∀ j, P 0 j = if j = 0 then 1 else 0
  absorbN : ∀ j, P (Fin.last N) j = if j = Fin.last N then 1 else 0
def InteriorHarmonic (P : Fin (N + 1) → Fin (N + 1) → ℝ) (g : Fin (N + 1) → ℝ) :
    Prop :=
  ∀ i : Fin (N + 1), 0 < (i : ℕ) → (i : ℕ) < N → ∑ j, P i j * g j = g i
theorem fixation_limit {P : Fin (N + 1) → Fin (N + 1) → ℝ} (hP : AbsorbingChain P) (hN
    : 0 < N)
    {h : Fin (N + 1) → ℝ} (hh : InteriorHarmonic P h) (h0 : h 0 = 0) (h1 : h (Fin.last
    N) = 1)
    (hvar : ∀ k : Fin (N + 1), 0 < (k : ℕ) → (k : ℕ) < N → 0 < ∑ j, P k j * (h j - h
     k) ^ 2)
    (i : Fin (N + 1)) :
    Tendsto (fun n => kPow P n i (Fin.last N)) atTop (𝓝 (h i)) ∧
    Tendsto (fun n => kPow P n i 0) atTop (𝓝 (1 - h i)) := by
```

`WrightFisherFixation.lean`, ll. 49–51:

```
theorem wright_fisher_fixation_limit {N : ℕ} (hN : 0 < N) (i : Fin (N + 1)) :
    Tendsto (fun n => kPow (wfP N) n i (Fin.last N)) atTop (𝓝 (((i : ℕ) : ℝ) / N)) ∧
    Tendsto (fun n => kPow (wfP N) n i 0) atTop (𝓝 (1 - ((i : ℕ) : ℝ) / N)) := by
```

Side by side: two different routes. The Lean proof does not use Lemma 4.10. It uses a *second-moment* argument, valid for every absorbing chain provided that a harmonic function h has a one-step variance $v(k) = \sum_j P(k,j)(h(j)-h(k))^2$ that is strictly positive at every interior state (hypothesis `hvar`). Here are the steps.

1. We have the identity $\sum_j P(k,j)h(j)^2 = h(k)^2 + v(k)$ (`second_moment`, l. 145).
2. With $\varepsilon = \min v > 0$, we obtain $P^{m+1}(h^2) \geq P^m(h^2) + \varepsilon I_m$. Summing, $\varepsilon \sum_{m<n} I_m(i) \leq \sum_j h(j)^2$.
3. The series $\sum I_m$ therefore has non-negative terms and bounded partial sums: it converges, and $I_m \to 0$ (`interiorMass_tendsto_zero`, l. 156).
4. We conclude by the squeeze above (`absorption_tendsto`, `mass_split`).

This route does not give the geometric rate. On the other hand, it applies as is to the Moran model (§4.3) and to graphs (§4.5), where a direct jump to 0 is impossible. For Wright–Fisher, the variance is positive thanks to the jump to 0 alone: $v(k) \geq P(k,0)(k/N)^2 > 0$ (`wf_var_pos`, l. 34).

Remark 4.11 (Controls **[Tier A]**). *Positive control.* We have $P^2(1,2) = \frac{3}{8}$ for $N = 2$ (`ctl_pos_wf_two_steps`, l. 62). The limit $\frac{1}{2}$ is therefore not yet reached after two generations. *Negative control.* The variance hypothesis cannot be dropped. The identity matrix on $\{0,1,2\}$ is stochastic and absorbing, and $j/2$ is harmonic for it; yet $P^n(1,2) = 0$ for every n (`ctl_neg_identity_no_absorption`, `Absorption.lean`, l. 363). *Uniqueness.* If g is another harmonic function with the same boundary values, the same argument gives $P^n(i,N) \to g(i)$. By uniqueness of limits, $g = h$ (`harmonic_unique_of_var`, l. 320). This replaces the maximum principle by the uniqueness of limits.

Remark 4.12 (The Genetic repository). The companion repository `SocrateAI-Scientific-Genetic` [59] contains an earlier version of these results, over $\mathbb{Q}$. It is in `formal/ANSE/AlixPopulationGenetics.lean`: the maximum principle `ilene_harmonic_le_zero` (l. 82), uniqueness `xav_wf_harmonic_solution_unique` (l. 199) and the bound without passing to the limit `xav_wf_fixation_by_time_le` (l. 268), which asserts $P^t(i,N) \le i/N$ for every t. The latter follows in one line from the martingale: $P^t(i,N)\cdot 1 \le \sum_j P^t(i,j)\, j/N = i/N$. The proof of this one-line bound is a human proof **[Tier C]**. The file was not recompiled for this book, and we assign it no tier. The statements of this chapter come from the AutoevolveAI modules H11 and P7. Between the two repositories, the link is only textual: no Lean term connects the two namespaces **[Tier C]**.

4.3 The Moran process with selection

4.3.1 The model

A population has N individuals, of which i are mutants with fitness $r > 0$; residents have fitness 1. At each time step, two independent draws take place.

- One individual is chosen to reproduce, with probability proportional to its fitness. A mutant is therefore chosen with probability $\frac{ri}{ri+N-i}$.
- One individual is chosen *uniformly* among the N to die, and the offspring takes its place.

The number of mutants goes up by 1 with probability p_i, goes down by 1 with probability q_i and stays unchanged otherwise:

$$p_i = \frac{ri}{ri+N-i}\cdot\frac{N-i}{N}, \qquad q_i = \frac{N-i}{ri+N-i}\cdot\frac{i}{N}.$$

These are the rates of Keller and Uğurlu, up to a one-line rewriting **[Tier L]** [49] (pp. 2–4) **[Tier C]**.

Lean 4 : the Moran model (`formal_cert/Programmes/PopGen/MoranModel.lean`, ll. 30–32, 74–83, 152–155)

```
noncomputable def bdP (N : ℕ) (p q : ℕ → ℝ) (i j : Fin (N + 1)) : ℝ :=
  (if (j : ℕ) = (i : ℕ) + 1 then p i else 0) + (if (j : ℕ) + 1 = (i : ℕ) then q i else
    0) +
    (if (j : ℕ) = (i : ℕ) then 1 - p i - q i else 0)
noncomputable def moranUp (N : ℕ) (r : ℝ) (i : ℕ) : ℝ :=
  r * i / (r * i + ((N : ℝ) - i)) * (((N : ℝ) - i) / N)
noncomputable def moranDown (N : ℕ) (r : ℝ) (i : ℕ) : ℝ :=
  ((N : ℝ) - i) / (r * i + ((N : ℝ) - i)) * ((i : ℝ) / N)
noncomputable def moranP (N : ℕ) (r : ℝ) : Fin (N + 1) → Fin (N + 1) → ℝ :=
  bdP N (moranUp N r) (moranDown N r)
noncomputable def geomS (r : ℝ) (n : ℕ) : ℝ := ∑ k ∈ range n, r⁻¹ ^ k
noncomputable def moranFix (N : ℕ) (r : ℝ) (i : ℕ) : ℝ := geomS r i / geomS r N
```

The independent auditor checked symbolically that `moranUp` and `moranDown` coincide with the Keller–Uğurlu rates (PG-C-0004). This is an AI judgement, not a human audit.

Lemma 4.13. *Let $N \ge 1$ and $r > 0$.*

(i) $p_i = r\, q_i$ *for every i: the ratio of the rates is the fitness.*

(ii) $1 - p_i - q_i = \dfrac{ri^2 + (N-i)^2}{(ri+N-i)N} \ge 0$, *so the matrix is stochastic. Moreover* $p_0 = q_0 = p_N = q_N = 0$, *so 0 and N are absorbing.*

(iii) For $0 < i < N$, we have $p_i > 0$ and $q_i > 0$.

[Tier A] *(`moranUp_eq`, `moran_stay_eq`, `moran_stay_nonneg`, `moranUp_pos`, `moranDown_pos`, `moran_absorbing`, `MoranModel.lean`, ll. 85–149, and `MoranFixation.lean`, l. 25).*

Human proof. (i) We have $p_i/q_i = \frac{ri(N-i)}{(N-i)i} = r$ if $0 < i < N$, and both rates vanish at the boundary. (ii) We compute $(ri+N-i)N-ri(N-i)-(N-i)i = riN-ri(N-i)+(N-i)(N-i) = ri^2 + (N-i)^2$. (iii) All factors are strictly positive when $0 < i < N$. □

4.3.2 Gambler's ruin

Theorem 4.14 (Fixation in the Moran model)**.** *Let $N \geq 1$, $r > 0$, and $S_n = \sum_{k<n} r^{-k}$. For every $0 \leq i \leq N$,*

$$P^n(i,N) \xrightarrow[n\to\infty]{} \frac{S_i}{S_N} = \begin{cases} \dfrac{1-r^{-i}}{1-r^{-N}} & \text{if } r \neq 1, \\ \dfrac{i}{N} & \text{if } r = 1, \end{cases} \qquad P^n(i,0) \longrightarrow 1 - \frac{S_i}{S_N}.$$

Moreover, S_i/S_N is the unique function harmonic in the interior that equals 0 at 0 and 1 at N, and $\frac{1-r^{-i}}{1-r^{-N}} \to \frac{i}{N}$ as $r \to 1$, $r \neq 1$. ***[Tier A]*** *(`moran_fixation_limit`, `moran_fixation_limit_closed`, `moran_fixation_limit_neutral`, `moran_fixation_unique`, `MoranFixation.lean`, ll. 72–97; `moranFix_closed`, `moran_closed_tendsto_neutral`, `MoranModel.lean`, ll. 203 and 231).*

Human proof. Step 1: the recurrence. Let h be harmonic in the interior, with $h(0) = 0$ and $h(N) = 1$. For $0 < i < N$, the first-step equation reads

$$p_i h(i+1) + q_i h(i-1) + (1-p_i-q_i)h(i) = h(i) \iff p_i\big(h(i+1)-h(i)\big) = q_i\big(h(i)-h(i-1)\big).$$

Since $p_i = rq_i$ and $q_i > 0$, the increments $\Delta_i = h(i+1) - h(i)$ satisfy $\Delta_i = r^{-1}\Delta_{i-1}$. This is a geometric sequence: $\Delta_i = r^{-i}\Delta_0$. Summing, $h(i) = \Delta_0\, S_i$. The condition $h(N) = 1$ then forces $\Delta_0 = 1/S_N$, with $S_N > 0$. There is therefore *at most* one solution, and conversely $h(i) = S_i/S_N$ is one, since the equivalences can be reversed. This is the classical "gambler's ruin" argument.

Step 2: closed form. If $r \neq 1$, set $\rho = 1/r \neq 1$. Then $S_n = \frac{1-\rho^n}{1-\rho}$, whence $\frac{S_i}{S_N} = \frac{1-r^{-i}}{1-r^{-N}}$. If $r = 1$, $S_n = n$. Each $S_n(r)$ is a finite sum of continuous functions of r and $S_N(1) = N \neq 0$, so the quotient is continuous at $r = 1$. This is the neutral limit.

Step 3: absorption. We repeat the proof of Theorem 4.9 with $h = S_\cdot/S_N$: $P^n h = h$, then $|h(i) - P^n(i,N)| \leq I_n(i)$, because $0 \leq h \leq 1$. It remains to show that $I_n(i) \to 0$. Lemma 4.10 no longer applies as is: from i, one cannot jump to 0 in one step if $i \geq 2$. We therefore argue in blocks. From every interior state k, the path $k \to k-1 \to \cdots \to 0$ has probability $q_k q_{k-1} \cdots q_1 > 0$ and lasts only $k \leq N-1$ steps. Once at 0, the chain stays there. Set $m = N-1$ and $\eta = \min_{0<k<N} q_k \cdots q_1 > 0$. Then $P^m(k,0) \geq \eta$ for every interior k, and the same computation as in Lemma 4.10, applied to the matrix P^m, gives $I_{n+m}(i) \leq (1-\eta)I_n(i)$. Since I_n is nonincreasing (absorbed mass stays absorbed), $I_n(i) \to 0$ **[Tier C]**. □

Lean 4 : the theorem and its closed form (`MoranFixation.lean`, ll. 72–74; `MoranModel.lean`, ll. 173–175, 203–204)

```
theorem moran_fixation_limit {N : ℕ} {r : ℝ} (hN : 0 < N) (hr : 0 < r) (i : Fin (N + 1)
    ) :
    Tendsto (fun n => kPow (moranP N r) n i (Fin.last N)) atTop (𝓝 (moranFix N r i)) ∧
    Tendsto (fun n => kPow (moranP N r) n i 0) atTop (𝓝 (1 - moranFix N r i)) :=
theorem moranFix_harmonic_nat {N : ℕ} {r : ℝ} (hr : 0 < r) {i : ℕ} (h1 : 0 < i) :
    moranUp N r i * moranFix N r (i + 1) + moranDown N r i * moranFix N r (i - 1)
```

```
      + (1 - moranUp N r i - moranDown N r i) * moranFix N r i = moranFix N r i := by
theorem moranFix_closed {N : ℕ} {r : ℝ} (hr1 : r ≠ 1) (i : ℕ) :
    moranFix N r i = (1 - r⁻¹ ^ i) / (1 - r⁻¹ ^ N) := by
```

Side by side. Lean does not *solve* the recurrence. It *checks* that the guessed solution S_i/S_N is harmonic (`moranFix_harmonic_nat`, with a single `linear_combination`). It then applies the general theorem `fixation_limit` of §4.2, with the variance $v(i) \geq p_i\,(r^{-i}/S_N)^2 > 0$ (`moran_var_pos_nat`, l. 189). The block argument of Step 3 is not formalised; the second moment replaces it. Uniqueness comes from the uniqueness of limits (`moran_fixation_unique`), not from Step 1.

Remark 4.15 (Controls **[Tier A]**). The following controls are all checked by the kernel.

- *Positive controls.* For $N = 2$, $r = 2$, we have $p_1 = \frac{1}{3}$, $q_1 = \frac{1}{6}$ and $h(1) = \frac{2}{3}$ (`ctl_pos_moran_rates`, `ctl_pos_moranFix_N2`). For $N = 3$, $r = 2$, we have $h = (0, \frac{4}{7}, \frac{6}{7}, 1)$, and the first-step equation holds (`ctl_pos_moranFix_N3`, `MoranModel.lean`, ll. 240–253).
- *Negative controls.* The "inverted-exponent" formula $(1 - r^i)/(1 - r^N)$ gives $\frac{1}{7} \neq \frac{4}{7}$ and violates the first-step equation. The neutral value i/N also violates it under selection (ll. 259–276). At the level of the chain, $P^n(1, 3)$ for $N = 3$, $r = 2$ tends neither to $\frac{1}{3}$ nor to $\frac{1}{7}$ (`ctl_neg_wrong_limits`, `MoranFixation.lean`, l. 102).
- *Planted controls.* In a separate file that nothing imports, a "theorem" proved by `sorry` and another derived from a planted axiom assert the false limits $\frac{1}{7}$ and $\frac{1}{3}$. This file (`PopGenProgControls.lean`) compiles with return code 0, but `#print axioms` shows `sorryAx`, resp. the planted axiom `ctl_cheat`. They are therefore detected (PG-A-0012).

Numerics : CVODE and the "fail-closed" certificate

[Tier X] Programme P7 also integrated the uniformised Kolmogorov equation $\dot{\pi} = \pi(P - I)$ with CVODE (BDF, exact Jacobian), in a new crate `qf-popgen` of rusty-SUNDIALS. The crate wraps itself in a certificate ("fail-closed": it refuses by default) that rejects any doubtful result: valid kernel, mass conservation, componentwise positivity, decay of the interior mass and agreement with the target. Outcome: 106 accepted runs agree with the closed forms to within 3.1×10^{-10}; the 9 negative controls are all rejected. The uniformised law is a Poisson mixture of the P^n **[Tier C]**. These computations therefore test the same formulas, but not the Lean matrix powers, and *they establish nothing.*

4.3.3 Interpretation: does a resistance mutation take over the population?

Physics and biology : biological reading

[Tier C] The Moran model is often read as a model of the following question: a bacterium or a cell becomes resistant to a drug; does its lineage take over? If the mutation confers, in the presence of the drug, a relative advantage $r > 1$, the probability that a single mutant fixes is

$$\rho_N(1) = \frac{1 - r^{-1}}{1 - r^{-N}} \xrightarrow[N\to\infty]{} 1 - \frac{1}{r}.$$

The closed form is **[Tier A]** (Theorem 4.14) and the limit in N is an elementary computation **[Tier C]**. Evaluated in floating point **[Tier X]**, it gives for $N = 100$:

	$r = 0.99$	$r = 1.01$	$r = 1.1$
$\rho_{100}(1)$	0.0058	0.0157	0.0909
neutral value $1/N$	0.01	0.01	0.01

Two qualitative lessons follow. First, even an advantageous mutant ($r = 1.1$) is lost most of the time, here in about 91% of cases, swept away by drift. Second, a disadvantageous mutant ($r < 1$) still has a small chance of fixing.
Caution. This is only a reading, and it remains at Tier C. The model assumes a population of fixed size, well mixed, a constant fitness, no new mutation, uniform death and generations that overlap step by step. A real infection has a varying size, changing drug concentrations, a cost of resistance, spatial structure and recurrent mutations. No data were used, and no Lean theorem certifies that a real population follows this chain.

4.4 The Price equation and the haploid corollary

4.4.1 An accounting identity

A finite population consists of types $i \in s$. Type i has a count n_i, a fitness w_i (mean number of offspring per individual), a parental trait z_i, and its offspring have mean trait z'_i. We write

$$M = \sum n_i, \quad A = \sum n_i w_i, \quad B = \sum n_i z_i, \quad C = \sum n_i w_i z_i, \quad D = \sum n_i w_i z'_i.$$

We then define $\bar{w} = A/M$ and $\bar{z} = B/M$. The mean of the next generation is $\bar{z}' = D/A$, since type i leaves $n_i w_i$ offspring: this is the *modelling assumption*. Finally we write $\mathrm{Cov}_n(w, z) = \frac{1}{M}\sum n_i (w_i - \bar{w})(z_i - \bar{z})$ and $\mathbb{E}_n[f] = \frac{1}{M}\sum n_i f_i$.

Theorem 4.16 (Price equation)**.** *Over an arbitrary field, if $M \neq 0$ and $A \neq 0$, then*

$$\bar{w}\,(\bar{z}' - \bar{z}) = \mathrm{Cov}_n(w, z) + \mathbb{E}_n\big[w\,(z' - z)\big].$$

[Tier A] *(`price_equation`, `formal_cert/Programmes/PopGen/Price.lean`, ll. 95–105; real version under positivity hypotheses `price_equation_pos`, l. 170).*

Human proof. On the one hand, $\mathrm{Cov}_n(w, z) = \frac{1}{M}\big(C - \bar{w}B - \bar{z}A + M\bar{w}\bar{z}\big) = \frac{C - AB/M}{M}$; on the other hand, $\mathbb{E}_n[w(z' - z)] = \frac{D - C}{M}$. The sum is $\frac{D}{M} - \frac{AB}{M^2}$. But

$$\bar{w}(\bar{z}' - \bar{z}) = \frac{A}{M}\Big(\frac{D}{A} - \frac{B}{M}\Big) = \frac{D}{M} - \frac{AB}{M^2}. \qquad \square$$

Lean 4 : the statement (`Price.lean`, ll. 95–98)

```
theorem price_equation (s : Finset ι) (n w z z' : ι → K)
    (hN : ∑ i ∈ s, n i ≠ 0) (hW : ∑ i ∈ s, n i * w i ≠ 0) :
    wmean s n w * (zbarNext s n w z' - wmean s n z)
      = wcov s n w z + wmean s n (fun i => w i * (z' i - z i)) := by
```

The Lean proof follows the human proof. It expands the two numerators into the five raw sums (`cov_num_expand`, `trans_num_expand`), then concludes with a scalar lemma in which M, A, B, C, D are atoms (`price_scalar`: `field_simp` then `ring`). The statement holds over an arbitrary field `K`, for example $\mathbb{Q}$ or $\mathbb{R}$, and for every finite set of types.

4.4.2 The haploid corollary

Corollary 4.17 (Haploid selection increment)**.** *Take as trait the fitness itself, $z = w$, and assume there is* no transmission bias*, $z' = w$: clonal types, constant fitnesses, no mutation. Then*

$$\bar{w}\,(\bar{w}' - \bar{w}) = \mathrm{Var}_n(w), \qquad \textit{hence} \qquad \Delta\bar{w} = \frac{\mathrm{Var}_n(w)}{\bar{w}} \geq 0 \quad (\textit{if } n_i > 0,\ w_i > 0).$$

[Tier A] *(`fisher_haploid`, `fisher_haploid_div`, `fisher_haploid_nondecreasing`, `fisher_haploid_pos`, `Price.lean`, ll. 125–185; `fisher_haploid_nondecreasing` is re-exported under the name `AURELIA`).*

Human proof. With $z = z' = w$, the transmission term $\mathbb{E}_n[w(w - w)]$ vanishes identically, and $\mathrm{Cov}_n(w, w) = \mathrm{Var}_n(w)$. We divide by $\bar{w} \neq 0$. The sign comes from $\mathrm{Var}_n(w) \geq 0$ and $\bar{w} > 0$. □

This is equation (2.27) of LaGatta, with a vanishing environmental term **[Tier L]** [51]. It is also the discrete analogue of Baez's remark: $d\bar{f}/dt = \mathrm{Var}(f)$ when the fitnesses are constant **[Tier L]** [52].

For two alleles with frequencies p, $1 - p$ and fitnesses w_1, w_2, one finds $\mathrm{Var}_n(w) = p(1 - p)(w_1 - w_2)^2$, whence $\Delta\bar{w} = p(1 - p)(w_1 - w_2)^2/\bar{w}$. Module P7 derives from it, in the case `Fin 2`, two statements whose text is identical to two lemmas of the Genetic repository (`Price.lean`, ll. 224 and 235) **[Tier A]**. The connection between the two repositories remains textual **[Tier C]**.

Remark 4.18 (Controls **[Tier A]**). *Positive control.* With three types, $n = (1, 2, 3)$, $w = (1, 2, 4)$, $z = (0, 1, 3)$ and $z' = (1, 1, 2)$, both sides of the Price equation equal $-\frac{13}{36}$ (`ctl_pos_three_types`, l. 249). *Transmission bias.* Take $n = (1, 1)$ and $w = z = (1, 2)$, but an offspring fitness $z' = (1, 1)$. Then $\Delta\bar{w} = -\frac{1}{2}$, whereas $\mathrm{Var}/\bar{w} = \frac{1}{6}$: the variance formula *fails*. The Price equation, however, still holds, with $-\frac{3}{4} = -\frac{3}{4}$ (`ctl_neg_transmission_bias`, l. 263). *The factor $1/\bar{w}$ matters.* Without bias, $\Delta\bar{w} = \frac{1}{6}$ but $\mathrm{Var}(w) = \frac{1}{4}$ (`ctl_neg_missing_wbar`, l. 280).

Warning : a negative result, announced as such

The preregistration of programme P7 included an item `fisher_general`: Fisher's fundamental theorem in Price's (1972) "partial change" reading, with a prior probability of 0.35. This item was **not reached**. Only the haploid case with constant fitnesses is proved. Nor is it the multi-allelic diploid case, where monotonicity is an inequality and not this equality. Price (1970, 1972) was not read; it is cited after LaGatta [51] (p. 69).

4.5 Evolutionary graph theory: the isothermal theorem

4.5.1 A structured population

In a tissue, a dividing cell does not replace just any other cell: it replaces a *neighbour*. Evolutionary graph theory describes this structure by a weighted directed graph. The individuals are the vertices $V = \{1, \ldots, n\}$. The *replacement matrix* W has entries ≥ 0, and is *row-stochastic*: $\sum_j W_{ij} = 1$. The entry W_{ij} is the probability that the offspring of i replaces j; loops $W_{ii} > 0$ are allowed.

A *state* is the set $S \subseteq V$ of mutant vertices. Mutants have fitness $r > 0$ and residents fitness 1; we write $f_S(i) = r$ if $i \in S$ and $f_S(i) = 1$ otherwise. Fix selection weights $\mu_i > 0$, and set $F(S) = \sum_k \mu_k f_S(k)$. One step of the *birth–death* update proceeds as follows.

- Vertex i is chosen to reproduce with probability $\mu_i f_S(i)/F(S)$.

- Its offspring replaces j with probability W_{ij}.
- The new state is $S \cup \{j\}$ if $i \in S$, and $S \setminus \{j\}$ otherwise.

The case $\mu \equiv 1$ is the process of Lieberman, Hauert and Nowak [54]. The chain has 2^n states; $\emptyset$ and V are absorbing. We write $P^+(S)$ and $P^-(S)$ for the probabilities that the *number* of mutants $|S|$ increases, resp. decreases, by one.

Definition 4.19. The matrix W is *isothermal* if it is, in addition, column-stochastic: $\sum_i W_{ij} = 1$. The "temperature" $\sum_i W_{ij}$ of each vertex then equals 1. It is *strongly connected* if, for all a, b, there is a path $a = x_0, \dots, x_m = b$, $m \geq 1$, with $W_{x_k x_{k+1}} > 0$.

Lean 4 : the model (`formal_cert/Programmes/EvoGraph/BirthDeath.lean`, ll. 44–61, 73–74, 90–91)

```
noncomputable def fit (r : ℝ) (S : Finset (Fin n)) (i : Fin n) : ℝ := if i ∈ S then r
    else 1
noncomputable def ind (S : Finset (Fin n)) (i : Fin n) : ℝ := if i ∈ S then 1 else 0
def upd (S : Finset (Fin n)) (i j : Fin n) : Finset (Fin n) := if i ∈ S then insert j S
    else S.erase j
noncomputable def totFit (μ : Fin n → ℝ) (r : ℝ) (S : Finset (Fin n)) : ℝ := ∑ k, μ k
     * fit r S k
noncomputable def coef (μ : Fin n → ℝ) (W : Fin n → Fin n → ℝ) (r : ℝ) (S : Finset (
    Fin n))
    (i j : Fin n) : ℝ :=
  μ i * fit r S i / totFit μ r S * W i j
noncomputable def bdK (μ : Fin n → ℝ) (W : Fin n → Fin n → ℝ) (r : ℝ) (S T : Finset
    (Fin n)) : ℝ :=
  ∑ i, ∑ j, coef μ W r S i j * (if upd S i j = T then 1 else 0)
noncomputable def upP (μ : Fin n → ℝ) (W : Fin n → Fin n → ℝ) (r : ℝ) (S : Finset (
    Fin n)) : ℝ :=
  ∑ T, bdK μ W r S T * (if T.card = S.card + 1 then 1 else 0)
def StronglyConnected (W : Fin n → Fin n → ℝ) : Prop :=
  ∀ a b, Relation.TransGen (fun x y => 0 < W x y) a b
```

The independent auditor judged that `upP`, `downP` and `bdK` correctly model the Lieberman–Hauert–Nowak process. This is an AI judgement, not a human audit **[Tier C]**.

4.5.2 Forward bias, configuration by configuration

For $S \subseteq V$, we define the *cuts*

$$\mathrm{out}(S) = \sum_{i \in S} \sum_{j \notin S} \mu_i W_{ij}, \qquad \mathrm{in}(S) = \sum_{i \notin S} \sum_{j \in S} \mu_i W_{ij}.$$

Lemma 4.20 (Moves are cuts)**.** *We have $P^+(S) = r\,\mathrm{out}(S)/F(S)$ and $P^-(S) = \mathrm{in}(S)/F(S)$. The chain is stochastic, and $\emptyset$, V are absorbing.* ***[Tier A]*** *(`up_eq`, `down_eq`, `bd_absorbing`, `BirthDeath.lean`, ll. 148–250).*

Human proof. The coefficients $c(S, i, j) = \mu_i f_S(i) W_{ij}/F(S)$ are ≥ 0, and their sum is 1, because $\sum_j W_{ij} = 1$ and $\sum_i \mu_i f_S(i) = F(S)$. There are four cases according to whether i and j are in S or not.

- $i \in S$ and $j \notin S$: a mutant invades a resident, and $|S|$ increases. The weight is $\mu_i r W_{ij}/F$.
- $i \notin S$ and $j \in S$: $|S|$ decreases. The weight is $\mu_i W_{ij}/F$.
- In the other two cases, the state does not change.

Summing, we obtain the two formulas. Finally, $\mathrm{upd}(\emptyset, i, j) = \emptyset$ and $\mathrm{upd}(V, i, j) = V$. □

Proposition 4.21 (Cut balance and forward bias)**.** *Assume μ is stationary: $\sum_i \mu_i W_{ij} = \mu_j$ for every j. Then* $\mathrm{out}(S) = \mathrm{in}(S)$ *for* every *configuration S, and therefore*

$$P^+(S) = r\,P^-(S) \qquad \text{for every } S \subseteq V.$$

In particular, this is the case if W is isothermal and $\mu \equiv 1$. ***[Tier A]*** *(`flow_balance`, `cut_balance`, `forward_bias`, `BirthDeath.lean`, ll. 176–256).*

Human proof. We compute the same quantity in two ways. On the one hand, $\sum_{i\in S}\sum_{j\in V}\mu_i W_{ij} = \mu(S)$, since $\sum_j W_{ij} = 1$. On the other hand, $\sum_{j\in S}\sum_{i\in V}\mu_i W_{ij} = \mu(S)$, by stationarity. Both double sums contain the common part $\sum_{i\in S}\sum_{j\in S}\mu_i W_{ij}$. Once it is removed, what remains is $\mathrm{out}(S) = \mathrm{in}(S)$. Lemma 4.20 then gives $P^+ = r\,\mathrm{out}/F = r\,\mathrm{in}/F = rP^-$. If W is isothermal, $\sum_i 1 \cdot W_{ij} = 1 = \mu_j$: this is the stationarity of $\mu \equiv 1$. □

Lean 4 : forward bias (`BirthDeath.lean`, ll. 251–253)

```
theorem forward_bias {μ : Fin n → ℝ} {W : Fin n → Fin n → ℝ} (r : ℝ) (hrow : ∀ i, ∑
    j, W i j = 1)
  (hstat : ∀ j, ∑ i, μ i * W i j = μ j) (S : Finset (Fin n)) :
  upP μ W r S = r * downP μ W r S := by
```

The forward bias of Pattni et al. **[Tier L]** [50] (Lemma 1, p. 11) and the ratio $p^-/p^+ = 1/r$ of Theorem 3.2 of Keller–Uğurlu **[Tier L]** [49] say the same thing. Note that the identity is written without division: it remains true even if $P^-(S) = 0$.

Proposition 4.22 (A harmonic function on configurations)**.** *Let $\rho_n(k) = S_k/S_n$ be the Moran fixation function of Theorem 4.14. If μ is stationary, the function $h(S) = \rho_n(|S|)$ is harmonic for the configuration chain at every $S \neq \emptyset$. If moreover an edge $i \in S \to j \notin S$ satisfies $\mu_i W_{ij} > 0$, the one-step variance of h at S is strictly positive.* ***[Tier A]*** *(`moranFix_card_harmonic`, `moranFix_card_var_pos`, `BirthDeath.lean`, ll. 285–328).*

Human proof. Set $k = |S| \geq 1$. By Step 1 of Theorem 4.14, h goes up by $\Delta^+ = r^{-k}/S_n$ on a $+1$ step, and goes down by $\Delta^- = r^{-(k-1)}/S_n = r\,\Delta^+$ on a -1 step. Otherwise, h does not change. Hence

$$\mathbb{E}\big[h(S') - h(S)\big] = P^+(S)\Delta^+ - P^-(S)\Delta^- = \frac{r^{-k}}{S_n}\big(P^+(S) - r\,P^-(S)\big) = 0$$

by Proposition 4.21. For the variance, the move $i \to j$ has probability > 0 and changes h by $\Delta^+ \neq 0$. □

This computation uses *only* the forward bias. It never asserts that the number of mutants is a Markov chain. We shall see in §4.5.4 that this assertion would be false.

Lemma 4.23 (Connectivity forces cuts)**.** *Assume W is strongly connected and $\mu > 0$. Then every nonempty subset S different from V has an outgoing edge $i \in S \to j \notin S$ with $\mu_i W_{ij} > 0$.* ***[Tier A]*** *(`cutPositive_of_strongly_connected`, `BirthDeath.lean`, l. 332).*

Human proof. Take $a \in S$, $b \notin S$, and a path $a = x_0, \ldots, x_m = b$ of weight > 0. Let ℓ be the smallest index such that $x_{\ell+1} \notin S$. It exists because $x_m \notin S$, and $x_\ell \in S$, because $x_0 \in S$. The edge $x_\ell \to x_{\ell+1}$ leaves S. In Lean, the proof proceeds by induction on the `TransGen` derivation. □

4.5.3 The theorem

Theorem 4.24 (Isothermal theorem, birth–death update)**.** *Let $n \geq 1$, $r > 0$, and W isothermal and strongly connected, with $\mu \equiv 1$. For* every *initial configuration S,*

$$P^t(S, V) \xrightarrow[t\to\infty]{} \rho_n(|S|) = \frac{1 - r^{-|S|}}{1 - r^{-n}} \quad (r \neq 1), \qquad \frac{|S|}{n} \quad (r = 1).$$

In particular, a single mutant placed at any *vertex v fixes with limiting probability $\frac{1-r^{-1}}{1-r^{-n}}$, exactly as in the well-mixed population. More generally, the same limit holds as soon as $\mu W = \mu$, $\mu > 0$, and every nonempty proper subset has an outgoing edge.* ***[Tier A]*** *(`isothermal_fixation`, `isothermal_fixation_single`, `isothermal_fixation_neutral`, `isothermal_fixation_uniform`, `stationary_fixation`, `BirthDeath.lean`, ll. 356–416; `isothermal_fixation_single` is re-exported under the name `ILENE`).*

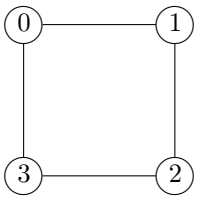
Human proof. Proposition 4.22 provides a function harmonic off $\{\emptyset, V\}$, equal to 0 at $\emptyset$ and 1 at V. Lemma 4.23 and Proposition 4.22 ensure that its variance is > 0 off the two absorbing states. The general absorption theorem of §4.2 applies, on an arbitrary finite state space. We transport it there by numbering the 2^n configurations from 0 to $2^n - 1$, with $\emptyset \mapsto 0$ and $V \mapsto 2^n - 1$. □

Lean 4 : the isothermal theorem (`BirthDeath.lean`, ll. 376–378)

```
theorem isothermal_fixation {W : Fin n → Fin n → ℝ} {r : ℝ} (hn : 0 < n) (hr : 0 < r)
    (hW : Isothermal W) (hsc : StronglyConnected W) (S : Finset (Fin n)) :
    Tendsto (fun t => sPow (bdK (fun _ => 1) W r) t S univ) atTop (𝓝 (moranFix n r S.
    card)) := by
```

Reuse. The module `Transport.lean` reproves nothing. The theorem `two_absorbing_limit` (l. 81) conjugates the configuration chain by a bijection with `Fin (M + 1)`, then applies the theorem `ProgPopGen.fixation_limit` of P7 *as is*. The files `Absorption.lean` and `MoranModel.lean` are reused without modification. *What is not proved.* The converse is not formalised: is a graph that gives the Moran value for every r necessarily isothermal? Proposition 2 of Pattni et al. is an equivalence; only the "sufficient" direction is formalised here **[Tier L]** [50]. The death–birth update is not formalised either.

4.5.4 Why the number of mutants does not "lump"



The 4-cycle: each vertex sends its offspring to one of its two neighbours, with probability $\frac{1}{2}$. W is doubly stochastic, hence isothermal.

Proposition 4.25 (The 4-cycle **[Tier A]**)**.** *On the 4-cycle, with $r = 2$ and $\mu \equiv 1$, the configurations $\{0, 1\}$ and $\{0, 2\}$ both have two mutants, but*

$$P^+(\{0, 1\}) = \tfrac{1}{3}, \qquad P^+(\{0, 2\}) = \tfrac{2}{3},$$

and the bias is $P^+ = 2P^-$ in both cases (`cycle4_not_lumpable`, `Examples.lean`, ll. 239–253).

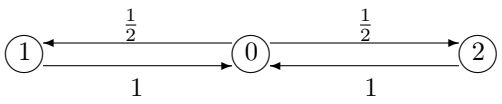
Human proof. For $S = \{0, 1\}$, we have $F = 2 + 2 + 1 + 1 = 6$. The outgoing edges are $0 \to 3$ and $1 \to 2$, each of weight $\frac{1}{2}$, so out $= 1$, whence $P^+ = 2 \cdot 1/6 = \frac{1}{3}$. Likewise in $= 1$ and $P^- = \frac{1}{6}$. For $S = \{0, 2\}$, the four edges leaving 0 and 2 all exit S, so out $=$ in $= 2$, $P^+ = \frac{4}{6} = \frac{2}{3}$ and $P^- = \frac{1}{3}$. □

Lean 4 : the counterexample (`Examples.lean`, ll. 230–231 and 239–243)

```
noncomputable def Wcyc : Fin 4 → Fin 4 → ℝ :=
  ![![0, 1 / 2, 0, 1 / 2], ![1 / 2, 0, 1 / 2, 0], ![0, 1 / 2, 0, 1 / 2], ![1 / 2, 0, 1 /
     2, 0]]
theorem cycle4_not_lumpable :
    ({0, 1} : Finset (Fin 4)).card = ({0, 2} : Finset (Fin 4)).card ∧
    upP (fun _ => 1) Wcyc 2 {0, 1} = 1 / 3 ∧ upP (fun _ => 1) Wcyc 2 {0, 2} = 2 / 3 ∧
    upP (fun _ => 1) Wcyc 2 {0, 1} = 2 * downP (fun _ => 1) Wcyc 2 {0, 1} ∧
    upP (fun _ => 1) Wcyc 2 {0, 2} = 2 * downP (fun _ => 1) Wcyc 2 {0, 2} := by
```

The transition probabilities of the count $|S_t|$ are therefore *not* functions of $|S_t|$. The number of mutants is not a birth–death chain with fixed rates, in the sense of the strong lumpability of Kemeny and Snell [58]. Two clarifications are in order **[Tier C]**. First, we do not claim that the number of mutants is non-Markovian for *every* initial distribution: only strong lumpability is refuted. Second, the *jump chain*, the one in which the steps where nothing happens are removed, does lump. On the 4-cycle, given that a move takes place, both configurations go up with the same probability $P^+/(P^+ + P^-) = \frac{2}{3}$, whereas their probabilities of staying put are $\frac{1}{2}$ and 0 (exact rational computation **[Tier X]**). This is the argument of Pattni et al. **[Tier L]** [50]. The Lean proof never lumps anything: it works configuration by configuration (Proposition 4.22).

4.5.5 The star graph: amplifying or suppressing selection



$$W_\star = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \qquad \text{(centre 0, leaves 1, 2).}$$

The star is row-stochastic, but its column 0 sums to 2: the centre has "temperature" 2, and the graph is *not* isothermal (`star_not_isothermal`) **[Tier A]**. Starting from a mutant at the centre, we have $P^+(\{0\}) = P^-(\{0\}) = \frac{1}{2}$. The ratio is 1, not $r = 2$ (`star_bias`) **[Tier A]**.

Proposition 4.26 (Exact values on the star **[Tier A]**). *For $n = 3$, $r = 2$ and $\mu \equiv 1$, the limiting fixation probabilities are*

$$\tfrac{5}{12} \text{ (mutant at the centre)}, \qquad \tfrac{2}{3} \text{ (mutant on a leaf)}, \qquad \tfrac{7}{12} \text{ (mutant placed uniformly)},$$

all different from the Moran value $\rho_3(1) = \frac{4}{7}$ (`star_fixation_centre`, `star_fixation_leaf`, `star_fixation_uniform`, `star_ne_moran`, `Examples.lean`, ll. 171–196). With the stationary weights $\mu = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$, one recovers $\frac{4}{7}$ starting from the centre (`starStationary_fixation`, l. 219).

The proof by hand is the subject of Exercise 4.11. In Lean, the table of the 2^3 values $h = (0, \frac{5}{12}, \frac{2}{3}, \frac{2}{3}, \frac{5}{6}, \frac{5}{6}, \frac{14}{15}, 1)$ is given explicitly (`hStar`, l. 139). One checks that it is harmonic (`star_harmonic`) and has positive variance, then feeds it into the *same* transported absorption machinery. An independent script in exact rational arithmetic gives the same table **[Tier X]**.

Physics and biology : amplifiers and suppressors

[Tier C] Compare with the Moran value $\frac{4}{7} \approx 0.571$. For a mutant placed uniformly at random, the star *amplifies* selection, since $\frac{7}{12} \approx 0.583 > \frac{4}{7}$. For a mutant born at the centre, it *suppresses* it, with $\frac{5}{12} \approx 0.417$. These conclusions must be read with care. They are only *three numbers*, for $n = 3$ and $r = 2$. They do not show that the star amplifies selection for every r or for every n. The preregistration, incidentally, called the star an "amplifier" without qualification. For $n = 4$, an unverified script gives in exact rationals $\frac{392}{1265}$ (centre), $\frac{168}{253}$ (leaf) and a mean $\frac{728}{1265} \neq \frac{8}{15}$. These values are **[Tier X]** and are not checked by the kernel. We do not quote the classical formula for the star, because the source (Lieberman–Hauert–Nowak) was not read.

4.5.6 Connectivity is necessary

Proposition 4.27 (Identity graph **[Tier A]**)**.** *On 2 vertices, the matrix $W = I$ is doubly stochastic, hence isothermal, but not strongly connected. With $r = 2$, we have $P^t(\{0\}, V) = 0$ for every t, which therefore does not tend to $\rho_2(1) = \frac{2}{3}$ (`identity_no_fixation`, `Examples.lean`, ll. 289–292).*

Each individual replaces only itself, and nothing ever moves. The strong connectivity hypothesis of Theorem 4.24 therefore cannot be dropped. This is a *counterexample*, not a proved converse **[Tier C]**.

4.5.7 A preregistration error, told honestly

Warning : what was announced, and what was proved

[Tier C] On 26 October 2026, item N4 of the preregistration announced: prove that if the graph is isothermal, "*the number of mutants is a birth-death chain with the Moran ratio r*", and deduce from it the fixation $(1 - r^{-1})/(1 - r^{-N})$ by reusing P7. Here is what happened.

- The *conclusion* is proved, and even more. It holds for every initial configuration, for a mutant at any vertex, for a uniformly placed mutant, and for $r = 1$ (Theorem 4.24).
- The italicised *intermediate clause* is **false** for the chain that includes the steps where nothing happens. It was therefore not formalised. The 4-cycle counterexample (Proposition 4.25) is checked by the kernel.
- What is true and proved is the forward bias *configuration by configuration*: $P^+(S) = rP^-(S)$ (Proposition 4.21).

This is not a discovery. Keller and Uğurlu observe that the Lieberman–Hauert–Nowak argument projects the chain onto the number of mutants, and that "the projection does not need to be Markovian" **[Tier L]** [49] (abstract, p. 4). The preregistration error thus reproduced a known gap in a published proof. Keller and Uğurlu also write that the theorem "remains unproven till today". We do not repeat this sentence, because Lemma 1 of Pattni et al., which is earlier, gives a valid proof of it via the jump chain **[Tier C]**. The lesson is one of method: state the goal, let the kernel decide, then publish the discrepancy.

Physics and biology : biological scope

[Tier C] In this model, a balanced (doubly stochastic) and strongly connected structure does not change the fixation probability of a mutant of fitness r, wherever it appears. A structure that amplifies or suppresses selection must therefore break this balance; the star is an example of this, for a single choice of the parameters. Tissues, where cells replace their neighbours, are the usual motivation. We make, however, no claim specific to any tissue, and use no data. The model omits mutation, other update rules, changing structures and frequency dependence.

4.6 Summary: what is verified, what is not

statement	Lean declaration (module)	tier
Hardy–Weinberg stationary	`hwStep_stationary` (H12)	A
increment identity, monotonicity of $\bar{w}$	`wbar_increment` (H12)	A
i/N harmonic and unique (maximum principle)	`harmonic_unique` (H11)	A
$P^n(i,N) \to h(i)$, general absorbing chain	`fixation_limit` (P7)	A
Wright–Fisher: $P^n(i,N) \to i/N$	`wright_fisher_fixation_limit` (P7)	A
geometric decay of I_n	— (Lemma 4.10)	C
Moran: $P^n(i,N) \to \frac{1-r^{-i}}{1-r^{-N}}$	`moran_fixation_limit_closed` (P7)	A
Price equation; $\Delta\bar{w} = \mathrm{Var}/\bar{w}$ (haploid)	`price_equation`, `fisher_haploid_div` (P7)	A
Fisher's theorem, "partial change" reading	not reached	—
forward bias $P^+ = rP^-$	`forward_bias` (N4)	A
isothermal theorem (sufficient direction)	`isothermal_fixation` (N4)	A
4-cycle: $\frac{1}{3} \neq \frac{2}{3}$	`cycle4_not_lumpable` (N4)	A
star $n = 3$: $\frac{5}{12}$, $\frac{2}{3}$, $\frac{7}{12}$; Moran $\frac{4}{7}$	`star_fixation_uniform` (N4)	A
star $n = 4$	exact script	X
"fixation probability" $= \lim P^n(i,N)$	no path space	C
biological reading (resistance, tissues)	—	C

Kernel replay performed for this chapter. On 2026-10-26, the writing agent ran `leanchecker` (Lean 4.34.1, pinned toolchain, two cores) on the ten modules cited. They are `Autoresearch.PopGenFisher`, `Autoresearch.PopGenWrightFisher`, `Programmes.PopGen.Absorption`, `WrightFisherFixation`, `MoranModel`, `MoranFixation`, `Price`, together with `Programmes.EvoGraph.Transport`, `BirthDeath` and `Examples`. All ten runs returned code 0, in 55 to 131 s each, with no output at all. The five modules previously without replay (`PopGenFisher`, `PopGenWrightFisher`, `WrightFisherFixation`, `MoranFixation`, `Price`) therefore now satisfy the definition of Tier A in this book. The `grep` search for `skipKernelTC`, `addDecl`, `native_decide`, `implemented_by`, `sorry`, `admit` and `axiom` in these ten sources finds only two mentions in documentation comments (`MoranModel.lean`, l. 237; `Examples.lean`, l. 6). Three limitations should be noted.

- The replay was run on the producer's `.olean` files, not on a clean recompilation. These files are all dated after their source.
- The replay is not `-fresh`: Mathlib and the Lean core were loaded, not replayed.
- The only negative control of this run is a nonexistent module, rejected with code 1. The control by falsification under `debug.skipKernelTC`, rejected by the same binary, is the

one from the N4 audit.

Like the rest of the chapter, this verification is the work of an AI agent; no human carried it out. The trace is `results/kernel_replay_2026-10-26/ch4_popgen_trace/leancheck_ch4.json`. The replay of the whole repository, run afterwards (`results/kernel_replay_2026-10-26/replay.jsonl`), also returned code 0 for `PopGenFisher`, `PopGenWrightFisher` and `PopGenImportBoth`.

What is not formalised. The following points remain beyond the scope of this chapter.

- There is no measure on path space. A natural next step would be to identify `kPow` with the law constructed by `ProbabilityTheory.Kernel.traj` (Ionescu–Tulcea, in Mathlib, after Marion **[Tier L]** [53]).
- Also not formalised: absorption times, Kimura's diffusion limit, mutation, multi-allelic diploid selection, frequency dependence, the converse of the isothermal theorem and the death–birth update.
- No human has audited the adequacy of the statements. The ledgers of P7 and N4 flag every Tier A line with `LEDGER_UNAUDITED_TIER_A`: there are 14 of them for P7 and 11 for N4. The Elenchus gate exits with code 1, and this is not hidden.

How to cite. Programme P7 (fixation, Moran, Price) is published as doi:10.5281/zenodo.23653850, and programme N4 (isothermal theorem) as doi:10.5281/zenodo.23660071 [46, 47]. Neither has been peer reviewed.

4.7 Exercises

Exercise 4.1 (Hardy–Weinberg)**.** A population has genotype frequencies $(x_{11}, x_{12}, x_{22}) = (0.5, 0.2, 0.3)$.

a) Compute p, then the genotype frequencies after one generation of random mating.

b) Check that a second generation changes nothing further. Which Lean theorem expresses this fact?

Answer. $p = 0.6$, then $(0.36, 0.48, 0.16)$. The theorem is `hwStep_stationary`.

Exercise 4.2 (Genic selection)**.** Take $w_{11} = (1+s)^2$, $w_{12} = 1+s$ and $w_{22} = 1$, with $s > -1$.

a) Show that $\bar{w}(p) = (1+ps)^2$ and that $p' = \dfrac{p(1+s)}{1+ps}$.

b) Deduce that, if $s > 0$ and $0 < p < 1$, the sequence of frequencies is strictly increasing and converges to 1.

Hint. For b), write $1 - p' = \frac{1-p}{1+ps}$ and iterate the inequality $1 - p_{t+1} \le \frac{1-p_t}{1+p_0 s}$.

Exercise 4.3 (The increment identity)**.** a) Prove the key identity $\bar{w} + a\,pq = p\,w_{11} + q\,w_{22}$, with $a = w_{11} - 2w_{12} + w_{22}$.

b) Deduce (4.1) without copying the proof given in the text.

c) For $w = (2, 1, 1)$ and $p = \frac{1}{2}$, compute $\bar{w}(p)$ and $\bar{w}(p')$. Check the identity on this example.

d) For $w = (-1, 1, 0)$ and $p = \frac{1}{2}$, show that $\bar{w}$ *decreases*. Which factor of (4.1) changes sign?

e) *Underdominance.* With $w = (1, \frac{1}{2}, 1)$, show that $p = \frac{1}{2}$ is an equilibrium. Compute p' and $\bar{w}(p')$ for $p = \frac{2}{5}$, and comment.

Answer. c) $\bar{w} = \frac{5}{4}$ and $\bar{w}(p') = \frac{34}{25}$. d) $\bar{w} = \frac{1}{4}$, $p' = 0$ and $\bar{w}(p') = 0$; the factor $\bar{w} + pw_{11} + qw_{22} = -\frac{1}{4}$ is negative. e) $p' = \frac{7}{19}$, and $\bar{w}$ goes from $\frac{19}{25}$ to $\frac{277}{361}$: the mean goes up, but p moves away from the equilibrium.

Exercise 4.4 (Wright–Fisher with two genes)**.** Take $N = 2$.

a) Write down the matrix P (3×3).

b) Show that $P^n(1,1) = 2^{-n}$ and that $P^n(1,2) = \frac{1}{2}\big(1 - 2^{-n}\big)$. Recover $P^2(1,2) = \frac{3}{8}$ (`ctl_pos_wf_two_steps`).

c) Compare the rate obtained with the bound of Lemma 4.10.

Answer. c) Here $\delta = \frac{1}{4}$, so the bound is $(\frac{3}{4})^n$, whereas the true decay is $(\frac{1}{2})^n$.

Exercise 4.5 (Wright–Fisher with three genes)**.** Take $N = 3$.

a) Compute rows 1 and 2 of P. Check that, from each interior state, the probability of staying in the interior is $\frac{2}{3}$.

b) Deduce that $I_n(1) = (\frac{2}{3})^n$ exactly, and that $\big|P^n(1,3) - \frac{1}{3}\big| \le \frac{2}{3}\,(\frac{2}{3})^n$.

Answer. a) Row 1 is $(\frac{8}{27}, \frac{4}{9}, \frac{2}{9}, \frac{1}{27})$ and row 2 is $(\frac{1}{27}, \frac{2}{9}, \frac{4}{9}, \frac{8}{27})$.

Exercise 4.6 (Moran, $N = 3$, $r = 2$)**.** a) Compute p_1, q_1, p_2, q_2 and check that $p_i = 2q_i$.

b) Solve the gambler's-ruin recurrence, and recover $h = (0, \frac{4}{7}, \frac{6}{7}, 1)$.

c) Show that the "inverted" formula $(1 - r^i)/(1 - r^N)$ gives $\frac{1}{7}$ at $i = 1$, and that it does not satisfy the first-step equation.

Answer. a) $p_1 = \frac{1}{3}$, $q_1 = \frac{1}{6}$, $p_2 = \frac{4}{15}$, $q_2 = \frac{2}{15}$.

Exercise 4.7 (Large population)**.** a) If $r > 1$, show that $\rho_N(1) = \frac{1 - r^{-1}}{1 - r^{-N}}$ decreases to $1 - \frac{1}{r}$ as $N \to \infty$.

b) If $0 < r < 1$, show that $\rho_N(1) \sim (r^{-1} - 1)\, r^N$. Compare with the neutral value $1/N$.

c) For $r = 1.1$, what proportion of advantageous mutants is lost in a very large population? Why does this reading remain at Tier C?

Exercise 4.8 (Price equation)**.** a) Take $n = (1,2,3)$, $w = (1,2,4)$, $z = (0,1,3)$ and $z' = (1,1,2)$. Compute $\bar{w}$, $\bar{z}$, $\bar{z}'$, $\mathrm{Cov}_n(w,z)$ and $\mathbb{E}_n[w\Delta z]$, and check that both sides equal $-\frac{13}{36}$.

b) For two types with frequencies p, $1-p$, show that $\mathrm{Var}_n(w) = p(1-p)(w_1 - w_2)^2$.

c) Take $n = (1,1)$, $w = z = (1,2)$ and $z' = (1,1)$. Show that $\Delta\bar{w} \neq \mathrm{Var}_n(w)/\bar{w}$, whereas the Price equation still holds. Which hypothesis of Corollary 4.17 is missing?

Answer. a) $\bar{w} = \frac{17}{6}$, $\bar{z} = \frac{11}{6}$ and $\bar{z}' = \frac{29}{17}$, so $\bar{z}' - \bar{z} = -\frac{13}{102}$; b) follows by a direct computation; c) $\Delta\bar{w} = -\frac{1}{2}$ and $\mathrm{Var}/\bar{w} = \frac{1}{6}$; what is missing is the absence of transmission bias.

Exercise 4.9 (The complete graph is the Moran model)**.** Take $W_{ij} = \frac{1}{n}$ for all i, j (loops included) and $\mu \equiv 1$.

a) Check that W is isothermal and strongly connected.

b) For $|S| = k$, compute $\mathrm{out}(S)$, $\mathrm{in}(S)$ and $F(S)$. Deduce that $P^+(S)$ and $P^-(S)$ are exactly the rates p_k, q_k of the Moran model (`wellMixed_up`, `wellMixed_down`, `Examples.lean`, ll. 62–81).

c) Deduce that, here, the configuration chain lumps exactly onto the Moran chain.

Answer. b) $\mathrm{out} = \mathrm{in} = k(n-k)/n$ and $F = rk + n - k$.

Exercise 4.10 (The 4-cycle, by hand)**.** a) Check that the matrix of the 4-cycle is doubly stochastic.

b) Recover $P^{\pm}(\{0,1\}) = (\frac{1}{3}, \frac{1}{6})$ and $P^{\pm}(\{0,2\}) = (\frac{2}{3}, \frac{1}{3})$.

c) Compute, for each of the two configurations, the probability of going up *given that a move takes place*, and comment.

d) What does the isothermal theorem give for a single mutant, when $r = 2$?

Answer. c) $\frac{2}{3}$ in both cases. d) $\rho_4(1) = \frac{1-1/2}{1-1/16} = \frac{8}{15}$.

Exercise 4.11 (Fixation probability on the star, by hand)**.** Take the three-vertex star (centre 0, leaves 1 and 2), with $W_{0j} = \frac{1}{2}$, $W_{j0} = 1$, $r = 2$ and $\mu \equiv 1$. By symmetry, set $x_0 = h(\{0\})$, $x_1 = h(\{1\}) = h(\{2\})$, $y = h(\{0,1\}) = h(\{0,2\})$ and $z = h(\{1,2\})$.

a) From $\{0\}$: show that $F = 4$, that the chain moves to $\{0, j\}$ with probability $\frac{1}{2}$ (in total) and to $\emptyset$ with probability $\frac{1}{2}$. Deduce that $x_0 = \frac{y}{2}$.

b) From $\{1\}$: show that the chain moves to $\{0,1\}$ with probability $\frac{1}{2}$, to $\emptyset$ with probability $\frac{1}{8}$, and stays put with probability $\frac{3}{8}$. Deduce that $x_1 = \frac{4}{5}y$.

c) From $\{0,1\}$: show that the chain moves to V with probability $\frac{1}{5}$ and to $\{1\}$ with probability $\frac{1}{5}$, and deduce that $2y = 1 + x_1$. From $\{1,2\}$: show that $z = \frac{4}{5} + \frac{x_1}{5}$.

d) Solve the system, and recover $x_0 = \frac{5}{12}$, $x_1 = \frac{2}{3}$, $y = \frac{5}{6}$ and $z = \frac{14}{15}$, then the value $\frac{7}{12}$ for a uniformly placed mutant. Compare with $\frac{4}{7}$.

e) Redo a) with $\mu = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$. Check that $\mu W_\star = \mu$, then that $x_0 = \frac{4}{7}$ (`starStationary_fixation`).

Answer. d) One obtains $y = \frac{1}{2} + \frac{2}{5}y$, whence $y = \frac{5}{6}$.

Exercise 4.12 (Reading a Lean statement)**.** a) Take $N = 2$. Show that the absorbing kernels for which $j \mapsto j/2$ is harmonic are exactly those of the form $P(1,0) = P(1,2) = a$, $P(1,1) = 1 - 2a$, with $a \in [0, \frac{1}{2}]$. Show that the hypothesis `hvar` of `fixation_limit` is equivalent to $a > 0$, and that in this case $P^n(1,2) = \frac{1}{2}\big(1 - (1-2a)^n\big)$. What happens for $a = 0$?

b) Why does a file containing `sorry` compile with return code 0? What are the three checks that make up the Tier A gate in this book?

c) Does the statement `isothermal_fixation` talk about probabilities? Write in one sentence the **[Tier C]** bridge that connects it to the "fixation probability".

Hint. a) The one-step variance at 1 is $2a \cdot \frac{1}{4} = \frac{a}{2}$. For $a = 0$, one recovers the identity kernel, that is, the negative control `ctl_neg_identity_no_absorption`.

Exercise 4.13 ((†) University exercise: Expected absorption time via the fundamental matrix)**.** Consider the neutral Moran chain ($r = 1$) with $N = 3$, so that $p_i = q_i = i(N-i)/N^2$.

a) Write down the interior block Q (states $1, 2$) and the interior-to-boundary block R (columns $0, 3$) of P.

b) Compute the fundamental matrix $(I - Q)^{-1}$, and deduce the expected absorption time $t_i = \mathbb{E}_i[T]$, $T = \min(T_0, T_3)$, from $i = 1$ and $i = 2$, using $t = (I - Q)^{-1}\mathbf{1}$.

c) Compute $B = (I - Q)^{-1}R$ and check that its column for state 3 is $(\frac{1}{3}, \frac{2}{3})$, in agreement with Theorem 4.14 for $r = 1$.

Answer. a) $Q = \begin{pmatrix} \frac{5}{9} & \frac{2}{9} \\ \frac{2}{9} & \frac{5}{9} \end{pmatrix}$, $R = \begin{pmatrix} \frac{2}{9} & 0 \\ 0 & \frac{2}{9} \end{pmatrix}$. b) $(I-Q)^{-1} = \begin{pmatrix} 3 & \frac{3}{2} \\ \frac{3}{2} & 3 \end{pmatrix}$, so $t_1 = t_2 = \frac{9}{2}$. c) $B = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix}$.

Exercise 4.14 ((†) University exercise: The Moran fixation formula via a martingale)**.** Let (X_n) be the Moran chain with $N \geq 1$ and $r > 0$, $r \neq 1$, started at $X_0 = i$, and let $T = \min(T_0, T_N)$.

a) Using $p_k = r\, q_k$ (Lemma 4.13), show that $\mathbb{E}\big[r^{-X_{n+1}} \mid X_n = k\big] = r^{-k}$ for every k, so that $M_n = r^{-X_n}$ is a martingale.

b) Justify that $T < \infty$ almost surely (for instance from $I_n(i) \to 0$ in the proof of Theorem 4.14), and that M is bounded.

c) Apply the optional stopping theorem to obtain $\mathbb{P}_i(X_T = N) = \dfrac{1 - r^{-i}}{1 - r^{-N}}$.

d) Which martingale plays the same role for $r = 1$?

Answer. a) $\mathbb{E}[r^{-X_{n+1}} \mid X_n = k] - r^{-k} = r^{-k}(r-1)\big(q_k - p_k/r\big) = 0$. c) $r^{-i} = \mathbb{E}_i[r^{-X_T}] = u\, r^{-N} + (1-u)$ with $u = \mathbb{P}_i(X_T = N)$. d) X_n itself, which gives i/N.

Chapter 5

Five use cases: data, rusty-SUNDIALS and LeanFlow

The previous chapters built statements: theorems checked by the Lean kernel, their human proofs and their hypotheses. This chapter puts them to work on five real systems:

- dipolar excitons in a transition-metal dichalcogenide heterobilayer;
- planar Yukawa crystals, as in dusty plasmas and colloids;
- the structure of superfluid helium;
- a shell model of turbulence;
- the fixation of a mutation in a population.

Every case follows the same plan, and you can redo all of it: each section ends with a script in `code/` that downloads or checks the data, runs the solver and produces the figure printed here.

Warning : what this chapter is, and what it is not

- The five cases *illustrate* theorems; they do not *prove* them. A simulation is Tier X: it never supports a claim.
- The datasets provide orders of magnitude, realistic initial conditions or spectra. None of them tests a Tier A theorem, and each time we say what it does not show.
- All results reused here were produced by AI agents (Claude) under the author's direction, then audited by a separate AI agent. The computations redone for this chapter are checked only by the agent that produced them: the reruns, the Python controls, the sequence statistics and the LeanFlow certificates. No human has audited these results.
- The statements about the triangular lattice (cases U1 and U2) are *conditional* on the upstream theorem `universal_energy_minimum` of openai/math, which is AI-generated and has not been peer reviewed.

5.1 Common method

The four ingredients. Each case combines four ingredients:

1. **A question** from physics or biology, asked before looking at the numbers.

2. **A theorem**, with its tier (A, L, C or X), its Lean name and its DOI. The tier refers to the Lean statement, never to its physical reading.

3. **Data**. For each dataset we give the identifier, the exact revision (a 40-character `git` hash), the licence as displayed on the dataset card, and the `sha256` digest of the file used. When no Hugging Face dataset exists, we say so.

4. **A numerical method**. It relies on rusty-SUNDIALS, a Rust rewrite of SUNDIALS: CVODE (BDF formulas) for stiff differential equations, plus physics modules (`qf-*`). When a value must be certified exactly, we go through LeanFlow, which turns a rational numerical claim into a Lean file, has it checked by the kernel and audits its axioms.

The control rule. Before a number is published, every experiment runs at least one *positive control* and one *negative control*:

- the positive control is a case whose answer is known, and the code must recover it;
- the negative control is a case that is wrong or deliberately broken, and the code must reject it.

A test that has never failed on a negative control has tested nothing. The five scripts terminate with exit code 1 if a control does not behave as expected.

Reproducing. The binaries come from a working copy of rusty-SUNDIALS. It contains the merged crates `qf-yukawa-crystal`, `qf-shell-cascade`, `qf-gpe2d` and `qf-popgen`:

- branch `textbook2/ch5-cas-usage-2026-10-26`;
- based on commit `cda9a9057ec16820f2e16b560cc0e63cf7eb05b9`.

They are built and used as follows:

```
CARGO_TARGET_DIR=~/rusty-target-ar cargo build --release -j 2 --offline \
   -p qf-yukawa-crystal -p qf-shell-cascade -p qf-popgen
python -I code/ch5_U1_dipolar.py  sortie_U1 --bin ~/rusty-target-ar/release
python -I code/ch5_U5_fixation.py sortie_U5 --bin ~/rusty-target-ar/release
```

The option `-I` isolates the interpreter: it ignores `PYTHONPATH` and the current directory, which protects against a planted file in a folder of downloaded data. Each script prints a summary and writes a file `ch5_U*_result.json`. The numbers in this chapter are those of these files.

We first reran the binaries inside the working copy itself: there, the outputs overwrite the published records. `git diff` then shows that the recomputed files are bit-for-bit identical to the records, only the run times differing:

- `lattice_gaps.jsonl` and `references.jsonl` (U1);
- `exact_refs.jsonl` and `exact_relax_kappa2_a0_f0.jsonl` (U2);
- `linear_response.json` of `qf-gpe2d` (U3);
- `sigma3_study.json` (U4);
- `results.jsonl` (U5).

Table 5.1: The five use cases at a glance.

case	theorem (tier)	data	numerics
U1 excitons	optimality of the triangular lattice for $2(1/r - 1/\sqrt{r^2+d^2})$ (A, conditional)	JARVIS-DFT (HF)	exact Ewald, CVODE
U2 Yukawa	same for $e^{-\kappa r}/r$ (A, conditional)	OC20 (HF), disorder only	exact energies, CVODE
U3 helium	$S_B(k) \le \sqrt{k/2k^*}$ (A)	Donnelly–Barenghi table (not HF)	arithmetic; Gross–Pitaevskii
U4 turbulence	enstrophy bound along trajectories, $\sigma = 3$ (A)	The Well (HF)	CVODE
U5 genetics	Moran fixation, isothermal theorem (A)	Genomic Benchmarks (HF), illustration	CVODE, LeanFlow certificate

5.2 U1 – Dipolar excitons in a heterobilayer

The question. In a $MoSe_2/WSe_2$ heterobilayer, the electrons stay in one layer and the holes in the other. They pair into *indirect* excitons, small dipoles perpendicular to the plane, of length close to the interlayer spacing d. If these dipoles are frozen, in which planar arrangement do they minimize their interaction energy, at a given density n?

Physics and biology : the potential

Two excitons at in-plane distance r involve four pairs of charges:

- the two electrons, at distance r, contribute $+1/r$;
- the two holes also contribute $+1/r$;
- the two crossed electron–hole pairs, at distance $\sqrt{r^2+d^2}$, each contribute $-1/\sqrt{r^2+d^2}$.

Hence $V(r) = 2\big(1/r - 1/\sqrt{r^2+d^2}\big)$, in units of $e^2/(4\pi\varepsilon_0\varepsilon_r)$. At large distances, $V(r) \approx d^2/r^3$: this is the dipole–dipole law. The charge count is a physical reading **[Tier C]**; the algebraic identity, by contrast, is checked in Lean.

The theorem and its tier. We write V as a function of the squared distance, $t = r^2$: $g_d(t) = 2\big(t^{-1/2} - (t+d^2)^{-1/2}\big)$. This potential is completely monotone. Indeed, each derivative reads

$$(-1)^m g_d^{(m)}(t) = 2\,\tfrac{1}{2}(\tfrac{1}{2}+1)\cdots(\tfrac{1}{2}+m-1)\big(t^{-1/2-m} - (t+d^2)^{-1/2-m}\big) \ge 0,$$

since $x \mapsto x^{-1/2-m}$ is decreasing. The upstream theorem of openai/math therefore applies **[Tier A]**, conditionally on `universal_energy_minimum`: the triangular lattice minimizes the lower energy among all locally finite configurations of density one. Module D2 extends this result to every density $\rho > 0$.

Lean 4 : `formal_cert/Autoresearch/DipolarExciton.lean` (statements, Lean 4.34.1)

```
theorem triangular_dipolar_optimal {d : ℝ} (hd : 0 < d) (C : Set Plane) (hC :
    LocallyFinite C)
    (hdens : DensityOne C) : latticeEnergy (g d) ≤ energy (g d) C
theorem latticeEnergy_dipolar_lt_top (d : ℝ) : latticeEnergy (g d) < ⊤
```

```
theorem h_not_admissible {d : ℝ} (hd : 0 < d) : ¬ AdmissiblePotential (h d)
```

The second statement is essential: it guarantees that the inequality is not the vacuous $\top \leq \top$. It comes from the bound $g_d(t) \leq d^2 t^{-3/2}$ and from the convergence of the Riesz sum with exponent 3. The third statement is a formal negative control: the potential $h_d(t) = t^{-1/2} - 2(t+d^2)^{-1/2}$, which counts the crossed attraction twice, is not admissible, since $h_d(d^2) = (1-\sqrt{2})/d < 0$. Paper: doi:10.5281/zenodo.23646103 [16].

The data. A search of the Hub (26 queries, including C2DB, *heterobilayer*, TMD and *interlayer*) found no heterobilayer dataset [16]. We therefore use a ColabFit mirror of JARVIS-DFT [61]:

- identifier: `colabfit/JARVIS_DFT_3D_12_12_2022`;
- revision `f4de0085edcf92885870c67459363d2c7080c90b`;
- file `co/co_0.parquet`, 54 027 813 bytes, sha256 `9f6c21a2...510b`;
- licence: "unknown" on the card. We reproduce only derived numbers, and the licence of the upstream source has not been checked.

The script reuses the paper's selection rule, fixed before the results were seen:

- semiconducting rows of $MoSe_2$ and WSe_2, with at least two metal planes;
- spacings between neighbouring metal planes between 4.5 and 8 Å, excluding vacuum gaps.

It obtains:

- for $MoSe_2$, 6 spacings in 3 rows, with mean 6.5827 Å (from 6.5818 to 6.5842 Å);
- for WSe_2, 3 spacings in 1 row, with mean 6.6151 Å.

The proxy for the heterobilayer is the mean of the two, $d = 6.599$ Å. The dimensionless parameter is $\tilde{d} = d\sqrt{n}$. It equals 0.0209, 0.0660 and 0.1143 for $n = 10^{11}$, 10^{12} and 3×10^{12} cm^{-2}, the density window of the experiments [62].

The numerical method. The module `dipolar_ewald.rs` of `qf-yukawa-crystal` computes the *exact* energy per particle, without truncation, of a periodic lattice. It uses an Ewald decomposition of

$$V(r) = \tfrac{4}{\sqrt{\pi}} \int_0^\infty e^{-r^2u^2} \left(1 - e^{-\tilde{d}^2u^2}\right) du.$$

Energies are expressed as $U = V/\tilde{d}^2$, which tends to $1/r^3$. The command `dipolar_crystal gaps` sweeps 41 values of $\tilde{d} \in [0.01, 3]$ for the triangular, square and honeycomb lattices. The relaxations of the paper integrate a gradient flow with CVODE (BDF, analytic Jacobian equal to minus the Hessian) until $\max|F| < 10^{-9}$, for 288 particles.

The script. `code/ch5_U1_dipolar.py OUTPUT [-bin DIR]`:

1. it checks the sha256 of the Parquet file;
2. it extracts the spacings between metal planes;
3. it runs `dipolar_crystal refs` and then `gaps`;
4. it runs the controls and computes the gap in meV;
5. it plots `figures/ch5_U1_dipolar.pdf`.

It takes about 11 s.

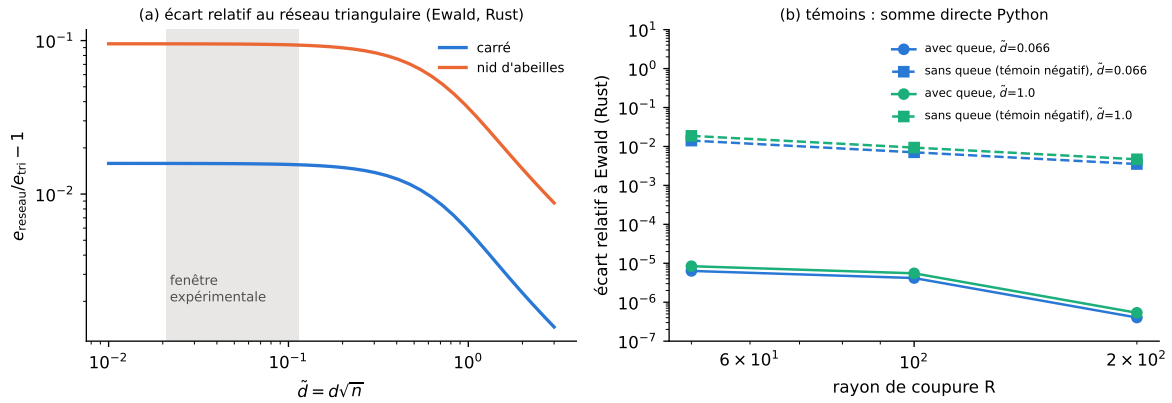


Figure 5.1: U1 **[Tier X]**. (a) Relative gap of the square and honeycomb lattices to the triangular lattice (exact Ewald energies, rusty-SUNDIALS), as a function of $\tilde{d}$. The grey band is the experimental window deduced from JARVIS-DFT. (b) Controls: a direct sum in Python, *with* the exact uniform tail, converges to the Ewald value; *without* the tail (negative control), the error stays of order 10^{-2}.

The result [Tier X]. Over the whole grid, both gaps remain positive:

- $e_{\text{square}}/e_{\text{tri}} - 1$ decreases from 1.58×10^{-2} to 1.36×10^{-3};
- the honeycomb gap decreases from 9.53×10^{-2} to 8.75×10^{-3}.

Converted to energy (unscreened, to be divided by ε_r), the square-minus-triangular gap is 0.0014, 0.044 and 0.224 meV per exciton at the three densities. These values are interpolated on the grid; the paper gave 0.223 meV at the nearest grid point. This is one to five orders of magnitude below the depth of moiré wells, "tens to 100 meV" [62]. In the published study, none of the 21 competitors relaxed by CVODE went below the triangular lattice [16].

The controls.

- *Positive.* An independent direct Python sum, plus the exact tail $2\pi/(\sqrt{R^2 + \tilde{d}^2} + R)$, recovers Rust's Ewald value. The relative deviation is 4.0×10^{-7} at $\tilde{d} = 0.066$ and 5.3×10^{-7} at $\tilde{d} = 1$, for $R = 200$. The Ewald energy is independent of the splitting parameter η to within 9.0×10^{-15}.
- *Negative.* Without the tail, the same sum is off by 3.5×10^{-3} and 4.7×10^{-3} at $R = 200$, and by 1.4×10^{-2} at $R = 50$.
- *Negative.* The script checks that $h_d(d^2) < 0$ for $d = 0.5$, 1 and 2. The square and honeycomb lattices keep a higher energy at each of the 41 grid points.

We also reran, separately, two relaxations of the published study at $\tilde{d} = 0.066$ (command `dipolar_crystal main <dir> 0.066 1 0 1`, about 8 minutes on two cores):

- the uniform start `uniform_00` ends at a relative gap of 1.382373×10^{-3}, with 4 defects;
- the incommensurate box, a negative control, ends at 2.000407×10^{-3}, with 8 defects.

Both values match the record, digit for digit.

What this does not show.

- The model is classical, at zero temperature, with rigid dipoles and no screening: no Rytova–Keldysh form, no moiré potential, no kinetic energy. In the experimental window, real excitons form a quantum gas, not a classical crystal [62].
- The JARVIS d is a proxy: that of two homogeneous stackings, not that of a heterobilayer. The dataset fixes the scale of $\tilde{d}$; it does not test the theorem.
- No relaxation proves optimality: 21 trials do not cover the infinitely many configurations.

5.3 U2 – Planar Yukawa crystals: dusty plasmas and colloids

The question. In a laboratory dusty plasma, charged microparticles levitate in a plane. They repel each other through a Coulomb potential screened by the ions, $V(r) = Q^2 e^{-r/\lambda_D}/r$. Charged colloids in suspension behave in the same way. At fixed density, which configuration minimizes the energy? Hexagonal crystals are observed; can this be proved within this model?

The theorem and its tier. With $t = r^2$ and $\kappa = \ell/\lambda_D$, where ℓ is the length scale set by the density, $g_\kappa(t) = e^{-\kappa\sqrt{t}}/\sqrt{t}$. The human proof **[Tier C]** proceeds in three steps:

- $t \mapsto \sqrt{t}$ is a Bernstein function and $x \mapsto e^{-\kappa x}$ is completely monotone, so their composite $e^{-\kappa\sqrt{t}}$ is completely monotone (this is the CM $\circ$ Bernstein rule of the P3 library, doi:10.5281/zenodo.23653842);
- $t^{-1/2}$ is completely monotone;
- a product of two completely monotone functions is again completely monotone (Leibniz rule: every term has the right sign).

Lean checks it **[Tier A]** (Lean 4.34.1), conditionally on the upstream result:

Lean 4 : `formal_cert/Autoresearch/Yukawa.lean` (statements)

```
theorem yukawa_admissible {κ : ℝ} (hκ : 0 < κ) : AdmissiblePotential (g κ)
theorem triangular_yukawa_optimal {κ : ℝ} (hκ : 0 < κ) (C : Set Plane) (hC :
    LocallyFinite C)
    (hdens : DensityOne C) : latticeEnergy (g κ) ≤ energy (g κ) C
theorem latticeEnergy_yukawa_lt_top {κ : ℝ} (hκ : 0 < κ) : latticeEnergy (g κ) < ⊤
```

The module `YukawaMix.lean` extends the result to every finite mixture $\sum_i w_i e^{-\kappa_i r}/r$ with weights $w_i \geq 0$. The formal negative control is the signed mixture $g_1 - 2g_2$: it equals $2e^{-1/2} - 4e^{-1} \approx -0.2585$ at $t = \frac{1}{4}$, so it is not admissible. The theorem does not apply to it, which does not mean that optimality fails. Paper: doi:10.5281/zenodo.23646105 [17].

University remark. **[Tier L]** By Bernstein's theorem, a function g on $(0, \infty)$ is completely monotone if and only if $g(t) = \int_{[0,\infty)} e^{-ts}\, d\mu(s)$ for a positive Borel measure μ on $[0, \infty)$ with finite integral for every $t > 0$ [6, 7]. In this language the three steps above are closure properties of Laplace transforms of positive measures: a product corresponds to the convolution of the measures, and composing with a Bernstein function corresponds to subordination. For instance $e^{-\kappa\sqrt{t}}$ is the Laplace transform of the one-sided $\frac{1}{2}$-stable (Lévy) density (standard).

The data. On 2026-10-25, a search of the Hub (*dusty plasma*, *complex plasma*, *colloid*, *yukawa*, *wigner crystal*) found no dataset of grain or colloid positions [17]. This is an absence observed by search, not a proof. The disordered initial conditions therefore come from an entirely different field, catalysis:

- identifier `colabfit/OC20_IS2RES_val_id`, licence CC-BY-4.0 [63];
- file `co/co_10.parquet`, 142 986 687 bytes; its local sha256 equals the Hub's LFS hash at revision `6a1dfef12aec878b4cda99edd54b16a6a254e135`;
- adsorbate structures, rotated in 3D, projected onto the plane, rescaled and filtered by a minimum distance.

The script checks the sha256 and finds in the file the OC20 configuration identifiers listed in `provenance.json`. These data provide realistic-looking *disorder*, nothing more: molecules adsorbed on a catalyst have nothing to do with screened charged grains.

The numerical method. The module `exact.rs` of `qf-yukawa-crystal` has three parts:

- lattice sums of the *untruncated* potential, with a rigorous bound on the tail;
- an independent Ewald decomposition;
- a periodic box summing all images within radius R, with analytic energy, force and Hessian.

The tail bound relies on Voronoi cells: for decreasing V, the tail beyond R is at most $(\pi/A)\big(1+\delta/(R-2\delta)\big)e^{-\kappa(R-2\delta)}/\kappa$. The relaxation (`exact-relax`) uses CVODE: BDF, Jacobian equal to minus the Hessian, rtol $= 10^{-10}$, atol $= 10^{-12}$ and force tolerance 10^{-9}. Each minimum is checked through the spectrum of the Hessian. This computation corrects a published erratum: an earlier version truncated the potential at $r_c = 8$ with a force shift, and this truncated potential is no longer completely monotone.

The script. `code/ch5_U2_yukawa.py OUTPUT`:

1. it copies the initial conditions into a working crate;
2. it runs `yukawa_crystal exact-refs`, then `exact-relax 2 1 0 0`. At $\kappa = 2$, this means three relaxations of about 1 to 2 minutes each on two cores: an OC20 start, a uniform start and an incommensurate box;
3. it runs the controls and plots `figures/ch5_U2_yukawa.pdf`.

The option `-skip-relax` rereads the record instead of rerunning the relaxations.

The result [Tier X]. Energies per particle at density one, in the $\frac{1}{2}\sum$ convention:

κ	triangular (exact)	square (exact)	gap	truncated–exact (tri.)
1	1.6253213396063	1.6348554953172	9.534×10^{-3}	-1.771×10^{-2}
2	0.3935428489324	0.4009742487651	7.431×10^{-3}	-9.48×10^{-6}

The tail bound is at most 5.6×10^{-17} ($\kappa = 1$, $R = 40$) and 1.2×10^{-16} ($\kappa = 2$, $R = 20$). The three relaxations at $\kappa = 2$ converge, with no negative mode of the Hessian:

- OC20 start: gap 1.742×10^{-3}, 24 defects, 157 CVODE steps;
- uniform start: gap 6.68×10^{-4}, 4 defects;
- incommensurate 17×17 box: gap 3.78×10^{-4}, 4 defects.

In the full study, none of the 24 stages relaxed under the exact potential went below the triangular lattice [17].

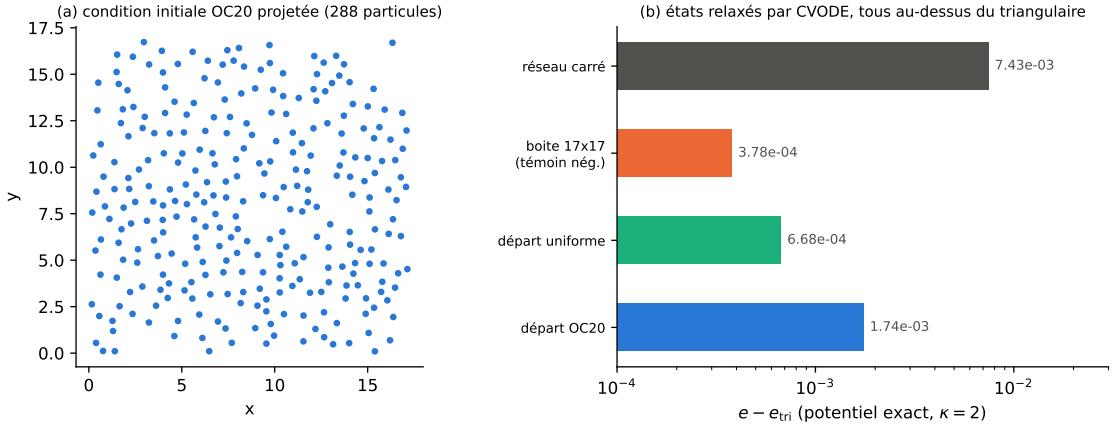


Figure 5.2: U2 **[Tier X]**. (a) Initial condition drawn from OC20 (288 particles in a box that exactly contains the triangular lattice). (b) Gap $e - e_{\text{tri}}$ of the states relaxed by CVODE under the exact potential ($\kappa = 2$), to be compared with the gap of the square lattice. No state lies below the triangular lattice.

The controls.

- *Positive.* A direct Python lattice sum recovers Rust's exact values to within 1.4×10^{-16} in relative terms at most (the required threshold is 10^{-12}). In addition, the periodic box containing the perfect lattice equals the infinite sum to within 2.2×10^{-16}.
- *Negative.* The energies truncated at $r_c = 8$ miss the exact value by far more than 10^{-9}. The 17×17 box, which cannot contain the triangular lattice, stays above. The perfect square lattice is a saddle of the exact potential, with 72 negative modes at $\kappa = 2$ and 64 at $\kappa = 1$ [17].

What this does not show.

- The OC20 data contain no plasma physics: they serve only as initial disorder.
- The potential $K_0(\kappa r)$, the 2D Debye–Hückel Green's function, is not treated. Neither are mixtures with negative weights, finite temperature or uniqueness.
- The annealing was not redone with the exact potential.

5.4 U3 – Superfluid helium: structure factor and the Feynman–Bijl test

The question. X-ray or neutron diffraction measures the static structure factor $S(k)$ of a liquid. For a weakly interacting Bose gas, Bogoliubov theory predicts $S_B(k) = \varepsilon_0(k)/\varepsilon_B(k)$, with $\varepsilon_0 = \hbar^2 k^2/2m$. Can one say, *from* $S(k)$ *alone*, that superfluid helium-4 is not a Bogoliubov gas?

The theorem and its tier. Set $k^* = 2mc/\hbar$, where c is the speed of sound. Then $S_B(k) = k/\sqrt{k^{*2} + k^2}$ and

$$S_B(k) \leq \sqrt{\frac{k}{2k^*}}, \qquad \text{with equality if and only if } k = k^*.$$

Human proof. Both sides are positive; we compare their squares. $S_B^2 \le k/2k^*$ is equivalent to $2k^*k \le k^{*2} + k^2$, that is, to $(k - k^*)^2 \ge 0$. □

The statement is **[Tier A without replay]**, in the QuantumFluids environment (Lean v4.34.0-rc2; no `leanchecker` replay recorded for this environment), under the name `sBog_le_sqrt`. Both controls are checked by the kernel: the test fires on the free value $S = 1$ for every $0 < k < 2k^*$ (`free_S_one_violates`), and never fires on S_B. The corollary `not_bogoliubov_of_structure_factor` gives the "diffraction test": if a measured S exceeds $\sqrt{k/2k^*}$, this S is not the structure factor of a Bogoliubov gas with that speed of sound. The Feynman–Bijl argument relates S to the excitation energy, $\varepsilon(k) \le \hbar^2k^2/(2mS(k))$. Its single-mode core is formalized in Lean 4.34.1, with the f-sum rule identity taken as a *hypothesis*. Papers: doi:10.5281/zenodo.23646117 [32] and doi:10.5281/zenodo.23653846 [33].

The data (not Hugging Face). No structure-factor dataset exists on the Hub: the searches for these two papers found none. We use Table 14.1 of the compilation by Donnelly and Barenghi [41], in its web edition (assumed, without verification, to reproduce the 1998 table):

- 99 points with $Q > 0$: 49 X-ray points (Robkoff–Hallock) and 50 neutron points (Svensson et al.);
- temperatures from 1.4 to 4 K, no error bars printed.

This is a secondary source at **[Tier L]**. The extracted file is `papers/prog_superfluid_structure_factor/data/he4_Sk_donnelly_table14_1.csv`. We take $c = 238.3 \pm 0.1$ m/s [42], whence $k^* = 3.00379$ Å^{-1} (between 3.0025 and 3.0051).

The method. For each point, the verdict is:

- COMPATIBLE if $S \le \sqrt{Q/2k^*}$;
- REFUTED if $S - \sqrt{Q/2k^*} > \Delta$;
- UNDECIDED otherwise.

The threshold $\Delta = 0.082$ is the gap between the maxima of the two datasets in the table. This threshold was fixed after the fact; it was not preregistered. On the rusty-SUNDIALS side, the module `linear_response` of `qf-gpe2d` gives a cross-check: one excites the Gross–Pitaevskii ground state (32×32 grid, Strang split step), measures the frequency of the response, and derives $\varepsilon_B(k)$ and $S(k)$ from it.

The script. `code/ch5_U3_helium.py OUTPUT [-gpe-json FILE]` applies the test and runs the controls. If it is given the output of the `bogoliubov_dispersion` example of `qf-gpe2d`, it also plots the linear response.

The result. Of the 99 points, 29 are compatible, 7 undecided and 63 refuted. The 21 points of the peak and roton region ($1.7 \le Q \le 2.4$ Å^{-1}) are all refuted: there S exceeds the envelope by a factor of 1.62 to 2.49. The helium-4 measurements are therefore not the S of a Bogoliubov gas with the same speed of sound. This textbook result is now a consequence of a checked statement and a published table.

We reran the `bogoliubov_dispersion` example (10 min 24 s on one core), and its output is bit-for-bit identical to the record. In this Gross–Pitaevskii linear response **[Tier X]**:

- the frequency measured for $g = 1$ recovers ε_B to within 4.1×10^{-5} in relative terms;
- the measured S stays below the envelope, since $\max(S/\sqrt{k/2k_s}) - 1 = -1.4 \times 10^{-3}$;

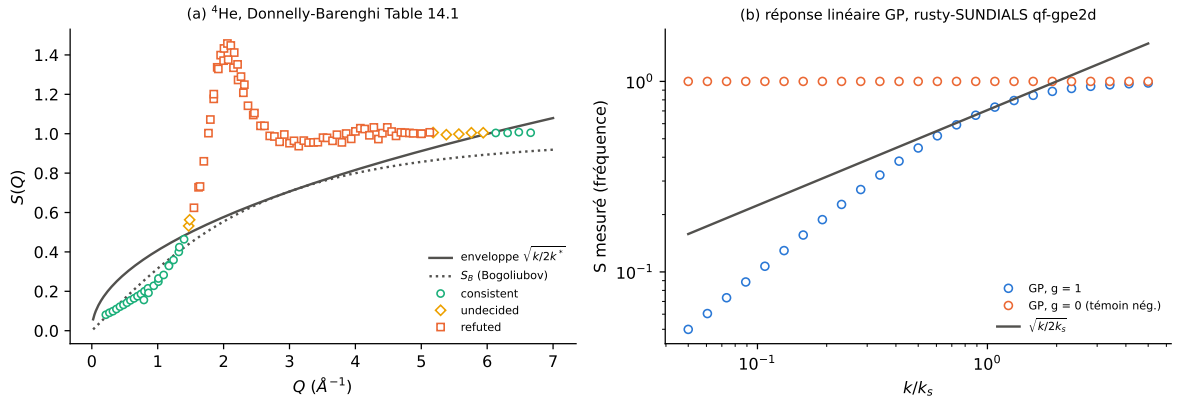


Figure 5.3: U3. (a) $S(Q)$ of helium-4 (Table 14.1, Tier L) against the envelope $\sqrt{Q/2k^*}$ (solid line) and S_B (dotted); the colour gives the verdict (arithmetic, Tier X). (b) Gross–Pitaevskii linear response computed by rusty-SUNDIALS **[Tier X]**: measured S for $g = 1$ below the envelope, control $g = 0$ ($S = 1$) above it for $k < 2k_s$.

- the control $g = 0$, a free gas where $S = 1$, exceeds the envelope at the 20 wavenumbers of the sweep with $k < 2k_s$.

The controls.

- *Positive.* S_B fires the test at none of the 99 points.
- *Negative.* The free value $S = 1$ fires it at 95 of the 95 points with $Q < 2k^*$.

Both controls are also Lean theorems.

What this does not show.

- The physical conclusion is conditional **[Tier C]**: the table was measured between 1.4 and 4 K, not at $T = 0$. We have not quantified how the peak of S varies with temperature.
- A compatible point is not *confirmed* "Bogoliubov": the envelope is only an upper bound, and at small k it has very little testing power.
- Beyond the peak, the failure resembles that of the control $S = 1$: this is no longer roton physics.

5.5 U4 – Turbulence: enstrophy bound in the shell model $\sigma = 3$

The question. Shell models replace a turbulent velocity field by one complex amplitude v_n per octave of wavenumbers, $k_n = 2^n$, $n = 0, \dots, N + 1$. Neighbouring shells exchange energy through a quadratic term. Add a dispersive term $-i\omega_n v_n$ with $\omega_n = Dk_n^3$ ("$\sigma = 3$"). Can one bound the enstrophy $\Omega = \frac{1}{2}\sum k_n^2 |v_n|^2$, which measures the small scales, along *every* trajectory and for *every* truncation N?

The theorem and its tier. The model conserves two quantities:

- the "mass" $S = \sum |v_n|^2 = 2E$;
- the invariant $H = \sum 2^{-n}\omega_n |v_n|^2 + \sum 2^{-n} k_n \operatorname{Im}(\bar{v}_n^2 v_{n+1})$, for *every* real dispersion law.

For $\omega_n = Dk_n^3$, the quadratic part of H equals exactly $2D\Omega$. The cubic part C satisfies $|C| \le S^{3/2}$.

Human proof. Write $H_3(0)$ for the value of H at time 0. Since H is conserved, $2D\Omega(t) = H_3(0) - C(t)$. Hence $2D\Omega(t) \le H_3(0) + |C(t)| \le H_3(0) + S(t)^{3/2}$. Since S is conserved, $2D\Omega(t) \le H_3(0) + (2E(0))^{3/2}$. □

Lean checks all of this **[Tier A without replay]** (Lean v4.34.0-rc2, the project's Mathlib; no `leanchecker` replay):

- the rate identity of H along the vector field;
- the chain rule;
- the constancy of H and S on $[0, T]$;
- the bound `enstrophy_bound_along_flow`, for every trajectory on $[0, T]$ and every N.

The statement assumes that the trajectory exists and is differentiable at every instant of $[0, T]$: existence is a *hypothesis*, not a theorem. The level N stays fixed. Paper: doi:10.5281/zenodo.23646111 [60].

The data. The Well [64] contains large-scale fluid simulations. We use:

- identifier: `polymathic-ai/turbulent_radiative_layer_2D`;
- revision `2ee7756575ff6f90981d0308cbcb6a2ab5995bdc`, licence CC-BY-4.0 according to the card;
- the file `data/test/turbulent_radiative_layer_tcool_0.32.hdf5`, sha256 `06cbd36c...f747`, checked by the script.

These are Athena++ simulations of the 2D compressible Euler equations with radiative cooling.

The spectrum is built as follows, for four developed snapshots (indices 25, 50, 75 and 100):

1. take the FFT of the velocity along the periodic axis x, without the mode $k_x = 0$;
2. average over y;
3. sum $k_x \in [2^n, 2^{n+1})$ into shell $n = 0, \ldots, 5$.

This step reads the HDF5 file with `h5py`, which is absent from AutoevolveAI's `.venv` environment: the script therefore rereads the file `well_amps.txt` that these steps produced. The $\log_2$ slopes of the spectra are -2.13, -1.56, -1.79 and -1.81 [60].

The numerical method. The module `sigma.rs` of `qf-shell-cascade` integrates the model with CVODE:

- BDF, analytic Jacobian, atol $= 10^{-3}$ rtol, $D = 0.05$;
- suprema taken at CVODE's internal step points, without interpolation.

The program `sigma3_study full` runs five blocks in sequence:

- a positive control, the convergence of conservation with the tolerance;
- a negative control, a perturbed coupling;

- a sweep of truncations $n_{\max} \leq 7$;
- the comparison $\sigma = 2$ versus $\sigma = 3$ on data concentrated in the last shell;
- 24 trajectories initialized from the moduli of The Well, with two draws of random phases and $n_{\max} = 5, 6, 7$.

The script. `code/ch5_U4_shell.py OUTPUT` runs `sigma3_study full` (from 2.5 to 6 minutes depending on the machine's load), checks the criteria and plots `figures/ch5_U4_shell.pdf`.

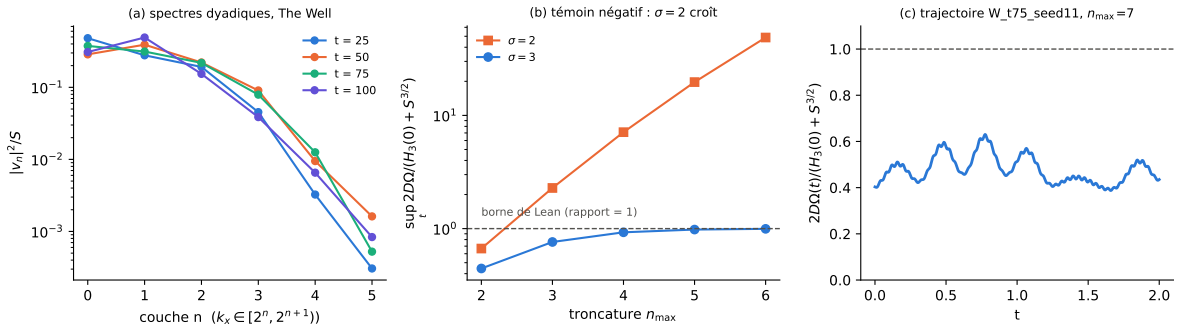


Figure 5.4: U4 **[Tier X]**. (a) Dyadic spectra of four snapshots of The Well, normalized to $S = 1$. (b) Ratio $\sup_t 2D\Omega/(H_3(0) + S^{3/2})$ on data concentrated in the last shell: it stays below 1 for $\sigma = 3$ and grows with $n_{\max}$ for $\sigma = 2$ (negative control). (c) The ratio along the trajectory initialized from The Well that comes closest to the bound.

The result [Tier X]. The 17 $\sigma = 3$ trajectories of the sweep and of the comparison all stay below the bound. The ratio $\sup_t 2D\Omega(t)/(H_3(0) + S(0)^{3/2})$ is at most 0.995; this maximum is reached on data concentrated in the last shell, where the ratio tends to 1 as $n_{\max}$ grows (0.444, 0.762, 0.928, 0.981 and 0.995 for $n_{\max} = 2, \ldots, 6$). The 24 trajectories initialized from The Well also stay below the bound, with a ratio of at most 0.631. Their relative drifts are at most 2.4×10^{-8} on S and 1.2×10^{-6} on H. The recomputed file `sigma3_study.json` is identical to the record of the paper [60], up to run times.

The controls.

- *Positive.* With the true coupling, the drift of H decreases with the tolerance. For $n_{\max} = 4$, it is 2.6×10^{-5}, 8.6×10^{-7} and then 1.9×10^{-8} for rtol $= 10^{-6}, 10^{-8}, 10^{-10}$.
- *Negative.* If the coupling leaving shell 1 is perturbed by 10^{-2}, the drift of H stays stuck near 2.3×10^{-3} ($n_{\max} = 4$) and 5.4×10^{-3} ($n_{\max} = 6$), whatever the tolerance. Conservation therefore comes from the structure of the model, not from the accuracy of the solver.
- *Negative.* For $\sigma = 2$, the same inequality has no proof, and the ratio is 0.667, 2.29, 7.11, 19.7 and then 48.8 for $n_{\max} = 2, \ldots, 6$. It exceeds 1 from $n_{\max} = 3$ on and grows faster than an exponential of ratio 2 over this range (Figure 5.4b).

What this does not show.

- The Well data give a realistic spectral shape, and nothing else. The flow is compressible and cooled; it does not conserve energy. The spectrum is one-dimensional and carries no phases (they are drawn at random).

- Over $T = 2$, the upper shells, initially empty, barely fill up. These trajectories therefore do not probe the dependence on N.
- There is no statement for infinitely many shells. Uniformity in N requires initial data whose $\Omega(0)$ and $S(0)$ are bounded uniformly in N (a Tier C corollary).
- BDF damping makes S drift downwards. The observed margins far exceed these drifts, but the simulation does not replace the theorem.

5.6 U5 – Genetics: fixation of a mutation

The question. A population has N individuals, i of whom carry a mutation of relative fitness r. What is the probability that the mutation eventually takes over the whole population (*fixation*)? And what becomes of this probability when the population lives on a graph?

The Moran model. At each step, one individual is chosen to reproduce with probability proportional to its fitness, and one individual drawn uniformly at random dies. The number of mutants is a birth–death chain on $\{0, \dots, N\}$. Its up and down probabilities are

$$p_i = \frac{ri}{ri + N - i} \cdot \frac{N - i}{N}, \qquad q_i = \frac{N - i}{ri + N - i} \cdot \frac{i}{N}, \qquad \text{hence } p_i = r\, q_i.$$

The theorem and its tier. Write φ_i for the fixation probability starting from i mutants. The first-step equation gives $p_i(\varphi_{i+1} - \varphi_i) = q_i(\varphi_i - \varphi_{i-1})$, with $\varphi_0 = 0$ and $\varphi_N = 1$.

Human proof (formula). Set $\delta_i = \varphi_{i+1} - \varphi_i$. The equation gives $\delta_i = \delta_{i-1}/r$, hence $\delta_i = r^{-i}\delta_0$ and $\varphi_i = \delta_0 S_i$, where $S_i = \sum_{k<i} r^{-k}$. The condition $\varphi_N = 1$ fixes $\delta_0 = 1/S_N$. For $r \neq 1$, we obtain

$$\varphi_i = \frac{S_i}{S_N} = \frac{1 - r^{-i}}{1 - r^{-N}}.$$

For $r = 1$, we obtain $\varphi_i = i/N$. □

Programme P7 checks more **[Tier A]** (Lean 4.34.1). It proves the *absorption limit*, $P^n(i, N) \to (1 - r^{-i})/(1 - r^{-N})$, and uniqueness: every function that is harmonic in the interior with values 0 and 1 on the boundary equals φ. Programme N4 proves the isothermal theorem (birth–death): on a strongly connected graph whose replacement matrix is doubly stochastic, fixation follows the same law.

Lean 4 : `PopGen/MoranFixation.lean` and `EvoGraph/BirthDeath.lean` (statements)

```
theorem moran_fixation_limit_closed {N : ℕ} {r : ℝ} (hN : 0 < N) (hr : 0 < r) (hr1 : r
    ≠ 1)
    (i : Fin (N + 1)) :
    Tendsto (fun n => kPow (moranP N r) n i (Fin.last N)) atTop
      (𝓝 ((1 - r⁻¹ ^ (i : ℕ)) / (1 - r⁻¹ ^ N)))
theorem moran_fixation_unique {N : ℕ} {r : ℝ} (hN : 0 < N) (hr : 0 < r) {g : Fin (N +
    1) → ℝ}
    (hg : InteriorHarmonic (moranP N r) g) (g0 : g 0 = 0) (g1 : g (Fin.last N) = 1) :
    ∀ i, g i = moranFix N r i
theorem isothermal_fixation {W : Fin n → Fin n → ℝ} {r : ℝ} (hn : 0 < n) (hr : 0 < r)
    (hW : Isothermal W) (hsc : StronglyConnected W) (S : Finset (Fin n)) :
    Tendsto (fun t => sPow (bdK (fun _ => 1) W r) t S univ) atTop (𝓝 (moranFix n r S.
    card))
```

The general absorption theorem of P7 and the single-mutant version of the isothermal theorem reappear under the names `KAL` and `ILENE` in Section 5.8.

University remark. **[Tier L]** For a finite absorbing Markov chain, the absorption probabilities form the unique bounded solution of the first-step equations that is harmonic at the transient states and takes the prescribed boundary values; equivalently, if h is such a solution then $h(X_n)$ is a bounded martingale and the optional stopping theorem at the absorption time gives $h(i) = \mathbb{P}_i(\text{absorption at } N)$ [68, 69].

The number of mutants on a graph is *not* a Markov chain in general. On the 4-cycle, the one-step up probability depends on the configuration, 1/3 versus 2/3 at $r = 2$. What is true, and proved, is the bias *per configuration* $P^+ = r\,P^-$. On the star with $n = 3$ nodes, which is not isothermal, the limits are 5/12, 2/3 and 7/12, against 4/7 for Moran. Papers: doi:10.5281/zenodo.23653850 [46] and doi:10.5281/zenodo.23660071 [47].

The data: what they illustrate, and what they do not estimate. The author's Genetic repository uses the datasets `katarinagresova/Genomic_Benchmarks_*` [65]. We take one of them:

- identifier `katarinagresova/Genomic_Benchmarks_demo_coding_vs_intergenomic_seqs`, revision `85458172cab6cffad46cb172e64ca94780506ed6`;
- the test split, `data/test-00000-of-00001-6fccc31405110e30.parquet`, 2 456 511 bytes, sha256 `dfe13715...9afe`.

The card states *no* licence and does not say what the labels 0 and 1 mean. We reproduce only aggregate statistics, and the licence of the upstream package has not been checked.

The script counts 25 000 sequences, 12 500 per label, all of length 200, with no symbol other than A, C, G and T. The GC fraction is 0.5381 for label 0 and 0.4004 for label 1 (Figure 5.5c).

This is what sequence data look like: a point mutation is the replacement of one letter. **But this dataset estimates no fitness** r, and nothing in this chapter derives one from it. The values of r used below (0.8, 1, 1.2, 1.5 and 2) are *chosen*, not measured.

The numerical method. The crate `qf-popgen` writes the absorbing chain as a linear differential equation, $\dot{\pi} = \pi(P - I)$: this is the forward Kolmogorov equation of the uniformized chain. It integrates it with CVODE (BDF, exact Jacobian, rtol $= 10^{-11}$, atol $= 10^{-15}$). An *absorption certificate* rejects any doubtful run, in a "fail-closed" spirit (rejecting by default): a run is accepted only if five conditions hold:

- the kernel is stochastic and absorbing;
- the total mass is conserved;
- no component becomes negative;
- the interior mass has dropped below 10^{-9};
- the mass at N equals the target to within 10^{-8}.

The passage from the differential equation to the n-step kernel (a Poisson mixture) is not formalized.

Finally, **LeanFlow** certifies a computed number. Its function `lean.certify` accepts a claim about rational enclosures and translates it into a statement about integers. The tactic `decide_kernel` (`decide +kernel`) has this statement decided by the kernel, with no extra axiom. LeanFlow then audits `#print axioms`, and an exact Python oracle must return the same verdict.

The script. `code/ch5_U5_fixation.py` `OUTPUT`:

1. it downloads the file at the pinned revision, checks its digest and computes the statistics;
2. it runs `fixation_ode`;
3. it compares the CVODE values with the exact fractions;
4. it submits five claims to LeanFlow (Lean 4.34.1, core only);
5. it plots `figures/ch5_U5_fixation.pdf`.

It takes about twenty seconds.

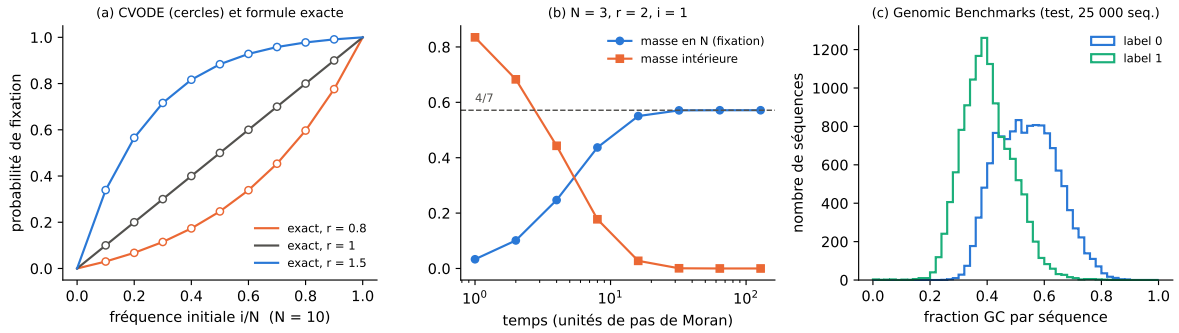


Figure 5.5: U5. (a) Fixation probability for $N = 10$: CVODE (circles, **[Tier X]**) and exact formula (lines), for $r = 0.8$, 1 and 1.5. (b) CVODE trajectory for $N = 3$, $r = 2$, $i = 1$: the mass absorbed at N tends to 4/7, the interior mass dies out. (c) GC fraction of the 25 000 test sequences of Genomic Benchmarks; this dataset illustrates sequence data and estimates no fitness.

The result. `fixation_ode` produces 115 rows: 106 accepted runs and 9 negative controls rejected, without a single exception. Over the accepted Moran runs, the maximal deviation between CVODE and the exact fraction S_i/S_N is 3.1×10^{-10} **[Tier X]**.

For $N = 3$, $r = 2$, $i = 1$, CVODE gives 0.5714285714284759, a deviation of 9.55×10^{-14} from 4/7. LeanFlow then certifies two claims **[Tier A]** (on the arithmetic statement only, empty axiom list; each claim is split by LeanFlow into elementary propositions):

- *proximity*: $4/7 - 10^{-11} < 0.5714285714284759 < 4/7 + 10^{-11}$;
- *harmonicity*: the vector $(0, \frac{4}{7}, \frac{6}{7}, 1)$ satisfies exactly the first-step equations of the chain $N = 3$, $r = 2$, with $p_1 = \frac{1}{3}$, $q_1 = \frac{1}{6}$, $p_2 = \frac{4}{15}$ and $q_2 = \frac{2}{15}$.

Here is one of the four generated propositions, as the kernel decided it:

```
-- harmonic at i=1
theorem cert_2 : (0 : Int) < ((((3 : Int) * (7 : Int)) * ((6 : Int) * (1 : Int))) * ((2 : Int)
    * (7 : Int))) ∧ ((((1 : Int) * (6 : Int)) * ((6 : Int) * (1 : Int)) + ((1 : Int) * (0 :
    Int)) * ((3 : Int) * (7 : Int))) * ((2 : Int) * (7 : Int)) + ((-1 : Int) * (4 : Int)) *
    (((3 : Int) * (7 : Int)) * ((6 : Int) * (1 : Int)))) * (1 : Int) = (0 : Int) * ((((3 : Int)
     * (7 : Int)) * ((6 : Int) * (1 : Int))) * ((2 : Int) * (7 : Int))) := by decide +kernel
```

LeanFlow audits only `#print axioms`, which can be fooled; we therefore completed the Tier A gate for these two certificates. The two generated sources were recompiled to `.olean` with Lean 4.34.1 and replayed by `leanchecker`: exit code 0 for both. The control for this step, a

missing module, exits with code 1. A search for `skipKernelTC`, `addDecl`, `native_decide` and `implemented_by` in the two sources finds nothing.

Combined with `moran_fixation_unique`, harmonicity says that $4/7$ *is* the fixation probability in this case. This link is a human argument **[Tier C]**: it identifies LeanFlow's whole statement with the definitions `moranP` of P7. P7 in fact already contains `cert_N3_r2_i1`, a hand-written Lean certificate for the same value.

The controls.

- *Negative, LeanFlow.* Three false claims are *rejected* (`false-statement`), and `#print axioms` reports `sorryAx` for them:
  - the neutral value $1/3$ in place of $4/7$;
  - a tolerance that is too tight, 10^{-14}, smaller than the actual error of 9.55×10^{-14};
  - the neutral vector $(0, \frac{1}{3}, \frac{2}{3}, 1)$, which violates both interior equations.
- *Negative, CVODE.* Nine controls are rejected, each for the intended reason:
  - the formula with inverted exponent and the neutral value i/N under selection (`ValueMismatch`, three each);
  - the identity kernel, which never absorbs, and a horizon that is too short (`NotAbsorbed`);
  - a row whose sum is 1.001 (`InvalidKernel`).
- *Positive.* The cases $N = 2$ and $N = 3$ at $r = 2$ recover $2/3$, $4/7$ and $6/7$, and the neutral case recovers i/N.

What this does not show.

- The Genomic Benchmarks dataset estimates neither r, nor N, nor any population-dynamics parameter. Its role is pedagogical.
- The LeanFlow certificate concerns the arithmetic of the stated numbers, not the correctness of CVODE.
- The Moran model assumes a constant size, a single locus and a constant r. The isothermal theorem is proved in the "sufficient" direction only, for the birth–death rule; the converse and the death–birth rule are not proved.

5.7 Summary: what the five cases teach

Three lessons emerge from the five cases.

Real data are not a proof. None of the five datasets tests a Tier A theorem. Each has a more modest, but real, role:

- JARVIS-DFT fixes the physical scale $\tilde{d}$;
- OC20 provides initial disorder;
- the Donnelly and Barenghi table *refutes* a modelling hypothesis ("helium is a Bogoliubov gas");
- The Well gives a spectral shape;
- Genomic Benchmarks shows what a sequence looks like.

Case U3 is the only one where the data say something physical. Even then it is *thanks to* the theorem: the refutation is meaningful only because the envelope is proved.

Table 5.2: Controls run for this chapter (script outputs of 2026-10-26).

case	positive control (must succeed)	negative control (must fail)
U1	direct Python sum + exact tail: 4×10^{-7} from the Rust Ewald value ($R = 200$)	without the tail: error 3.5×10^{-3}; $h_d(d^2) < 0$; incommensurate box above
U2	direct Python sum: $\leq 1.4 \times 10^{-16}$ from Rust	energies truncated at $r_c = 8$: error $\geq 9.4 \times 10^{-6}$; 17×17 box above
U3	S_B never fires the test (0 of 99); GP $g = 1$ below the envelope	$S = 1$ fires it (95 of 95); GP $g = 0$ above (20 of 20)
U4	drift of H divided by $\approx 10^3$ when rtol goes from 10^{-6} to 10^{-10}	perturbed coupling: drift stuck near 2–5×10^{-3}; $\sigma = 2$ exceeds the bound
U5	106 accepted runs; two LeanFlow certificates `ok`	9 CVODE controls rejected; 3 LeanFlow claims rejected

Disagreement is more instructive than agreement. Agreement between CVODE and an exact formula, to within 3×10^{-10} (U5) or 10^{-16} (U2), is reassuring about the code, but it says nothing about the model. It is the negative controls that give the positive controls their value:

- in U2, an innocent truncation changes the energy by 1.8×10^{-2}, twice the gap between square and triangular;
- in U4, a coupling perturbed by 1% destroys a conservation law that no accuracy restores;
- in U5, a tolerance of 10^{-14} is rejected because the actual error is 9.55×10^{-14}.

Each tool has its scope. The three tools share the work:

- Lean checks statements about definitions;
- rusty-SUNDIALS computes in floating point;
- LeanFlow bridges the two, by certifying that a *stated* number satisfies an exact relation.

None of the three says that the model describes nature. Each section therefore ends with "what this does not show", and that heading counts as much as the result.

5.8 The hidden acronyms

A few theorems of this book carry, in the Lean code, a second name made of initials. These names prove nothing new: they are re-exports of results already checked by the kernel, each accompanied by a backronym in its documentation. They are collected in the file `Acronymes.lean` of `formal_cert/Programmes/`, listed below.

```
import Triangular.AnyDensity
import Programmes.PopGen.Absorption
import Programmes.EvoGraph.BirthDeath
import Autoresearch.KeldyshCM
import Programmes.PopGen.Price

/-!
# Acronymes : cinq ré-exportations sous des noms courts

Each theorem below is a **re-export** of a result that is already kernel-checked elsewhere in
`formal_cert/`. The statement is copied verbatim from the source theorem and the proof is the
    source
theorem applied to its arguments; nothing new is proved here. The `example`s after each theorem
     check,
```

```
by `rfl`, that the re-exported constant has exactly the type of its source (same statement, not
     a
weaker or stronger one).

Each name is an acronym; its docstring gives the backronym.

| name      | source                                                          |
|-----------|-----------------------------------------------------------------|
| `ALIX`    | `TriangularDensity.triangular_optimal_any_density`              |
| `KAL`     | `ProgPopGen.fixation_limit`                                     |
| `ILENE`   | `EvoGraph.isothermal_fixation_single`                           |
| `ALEXIS`  | `AutoresearchKeldysh.laplace_admissible`                        |
| `AURELIA` | `ProgPopGen.Price.fisher_haploid_nondecreasing`                 |

`ALIX` rests on upstream openai/math `universal_energy_minimum` (AI-generated, not peer
    reviewed);
everything is conditional on the fidelity of the definitions it uses.
-/

namespace Acronymes

section Lattice

open OAI.AtomicTriangular TriangularDensity Filter Topology Set
open scoped Pointwise

/-- **A** **L**attice **I**s e**X**tremal.

At every density `ρ > 0` the rescaled triangular lattice `ρ^{-1/2} A` is locally finite, has
    density
`ρ`, and its lower energy is `≤` that of every locally finite density-`ρ` configuration, for
    every
admissible (completely monotone in `r^2`) potential `h`. Re-export of
`TriangularDensity.triangular_optimal_any_density` (conditional on upstream). -/
theorem ALIX (h : ℝ → ℝ) (hh : AdmissiblePotential h) {ρ : ℝ} (hρ : 0 < ρ) :
    LocallyFinite ((Real.sqrt ρ)⁻¹ • A) ∧ DensityRho ρ ((Real.sqrt ρ)⁻¹ • A) ∧
      ∀ C : Set Plane, LocallyFinite C → DensityRho ρ C → energy h ((Real.sqrt ρ)⁻¹ • A) ≤
    energy h C :=
  triangular_optimal_any_density h hh hρ

example : @ALIX = @triangular_optimal_any_density := rfl

end Lattice

section Absorption

open Filter Topology ProgPopGen

/-- **K**olmogorov **A**bsorption **L**imit.

For a stochastic kernel on `{0,...,N}` with absorbing ends, whose `n`-step kernel is the
Chapman-Kolmogorov power `kPow P n`: if `h` is interior-harmonic with `h 0 = 0`, `h N = 1` and
    the
one-step variance of `h` is positive at every interior state, then `P^n(i,N) → h i` and
`P^n(i,0) → 1 - h i`. Re-export of `ProgPopGen.fixation_limit`. -/
theorem KAL {N : ℕ} {P : Fin (N + 1) → Fin (N + 1) → ℝ} (hP : AbsorbingChain P) (hN : 0 < N)
    {h : Fin (N + 1) → ℝ} (hh : InteriorHarmonic P h) (h0 : h 0 = 0) (h1 : h (Fin.last N) = 1)

    (hvar : ∀ k : Fin (N + 1), 0 < (k : ℕ) → (k : ℕ) < N → 0 < ∑ j, P k j * (h j - h k) ^
    2)
    (i : Fin (N + 1)) :
    Tendsto (fun n => kPow P n i (Fin.last N)) atTop (𝓝 (h i)) ∧
    Tendsto (fun n => kPow P n i 0) atTop (𝓝 (1 - h i)) :=
```

```
  fixation_limit hP hN hh h0 h1 hvar i

example : @KAL = @fixation_limit := rfl

end Absorption

section Isothermal

open Finset Filter Topology EvoGraph

/-- **I**sothermal **L**aw: **E**very **N**ode **E**quivalent.

Birth-death updating with uniform selection on a strongly connected isothermal graph (`W`
    doubly
stochastic): a single mutant of fitness `r ≠ 1` placed at *any* node `v` fixes with limiting
probability `(1 - r^(-1))/(1 - r^(-n))`, the same for every node and equal to Moran's value.
Re-export of `EvoGraph.isothermal_fixation_single`. -/
theorem ILENE {n : ℕ} {W : Fin n → Fin n → ℝ} {r : ℝ} (hn : 0 < n) (hr : 0 < r)
    (hr1 : r ≠ 1) (hW : Isothermal W) (hsc : StronglyConnected W) (v : Fin n) :
    Tendsto (fun t => sPow (bdK (fun _ => 1) W r) t {v} univ) atTop
      (𝓝 ((1 - r⁻¹) / (1 - r⁻¹ ^ n))) :=
  isothermal_fixation_single hn hr hr1 hW hsc v

example : @ILENE = @isothermal_fixation_single := rfl

end Isothermal

section Laplace

open MeasureTheory OAI.AtomicTriangular AutoresearchKeldysh

/-- **A**veraged **L**aplace **EX**ponentials **I**nherit **S**igns.

A positive mixture of decaying exponentials `fun t => ∫ e^{-t k(x)} m(x) dμ(x)` (with `k, m ≥
    0` and all
moments `∫ k^n e^{-c k} m dμ` finite for `c > 0`) is `AdmissiblePotential`: `C^∞` on `(0, ∞)
    `,
nonnegative, with alternating signs `(-1)^r g^{(r)} ≥ 0`, each exponential's signs passing to
    the
average. Re-export of `AutoresearchKeldysh.laplace_admissible`. -/
theorem ALEXIS {α : Type*} [MeasurableSpace α] {μ : Measure α} {k m : α → ℝ}
    (h : MixtureData μ k m) :
    AdmissiblePotential (fun t => ∫ x, Real.exp (-(t * k x)) * m x ∂μ) :=
  laplace_admissible h

example : @ALEXIS.{0} = @laplace_admissible.{0} := rfl

end Laplace

section Fisher

open ProgPopGen.Price

/-- **A**verage **U**nder **R**eplication: **E**volution **L**ifts **I**t, **A**lways.

Haploid selection with constant type fitnesses and no transmission bias, nonnegative abundances
    ,
positive total abundance and positive mean fitness: the mean fitness does not decrease,
`wbar ≤ wbar'`. This is a special case of Fisher's fundamental theorem (haploid types,
    constant
fitnesses, no transmission bias), obtained from the exact identity `wbar' - wbar = Var(w)/wbar`
derived from the Price equation. Re-export of `ProgPopGen.Price.fisher_haploid_nondecreasing`.
    -/
```

```
theorem AURELIA {ι : Type*} (s : Finset ι) (n w : ι → ℝ)
    (hn : ∀ i ∈ s, 0 ≤ n i) (hN : 0 < ∑ i ∈ s, n i) (hW : 0 < ∑ i ∈ s, n i * w i) :
    wmean s n w ≤ wbarNext s n w :=
  fisher_haploid_nondecreasing s n w hn hN hW

example : @AURELIA.{0} = @fisher_haploid_nondecreasing.{0} := rfl

end Fisher

end Acronymes

#print axioms Acronymes.ALIX
#print axioms Acronymes.KAL
#print axioms Acronymes.ILENE
#print axioms Acronymes.ALEXIS
#print axioms Acronymes.AURELIA
```

5.9 Exercises

Exercises marked (S) require running a script; allow at most two cores, for example with `taskset -c 6,7`.

Exercise 5.1 (U1, computation). 1. Show that $h_d(t) = t^{-1/2} - 2(t+d^2)^{-1/2}$ satisfies $h_d(d^2) = (1-\sqrt{2})/d$. Deduce that h_d is not completely monotone.

2. With $a = t^{-1/2}$ and $b = (t+d^2)^{-1/2}$, establish $(a-b)(a+b) = d^2a^2b^2$, and then $g_d(t) \le d^2 t^{-3/2}$. Why does this inequality guarantee that `latticeEnergy (g d)` is finite?

Exercise 5.2 (U1, (S)). Run `ch5_U1_dipolar.py`.

1. Check in the JSON that $\tilde{d} = 0.0660$ at $n = 10^{12}$ cm^{-2}.

2. Add the density $n = 5 \times 10^{11}$ cm^{-2} and give the square-minus-triangular gap in meV for $\varepsilon_r = 4$.

3. Replace the exact tail by 0 for $R = 400$. Is the error halved compared with $R = 200$? Explain this with the tail formula.

Exercise 5.3 (U2, computation). Check by hand that $g_1 - 2g_2$ is negative at $t = \frac{1}{4}$. Then show that a mixture $\sum_i w_i e^{-\kappa_i \sqrt{t}}/\sqrt{t}$ with positive weights is completely monotone, taking the single-term case for granted.

Exercise 5.4 (U2, (S)). In `exact_refs.jsonl`, produced by the script, the rows `"kind":"sweep"` give the error of the truncated energies (`sf_err`) as a function of r_c.

1. Plot $|$`sf_err`$|$ at $\kappa = 1$. From which r_c on does it become smaller than the square-minus-triangular gap, 9.5×10^{-3}?

2. Compare with the Voronoi tail bound.

Exercise 5.5 (U3, proof). 1. Show that $S_B(k) = \varepsilon_0/\varepsilon_B$ with $\varepsilon_B = \sqrt{\varepsilon_0(\varepsilon_0 + 2gn)}$ can be written $k/\sqrt{k^{*2}+k^2}$, where $k^* = 2mc/\hbar$ and $mc^2 = gn$.

2. Establish the envelope $S_B \le \sqrt{k/2k^*}$ and its equality case.

3. Show that $S = 1$ exceeds the envelope for every $0 < k < 2k^*$.

Exercise 5.6 (U3, (S))**.** Modify `ch5_U3_helium.py` to use $c = 238.2$ and then 238.4 m/s. Do the counts 29 / 7 / 63 change? Repeat with $\Delta = 0.04$. Which part of the result depends on the threshold fixed after the fact?

Exercise 5.7 (U4, proof and (S))**.** 1. For $\omega_n = Dk_n^3$, check that $\sum 2^{-n}\omega_n|v_n|^2 = 2D\Omega$.

2. Taking the conservation of H and S for granted, prove $2D\Omega(t) \le H_3(0) + (2E(0))^{3/2}$.

3. Run `sigma3_study probe 6 3 0.05 0.5 1e-10`, then `probe 6 2 0.05 0.5 1e-10`. Compare the `sup_ratio` fields. Which of the two models is not covered by the Lean bound, and why?

Exercise 5.8 (U5, computation)**.** 1. Compute φ_1 exactly for $N = 10$ and $r = 3/2$. Compare with the CVODE value, 0.33921585523416337, printed in `results.jsonl`.

2. Show that $p_i = r\,q_i$ and deduce the formula $\varphi_i = S_i/S_N$.

3. For a neutral population ($r = 1$), interpret $\varphi_i = i/N$.

Exercise 5.9 (U5, LeanFlow, (S))**.** Following the `harmonic` function of the script, write the LeanFlow claim that certifies that $(\varphi_0, \dots, \varphi_{10})$ is harmonic for the Moran chain $N = 10$, $r = 3/2$.

1. Submit it: the status must be `ok` and the axiom list empty.

2. Change a single value φ_5 by one part in 10^{12} and check that exactly three assertions are rejected. Why three?

Exercise 5.10 (U5, data, (S))**.** 1. Download the *train* split of the same dataset, at the same revision, and check that it contains 75 000 sequences, as the card states.

2. Compute the GC fraction per label.

3. Write three sentences explaining why this difference in composition does not provide a fitness r for the Moran model.

Exercise 5.11 (Controls, (S))**.** Choose one of the five scripts and deliberately break a positive control. For example, in `ch5_U1_dipolar.py`, forget the factor $\frac{1}{2}$ in the direct sum. Check that the script exits with code 1. Explain why a script that accepted this error would test nothing.

Exercise 5.12 ((†) University exercise: Fixation by optional stopping)**.** (This restates Exercise 4.14 for readers of this chapter alone.) Let (X_n) be the Moran chain on $\{0, \dots, N\}$ with $p_i = r\,q_i$, $r > 0$, $r \neq 1$, and let $\tau = \inf\{n : X_n \in \{0, N\}\}$.

1. Show that $M_n = r^{-X_n}$ is a martingale with respect to the natural filtration of (X_n).

2. Show that $\tau < \infty$ almost surely, and apply the optional stopping theorem to the bounded martingale $M_{n\wedge\tau}$ to obtain $\mathbb{P}_i(X_\tau = N) = (1 - r^{-i})/(1 - r^{-N})$.

3. For $r = 1$, find the analogous martingale and recover $\varphi_i = i/N$.

Exercise 5.13 ((†) University exercise: The Yukawa factor as a Laplace transform)**.** Let $\kappa > 0$.

1. Show that, for $t > 0$,

$$e^{-\kappa\sqrt{t}} = \int_0^\infty e^{-ts}\,\frac{\kappa}{2\sqrt{\pi}}\,s^{-3/2}e^{-\kappa^2/(4s)}\,ds.$$

(Hint: use the substitution $s = \kappa^2/(4u^2)$ and the Gaussian integral $\int_0^\infty e^{-(u-a/u)^2}\,du = \sqrt{\pi}/2$ for $a > 0$.)

2. Deduce, by differentiating under the integral sign (justify it by dominated convergence), that $e^{-\kappa\sqrt{t}}$ is completely monotone on $(0,\infty)$.

3. Writing $t^{-1/2} = \int_0^\infty e^{-ts}(\pi s)^{-1/2}\,ds$, express $g_\kappa(t) = e^{-\kappa\sqrt{t}}/\sqrt{t}$ as the Laplace transform of a positive function, and conclude again that g_κ is completely monotone.

Answer. c) $g_\kappa(t) = \int_0^\infty e^{-ts}\,(\pi s)^{-1/2} e^{-\kappa^2/(4s)}\,ds$.

The five images

Each of the five short names of Section 5.8 receives an image here. The tier of the theorem (**[Tier A]**) refers to the re-exported Lean statement. The curves, colours and displayed values are numerical computations **[Tier X]**: they illustrate a special case of the statement, without proving or extending it.

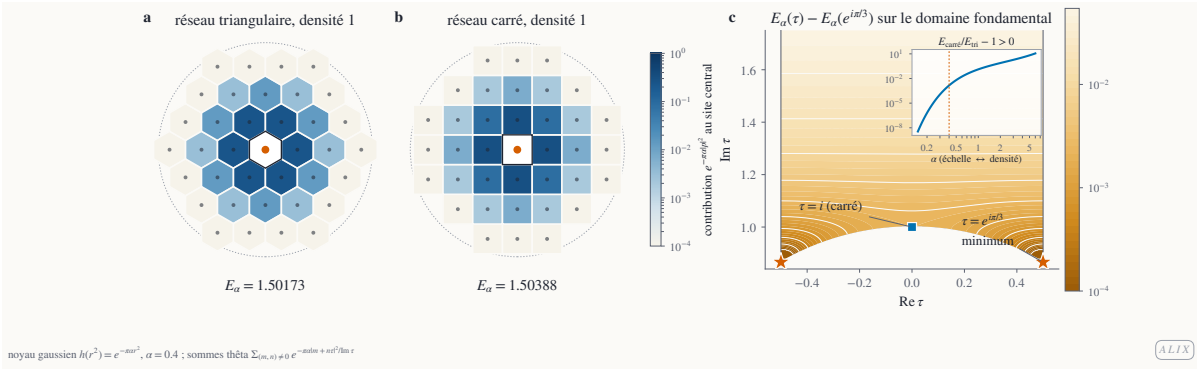


Figure 5.6: `ALIX`, *A Lattice Is eXtremal*. The theorem **[Tier A]** (conditional on the upstream theorem `universal_energy_minimum` of openai/math, AI-generated and not peer reviewed) states that at every density $\rho > 0$ the triangular lattice minimizes the energy of every admissible potential, among all locally finite configurations of density ρ. The image **[Tier X]** takes the Gaussian kernel $e^{-\pi\alpha r^2}$, $\alpha = 0.4$, at density 1, and compares *lattices* only. (a) and (b): Voronoi cells coloured by the contribution $e^{-\pi\alpha|p|^2}$ of each site p to the energy of the central site; $E_\alpha = 1.50173$ (triangular) versus 1.50388 (square). (c) The difference $E_\alpha(\tau) - E_\alpha(e^{i\pi/3})$ over the fundamental domain of lattice shapes, computed by theta sums; the grid minimum (at the equivalent point $-\frac{1}{2} + i\frac{\sqrt{3}}{2}$, to within one grid step) and five Nelder–Mead minimizations (to within 10^{-7}) land back on $\tau = e^{i\pi/3}$. Inset: $E_{\text{square}}/E_{\text{tri}} - 1 > 0$ for $\alpha \in [0.15, 6]$, that is, for a whole range of densities.

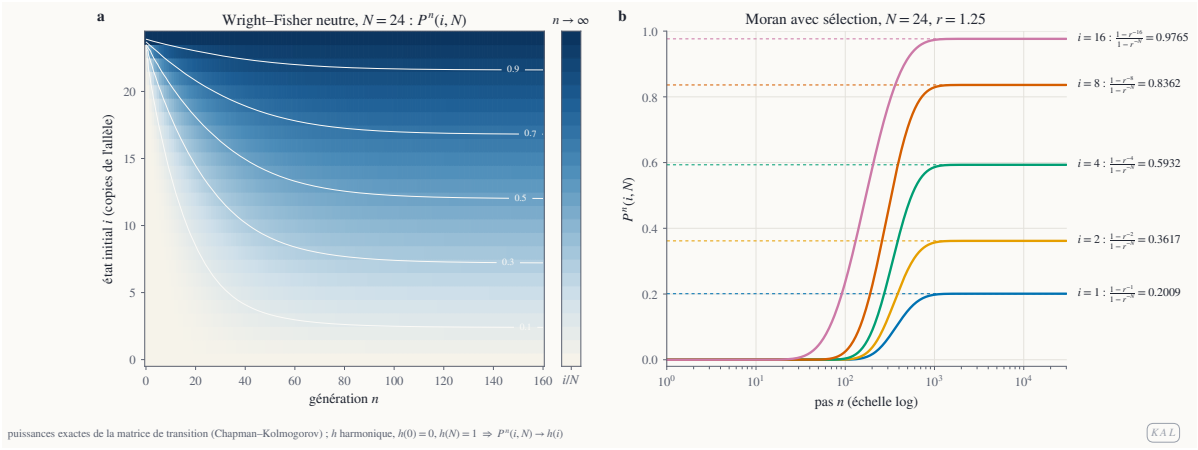


Figure 5.7: KAL, *Kolmogorov Absorption Limit.* The theorem **[Tier A]**: for a chain on $\{0, \dots, N\}$ with absorbing boundaries, if h is harmonic at the interior states, $h(0) = 0$, $h(N) = 1$, and the one-step variance of h is strictly positive, then $P^n(i, N) \to h(i)$ and $P^n(i, 0) \to 1 - h(i)$. Images **[Tier X]**, by exact powers of the transition matrix. (a) Neutral Wright–Fisher, $N = 24$: $P^n(i, N)$ converges to $h(i) = i/N$ (right-hand band); maximal deviation at $n = 4096$: $6 \cdot 10^{-16}$. (b) Moran with selection, $N = 24$, $r = 1.25$: $P^n(i, N)$ tends to $(1 - r^{-i})/(1 - r^{-N})$ (dotted), deviation $7 \cdot 10^{-15}$ at $n = 2^{22}$. The harmonicity of both functions h is checked in exact rationals (for Wright–Fisher, at $N = 8$).

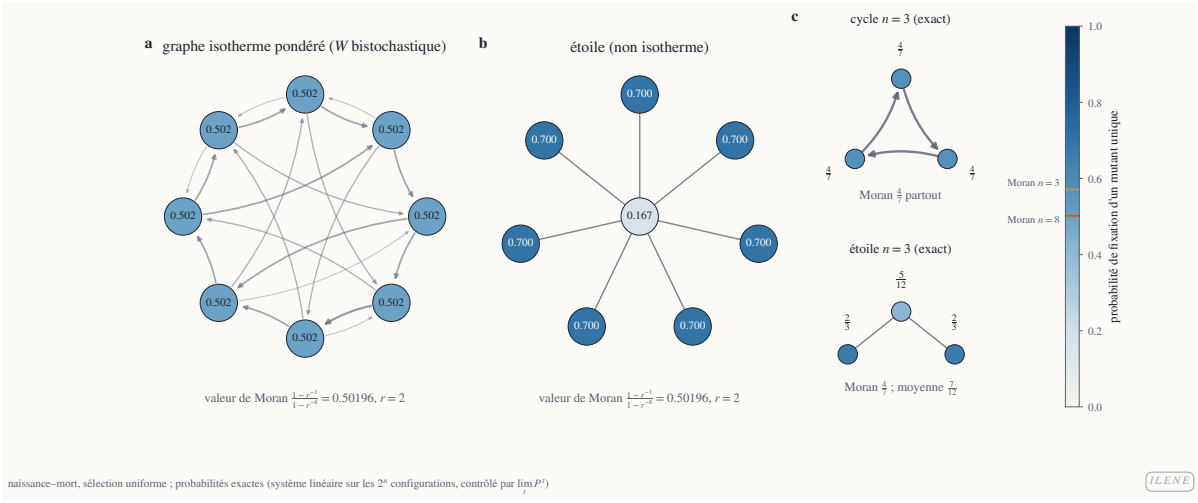


Figure 5.8: ILENE, *Isothermal Law: Every Node Equivalent.* The theorem **[Tier A]**: on a strongly connected isothermal graph (W doubly stochastic), under birth–death with uniform selection, a single mutant of fitness $r \neq 1$ fixes with probability $(1 - r^{-1})/(1 - r^{-n})$, whatever its starting vertex. Images **[Tier X]**, $r = 2$, probabilities obtained by solving the linear system on the 2^n configurations (in floating point for $n = 8$, in rationals for $n = 3$), and checked against the limit of P^t. (a) Weighted graph with $n = 8$ vertices, $W = \frac{1}{2}\Pi_1 + \frac{1}{3}\Pi_2 + \frac{1}{6}\Pi_3$ (three fixed-point-free permutations): all eight vertices give the Moran value 0.50196 to within $2 \cdot 10^{-15}$. (b) The star with 7 leaves is not isothermal and lies outside the hypotheses: centre 0.167, leaves 0.700. (c) Rational computation for $n = 3$: the 3-cycle gives $\frac{4}{7}$ everywhere; the star gives $\frac{5}{12}$ at the centre and $\frac{2}{3}$ at the leaves, with mean $\frac{7}{12} \neq \frac{4}{7}$.

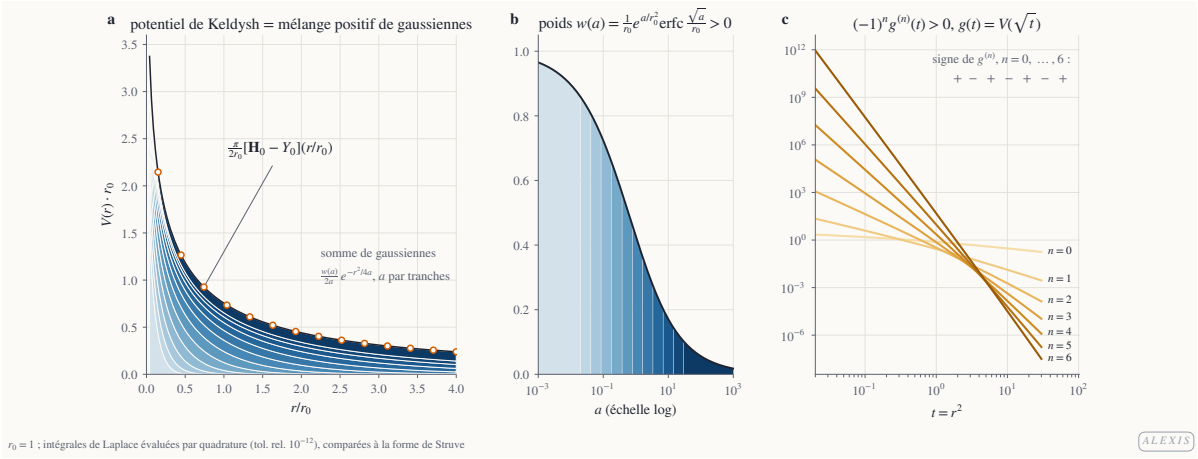


Figure 5.9: `ALEXIS`, *Averaged Laplace EXponentials Inherit Signs.* The theorem **[Tier A]**: a positive mixture of exponentials $t \mapsto \int e^{-t\,k(x)} m(x)\,d\mu(x)$ is an admissible potential, of class C^∞ on $(0,+\infty)$, positive, with $(-1)^n g^{(n)} \geq 0$. The image **[Tier X]** shows one instance, the Rytova–Keldysh potential ($r_0 = 1$), written as a Gaussian mixture $V(r) = \int_0^\infty \frac{w(a)}{2a}\, e^{-r^2/4a}\, da$ with weight $w(a) = \frac{1}{r_0} e^{a/r_0^2} \operatorname{erfc}(\sqrt{a}/r_0) > 0$. (a) The contributions of twelve slices in a, stacked, reproduce the Struve form $\frac{\pi}{2r_0}[\mathbf{H}_0 - Y_0](r/r_0)$ (dots) to within $3 \cdot 10^{-13}$ in relative terms; the identification with the Struve form is a human proof, not machine-checked **[Tier C]**. (b) The weight w, coloured by slice. (c) $(-1)^n g^{(n)}(t)$, $g(t) = V(\sqrt{t})$, for $n = 0, \ldots, 6$: all curves are positive, so the derivatives $g^{(n)}$ alternate in sign; $n = 1$ is cross-checked by finite differences of the Struve form.

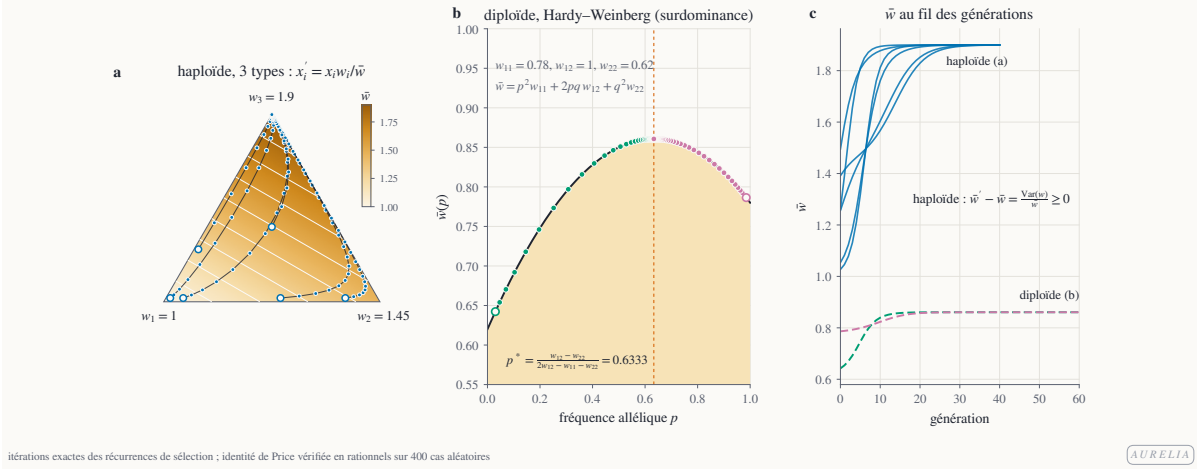


Figure 5.10: `AURELIA`, *Average Under Replication: Evolution Lifts It, Always.* The theorem **[Tier A]**: under haploid selection with constant fitnesses and no transmission bias, mean fitness does not decrease, $\bar{w} \leq \bar{w}'$; this is a special case of Fisher's fundamental theorem, derived from the Price equation. Images **[Tier X]**. (a) Three types, $w = (1, 1.45, 1.9)$: on the simplex coloured by $\bar{w}$, six trajectories of $x_i' = x_i w_i / \bar{w}$ climb towards the best type. (b) Diploid case under Hardy–Weinberg, with overdominance ($w_{11} = 0.78$, $w_{12} = 1$, $w_{22} = 0.62$): both trajectories climb $\bar{w}(p)$ up to $p^* = 0.6333$. This panel falls under the diploid theorem `wbar_increment` of Chapter 4 (H12) **[Tier A]**, not under `AURELIA`. (c) $\bar{w}$ over the generations, never decreasing. The identity $\bar{w}' - \bar{w} = \mathrm{Var}(w)/\bar{w}$ is checked in exact rationals on 400 random cases; negative control: with a mutation bias from the best type towards the worst, $\bar{w}$ drops by 0.35.

Appendix A

Analysis background

This appendix collects the analytical tools that the chapters use without always re-proving them:

- interchanging a limit with an integral or a series (§A.1);
- the Fourier and Laplace transforms (§A.2);
- the Poisson summation formula, in the general case (§A.3);
- the special functions that occur: Γ, erfc, J_0, K_0, Y_0 and the Struve function $\mathbf{H}_0$ (§A.4);
- lattices in the plane and the fundamental domain of $SL_2(\mathbb{Z})$ (§A.5).

The level is that of a second-year undergraduate analysis course. The integral is Lebesgue's, but it suffices to be able to apply the dominated convergence theorem and term-by-term integration of series, both recalled below.

Warning : tiers of the statements in this appendix

The proofs written here are human proofs **[Tier C]**: none of them is checked by Lean in this book. The statements taken as given (dominated convergence, Fejér's theorem, properties of the Bessel and Struve functions taken from the DLMF [66]) are **[Tier L]**. The pinned Mathlib contains formal versions of several of them: dominated convergence (`MeasureTheory.tendsto_integral_of_dominated_convergence`, file `MeasureTheory/Integral/DominatedConvergence.lean`) and the Poisson summation formula (files `Analysis/Fourier/PoissonSummation.lean` and `Analysis/SpecialFunctions/Gaussian/PoissonSummation.lean`). We have only checked that these files exist; we have neither compared their statements with those of this appendix, nor connected those statements to the proofs below.

A.1 Series and integrals depending on a parameter

A.1.1 The dominated convergence theorem

Theorem A.1 (Dominated convergence **[Tier L]**)**.** *Let I be an interval of $\mathbb{R}$ and (f_n) a sequence of piecewise continuous functions on I. Assume that:*

1. *(f_n) converges pointwise on I to a piecewise continuous function f;*
2. *there exists $\varphi : I \to \mathbb{R}_+$, integrable on I, such that $|f_n(x)| \leq \varphi(x)$ for all n and all x (domination hypothesis).*

Then the f_n and f are integrable on I, and $\int_I f_n \to \int_I f$.

The essential point is that the dominating function φ does not depend on n. Without it, the conclusion can fail: $f_n = n\,\mathbf{1}_{(0,1/n]}$ tends pointwise to 0 on $(0,1]$, whereas $\int f_n = 1$ for all n.

University remark. **[Tier L]** The general form of Lebesgue's theorem holds on any measure space $(\Omega, \mathcal{F}, \mu)$: if the f_n are measurable, $f_n \to f$ μ-almost everywhere, and $|f_n| \le g$ μ-a.e. for every n with $g \in L^1(\mu)$, then $f \in L^1(\mu)$ and $\int |f_n - f|\, d\mu \to 0$, hence $\int f_n\, d\mu \to \int f\, d\mu$. No continuity is needed; the piecewise-continuous version above is the special case of Lebesgue measure on an interval. The probabilistic form (expectations of a.s. convergent, dominated random variables) is the case $\mu(\Omega) = 1$ [69].

Theorem A.2 (Interchanging series and integral **[Tier L]**)**.** *Let (u_n) be a sequence of integrable functions on I such that $\sum_n u_n$ converges pointwise to a piecewise continuous function and $\sum_n \int_I |u_n| < \infty$. Then $\sum_n u_n$ is integrable and $\int_I \sum_n u_n = \sum_n \int_I u_n$.*

A.1.2 Continuity and differentiation under the integral sign

Theorem A.3 (Continuity under the integral sign)**.** *Let A be a subset of $\mathbb{R}^p$ and $f : A \times I \to \mathbb{C}$ such that $f(\cdot, x)$ is continuous for every x, $f(a, \cdot)$ is piecewise continuous for every a, and $|f(a,x)| \le \varphi(x)$ with φ integrable. Then $F(a) = \int_I f(a,x)\, dx$ is continuous on A.*

Human proof. Let $a_n \to a$ in A. The functions $x \mapsto f(a_n, x)$ converge pointwise to $f(a, \cdot)$ and are dominated by φ. Theorem A.1 gives $F(a_n) \to F(a)$, and the sequential characterisation of continuity concludes. □

Theorem A.4 (Differentiation under the integral sign)**.** *Let J be an interval and $f : J \times I \to \mathbb{C}$. Assume that $f(a, \cdot)$ is integrable for every $a \in J$, that $\partial f/\partial a$ exists, is continuous in a and piecewise continuous in x, and that there exists an integrable ψ with $|\partial_a f(a,x)| \le \psi(x)$ for all a, x. Then $F(a) = \int_I f(a,x)\, dx$ is of class $\mathcal{C}^1$ on J, and $F'(a) = \int_I \partial_a f(a,x)\, dx$.*

Human proof. Fix $a \in J$ and a sequence $h_n \to 0$, $h_n \neq 0$, with $a + h_n \in J$. Set $g_n(x) = \big(f(a+h_n, x) - f(a,x)\big)/h_n$. By definition of the partial derivative, $g_n(x) \to \partial_a f(a,x)$ for every x. By the mean value inequality applied to $t \mapsto f(t,x)$ between a and $a + h_n$, $|g_n(x)| \le \sup_t |\partial_a f(t,x)| \le \psi(x)$. Theorem A.1 gives $\big(F(a+h_n) - F(a)\big)/h_n = \int g_n \to \int \partial_a f(a,x)\, dx$. The continuity of F' follows from Theorem A.3 applied to $\partial_a f$. □

Remark A.5 (Local domination). In practice it suffices that the domination hold on every compact interval $[\alpha, \beta] \subset J$: differentiability is a local property. This is how the Laplace transforms below are handled, where the domination is uniform only for $t \ge \alpha > 0$.

University remark. **[Tier L]** In measure-theoretic form: let $(\Omega, \mathcal{F}, \mu)$ be a measure space, $J \subset \mathbb{R}$ an open interval, and $f : J \times \Omega \to \mathbb{C}$ with $f(a, \cdot) \in L^1(\mu)$ for every $a \in J$. If, for μ-a.e. ω, the map $a \mapsto f(a, \omega)$ is differentiable on J and $|\partial_a f(a,\omega)| \le g(\omega)$ for all $a \in J$, with $g \in L^1(\mu)$, then $F(a) = \int f(a,\omega)\, \mu(d\omega)$ is differentiable on J and $F'(a) = \int \partial_a f(a,\omega)\, \mu(d\omega)$. The proof is the one above, with the general dominated convergence theorem in place of Theorem A.1 (standard).

Example A.6 (Gaussian moments)**.** For $\alpha > 0$, $G(\alpha) = \int_{\mathbb{R}} e^{-\pi\alpha x^2} dx = \alpha^{-1/2}$ (Gaussian integral). On $[\alpha_0, \infty)$ with $\alpha_0 > 0$, $|\partial_\alpha e^{-\pi\alpha x^2}| = \pi x^2 e^{-\pi\alpha x^2} \le \pi x^2 e^{-\pi\alpha_0 x^2}$, which is integrable. Theorem A.4 gives $\int \pi x^2 e^{-\pi\alpha x^2} dx = -G'(\alpha) = \frac{1}{2}\alpha^{-3/2}$. Differentiating m times, $\int (\pi x^2)^m e^{-\pi\alpha x^2} dx = \frac{(2m)!}{4^m m!}\, \alpha^{-1/2-m}$ (Exercise A.1). The same technique computes, in Chapter 1, the Fourier transform of a Gaussian (Theorem 1.10, step 1).

A.1.3 Series of functions

Proposition A.7 (Normal convergence [Tier L]). *Let (u_n) be continuous functions on a subset A of $\mathbb{R}^d$ with $\sum_n \sup_A |u_n| < \infty$. Then $\sum_n u_n$ converges uniformly on A and its sum is continuous. If moreover the u_n are $\mathcal{C}^1$ on an open set and $\sum_n \sup |\nabla u_n| < \infty$, the sum is $\mathcal{C}^1$ and may be differentiated term by term.*

Example A.8 (Theta series). For a lattice Λ in the plane and $\alpha \geq \alpha_0 > 0$, one has $|x|^2 \geq c\,(m^2+n^2)$ for $x = mu + nv$ (Chapter 1), hence $e^{-\pi\alpha|x|^2} \leq e^{-\pi\alpha_0 c(m^2+n^2)}$, the general term of a convergent series. The series $\Theta_\Lambda(\alpha)$ therefore converges normally on $[\alpha_0, \infty)$, and it is $\mathcal{C}^\infty$ in α there: one may differentiate term by term as often as one likes. The lattice sums $S(\alpha) = \sum |P|^2 e^{-\pi\alpha|P|^2}$ of Chapter 3 are precisely $-\Theta'_\Lambda(\alpha)/\pi$.

A.2 Fourier and Laplace transforms

A.2.1 The Fourier transform

Definition A.9. For integrable $f : \mathbb{R}^d \to \mathbb{C}$, the *Fourier transform* of f is

$$\hat{f}(\xi) = \int_{\mathbb{R}^d} f(x)\, e^{-2\pi i\, \xi\cdot x}\, dx, \qquad \xi \in \mathbb{R}^d.$$

This is the convention of Chapter 1: the factor 2π sits in the exponent, and the inversion formula then has no factor in front of the integral. Other books, especially in physics, write $\int f(x) e^{-ik\cdot x} dx$; one passes from one to the other via $k = 2\pi\xi$.

Proposition A.10 (Elementary properties). *Let f be integrable, $a \in \mathbb{R}^d$, $\lambda > 0$ and M an invertible matrix.*

1. *$\hat{f}$ is continuous and bounded by $\|f\|_1$.*
2. *Translation: $\widehat{f(\cdot - a)}(\xi) = e^{-2\pi i\, a\cdot\xi}\hat{f}(\xi)$.*
3. *Linear change of variables: $\widehat{f \circ M}(\xi) = |\det M|^{-1}\hat{f}(M^{-\mathsf{T}}\xi)$.*
4. *If f is $\mathcal{C}^1$ and f and $\partial_j f$ are integrable, then $\widehat{\partial_j f}(\xi) = 2\pi i\, \xi_j \hat{f}(\xi)$.*

Human proof. (1) follows from Theorem A.3, with $|f(x)e^{-2\pi i\xi\cdot x}| = |f(x)|$ (the version of Theorem A.3 for a function that is only integrable in x has the same proof). (2) and (3) are the changes of variables $y = x - a$ and $y = Mx$ (Jacobian $|\det M|$, and $\xi \cdot M^{-1}y = (M^{-\mathsf{T}}\xi)\cdot y$). (4) is an integration by parts in x_j; the boundary term tends to 0 along a sequence of bounds, because f is integrable and has an integrable derivative. □

Example A.11 (Gaussians). For $\alpha > 0$, $\widehat{e^{-\pi\alpha|x|^2}}(\xi) = \alpha^{-d/2} e^{-\pi|\xi|^2/\alpha}$. The proof in dimension 1 (a differential equation for $\hat{f}$, obtained by differentiation under the integral sign) is step 1 of the proof of Theorem 1.10; one passes to dimension d by Fubini, since the exponential factorises. In particular $e^{-\pi|x|^2}$ is its own Fourier transform. For a Gaussian with arbitrary covariance, item (3) of Proposition A.10 gives the general formula.

Theorem A.12 (Inversion [Tier L]). *If f and $\hat{f}$ are integrable, then $f(x) = \int \hat{f}(\xi) e^{2\pi i\, \xi\cdot x} d\xi$ at every point where f is continuous.*

A.2.2 Radial functions in the plane and the Bessel function J_0

The potentials of Chapters 2 and 3 depend only on the distance $r = |x|$. Their planar Fourier transform is then radial too, and it is expressed with the Bessel function J_0, defined by

$$J_0(z) = \frac{1}{2\pi}\int_0^{2\pi} e^{iz\cos\theta}\,d\theta = \frac{1}{\pi}\int_0^{\pi}\cos(z\cos\theta)\,d\theta. \tag{A.1}$$

Proposition A.13 (Hankel transform of order 0)**.** *If $f(x) = F(|x|)$ is integrable on $\mathbb{R}^2$, then $\hat f(\xi) = 2\pi\int_0^\infty F(r)\,J_0(2\pi|\xi|r)\,r\,dr$.*

Human proof. In polar coordinates $x = r(\cos\theta, \sin\theta)$, and choosing the axes so that $\xi = (|\xi|, 0)$ (the integral is rotation invariant), $\xi\cdot x = |\xi| r\cos\theta$. By Fubini,

$$\hat f(\xi) = \int_0^\infty F(r)\Big(\int_0^{2\pi} e^{-2\pi i|\xi| r\cos\theta}d\theta\Big) r\,dr = 2\pi\int_0^\infty F(r)J_0(2\pi|\xi|r)\,r\,dr,$$

since J_0 is even (replace θ by $\theta+\pi$ in (A.1)). □

Remark A.14 (Check on the Gaussian). With $F(r) = e^{-\pi a r^2}$, the proposition and Example A.11 give $2\pi\int_0^\infty e^{-\pi a r^2}J_0(2\pi k r)\,r\,dr = a^{-1}e^{-\pi k^2/a} > 0$: this is the formula used in Chapter 3 to contrast the Gaussian, whose transform is positive everywhere, with the soft-shoulder potential of Rydberg-dressed atoms, whose planar transform takes negative values. The numerical computation of the latter is the Hankel transform above, evaluated by quadrature **[Tier X]**.

A.2.3 The Laplace transform

Definition A.15. Let μ be a positive measure on $[0,\infty)$ (for example $\mu(da) = w(a)\,da$ with $w \ge 0$). Its *Laplace transform* is $\mathcal{L}\mu(t) = \int_0^\infty e^{-ta}\,\mu(da)$, for those t at which the integral converges.

Proposition A.16 (A Laplace transform is completely monotone)**.** *If $\mathcal{L}\mu(t) < \infty$ for every $t > 0$, then $\mathcal{L}\mu$ is $\mathcal{C}^\infty$ on $(0,\infty)$ and $(-1)^m(\mathcal{L}\mu)^{(m)}(t) = \int_0^\infty a^m e^{-ta}\mu(da) \ge 0$.*

Human proof. On $[t_0,\infty)$ with $t_0 > 0$, one has $a^m e^{-ta} \le a^m e^{-t_0 a} \le C_m\,e^{-t_0 a/2}$, where $C_m = \sup_{a\ge 0} a^m e^{-t_0 a/2} < \infty$. This function is μ-integrable since $\mathcal{L}\mu(t_0/2) < \infty$. Theorem A.4 (measure version, same proof) applies m times. □

This is the "easy" direction of Bernstein's theorem (Theorem 1.22, Proposition 1.23); the converse, that every completely monotone function is a Laplace transform, is **[Tier L]**.

University remark. **[Tier L]** Bernstein's theorem in full: a function $g : (0,\infty)\to\mathbb{R}$ is completely monotone, i.e. $\mathcal{C}^\infty$ with $(-1)^m g^{(m)} \ge 0$ for all $m \ge 0$, if and only if $g = \mathcal{L}\mu$ for a positive Borel measure μ on $[0,\infty)$ with $\mathcal{L}\mu(t) < \infty$ for all $t > 0$; the measure μ is then unique [6, 7].

Example A.17 (Three Laplace transforms used in the book)**.** 1. *Powers.* For $s > 0$ and $t > 0$, $t^{-s} = \frac{1}{\Gamma(s)}\int_0^\infty a^{s-1}e^{-ta}\,da$. Indeed, the change of variable $u = ta$ gives

$$\int_0^\infty a^{s-1}e^{-ta}\,da = t^{-s}\int_0^\infty u^{s-1}e^{-u}\,du = t^{-s}\Gamma(s).$$

With $t = r^2$ and $s = 1/2$, one obtains $1/r = \pi^{-1/2}\int_0^\infty a^{-1/2}e^{-ar^2}da$: the Coulomb potential is a mixture of Gaussians. The substitution $a = u^2$ gives the form $1/r = \frac{2}{\sqrt\pi}\int_0^\infty e^{-r^2u^2}du$, the starting point of the Ewald decomposition of case U1 (Chapter 5).

2. *Dipolar exciton.* Subtracting the same formula written for r^2+d^2, $\frac{1}{r} - \frac{1}{\sqrt{r^2+d^2}} = \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-r^2u^2}\big(1-e^{-d^2u^2}\big)du$, with a positive weight: this is the formula of Chapter 5 (case U1) up to the factor 2.

3. *Subordination.* For $\rho > 0$, $e^{-\rho\sqrt{t}} = \int_0^\infty e^{-ta}\, \frac{\rho}{2\sqrt{\pi}}\, a^{-3/2} e^{-\rho^2/(4a)}\, da$ **[Tier L]**; this is the formula that Chapter 2 uses for the Yukawa potential and for the Laplace form of the Keldysh potential.

A.3 The Poisson summation formula

Chapter 1 proves the Poisson formula for Gaussians. Here is the general statement of which it is a special case, first in dimension 1, then for an arbitrary lattice.

Theorem A.18 (Poisson formula in dimension 1)**.** *Let $f : \mathbb{R} \to \mathbb{C}$ be continuous. Assume there exist $C > 0$ and $\delta > 0$ such that*

$$|f(x)| \le C(1+|x|)^{-1-\delta} \quad \text{and} \quad |\hat{f}(\xi)| \le C(1+|\xi|)^{-1-\delta} \qquad (x, \xi \in \mathbb{R}).$$

Then, for every $x \in \mathbb{R}$, $\displaystyle\sum_{n\in\mathbb{Z}} f(x+n) = \sum_{k\in\mathbb{Z}} \hat{f}(k)\, e^{2\pi i k x}$, both series converging absolutely. In particular $\sum_n f(n) = \sum_k \hat{f}(k)$.

Human proof. Step 1: the periodisation. Set $F(x) = \sum_{n\in\mathbb{Z}} f(x+n)$. On $[0,1]$, $|f(x+n)| \le C(1+|n|-1)^{-1-\delta}$ for $|n| \ge 2$, the general term of a convergent series; the series therefore converges normally on $[0,1]$, and by periodicity on $\mathbb{R}$. The function F is continuous (Proposition A.7) and 1-periodic.

Step 2: its Fourier coefficients. For $k \in \mathbb{Z}$, by Theorem A.2 (the series of the integrals of the moduli converges, since it equals $\int_{\mathbb{R}} |f| < \infty$),

$$c_k(F) = \int_0^1 F(x) e^{-2\pi i k x} dx = \sum_{n\in\mathbb{Z}} \int_n^{n+1} f(y) e^{-2\pi i k(y-n)} dy = \int_{\mathbb{R}} f(y) e^{-2\pi i k y} dy = \hat{f}(k),$$

since $e^{2\pi i k n} = 1$.

Step 3: the Fourier series converges to F. The hypothesis on $\hat{f}$ gives $\sum_k |c_k(F)| < \infty$. The series $G(x) = \sum_k \hat{f}(k) e^{2\pi i k x}$ therefore converges normally to a continuous 1-periodic function, and $c_k(G) = \hat{f}(k)$ (integrate term by term). The function $H = F - G$ is continuous, periodic, and all its Fourier coefficients vanish. By Fejér's theorem **[Tier L]**, the Cesàro means of the Fourier series of a continuous periodic function converge uniformly to it; here they are all zero, hence $H = 0$. □

Remark A.19 (Why two decay hypotheses). The decay of f makes F continuous; that of $\hat{f}$ makes the Fourier series absolutely convergent. Without the second, the formula can fail: there exist continuous integrable functions for which the series $\sum f(n)$ converges but does not equal $\sum \hat{f}(k)$. Gaussians, and more generally Schwartz functions, satisfy both hypotheses with any δ.

Theorem A.20 (Poisson formula for a lattice)**.** *Let $\Lambda = M\mathbb{Z}^d$ be a lattice in $\mathbb{R}^d$, of covolume $V = |\det M|$, and $\Lambda^* = M^{-\mathsf{T}}\mathbb{Z}^d$ its dual lattice. If f is continuous with $|f(x)| \le C(1+|x|)^{-d-\delta}$ and $|\hat{f}(\xi)| \le C(1+|\xi|)^{-d-\delta}$, then*

$$\sum_{x\in\Lambda} f(x+y) = \frac{1}{V} \sum_{\xi\in\Lambda^*} \hat{f}(\xi)\, e^{2\pi i\, \xi\cdot y} \qquad (y \in \mathbb{R}^d).$$

Human proof. Apply the d-dimensional version of the preceding proof (the torus $\mathbb{R}^d/\mathbb{Z}^d$ in place of $\mathbb{R}/\mathbb{Z}$, and Fejér's theorem on the torus **[Tier L]**) to $g = f \circ M$. The decay hypotheses pass from f to g, since $|Mx| \geq \|M^{-1}\|^{-1}|x|$. One obtains $\sum_{n\in\mathbb{Z}^d} g(n+z) = \sum_{k\in\mathbb{Z}^d} \hat{g}(k)e^{2\pi ik\cdot z}$. Next, $\hat{g}(k) = V^{-1}\hat{f}(M^{-\mathsf{T}}k)$ (Proposition A.10, item 3). Finally set $y = Mz$: $k \cdot z = (M^{-\mathsf{T}}k) \cdot y$, and $M^{-\mathsf{T}}k$ ranges over Λ^*. □

With $f(x) = e^{-\pi\alpha|x|^2}$ and $y = 0$, one recovers exactly Theorem 1.10.

A.3.1 Application: the idea of Ewald summation

Planar lattice sums of a long-range potential, such as $\sum_{x\in\Lambda\setminus\{0\}} |x|^{-3}$, converge slowly: the truncation error at distance R decays like $1/R$. Ewald's method splits the potential in two using a Gaussian-mixture representation $V(r) = \int_0^\infty e^{-r^2u^2} w(u)\, du$, with $w \geq 0$ (Example A.17). Choose a parameter $\eta > 0$ and write

$$V(r) = \underbrace{\int_\eta^\infty e^{-r^2u^2} w(u)\, du}_{\text{short range}} + \underbrace{\int_0^\eta e^{-r^2u^2} w(u)\, du}_{\text{smooth, long range}}.$$

- The short-range part decays like $e^{-\eta^2r^2}$: its sum over the lattice converges very fast in direct space.
- The smooth part is a mixture of wide Gaussians. Theorem A.20 turns its sum into a sum over the dual lattice of narrow Gaussians $e^{-\pi^2|\xi|^2/u^2}$, which also converges very fast.

The result does not depend on η: this is a natural check of the code. Case U1 of Chapter 5 reports precisely that the Ewald energy of rusty-SUNDIALS does not vary with this splitting parameter by more than 9.0×10^{-15} **[Tier X]**.

A.4 Special functions

We prove only the elementary properties here. The others are cited from the NIST *Digital Library of Mathematical Functions* [66] (Chapters 5 for Γ, 7 for erf, 10 for the Bessel functions, 11 for the Struve functions) **[Tier L]**.

A.4.1 The function Γ

For $s > 0$, $\Gamma(s) = \int_0^\infty u^{s-1}e^{-u}du$. An integration by parts gives $\Gamma(s+1) = s\Gamma(s)$, hence $\Gamma(n+1) = n!$. The substitution $u = \pi x^2$ gives $\Gamma(1/2) = \int_{\mathbb{R}} e^{-\pi x^2}dx \cdot \sqrt{\pi} = \sqrt{\pi}$, by the Gaussian integral.

A.4.2 The error function and the complementary error function

Definition A.21. $\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-s^2} ds$, $\operatorname{erfc}(x) = 1 - \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty e^{-s^2} ds$, and $\operatorname{erfcx}(x) = e^{x^2} \operatorname{erfc}(x)$.

Proposition A.22. *For $x > 0$:* $\dfrac{2}{\sqrt{\pi}} \dfrac{e^{-x^2}}{x + \sqrt{x^2+2}} \leq \operatorname{erfc}(x) \leq \dfrac{e^{-x^2}}{x\sqrt{\pi}}$. *In particular* $\operatorname{erfcx}(x) \leq \frac{1}{x\sqrt{\pi}}$ *and* $\operatorname{erfc}(x) \sim \frac{e^{-x^2}}{x\sqrt{\pi}}$ *as $x \to \infty$.*

Human proof. Upper bound. For $s \geq x > 0$, one has $1 \leq s/x$, hence

$$\int_x^\infty e^{-s^2} ds \leq \frac{1}{x} \int_x^\infty s e^{-s^2} ds = \frac{e^{-x^2}}{2x}.$$

Multiply by $2/\sqrt{\pi}$.

Lower bound. We take it as given **[Tier L]** (it is proved by studying the difference of the two sides, which tends to 0 at infinity and whose derivative has a controlled sign); it is used only for the asymptotic equivalent, which the upper bound and an integration by parts also suffice to establish (Exercise A.2). □

Chapter 2 uses the bounds $0 < \operatorname{erfcx}(x) \le 1$ for $x \ge 0$ and $\operatorname{erfcx}(x) \le 1/(\sqrt{\pi}x)$ for $x > 0$, proved there via the representation $\operatorname{erfcx}(x) = \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-(y^2+2xy)} dy$. The proposition also serves to bound the tail of a Gaussian series: by series–integral comparison, $\sum_{n>R} e^{-\pi n^2} \le \int_{R-1}^\infty e^{-\pi x^2} dx = \frac{1}{2} \operatorname{erfc}\big(\sqrt{\pi}(R-1)\big)$ for $R \ge 1$ (Exercise A.2).

A.4.3 The Bessel functions J_0, Y_0 and K_0

Proposition A.23 (J_0)**.** *The function J_0 defined by* (A.1) *satisfies:*

1. $J_0(z) = \sum_{m \ge 0} \frac{(-1)^m}{(m!)^2} \big(\frac{z}{2}\big)^{2m}$ *for every* $z \in \mathbb{R}$*;*
2. $|J_0(z)| \le 1$ *and* $J_0(0) = 1$*;*
3. J_0 *solves Bessel's equation of order* 0*:* $zy'' + y' + zy = 0$*;*
4. ***[Tier L]*** $J_0(z) = \sqrt{2/(\pi z)}\,\big(\cos(z - \pi/4) + O(1/z)\big)$ *as* $z \to +\infty$.

Human proof. (1) Expand $\cos(z\cos\theta) = \sum_m (-1)^m z^{2m} \cos^{2m}\theta/(2m)!$; the series converges normally in θ, and one integrates term by term. The Wallis integrals give $\frac{1}{\pi}\int_0^\pi \cos^{2m}\theta\, d\theta = \binom{2m}{m} 4^{-m}$, whence the coefficient $\frac{1}{(2m)!}\binom{2m}{m}4^{-m} = \frac{1}{(m!)^2 4^m}$. (2) The modulus of the integrand of (A.1) equals 1. (3) By differentiation under the integral sign, $J_0'(z) = -\frac{1}{\pi}\int_0^\pi \cos\theta \sin(z\cos\theta) d\theta$ and $J_0''(z) = -\frac{1}{\pi}\int_0^\pi \cos^2\theta \cos(z\cos\theta) d\theta$. Hence $zJ_0'' + zJ_0 = \frac{z}{\pi}\int_0^\pi \sin^2\theta \cos(z\cos\theta) d\theta$. An integration by parts, with $\frac{d}{d\theta}\sin(z\cos\theta) = -z\sin\theta\cos(z\cos\theta)$, turns this integral into $\frac{1}{\pi}\int_0^\pi \cos\theta\sin(z\cos\theta)d\theta = -J_0'(z)$. □

Bessel's equation has a second solution, singular at 0: the Neumann function (Bessel function of the second kind) Y_0, with $Y_0(z) \sim \frac{2}{\pi}\ln z$ as $z \to 0^+$ **[Tier L]**. The *modified Bessel function* K_0 is defined for $x > 0$ by

$$K_0(x) = \int_0^\infty e^{-x\cosh t}\, dt.$$

It is positive and decreasing, behaves like $-\ln x$ at 0 and like $\sqrt{\pi/(2x)}\, e^{-x}$ at infinity **[Tier L]**. It is the Green's function of the operator $-\Delta + \kappa^2$ in the plane: $(-\Delta+\kappa^2)\frac{1}{2\pi}K_0(\kappa|x|) = \delta_0$ **[Tier L]**. Chapters 2 and 5 note that this two-dimensional Debye–Hückel potential is *not* treated in this book, which studies the three-dimensional Yukawa potential restricted to the plane, $e^{-\kappa r}/r$.

A.4.4 The Struve function $\mathbf{H}_0$ and the Rytova–Keldysh potential

Definition A.24. The Struve function of order 0 is $\mathbf{H}_0(z) = \frac{2}{\pi}\sum_{m\ge 0} \frac{(-1)^m z^{2m+1}}{\big(1\cdot 3\cdot 5\cdots(2m+1)\big)^2}$ for $z \in \mathbb{R}$.

The series converges for every z (ratio test). One also has the integral representation $\mathbf{H}_0(z) = \frac{2}{\pi}\int_0^{\pi/2} \sin(z\cos\theta)\, d\theta$ **[Tier L]**, and $\mathbf{H}_0(z) - Y_0(z) \sim \frac{2}{\pi z}$ as $z \to +\infty$ **[Tier L]**. Mathlib, at the pinned version, knows neither $\mathbf{H}_0$ nor Y_0: this is why Chapter 2 formalises the Keldysh potential in a Laplace form, and connects this form to the combination $\mathbf{H}_0 - Y_0$ by a **[Tier C]** argument that relies on the representation $\mathbf{K}_0(z) = \mathbf{H}_0(z) - Y_0(z) = \frac{2}{\pi}\int_0^\infty e^{-zu}(1+u^2)^{-1/2} du$ for $\operatorname{Re} z > 0$ (DLMF 11.5.2, cited in Chapter 2 [27]). The behaviour $\sim 2/(\pi z)$ is the Coulomb law $1/r$ recovered at large distances.

A.5 Lattices in the plane and the fundamental domain of $SL_2(\mathbb{Z})$

A.5.1 Bases of a lattice

A lattice in the plane is $\Lambda = \mathbb{Z}u \oplus \mathbb{Z}v$ with (u, v) a basis of $\mathbb{R}^2$ (Definition 1.1). Two bases (u, v) and (u', v') generate the same lattice if and only if $\binom{u'}{v'} = P\binom{u}{v}$ with $P \in GL_2(\mathbb{Z})$, that is, P has integer entries and $\det P = \pm 1$.

Human proof. If $\mathbb{Z}u' \oplus \mathbb{Z}v' = \mathbb{Z}u \oplus \mathbb{Z}v$, each basis is expressed in the other with integer coefficients: P and P' are integer matrices with $PP' = I$, so $\det P \det P' = 1$ in $\mathbb{Z}$, whence $\det P = \pm 1$. Conversely, if $P \in GL_2(\mathbb{Z})$, then $P^{-1} = \pm \operatorname{com}(P)^{\mathsf{T}}$ (the transposed cofactor matrix) has integer entries, and each of the two lattices is contained in the other. □

The covolume $|\det(u, v)|$ therefore does not depend on the basis.

A.5.2 The shape parameter τ

Identify $\mathbb{R}^2$ with $\mathbb{C}$. After an orientation-preserving similarity (a rotation and a dilation, which do not change the "shape"), we may assume $u = 1$. After replacing v by $-v$ if necessary, we may assume $\operatorname{Im} v > 0$. The lattice is then $\mathbb{Z} + \mathbb{Z}\tau$ with τ in the Poincaré upper half-plane $\mathbb{H} = \{\operatorname{Im} \tau > 0\}$. An orientation-preserving change of basis $\binom{u'}{v'} = \begin{pmatrix} d & c \\ b & a \end{pmatrix}\binom{u}{v}$ with $ad - bc = 1$ replaces τ by

$$\gamma \cdot \tau = \frac{a\tau + b}{c\tau + d}, \qquad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z}),$$

after renormalising u' to 1. Lattice shapes up to similarity therefore correspond to the orbits of $SL_2(\mathbb{Z})$ in $\mathbb{H}$. The key computation is

$$\operatorname{Im}(\gamma \cdot \tau) = \frac{\operatorname{Im} \tau}{|c\tau + d|^2}, \tag{A.2}$$

obtained by multiplying numerator and denominator by $\overline{c\tau + d}$ and using $ad - bc = 1$.

A.5.3 The fundamental domain

Theorem A.25. *Let* $\mathcal{F} = \{\tau \in \mathbb{H} : |\operatorname{Re} \tau| \le \frac{1}{2},\ |\tau| \ge 1\}$.

1. *Every* $\tau \in \mathbb{H}$ *has a point of its* $SL_2(\mathbb{Z})$*-orbit in* $\mathcal{F}$.
2. *If two distinct points of* $\mathcal{F}$ *lie in the same orbit, they lie on the boundary of* $\mathcal{F}$ *(symmetric with respect to the imaginary axis, on the lines* $\operatorname{Re} \tau = \pm\frac{1}{2}$ *or on the circle* $|\tau| = 1$*).*

Human proof of item 1. Write $T : \tau \mapsto \tau + 1$ and $S : \tau \mapsto -1/\tau$; they correspond to $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. For fixed τ, the numbers $c\tau + d$, $(c, d) \in \mathbb{Z}^2 \setminus \{0\}$, form a lattice in $\mathbb{C}$; only finitely many of them have modulus ≤ 1. By (A.2), $\operatorname{Im}(\gamma \cdot \tau)$ therefore takes only finitely many values $\ge \operatorname{Im} \tau$, and there exists γ_0 such that $\tau' = \gamma_0 \cdot \tau$ has maximal imaginary part in the orbit. A power of T brings $\operatorname{Re} \tau'$ into $[-\frac{1}{2}, \frac{1}{2}]$ without changing its imaginary part. If we had $|\tau'| < 1$, then $\operatorname{Im}(S\tau') = \operatorname{Im} \tau'/|\tau'|^2 > \operatorname{Im} \tau'$, which would contradict maximality. Hence $\tau' \in \mathcal{F}$. □

Item 2, whose proof is a case analysis based on (A.2), is **[Tier L]** [67]. The domain $\mathcal{F}$ contains two distinguished points:

- $\tau = i$, the square lattice. Its stabiliser in $SL_2(\mathbb{Z})$ is generated by S, of order 4 in $SL_2(\mathbb{Z})$ (and 2 in the quotient by $\pm I$): near i, S acts as a rotation by angle π.

- $\tau = \rho = e^{i\pi/3} = \frac{1}{2} + i\frac{\sqrt{3}}{2}$, the triangular lattice. It is fixed by $\gamma(\tau) = (\tau - 1)/\tau$, of order 3 modulo $\pm I$, which acts near ρ as a rotation by angle $\pm 120°$.

This is the argument of §3.1.4: a function of the lattice shape that is $SL_2(\mathbb{Z})$-invariant and of class $\mathcal{C}^2$ has zero gradient at ρ and a scalar Hessian, because the only symmetric matrices that commute with a rotation by 120° are multiples of the identity. At i, the rotation by angle π forces only the gradient to vanish: the Hessian may be anisotropic there.

A.5.4 Gauss reduction of a basis

Item 1 of the theorem is also an algorithm. Starting from a basis (u, v):

1. if $|v| < |u|$, swap u and v;
2. replace v by $v - ku$, where k is the integer nearest to $\mathrm{Re}(v\bar{u})/|u|^2$;
3. repeat as long as $|v| < |u|$.

The algorithm terminates, because $|u|$ decreases strictly at each swap and takes only finitely many values (they are norms of lattice vectors bounded by the initial norm). At the end, $|u| \leq |v| \leq |v \pm u|$: u is a shortest nonzero vector and $\tau = v/u$ (up to sign) lies in $\mathcal{F}$. With a reduced basis, truncating a lattice sum to the disk of radius R is convenient: all points $mu + nv$ of the disk satisfy $|m|, |n| \leq CR$ for a constant C depending only on the shape (exercise). We have not checked which codes in the book reduce their basis; this is a methodological remark, not a description of those codes.

A.6 Exercises

Exercise A.1. Prove by induction on m the formula of Example A.6. Deduce $\int_{\mathbb{R}} x^{2m} e^{-x^2} dx = \Gamma(m + \frac{1}{2})$.

Exercise A.2. Show, by an integration by parts, that $\mathrm{erfc}(x) = \frac{e^{-x^2}}{x\sqrt{\pi}}\big(1 - \frac{1}{2x^2} + O(x^{-4})\big)$ as $x \to \infty$. How many terms of $\sum_{n\geq 1} e^{-\pi n^2}$ are needed for an error below 10^{-30}?

Exercise A.3. Apply Theorem A.18 to $f(x) = e^{-2\pi a|x|}$, $a > 0$, whose Fourier transform is $\hat{f}(\xi) = \frac{a}{\pi(a^2+\xi^2)}$ (check this). Deduce $\sum_{k\in\mathbb{Z}} \frac{1}{a^2+k^2} = \frac{\pi}{a}\coth(\pi a)$, then $\sum_{k\geq 1} k^{-2} = \pi^2/6$ by letting a tend to 0.

Exercise A.4. Check Proposition A.13 numerically for $F(r) = e^{-\pi r^2}$ by evaluating J_0 through its power series (with `mpmath`, for example). Why is the power series of J_0 a poor way to compute $J_0(50)$ in floating point? (Think of the cancellation of terms of order e^{50}.)

Exercise A.5. Reduce the basis $u = (1, 0)$, $v = (7.5, 0.2)$ by Gauss's algorithm, and give the resulting parameter $\tau \in \mathcal{F}$ (in case of a tie, take the lower integer; the result is $\tau = -\frac{1}{2} + 1.25\,i$, equivalent under T to $\frac{1}{2} + 1.25\,i$, on the boundary of $\mathcal{F}$). Compute the stabiliser of ρ in $SL_2(\mathbb{Z})$ and check that it has six elements.

Appendix B

Finite Markov chains

Chapter 4 studies three chains: Wright–Fisher, Moran, and the chain of configurations on a graph. All three are Markov chains on a finite set, with absorbing states. This appendix collects, with their proofs, the general results about such chains: fundamental matrix, absorption probabilities and mean absorption times, harmonic functions, maximum principle, martingales and the optional stopping theorem. Chapter 4 proves special cases of them; cross-references are given along the way. The classical references are Kemeny and Snell [58], Norris [68] and Williams [69].

Warning : tiers

The proofs in this appendix are human proofs **[Tier C]**. The special cases formalised in Chapter 4 (absorption for Wright–Fisher and Moran, uniqueness of the harmonic function, isothermal theorem) are **[Tier A]** in the files `formal_cert/Programmes/PopGen/` and `EvoGraph/`; the general statements below are not.

B.1 Stochastic matrices and Markov chains

Definition B.1. Let E be a finite set. A matrix $P = (P(i,j))_{i,j\in E}$ is *stochastic* if $P(i,j) \geq 0$ for all i, j and $\sum_j P(i,j) = 1$ for every i. A sequence of random variables $(X_n)_{n\geq 0}$ with values in E is a *Markov chain with transition matrix* P if, for every n and all $i_0, \dots, i_{n+1}$ such that the conditioning event has nonzero probability,

$$\mathbb{P}(X_{n+1} = i_{n+1} \mid X_0 = i_0, \dots, X_n = i_n) = P(i_n, i_{n+1}).$$

We write $\mathbb{P}_i$ and $\mathbb{E}_i$ for probability and expectation when $X_0 = i$. A stochastic matrix has the eigenvalue 1, with eigenvector $\mathbf{1} = (1, \dots, 1)^{\mathsf{T}}$, since $P\mathbf{1} = \mathbf{1}$.

Proposition B.2 (Chapman–Kolmogorov)**.** *For all $n \geq 0$ and $i, j \in E$, $\mathbb{P}_i(X_n = j) = P^n(i,j)$. More generally, if the law of X_0 is the row vector π_0, the law of X_n is $\pi_0 P^n$.*

Human proof. By induction on n. For $n = 0$, $P^0 = I$. If it holds at step n, the law of total probability and the Markov property give $\mathbb{P}_i(X_{n+1} = j) = \sum_k \mathbb{P}_i(X_n = k)\, P(k,j) = \sum_k P^n(i,k)P(k,j) = P^{n+1}(i,j)$. □

Example B.3 (The chains of Chapter 4)**.**

- *Neutral Wright–Fisher*, $E = \{0, \dots, N\}$: the law of X_{n+1} given $X_n = i$ is binomial, $P(i,j) = \binom{N}{j}(i/N)^j(1 - i/N)^{N-j}$.
- *Moran with selection*, $E = \{0, \dots, N\}$: $P(i, i+1) = p_i$, $P(i, i-1) = q_i$, $P(i,i) = 1 - p_i - q_i$, with $p_i = rq_i$ (Chapter 5, §5.6). This is a *birth–death chain*.
- *Configurations on a graph*, $E = \mathcal{P}(V)$: the state is the set of mutant vertices.

In all three cases, the states "all residents" and "all mutants" are absorbing.

B.2 Absorbing states and the fundamental matrix

Definition B.4. A state a is *absorbing* if $P(a,a) = 1$. We write A for the set of absorbing states and $T = E \setminus A$ for the set of the other states, called *transient* in this context. The chain is *absorbing* if $A \neq \emptyset$ and, from every state of T, some state of A is accessible: for every $i \in T$ there exist n and $a \in A$ with $P^n(i,a) > 0$.

Ordering the states of T before those of A, the matrix takes the *canonical form*

$$P = \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix}, \qquad P^n = \begin{pmatrix} Q^n & (I + Q + \cdots + Q^{n-1})R \\ 0 & I \end{pmatrix},$$

where Q is the restriction of P to $T \times T$ and R its restriction to $T \times A$. The second formula is proved by induction by carrying out the block product.

Lemma B.5 (Geometric decay)**.** *If the chain is absorbing, there exist $m \geq 1$ and $\alpha < 1$ such that $\sum_{j \in T} Q^m(i,j) \leq \alpha$ for every $i \in T$. Consequently $\sum_{j \in T} Q^n(i,j) \leq \alpha^{\lfloor n/m \rfloor}$ and $Q^n \to 0$.*

Human proof. For each $i \in T$, there exists n_i with $\mathbb{P}_i(X_{n_i} \in A) > 0$. Since A is absorbing, $\mathbb{P}_i(X_n \in A)$ is nondecreasing in n. With $m = \max_i n_i$, we therefore have $\mathbb{P}_i(X_m \in A) > 0$ for every $i \in T$, and $\sum_{j \in T} Q^m(i,j) = \mathbb{P}_i(X_m \in T) = 1 - \mathbb{P}_i(X_m \in A)$. Set $\alpha = \max_{i \in T}\big(1 - \mathbb{P}_i(X_m \in A)\big) < 1$ (T is finite). Write $s_n = \max_i \sum_j Q^n(i,j)$; since the entries of Q are nonnegative, $s_{n+m} \leq s_n \, s_m \leq \alpha \, s_n$, and $s_n \leq 1$. □

This is the general version of Lemma 4.10 of Chapter 4.

Theorem B.6 (Fundamental matrix)**.** *If the chain is absorbing, the matrix $I - Q$ is invertible, and*

$$N = (I - Q)^{-1} = \sum_{n \geq 0} Q^n.$$

The entry $N(i,j)$ is the expected number of visits to j starting from i: $N(i,j) = \mathbb{E}_i\big[\sum_{n \geq 0} \mathbf{1}_{X_n = j}\big]$.

Human proof. By the lemma, the series $\sum_n Q^n$ converges (its nonnegative entries are bounded by a geometric series). Its sum N satisfies $(I - Q)N = N(I - Q) = \lim_n (I - Q^{n+1}) = I$. Finally, by Chapman–Kolmogorov and the monotone convergence theorem, $\mathbb{E}_i\big[\sum_n \mathbf{1}_{X_n = j}\big] = \sum_n P^n(i,j) = \sum_n Q^n(i,j)$ for $i, j \in T$. □

Theorem B.7 (Absorption: probabilities and mean time)**.** *Consider an absorbing chain and let $\tau = \inf\{n \geq 0 : X_n \in A\}$ be the absorption time.*

1. *$\tau < \infty$ almost surely, and even $\mathbb{E}_i[\tau] < \infty$.*
2. *The vector of mean times $t(i) = \mathbb{E}_i[\tau]$, $i \in T$, equals $t = N\mathbf{1}$.*
3. *The matrix $B(i,a) = \mathbb{P}_i(X_\tau = a)$, $i \in T$, $a \in A$, equals $B = NR$.*
4. *$\lim_{n \to \infty} P^n(i,a) = B(i,a)$ for $i \in T$, $a \in A$.*

Human proof. (1) $\mathbb{P}_i(\tau > n) = \mathbb{P}_i(X_n \in T) = \sum_{j \in T} Q^n(i,j) \leq \alpha^{\lfloor n/m \rfloor}$ by Lemma B.5. Hence $\mathbb{P}_i(\tau = \infty) = 0$ and $\mathbb{E}_i[\tau] = \sum_{n \geq 0} \mathbb{P}_i(\tau > n) < \infty$.

(2) Before absorption, each step is a visit to a state of T: $\tau = \sum_{n \geq 0} \sum_{j \in T} \mathbf{1}_{X_n = j}$. Taking expectations, $t(i) = \sum_{j \in T} N(i,j)$.

(3) Absorption at a occurs at step $n+1$ if $X_n = j \in T$ and $X_{n+1} = a$: $\mathbb{P}_i(X_\tau = a) = \sum_{n \geq 0} \sum_{j \in T} Q^n(i,j) R(j,a) = (NR)(i,a)$.

(4) The canonical form of P^n gives $P^n(i,a) = \big((I + \cdots + Q^{n-1})R\big)(i,a) \to (NR)(i,a)$. □

University remark. **[Tier L]** Without assuming the chain absorbing, and on a countable state space, the vector of hitting probabilities $h(i) = \mathbb{P}_i(\text{hit } A)$ is the *minimal non-negative solution* of the system $h = 1$ on A, $h(i) = \sum_j P(i,j)h(j)$ for $i \notin A$; likewise the mean hitting times $k(i) = \mathbb{E}_i[\tau]$ form the minimal non-negative solution of $k = 0$ on A, $k(i) = 1 + \sum_{j \notin A} P(i,j)k(j)$ for $i \notin A$ [68]. For a finite absorbing chain and $a \in A$, the minimal solution for the target $\{a\}$ vanishes on $A \setminus \{a\}$; on T it solves $(I - Q)h_T = R(\cdot, a)$, so $h_T = (NR)(\cdot, a) = B(\cdot, a)$. The mean-time system on T is $(I - Q)k_T = \mathbf{1}$, whose unique solution is $t = N\mathbf{1}$ [68] (standard).

Example B.8 (Moran, $N = 3$, $r = 2$: the full computation)**.** The transition probabilities are $p_1 = \frac{1}{3}$, $q_1 = \frac{1}{6}$, $p_2 = \frac{4}{15}$, $q_2 = \frac{2}{15}$ (Chapter 5, §5.6). With $T = \{1, 2\}$ and $A = \{0, 3\}$,

$$Q = \begin{pmatrix} \frac{1}{2} & \frac{1}{3} \\ \frac{2}{15} & \frac{3}{5} \end{pmatrix}, \qquad I - Q = \begin{pmatrix} \frac{1}{2} & -\frac{1}{3} \\ -\frac{2}{15} & \frac{2}{5} \end{pmatrix}, \qquad \det(I - Q) = \frac{1}{5} - \frac{2}{45} = \frac{7}{45},$$

$$N = \frac{45}{7}\begin{pmatrix} \frac{2}{5} & \frac{1}{3} \\ \frac{2}{15} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{18}{7} & \frac{15}{7} \\ \frac{6}{7} & \frac{45}{14} \end{pmatrix}, \qquad t = N\mathbf{1} = \begin{pmatrix} \frac{33}{7} \\ \frac{57}{14} \end{pmatrix}, \qquad B(\cdot, 3) = N\begin{pmatrix} 0 \\ \frac{4}{15} \end{pmatrix} = \begin{pmatrix} \frac{4}{7} \\ \frac{6}{7} \end{pmatrix}.$$

We recover the fixation probabilities $\frac{4}{7}$ and $\frac{6}{7}$ of the formula $(1 - r^{-i})/(1 - r^{-N})$, and we obtain in addition the mean absorption times: $33/7 \approx 4.71$ steps from one mutant, $57/14 \approx 4.07$ from two. These fractions were recomputed in exact arithmetic (Python's `fractions.Fraction`) for this appendix **[Tier X]**: $N = \left(\begin{smallmatrix} 18/7 & 15/7 \\ 6/7 & 45/14 \end{smallmatrix}\right)$, $t = (33/7,\ 57/14)$, $B(\cdot, 3) = (4/7,\ 6/7)$ and $B(\cdot, 0) = (3/7,\ 1/7)$.

B.3 Harmonic functions and the maximum principle

Definition B.9. A function $h : E \to \mathbb{R}$ is *harmonic* at i if $h(i) = \sum_j P(i,j)h(j)$, that is, if $(Ph)(i) = h(i)$. It is *harmonic on* T if it is harmonic at every point of T.

In other words, $h(i)$ is the average of the values of h one step later. For a birth–death chain, harmonicity at i reads $p_i\big(h(i+1) - h(i)\big) = q_i\big(h(i) - h(i-1)\big)$: this is the first-step equation of Chapter 5.

Theorem B.10 (Maximum principle)**.** *Consider an absorbing chain and h harmonic on T. Then $\max_E h = \max_A h$ and $\min_E h = \min_A h$.*

Human proof. Let $M = \max_E h$, attained at a state i_0. If $i_0 \in A$, we are done. Otherwise, $i_0 \in T$ and $M = h(i_0) = \sum_j P(i_0, j)h(j)$ is an average of values $\le M$, with nonnegative weights summing to 1: all the values $h(j)$ with $P(i_0, j) > 0$ equal M. Step by step, $h = M$ at every state accessible from i_0, in particular at some state of A (the chain is absorbing). For the minimum, apply the result to $-h$. □

Corollary B.11 (Dirichlet problem)**.** *For every function $f : A \to \mathbb{R}$, there exists a unique function h harmonic on T and equal to f on A: it is $h(i) = \mathbb{E}_i[f(X_\tau)] = \sum_{a \in A} B(i,a)f(a)$ for $i \in T$.*

Human proof. Uniqueness. The difference d of two solutions is harmonic on T and zero on A; by Theorem B.10, $0 \le d \le 0$. *Existence.* On T, the equation reads $h_T = Qh_T + Rf$, i.e. $(I - Q)h_T = Rf$, which has the solution $h_T = NRf = Bf$ (Theorem B.6). □

Taking $A = \{0, N\}$, $f(0) = 0$ and $f(N) = 1$, we obtain: *the fixation probability is the unique function harmonic in the interior that equals 0 at 0 and 1 at N.* This is exactly the structure of the proofs of Chapter 4: Theorem 4.8 for Wright–Fisher, where $h(j) = j/N$; Theorem 4.14 for Moran, where $h(i) = S_i/S_N$; Proposition 4.22 for graphs. In Lean, the step "the fixation probability is harmonic" is replaced by the limit $P^n(i, N) \to h(i)$ (item 4 of Theorem B.7), which does not require defining τ.

Remark B.12 (Connectivity matters). Without the "absorbing" hypothesis, the maximum principle is false. On $E = \{0, 1, 2, 3\}$, with 0 and 3 absorbing and $P(1, 2) = P(2, 1) = 1$, no absorbing state is accessible from $T = \{1, 2\}$: every function with $h(1) = h(2)$ is harmonic on T, the common value is arbitrary, and the Dirichlet problem has infinitely many solutions. This is the reason for the hypotheses of positive variance (Chapter 4, §4.2) and of strong connectivity (Lemma 4.23) in the Lean statements.

B.4 Discrete-time martingales

We stay within the elementary setting of a finite Markov chain, where all the random variables are bounded.

Definition B.13. A sequence (M_n) of bounded real random variables, with M_n a function of $(X_0, \dots, X_n)$, is a *martingale* (with respect to (X_n)) if $\mathbb{E}[M_{n+1} \mid X_0, \dots, X_n] = M_n$ for every n. Equivalently: $\mathbb{E}[(M_{n+1} - M_n)\, Z] = 0$ for every bounded random variable Z that is a function of $(X_0, \dots, X_n)$.

Proposition B.14. *If h is harmonic on T (it is then automatically harmonic on A, since $P(a, a) = 1$), then $M_n = h(X_n)$ is a martingale.*

Human proof. By the Markov property,

$$\mathbb{E}[h(X_{n+1}) \mid X_0, \dots, X_n] = \sum_j P(X_n, j) h(j) = (Ph)(X_n) = h(X_n).$$

At an absorbing state a, $(Ph)(a) = h(a)$ automatically, since $P(a, a) = 1$. □

Example B.15.

- *Wright–Fisher.* The mean of the binomial law $\mathcal{B}(N, i/N)$ is i, so $h(j) = j$ is harmonic: the number of mutants X_n is a martingale (Chapter 4, §4.2).
- *Moran with selection.* $h(i) = r^{-i}$ is harmonic at $0 < i < N$. Indeed, harmonicity reads $p_i\big(h(i+1) - h(i)\big) + q_i\big(h(i-1) - h(i)\big) = 0$, and with $p_i = rq_i$: $rq_i(r^{-i-1} - r^{-i}) + q_i(r^{-i+1} - r^{-i}) = q_i r^{-i}\big((1 - r) + (r - 1)\big) = 0$. Hence r^{-X_n} is a martingale. This is the martingale that Keller and Uğurlu use for graphs [49].

Theorem B.16 (Optional stopping theorem)**.** *Let (M_n) be a martingale and τ a* stopping time*: for every n, the event $\{\tau = n\}$ depends only on $(X_0, \dots, X_n)$.*

1. *If $\tau \le K$ almost surely for some integer K, then $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$.*
2. *If $\tau < \infty$ almost surely and $|M_n| \le C$ for every n, then $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$.*

Human proof. (1) Write $M_\tau = M_0 + \sum_{k=0}^{K-1} (M_{k+1} - M_k) \mathbf{1}_{\tau > k}$. The event $\{\tau > k\}$, the complement of $\{\tau \le k\}$, depends only on $(X_0, \dots, X_k)$; by definition of a martingale, each term of the sum has zero expectation.

(2) For fixed K, $\tau \wedge K = \min(\tau, K)$ is a bounded stopping time, so $\mathbb{E}[M_{\tau \wedge K}] = \mathbb{E}[M_0]$ by (1). As $K \to \infty$, $M_{\tau \wedge K} \to M_\tau$ almost surely (since $\tau < \infty$) and $|M_{\tau \wedge K}| \le C$: dominated convergence (Theorem A.1, in its probabilistic form) concludes. □

University remark. **[Tier L]** In the general setting of a filtration $(\mathcal{F}_n)$, Doob's optional stopping theorem gives $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$ for a martingale M and a stopping time τ under any one of three conditions: τ is bounded; or M is bounded and $\tau < \infty$ a.s.; or $\mathbb{E}[\tau] < \infty$ and the increments $|M_{n+1} - M_n|$ are bounded by a constant. In the same setting, Doob's martingale convergence theorem states that a supermartingale bounded in L^1 converges almost surely to a finite limit [69].

Corollary B.17 (Gambler's ruin and Moran's formula)**.** *For the Moran chain with $r \neq 1$, starting from i, $\mathbb{P}_i(X_\tau = N) = \dfrac{1 - r^{-i}}{1 - r^{-N}}$.*

Human proof. The chain is absorbing ($p_i > 0$ for $0 < i < N$), so $\tau < \infty$ almost surely (Theorem B.7). The martingale r^{-X_n} is bounded by $\max(1, r^{-N})$. The optional stopping theorem gives $r^{-i} = \mathbb{E}_i[r^{-X_\tau}] = \varphi_i\, r^{-N} + (1 - \varphi_i) \cdot 1$, with $\varphi_i = \mathbb{P}_i(X_\tau = N)$. Solving: $\varphi_i = (1 - r^{-i})/(1 - r^{-N})$. □

With the Wright–Fisher martingale X_n, the same computation gives $\varphi_i = i/N$. These are two probabilistic proofs of the formulas that Chapter 4 proves by linear algebra (induction on the differences δ_i, then the limit of P^n).

B.5 Irreducible chains and the stationary distribution

Absorbing chains are not the only ones in the book: the replacement matrix W of a graph (Chapter 4, §4.5) is itself a stochastic matrix, in general *irreducible*, and the Keller–Uğurlu theorem [49] involves its stationary distribution.

Definition B.18. A stochastic matrix P is *irreducible* if, for all i, j, there exists $n \geq 0$ with $P^n(i, j) > 0$. A *stationary distribution* is a row vector $\pi \geq 0$, summing to 1, such that $\pi P = \pi$.

Proposition B.19 (Existence)**.** *Every stochastic matrix on a finite set has at least one stationary distribution.*

Human proof. Start from an arbitrary distribution π_0 and set $\mu_n = \frac{1}{n}\sum_{k=0}^{n-1} \pi_0 P^k$. These are probability distributions; they live in the simplex, which is compact. A subsequence (μ_{n_l}) converges to a distribution μ. Now $\mu_n P - \mu_n = \frac{1}{n}(\pi_0 P^n - \pi_0)$, each entry of which is bounded by $1/n$ in absolute value. In the limit, $\mu P = \mu$. □

Proposition B.20 (Uniqueness and positivity)**.** *If P is irreducible, the stationary distribution π is unique and $\pi(j) > 0$ for every j.*

Human proof. Harmonic functions are constant. If $Ph = h$ on all of E, the argument of Theorem B.10 shows that h equals its maximum at every state accessible from a point where the maximum is attained, hence everywhere, by irreducibility. The kernel of $P - I$ is therefore the line spanned by $\mathbf{1}$. *Uniqueness.* The left kernel, $\{x : x(P - I) = 0\}$, has the same dimension as the right kernel (rank of a matrix and of its transpose), so it has dimension 1; exactly one vector of this line has entries summing to 1 (the line contains the stationary law of Proposition B.19, whose sum is 1, so the sum is not identically zero on it). *Positivity.* If $\pi(i) > 0$ (such an i exists) and $P^n(i, j) > 0$, then $\pi(j) = (\pi P^n)(j) \geq \pi(i) P^n(i, j) > 0$. □

Theorem B.21 (Geometric convergence)**.** *Assume there exist $m \geq 1$ and $\delta > 0$ such that $P^m(i, j) \geq \delta$ for all i, j (a* primitive *matrix). Let $n = |E|$. For all distributions μ, ν, $\|\mu P^m - \nu P^m\|_1 \leq (1 - n\delta)\|\mu - \nu\|_1$. In particular $\pi_0 P^k \to \pi$ for every initial distribution π_0, at a geometric rate.*

Human proof. One has $n\delta \leq 1$, since a row of P^m has n entries $\geq \delta$ summing to 1. If $n\delta = 1$, all rows of P^m are uniform and $\mu P^m = \nu P^m$. Otherwise, write $P^m = n\delta\, U + (1 - n\delta) R$, where all rows of U equal $(\frac{1}{n}, \ldots, \frac{1}{n})$; the matrix $R = (P^m - n\delta U)/(1 - n\delta)$ then has nonnegative entries and rows summing to 1. The vector $x = \mu - \nu$ sums to zero, so $xU = 0$, and $xP^m = (1 - n\delta)\, xR$. Finally $\|xR\|_1 \leq \sum_i |x_i| \sum_j R(i, j) = \|x\|_1$. Apply the inequality to $\mu = \pi_0 P^k$ and $\nu = \pi$ (which satisfies $\pi P^m = \pi$); the intermediate powers do not increase the norm, by the same computation with P in place of R. □

Example B.22 (The star graph on three vertices)**.** Take the centre 0 and the leaves 1, 2, with $W(0,1) = W(0,2) = \frac{1}{2}$ and $W(1,0) = W(2,0) = 1$ (Exercise 4.11). The equation $\pi W = \pi$ gives $\pi(0) = \pi(1) + \pi(2)$ and $\pi(1) = \pi(2) = \pi(0)/2$, whence $\pi = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$. This matrix is irreducible but *not* primitive: it is periodic with period 2 (one alternates between the centre and the leaves), and $\pi_0 W^k$ does not converge if one starts from the centre. Theorem B.21 therefore does not apply; the stationary distribution nevertheless exists and is unique. The star is not isothermal (W is not doubly stochastic: the centre's column sums to 2), which explains why its fixation probability differs from Moran's (Chapter 4). With selection weights stationary in the sense of the Lean statement `stationary_fixation` (`EvoGraph/BirthDeath.lean`), the N4 record [47] recovers the Moran value 4/7 from each of the three vertices (exact-fraction computation, **[Tier X]**).

Remark B.23 (What about absorbing chains?)*.* An absorbing chain is not irreducible as soon as it has a transient state. Its stationary distributions are the mixtures of Dirac masses on the absorbing states, and the limiting distribution depends on the starting point: it is the matrix B of Theorem B.7.

B.6 From discrete to continuous time: uniformisation

The crate `qf-popgen` (Chapter 5, case U5) does not iterate P^n: it integrates the linear differential equation $\dot\pi(t) = \pi(t)(P - I)$ with CVODE. Here is why the two computations have the same limit.

Proposition B.24 (Poisson mixture)**.** *The solution of $\dot\pi = \pi(P - I)$, $\pi(0) = \pi_0$, is*

$$\pi(t) = \pi_0 e^{t(P-I)} = \sum_{n \geq 0} e^{-t} \frac{t^n}{n!} \pi_0 P^n.$$

If P is absorbing, $\pi(t)(a) \to \sum_i \pi_0(i) B(i,a)$ as $t \to \infty$, for every $a \in A$.

Human proof. The matrices tP and $-tI$ commute, hence

$$e^{t(P-I)} = e^{-t} e^{tP} = e^{-t} \sum_{n \geq 0} \frac{t^n P^n}{n!}.$$

For the limit, write $u_n = \pi_0 P^n(a)$, which converges to $\ell = \sum_i \pi_0(i) B(i,a)$ (Theorem B.7, item 4, and $P^n(a,a) = 1$). The weights $e^{-t}t^n/n!$ form a Poisson distribution with parameter t, whose mass on $\{n \leq n_0\}$ tends to 0 as $t \to \infty$, for every fixed n_0. Split the sum into $n \leq n_0$ and $n > n_0$, with n_0 such that $|u_n - \ell| \leq \varepsilon$ for $n > n_0$; since $|u_n - \ell| \leq 2$ for every n,

$$|\pi(t)(a) - \ell| \leq 2\,\mathbb{P}\big(\mathrm{Poisson}(t) \leq n_0\big) + \varepsilon.$$

□

This proposition is the step "the differential equation and the n-step kernel have the same limit" that Chapter 5 flags as not formalised: it is **[Tier C]** in this book.

B.7 Exercises

Exercise B.1. For the neutral Moran chain ($r = 1$) with N individuals, show that $\delta_i = \varphi_{i+1} - \varphi_i$ is constant and recover $\varphi_i = i/N$. Compute the fundamental matrix for $N = 3$ and the mean absorption time from $i = 1$.

Exercise B.2. Show that, for a birth–death chain absorbing at 0 and N, the vector t of mean times satisfies $t(i) = 1 + p_i t(i+1) + q_i t(i-1) + (1 - p_i - q_i)t(i)$ for $0 < i < N$, with $t(0) = t(N) = 0$. Check on Example B.8 that $(33/7, 57/14)$ is a solution.

Exercise B.3. Let (X_n) be the neutral Wright–Fisher chain. Show that $X_n(N - X_n)\,(1 - 1/N)^{-n}$ is a martingale. Deduce that $\mathbb{E}_i[X_n(N - X_n)] = i(N - i)(1 - 1/N)^n$, and then an upper bound on $\mathbb{P}_i(\tau > n)$.

Exercise B.4 (The negative control for the optional stopping theorem)**.** On $\mathbb{Z}$, the simple symmetric random walk S_n started at 0 is a martingale, and $\tau = \inf\{n : S_n = 1\}$ is finite almost surely (take this as given). Show that $\mathbb{E}[S_\tau] = 1 \neq \mathbb{E}[S_0]$. Which hypothesis of Theorem B.16 is violated?

Appendix C

Lean 4 for beginners

Section 1.4 presented Lean from the reader's side: how to read a statement and what the kernel guarantees. This appendix is for those who want to *write* their first proofs. It follows six small files, stored in `code/lean/appC_1_logique.lean` through `appC_6_contrefacon.lean`. All of them were actually compiled, on 2026-10-26, with the book's pinned toolchain (Lean 4.34.1, Mathlib at commit `d13f23b723b8a846827a245b89c10fc7d3f11612`). Every output reproduced here is copied from the compilation, without editing, except that absolute file paths have been shortened.

C.1 Installing Lean, Lake and Mathlib

The three tools.

- **elan** is the version manager: it installs Lean toolchains and selects the right one for each project, according to the project's `lean-toolchain` file.
- **lake** is the build tool, shipped with Lean. It reads `lakefile.lean` (or `lakefile.toml`), downloads the dependencies and compiles the modules in the order of their `import`s.
- **Mathlib** is the community mathematics library [9]. One depends on it as a Lake package, pinned to a precise commit.

Typical installation (Linux or macOS). The following commands are the procedure documented by the elan and Mathlib projects [72, 73]; we did *not* replay them on a fresh machine for this book (the author's machine already had its toolchains), and a more recent version of these tools may change the details.

```
curl https://raw.githubusercontent.com/leanprover/elan/master/elan-init.sh -sSf | sh
elan toolchain install leanprover/lean4:v4.34.1
lake new mon_projet math          # gabarit « math » : projet qui dépend de Mathlib
cd mon_projet
lake exe cache get                # télécharge les .olean précompilés de Mathlib
lake build
```

(The comments in this listing are kept as in the French edition: `math` is the template of a project that depends on Mathlib, and `lake exe cache get` downloads Mathlib's precompiled `.olean` files.) To reproduce the book's version exactly, pin Mathlib in `lakefile.lean`:

```
require mathlib from git "https://github.com/leanprover-community/mathlib4.git" @ "
    d13f23b723b8a846827a245b89c10fc7d3f11612"
```

and put `leanprover/lean4:v4.34.1` in `lean-toolchain`. The precompiled cache exists only for published Mathlib commits; otherwise `lake build` recompiles Mathlib, which takes hours.

The book's toolchain. The Lean results of Chapters 1 to 5 were checked in the copy of the `lean/` directory of openai/math (with its Mathlib and its thirty or so dependencies), not in a fresh project (Appendix E, §E.1.3). The six examples of this appendix import only modules of Mathlib and of the Lean core; a `lake new ... math` project at the same Mathlib revision should therefore compile them too, but we have not checked this.

Compiling a file. Inside a project, `lake env lean File.lean` compiles a file against the right libraries; in an editor (VS Code with the Lean 4 extension), the proof state is displayed continuously. For this book we used the repository's script:

```
taskset -c 6,7 .venv/bin/python -I scripts/certified_numerics/lean_pinned.py compile FICHIER.
    lean --json-out SORTIE.json
```

It calls `lean -DautoImplicit=false` with the `LEAN_PATH` of the pinned stack and writes the return code, the duration and the whole output to `SORTIE.json` (`FICHIER` = file, `SORTIE` = output).

Warning : no global `import Mathlib`

Import specific modules (`Mathlib.Tactic.Ring`, `Mathlib.Tactic.Linarith`, ...), as our examples do. An `import Mathlib` loads the whole library: it is slow, and, in a partial build of Mathlib (such as the one in the `formal/` directory of the AutoevolveAI repository), it fails on the modules that have not been built.

C.2 First theorems: logic and quantifiers

The first file imports nothing: it uses only the Lean core.

code/lean/appC_1_logique.lean

```
/-
Annexe C, exemple 1 : logique propositionnelle et quantificateurs, Lean 4 sans bibliothèque.
Tactiques : intro, exact, apply, constructor, cases (via obtain), rfl.
-/

-- Un terme de preuve : on écrit directement l'objet (une paire).
theorem et_comm_terme (p q : Prop) (h : p ∧ q) : q ∧ p := ⟨h.2, h.1⟩

-- La même chose en mode tactique.
theorem et_comm_tactique (p q : Prop) : p ∧ q → q ∧ p := by
  intro h
  obtain ⟨hp, hq⟩ := h
  constructor
  · exact hq
  · exact hp

-- apply : on raisonne « en arrière » à partir de la conclusion.
theorem modus_ponens_chaine (p q r : Prop) (hpq : p → q) (hqr : q → r) : p → r := by
  intro hp
  apply hqr
  apply hpq
  exact hp

-- Quantificateurs : un « pour tout » est une fonction, un « il existe » une paire (témoin,
    preuve).
theorem existe_successeur : ∀ n : Nat, ∃ m : Nat, m = n + 1 := by
  intro n
  exact ⟨n + 1, rfl⟩
```

```
#print axioms et_comm_terme
#print axioms et_comm_tactique
#print axioms modus_ponens_chaine
#print axioms existe_successeur
```

Compilation output (return code 0, 2.0 s):

```
'et_comm_terme' does not depend on any axioms
'et_comm_tactique' does not depend on any axioms
'modus_ponens_chaine' does not depend on any axioms
'existe_successeur' does not depend on any axioms
```

Key points.

- After `:=`, one gives either a *term* (⟨`h.2, h.1`⟩) or `by` followed by *tactics*.
- `intro h` turns the goal $P \to Q$ into the goal Q, with the hypothesis `h : P` in the context.
- `exact e` closes the goal if `e` has exactly the type of the goal.
- `apply f` reasons backwards: if `f : A → B` and the goal is `B`, the new goal is `A`.
- `constructor` splits a goal $P \wedge Q$ into two goals; the bullets · handle them one at a time.
- `obtain ⟨hp, hq⟩ := h` breaks a hypothesis `h : P ∧ Q` into its two pieces.
- An existential is proved by a pair ⟨witness, proof⟩; here the witness is $n + 1$ and the proof is `rfl` ("equal by definition").

None of these four theorems uses an axiom: intuitionistic logic suffices.

C.3 The common tactics

Table C.1: The tactics used in the examples, and what they do.

tactic	effect
`intro`	introduces the hypothesis of an implication or the variable of a ∀
`exact`	closes the goal with a term of the right type
`apply`	reduces the goal to the hypothesis of a known implication
`constructor`	splits a ∧ (or a ↔) into two goals
`obtain`, `rcases`	destructures a hypothesis (∧, ∃, ∨)
`rfl`	closes a goal $a = a$ "by definition"
`simp`	rewrites with the lemmas tagged `@[simp]` (and those supplied to it) until a normal form is reached
`ring`	proves an equality in a commutative (semi)ring (polynomial identity)
`linarith`	proves a linear inequality that is a combination of the hypotheses (linear order, ordered field)
`nlinarith`	like `linarith`, also multiplying hypotheses together; one helps it by supplying squares
`norm_num`	computes exactly on numeric literals (integers, rationals)
`positivity`	proves $0 < e$ or $0 \le e$ from the syntactic form of e
`decide`	evaluates a closed decidable proposition (the kernel does the computation)
`omega`	linear arithmetic over $\mathbb{N}$ and $\mathbb{Z}$
`induction`	induction on a natural number (or any inductive structure)

C.3.1 Computing in the reals

code/lean/appC_2_calcul.lean

```
/-
Annexe C, exemple 2 : calcul dans les réels avec Mathlib.
Tactiques : ring, linarith, nlinarith, norm_num, positivity.
Imports épinglés sur des modules construits (jamais `import Mathlib` en entier).
-/
import Mathlib.Basic.Real.Basic
import Mathlib.Tactic.Ring
import Mathlib.Tactic.Linarith
import Mathlib.Tactic.NormNum
import Mathlib.Tactic.Positivity

-- ring : identité de polynômes, dans tout anneau commutatif.
theorem carre_somme (a b : ℝ) : (a + b) ^ 2 = a ^ 2 + 2 * a * b + b ^ 2 := by
  ring

-- linarith : combinaison linéaire d'hypothèses (inégalités linéaires).
theorem moyenne_entre (x y : ℝ) (h : x ≤ y) : x ≤ (x + y) / 2 := by
  linarith

-- nlinarith : on lui fournit le carré positif qui sert d'indice.
theorem am_gm_deux (a b : ℝ) : 2 * a * b ≤ a ^ 2 + b ^ 2 := by
  nlinarith [sq_nonneg (a - b)]

-- norm_num : arithmétique exacte sur des littéraux (ici, des rationnels).
theorem quatre_septiemes : (4 : ℝ) / 7 - 1 / 3 = 5 / 21 := by
  norm_num

-- positivity : prouve 0 < e ou 0 ≤ e à partir de la forme de l'expression.
theorem positif (x : ℝ) : 0 < x ^ 2 + 1 := by
  positivity

#print axioms carre_somme
#print axioms moyenne_entre
#print axioms am_gm_deux
#print axioms quatre_septiemes
#print axioms positif
```

Output (return code 0, 6.3 s):

```
'carre_somme' depends on axioms: [propext, Classical.choice, Quot.sound]
'moyenne_entre' depends on axioms: [propext, Classical.choice, Quot.sound]
'am_gm_deux' depends on axioms: [propext, Classical.choice, Quot.sound]
'quatre_septiemes' depends on axioms: [propext, Classical.choice, Quot.sound]
'positif' depends on axioms: [propext, Classical.choice, Quot.sound]
```

Mathlib's real numbers are constructed from Cauchy sequences, using the axiom of choice: in practice, every theorem about $\mathbb{R}$ depends on the three axioms of the whitelist. This is expected.

Remark C.1 (A library that moves). The first version of this file imported `Mathlib.Data.Real.Basic`. It compiled, with this warning:

```
warning: 'Mathlib.Data.Real.Basic' has been deprecated: please replace this import by
import Mathlib.Basic.Real.Basic
```

We followed the advice. Module and lemma names change from one Mathlib version to the next: this is why one pins a commit, and why a file written for another version may no longer compile.

C.3.2 Induction, `simp` and `decide`

code/lean/appC_3_recurrence.lean

```
/-
Annexe C, exemple 3 : récurrence, simp et decide.
La somme 0 + 1 + ... + (n-1), multipliée par 2, vaut n (n - 1), et un calcul fini décidé par le
     noyau.
-/
import Mathlib.Algebra.BigOperators.Group.Finset.Basic
import Mathlib.Tactic.Ring

open Finset

-- Récurrence sur n ; `simp` déplie la somme, `ring` conclut.
theorem double_somme (n : ℕ) : 2 * (∑ i ∈ range n, i) + n = n * n := by
  induction n with
  | zero => simp
  | succ k ih =>
    rw [sum_range_succ]
    ring_nf
    ring_nf at ih
    omega

-- decide : la proposition est décidable et close ; le noyau l'évalue.
theorem petit_calcul : (∑ i ∈ range 11, i) = 55 := by
  decide

-- simp avec une liste de lemmes : réécriture jusqu'à une forme normale.
theorem simp_exemple (l : List ℕ) : (l ++ []).length = l.length := by
  simp

#print axioms double_somme
#print axioms petit_calcul
#print axioms simp_exemple
```

Output (return code 0, 4.3 s):

```
'double_somme' depends on axioms: [propext, Classical.choice, Quot.sound]
'petit_calcul' depends on axioms: [propext, Classical.choice, Quot.sound]
'simp_exemple' depends on axioms: [propext]
```

In `double_somme`, the case `succ k` begins with `rw [sum_range_succ]`, which splits off the last term: $\sum_{i<k+1} i = \sum_{i<k} i + k$. After normalisation by `ring_nf`, the induction hypothesis `ih` and the goal are two equalities of integers that `omega` connects. There are often several routes; this one is not the shortest.

C.4 Reading a Mathlib statement

Before using a lemma, one reads its exact statement with `#check`. The prefix `@` also displays the implicit arguments.

code/lean/appC_4_lire_mathlib.lean

```
/-
Annexe C, exemple 4 : lire un énoncé de Mathlib, puis s'en servir.
#check affiche l'énoncé exact (hypothèses implicites comprises) ; on l'applique ensuite comme
     une fonction.
-/
import Mathlib.Analysis.Complex.Exponential
import Mathlib.Tactic.Linarith

#check @Real.exp_pos
#check @Real.add_one_le_exp
```

```
#check @abs_sub_comm

-- On applique les énoncés lus ci-dessus.
theorem exp_moins_positif (x : ℝ) : 0 < Real.exp (-x) := Real.exp_pos (-x)

theorem exp_minore (x : ℝ) (hx : 0 ≤ x) : 1 ≤ Real.exp x := by
  have h := Real.add_one_le_exp x
  linarith

#print axioms exp_moins_positif
#print axioms exp_minore
```

Output (return code 0, 5.3 s):

```
Real.exp_pos : ∀ (x : ℝ), 0 < Real.exp x
Real.add_one_le_exp : ∀ (x : ℝ), x + 1 ≤ Real.exp x
@abs_sub_comm : ∀ {α : Type u_1} [inst : Lattice α] [inst_1 : AddGroup α] (a b : α), |a - b| =
     |b - a|
'exp_moins_positif' depends on axioms: [propext, Classical.choice, Quot.sound]
'exp_minore' depends on axioms: [propext, Classical.choice, Quot.sound]
```

How to read the third line.

- `{α : Type u_1}` in braces: an *implicit* argument, which Lean infers (here the type of the elements).
- `[inst : Lattice α]` in square brackets: an *instance*, a structure that Lean finds by itself (here: α is a lattice, which provides max and min, hence the absolute value $|a| = \max(a, -a)$).
- `(a b : α)` in parentheses: the explicit arguments, supplied at the call.
- After the comma, the conclusion.

The statement is therefore: in every lattice-ordered additive group, $|a-b| = |b-a|$. One applies it like a function: the term `abs_sub_comm x y` has type `|x - y| = |y - x|`. This is exactly what `exp_moins_positif` does with `Real.exp_pos (-x)`. The book's statements are read in the same way; for example, in `triangular_yukawa_optimal {κ : ℝ} (hκ : 0 < κ) (C : Set Plane) ...` (Chapter 5), κ is implicit because it can be read off the type of `hκ`.

Finding a lemma. Mathlib names follow a convention: `add_one_le_exp` reads "$x+1 \le \exp x$". The tactic `exact?` searches for a lemma that closes the goal, and the Mathlib documentation site allows search by name. The LeanMaster server used while writing this book also offers a search (`search_theorems`), but over its own library.

C.5 Checking the axioms

A file that compiles is not a file that is proved. The fifth example contains two false theorems:

code/lean/appC_5_temoins.lean

```
/-
Annexe C, exemple 5 : témoins négatifs de la porte d'axiomes.
Ce fichier compile avec le code de retour 0 ; ce sont les lignes `#print axioms` qui révèlent
    la triche.
-/

theorem vrai_controle_positif (n : Nat) : n + 0 = n := rfl
```

```
theorem faux_par_sorry : 1 = 2 := by sorry

axiom triche : False
theorem faux_par_axiome : 1 = 2 := triche.elim

#print axioms vrai_controle_positif
#print axioms faux_par_sorry
#print axioms faux_par_axiome
```

Output (return code 0!, 1.0 s):

```
appC_5_temoins.lean:8:8: warning: declaration uses `sorry`
'vrai_controle_positif' does not depend on any axioms
'faux_par_sorry' depends on axioms: [sorryAx]
'faux_par_axiome' depends on axioms: [triche]
```

The compiler accepts the file: `sorry` produces only a warning, and an `axiom` is a legal declaration. It is the `#print axioms` lines that reveal the cheating: `sorryAx` in the first, the user axiom `triche` in the second. The rule of this book is therefore:

> *a theorem is accepted only if `#print axioms` lists only axioms from* {`propext`, `Classical.choice`, `Quot.sound`}.

The theorem `vrai_controle_positif` is the positive control: the gate lets it through.

C.6 Why the kernel must also be replayed

C.6.1 A forgery invisible to `#print axioms`

Lean's kernel is the only judge, but the compiler can ask it not to judge: the option `debug.skipKernelTC` disables the kernel's type checking for the declarations added. The sixth file uses it to manufacture a "proof" of `False`:

code/lean/appC_6_contrefacon.lean

```
/-
Annexe C, exemple 6 : une contrefaçon que `#print axioms` ne voit pas, et que le rejeu du noyau
     refuse.
On ajoute directement au noyau (addDecl) une « preuve » de False dont le terme est True.intro,
     en lui
demandant de ne pas vérifier les types (debug.skipKernelTC). Le compilateur accepte ; `#print
     axioms`
n'affiche que la liste blanche ; seul `leanchecker`, qui rejoue chaque déclaration, la refuse.
NE JAMAIS utiliser cette option dans un vrai développement : c'est un témoin négatif.
-/
import Lean
open Lean Elab Command

set_option debug.skipKernelTC true in
run_cmd liftCoreM <| addDecl (.thmDecl
  { name := `contrefacon, levelParams := [], type := mkConst ``False,
    value := mkConst ``True.intro })

theorem un_egale_deux : 1 = 2 := contrefacon.elim

#print axioms contrefacon
#print axioms un_egale_deux
```

Output (return code 0; 51.6 s on the first compilation with a cold disk cache, 4.4 s on the second):

```
'contrefacon' does not depend on any axioms
'un_egale_deux' does not depend on any axioms
```

The file compiles, and `#print axioms` declares that $1 = 2$ depends on *no* axiom. The gate of the previous section would let it through.

Remark C.2 (A first attempt, rejected). Our first version simply wrote `set_option debug.skipKernelTC true in theorem contrefacon : False := True.intro`. It was rejected (return code 1) with the message `Type mismatch: True.intro has type True but is expected to have type False`: before the kernel, the *elaborator* also checks types. The option disconnects only the kernel; for the forgery one must therefore add the declaration directly to the kernel (`addDecl`), as the N4 auditor did. This detail is no comfort: a hostile file can do exactly that.

C.6.2 Replay by `leanchecker`

The tool `leanchecker`, shipped with Lean, rereads the compiled `.olean` file and *replays* every declaration of the module in a fresh kernel, without trusting what the compiler recorded. One must therefore first produce an `.olean`. The script `code/appC_leanchecker.py` compiles each example as a module `AppC.E<n>_<nom>` in a working directory, then runs `leanchecker` with the same `LEAN_PATH`:

```
taskset -c 6,7 .venv/bin/python -I papers/textbook_formalisation_2026/code/appC_leanchecker.py
    \
    DOSSIER_DE_TRAVAIL SORTIE.json appC_1_logique appC_2_calcul appC_3_recurrence \
    appC_4_lire_mathlib appC_5_temoins appC_6_contrefacon
```

Result of 2026-10-26, for the six examples and then a nonexistent module (compilation and replay, two cores):

module	compile	replay	duration	replay output
`AppC.E1_logique`	0	0	48.4 s	(empty)
`AppC.E2_calcul`	0	0	54.3 s	(empty)
`AppC.E3_recurrence`	0	0	56.7 s	(empty)
`AppC.E4_lire_mathlib`	0	0	58.4 s	(empty)
`AppC.E5_temoins`	0	0	53.6 s	(empty)
`AppC.E6_contrefacon`	0	**1**	58.6 s	see below
`AppC.ModuleInexistant`	–	**1**	–	`Could not find any oleans`

The replay output for Example 6 is:

```
leanchecker found a problem in AppC.E6_contrefacon
uncaught exception: while replaying declaration 'contrefacon':
(kernel) declaration type mismatch, 'contrefacon' has type
  True
but it is expected to have type
  False
```

The script ends with the line `{"all_controls_pass": true}`: every module returned the expected code.

Reading.

- *Positive controls.* The honest modules are replayed without error (code 0, no output).
- *Main negative control.* The forgery is rejected (code 1): when replaying `contrefacon`, the kernel finds that the term `True.intro` has type `True` and not `False`.

- *Trivial negative control.* A nonexistent module is rejected (code 1): the code 0 of the others is not that of a tool that does nothing.
- About one minute per module, even for Examples 1 and 5, which import nothing: the time is that of starting `leanchecker`, not that of the module (we have not measured the cause). The replay covers only the module's declarations: Mathlib and the Lean core are loaded, not replayed (that would require the option `-fresh`, which is much more expensive).

Warning : each check catches what the other lets through

`leanchecker` does not replace `#print axioms`: a `sorry` or a declared `axiom` is *legal* for the kernel, and the replay of the module of Example 5 is accepted (table above: code 0; this is the expected behaviour). Conversely, `#print axioms` does not see the forgery of Example 6. Hence the book's Tier A gate, with four checks (§1.4):

1. compilation without error;
2. the list of axioms contained in the whitelist;
3. the `leanchecker` replay of the module, with code 0;
4. a text search for `skipKernelTC`, `addDecl`, `native_decide` and `implemented_by` in the sources (it would find Example 6 immediately).

When the third check is missing, the book writes **[Tier A without replay]** instead of **[Tier A]**.

And `native_decide`? The tactic `native_decide` has a proposition evaluated by compiled code, outside the kernel. Lean 4.34.1 records it as an axiom specific to each use (of the form `_native.native_decide.ax_...`), which `#print axioms` displays and the whitelist rejects; LeanFlow classifies it separately (Appendix D). To make the kernel itself compute, one writes `decide +kernel`.

C.7 Exercises

Exercise C.1. Write and compile a proof of $\forall\, p\, q : \mathrm{Prop},\ (p \vee q) \to (q \vee p)$, first as a term, then with the tactics `intro`, `rcases` and `exact` (hints: `Or.inl`, `Or.inr`).

Exercise C.2. Prove in Lean that $\sum_{i<n}(2i+1) = n^2$, following the model of `double_somme`. Check with `#print axioms` that your proof uses only the whitelist.

Exercise C.3. In `appC_2_calcul.lean`, replace `nlinarith [sq_nonneg (a - b)]` by `linarith`. What happens, and why?

Exercise C.4 (Negative control)**.** Change `quatre_septiemes` to $4/7 - 1/3 = 1/4$ and compile. What is the message, and what is the return code? Compare with Example 5: why does a false proof make the compilation fail, whereas a `sorry` does not?

Appendix D

Certified numerical computation

A number computed in floating point is an approximation whose error is unknown. A *certified* number comes with a proved enclosure. Between the two, this book distinguishes several degrees (§1.4.6):

- a floating-point computation, however precise, is **[Tier X]**;
- an enclosure in ball arithmetic (Arb) is rigorous *if the code is correct*; since that code is not verified by Lean, the book classifies it as **[Tier C]** ("C-ball" in Chapter 3);
- a claim about *stated rationals*, decided by Lean's kernel, is **[Tier A without replay]** for that arithmetic claim alone, and **[Tier A]** when the module has in addition been replayed by `leanchecker` with code 0.

This appendix explains these three levels, then the solution of stiff differential equations with CVODE, and why a CVODE result remains Tier X. A complete example, `code/appD_certifie.py`, goes through the three levels with its controls.

D.1 Floating point

A binary64 floating-point number (IEEE 754 standard [70]) is written $\pm m \cdot 2^e$ with a 53-bit significand. The result of an operation $(+, -, \times, \div, \sqrt{})$ is the rounding of the exact result, and the standard guarantees

$$\mathrm{fl}(x \circ y) = (x \circ y)(1+\varepsilon), \qquad |\varepsilon| \le u = 2^{-53} \approx 1.1 \times 10^{-16},$$

in round-to-nearest (barring overflow). The standard also offers *directed roundings*: towards $-\infty$ and towards $+\infty$.

University remark. **[Tier L]** The bound above is an instance of the standard model of floating-point arithmetic: for every real x in the normal range of binary64, round-to-nearest satisfies $\mathrm{fl}(x) = x(1+\varepsilon)$ with $|\varepsilon| \le u$, i.e. $|\mathrm{fl}(x) - x| \le u|x|$; correct rounding of $+, -, \times, \div, \sqrt{}$ means $\mathrm{fl}(x \circ y)$ is fl applied to the exact $x \circ y$, which gives the displayed bound [70].

Example D.1 (Ten times one tenth)**.** 1/10 is not representable in binary. The floating-point number `0.1` is exactly 0.1000000000000000055511…, slightly more than 1/10. In Python, `sum([0.1]*10)` gives `0.9999999999999999` and not 1: the rounding error accumulates, and the sign of the final error is not that of the initial error.

The elementary functions (exp, sin…) of the system's mathematical library have in general *no* guarantee of correct rounding: the IEEE 754 standard recommends it without requiring it. A rigorous enclosure therefore cannot rely on `math.exp`.

D.2 Interval arithmetic

Definition D.2. An unknown real x is represented by an interval $[\underline{x}, \overline{x}]$ that contains it. The operations are defined by

$$[a,b] + [c,d] = [a+c,\, b+d], \qquad [a,b] - [c,d] = [a-d,\, b-c],$$

$$[a,b] \times [c,d] = [\min S, \max S], \quad \text{where } S = \{ac, ad, bc, bd\}.$$

Theorem D.3 (Inclusion property)**.** *If $x \in [a,b]$ and $y \in [c,d]$, then $x+y \in [a,b]+[c,d]$, $x-y \in [a,b]-[c,d]$ and $xy \in [a,b]\times[c,d]$. By induction, every expression built from these operations, evaluated on intervals containing the arguments, yields an interval that contains the exact value.*

Human proof. For the sum: $a \le x \le b$ and $c \le y \le d$ give $a+c \le x+y \le b+d$. For the difference, add $-d \le -y \le -c$. For the product, for fixed y the map $x \mapsto xy$ is affine, so xy lies between ay and by; then $y \mapsto ay$ and $y \mapsto by$ are affine, so ay and by lie between their values at c and d. Hence xy lies between the smallest and the largest of the four products. □

Directed rounding. On a machine, the endpoints are floating-point numbers and every operation rounds. To preserve the inclusion property, one rounds the lower endpoint *downwards* and the upper endpoint *upwards*: this is *outward rounding*. Without access to the rounding modes, one can widen each endpoint by one ulp (the gap between two consecutive floating-point numbers) with `math.nextafter`: this is wider than necessary, but safe, since round-to-nearest is off by at most half an ulp.

The dependency problem. Interval arithmetic does not know that two occurrences of the same variable are equal: for $x \in [0,1]$, $x - x$ gives $[0,1]-[0,1] = [-1,1]$, and not $\{0\}$. Enclosures grow quickly unless one takes care; the remedies are to rewrite the expressions, to subdivide the domain (the 206 468 cells of B2 in Chapter 3) or to use Taylor expansions.

Numerics : step 1 of `code/appD_certifie.py`

Ten additions of the interval [`nextafter(0.1,-inf)`, `0.1`], which contains 1/10, widened by one ulp at each addition. Actual output (2026-10-26):

```
"naive_sum": "0.9999999999999999",   "naive_equals_1": false,
"outward": ["0.9999999999999991", "1.0000000000000009"],   "outward_contains_1": true,
"nearest": ["0.9999999999999999", "0.9999999999999999"],   "nearest_contains_1": false
```

Positive control: the outward-rounded interval contains exactly 1 (the membership test is done in exact rationals, `fractions.Fraction`). *Negative control:* the same computation, with endpoints rounded to nearest, gives an "interval" that excludes the true value.

D.3 Ball arithmetic: Arb and python-flint

Arb [36] represents a real number by a *ball* $[m \pm r]$: a midpoint m in arbitrary precision (a dyadic number $M \cdot 2^E$) and a radius r in low precision, rounded upwards. It is interval arithmetic in midpoint–radius form, which is more economical when the precision is high.

Proposition D.4 (Product of two balls)**.** *If $|x-m_1| \le r_1$ and $|y-m_2| \le r_2$, then $|xy - m_1 m_2| \le |m_1| r_2 + |m_2| r_1 + r_1 r_2$.*

Human proof. $xy - m_1 m_2 = m_1(y-m_2) + m_2(x-m_1) + (x-m_1)(y-m_2)$; apply the triangle inequality. □

On a machine, Arb computes the rounded midpoint $m_1 m_2$, and adds to the radius the error of that rounding, bounded from above. For functions (exp, erf, integrals…), the library uses expansions with an explicit bound on the remainder. The result contains the exact value *if Arb's code is correct*: it is tested and published software, not a formally verified program.

In Python. The `python-flint` library gives access to Arb. The B2 study of Chapter 3 used version 0.6.0, installed in `.scratchpad/b2_pylib` of the repository for the Python 3.11 of the `.venv` environment; the system `python3` (3.10) carries version 0.9.0 instead (Appendix E). The example in this appendix uses version 0.6.0.

Numerics : step 2 of `code/appD_certifie.py`

Enclosure of $I = \int_0^1 e^{-x^2} dx$ by `acb.integral` at 96 bits, checked against the independent ball $\frac{\sqrt{\pi}}{2}\,\mathrm{erf}(1)$. Actual output:

```
"python_flint": "0.6.0",  "prec_bits": 96,
"ball":     "[0.74682413281242702539946743 6 +/- 2.30e-28]",
"erf_ball": "[0.74682413281242702539946743 61 +/- 5.34e-29]",   "erf_overlaps": true,
"midpoint_float_n10": "0.7471308777479975",   "midpoint_inside_ball": false
```

Positive control: the two balls, computed along two routes (rigorous quadrature, error function), overlap. *Negative control:* the 10-point midpoint rule, in floating point, gives 0.74713: a plausible value, wrong in the fourth digit, and outside the ball. A floating-point number is not an enclosure.

D.4 Rational certificates checked by Lean: LeanFlow

D.4.1 The principle

Lean's kernel can decide equalities and inequalities between explicit *integers*: this is a finite computation. LeanFlow (`leanflow_mcp.lean`, function `certify`) turns a numerical claim into such a statement, in four steps:

1. **The claim.** Named rational enclosures, `x = [lo, hi]`, and linear inequalities between their endpoints. Numbers are written `p/q` or in decimal; a binary floating-point number is rejected (it is Tier X).

2. **The template.** Each inequality becomes a theorem over $\mathbb{Z}$, with denominators cleared symbolically, of the form `theorem cert_i : (0 : Int) < D ∧ N * RD ≤ RN * D := by decide +kernel`. Python only assembles text from integer literals; all multiplications and comparisons are done by Lean.

3. **The kernel decides.** The tactic `decide +kernel` (LeanFlow name: `decide_kernel`) has the proposition evaluated by the kernel itself, with no additional axiom.

4. **The audit.** LeanFlow appends to the source one `#print axioms` per theorem and a random sentinel, then requires exactly one report per theorem, within the whitelist, and the sentinel at the end. An exact oracle in Python (`Fraction`) must return the same verdict.

Fail-closed. "Closed when in doubt": every abnormal situation yields `error`, never `ok`. LeanFlow's documentation (`docs/LEAN_CERTIFY.md`) lists the cases, each with a test: missing Lean toolchain, imported module without `.olean`, nonzero return code, no axiom report, `sorry`, user axiom, `native_decide` (which introduces an axiom specific to the use), forged axiom report,

`#exit`, invalid claim, disagreement with the oracle. A *false* claim yields `rejected` with the code `false-statement`.

What Tier A covers. The status `ok` concerns the arithmetic statement *about the stated numbers*, and nothing else: neither the provenance of these numbers, nor the fact that they really enclose the integral. This is why, in Chapter 3, the B2 chain is **[Tier C]** overall: the arithmetic is verified, the ball is not. LeanFlow audits only `#print axioms`; the book adds the `leanchecker` replay when it claims **[Tier A]** (Appendix C, §C.6, and case U5 of Chapter 5).

D.4.2 The complete example

The ball of step 2 is rounded outwards to two decimals with 30 digits, in exact rationals: $\texttt{lo} = \lfloor 10^{30}(m-r)\rfloor/10^{30}$ and $\texttt{hi} = \lceil 10^{30}(m+r)\rceil/10^{30}$. LeanFlow then receives two assertions:

```
{"name": "gaussI", "tactic": "decide_kernel",
 "enclosures": {"I": ["1867060332031067563498668589 69/250000000000000000000000000000",
                      "7468241328124270253994674362 3/100000000000000000000000000000"]},
 "assertions": [{"label": "I.lo > 0.7468", "endpoint": "I.lo", "op": ">", "rhs": "7468/10000"},
                {"label": "I.hi < 0.7469", "endpoint": "I.hi", "op": "<", "rhs":
    "7469/10000"}]}
```

and, as a negative control, the same ball with the false assertion `I.hi < 7468/10000`.

Numerics : step 3 of `code/appD_certifie.py`

Actual output (Lean 4.34.1, commit `5045d0056413266e57c625dcd7c365b10e377c52`, core only):

```
"pos_I_between_0.7468_and_0.7469": {"status": "ok", "code": "verified", "tier": "A", "
    failed": [],
                                    "seconds": 0.64}
"neg_I_below_0.7468":              {"status": "rejected", "code": "false-statement", "
    tier": null,
                                    "failed": ["cert_0"], "seconds": 0.49}
"all_controls_pass": true
```

The script's seven controls (three for step 1, two for each of steps 2 and 3) passed; the whole script took 1.7 s.

Tier summary: the statement "$0.7468 <$ `lo` and `hi` < 0.7469" is **[Tier A without replay]**: LeanFlow returns `"tier": "A"`, but we did not replay these certificates with `leanchecker`, and the book's convention reserves **[Tier A]** for modules replayed with code 0; the claim "$I \in [\texttt{lo}, \texttt{hi}]$" is **[Tier C]** (it rests on Arb); the conclusion "$0.7468 < I < 0.7469$" inherits the weaker of the two, **[Tier C]**.

D.5 Stiff differential equations and CVODE

D.5.1 Stiffness

Consider $y' = -\lambda y$ with $\lambda > 0$ large. The solution $e^{-\lambda t}$ is trivial, but the explicit Euler method $y_{n+1} = y_n + hf(y_n) = (1-h\lambda)y_n$ stays bounded only if $|1-h\lambda| \le 1$, that is, $h \le 2/\lambda$. A system is *stiff* when it mixes very different time scales: the stability of an explicit method then imposes a step dictated by the fastest one, much smaller than what accuracy on the slowest one requires. The *implicit* Euler method, $y_{n+1} = y_n + hf(y_{n+1})$, gives $y_{n+1} = y_n/(1+h\lambda)$: it is stable for every $h > 0$.

D.5.2 BDF formulas

CVODE [71] integrates stiff systems with the *backward differentiation formulas* (BDF) of order $q = 1, \dots, 5$. One approximates $y'(t_{n+1})$ by the derivative of the polynomial interpolating $y_{n+1}, y_n, \dots, y_{n+1-q}$:

$$\text{BDF1: } y_{n+1} - y_n = hf(y_{n+1}), \qquad \text{BDF2: } y_{n+1} - \tfrac{4}{3}y_n + \tfrac{1}{3}y_{n-1} = \tfrac{2}{3}hf(y_{n+1}).$$

Each step requires solving a *nonlinear* equation in y_{n+1}, of the form $G(y) = y - \gamma h f(y) - a_n = 0$, where a_n collects the past values and γ depends on the order.

D.5.3 Newton and the Jacobian

CVODE solves $G(y) = 0$ by a modified Newton method: $\big(I - \gamma h J\big)\,\delta^{(k)} = -G(y^{(k)})$, $y^{(k+1)} = y^{(k)} + \delta^{(k)}$, where $J = \partial f/\partial y$ is the Jacobian. The matrix $I - \gamma h J$ is recomputed and factorised only from time to time. The Jacobian is either supplied analytically by the user or approximated by finite differences, column by column (one extra evaluation of f per column). The book's `qf-*` crates supply analytic Jacobians: for a gradient flow $\dot{x} = -\nabla E(x)$ (relaxations in cases U1 and U2), $J = -\operatorname{Hess} E$; for the Kolmogorov equation $\dot{\pi} = \pi(P - I)$ (case U5), the system is linear and J is exact.

D.5.4 Error control, and what it does not guarantee

At each step, CVODE estimates the *local* error e (the error made in one step, starting from values assumed exact) and accepts the step if

$$\|e\|_{\mathrm{WRMS}} = \Big(\frac{1}{n}\sum_{i=1}^{n}\Big(\frac{e_i}{\mathrm{rtol}\,|y_i| + \mathrm{atol}_i}\Big)^2\Big)^{1/2} \le 1;$$

otherwise it reduces the step. It also adjusts the order q. This control is *heuristic*: the global error, after thousands of steps, is not bounded by rtol; the local error estimate rests on asymptotic expansions; and everything is done in floating point. This is why **a CVODE result is Tier X**. The book never presents it as a proof: it compares it with an exact value (U5), with a conservation law (U4), or with a theorem (U1, U2), with controls.

The `qf-popgen` certificate is not a proof. The "absorption certificate" of case U5 (stochastic kernel, conserved mass, no negative component, interior mass $< 10^{-9}$, final value within 10^{-8} of the target) is a set of *checks* that reject dubious runs. It makes a run more reliable; it does not prove it. The only Tier A part of case U5 is the LeanFlow statement about the number `0.5714285714284759` and about the vector $(0, \frac{4}{7}, \frac{6}{7}, 1)$.

D.5.5 A stiff example: Robertson kinetics

The Robertson problem is the classic stiffness test, with rate constants from 0.04 to 3×10^7:

$$y_1' = -0.04y_1 + 10^4 y_2 y_3, \quad y_2' = 0.04y_1 - 10^4 y_2 y_3 - 3\times 10^7 y_2^2, \quad y_3' = 3\times 10^7 y_2^2,$$

with $y(0) = (1, 0, 0)$ and $t \in [0, 4\times 10^{10}]$. The sum $y_1 + y_2 + y_3$ is conserved exactly by the system, which provides a check. rusty-SUNDIALS ships it as the example `examples/robertson.rs` (BDF up to order 5, rtol $= 10^{-4}$, atol $= 10^{-8}$, analytic Jacobian). Command, from the working copy `rusty-SUNDIALS-wt-tb2ch5`:

```
cargo run --manifest-path ~/rusty-SUNDIALS-wt-tb2ch5/Cargo.toml --target-dir ~/rusty-target-ar
    \
    --release -j 2 --offline -p examples --example robertson
```

Numerics : actual output (2026-10-26, 43 s including compilation)

```
           t             y1             y2             y3
-------------------------------------------------------------
   4.0000e-1    9.851721e-1    3.386396e-5    1.479401e-2
    4.0000e0    9.055204e-1    2.240497e-5    9.445716e-2
    4.0000e2    4.505200e-1    3.222938e-6    5.494767e-1
    4.0000e6    5.169270e-4    2.068765e-9    9.994831e-1
   4.0000e10    5.155066e-8   2.062027e-13    9.999999e-1

Solver statistics:
  Steps:      633  (C ref: 1070, ratio: 0.6x)
  RHS evals:  1762  (C ref: 1537, ratio: 1.1x)
  Final order: 5
  Conservation (y1+y2+y3): 1.0000000000000000e0 (should be 1.0)
  Conservation error: 4.44e-16  (threshold 1e-12)
```

(Intermediate lines omitted; the program's last line, `Conservation:  PASS` followed by a check mark, is cut here because the check mark is not printable.)

Reading: in 633 steps, CVODE traverses eleven orders of magnitude in time; the step size grows accordingly. An explicit method would be limited in stability, over the whole interval, by the fast eigenvalues of the Jacobian. The program compares its statistics with the C version of SUNDIALS (`cvRoberts_dns`, reference values written in the example's code, not rerun here; with an absolute tolerance different from that of the Rust example, so the comparison of step counts is only indicative). Conservation to within 4.4×10^{-16} is a *necessary* check; it proves nothing about y_2, which could be wrong without violating conservation. This is a **[Tier X]** computation.

D.6 Exercises

Exercise D.1. Prove that, for $x \in [a, b]$ with $0 < a$, the interval $[1/b, 1/a]$ contains $1/x$. What should be done if $0 \in [a, b]$?

Exercise D.2. Evaluate $f(x) = x^2 - 2x$ on $[0, 2]$ in interval arithmetic, then the form $f(x) = (x-1)^2 - 1$. Compare with the exact image $[-1, 0]$ and explain the discrepancy by the dependency problem.

Exercise D.3. For the BDF2 method applied to $y' = -\lambda y$, write the linear recurrence satisfied by y_n and show that its characteristic roots have modulus < 1 for every $h\lambda > 0$ (A-stability of BDF2).

Exercise D.4 (LeanFlow, negative control)**.** In `code/appD_certifie.py`, change the precision `ctx.prec` to 20 bits. Does the ball still contain the value? Do the two LeanFlow certificates still pass? Why does a wider enclosure make the assertion `I.lo > 0.7468` harder to certify?

Appendix E

Reproducing the book's computations

This appendix is a step-by-step guide to redoing every numerical solve and every simulation in the book. It gives the exact environment (§E.1), the datasets (§E.2), the commands chapter by chapter with their expected outputs (§E.3 to §E.8), the computations rerun for this appendix (§E.9) and the known failures (§E.10).

Warning : what "reproduce" means here

- The commands were written and run on the author's machine (Linux 6.8, 8 cores, 29 GB of memory), with absolute paths under `/home/callensxavier_gmail_com` and a second disk mounted under `/mnt/disks/disk-socrateai-local-1`. On another machine, replace these prefixes. We write `~` for the home directory, `W` for the root of the AutoevolveAI working copy (`~/AutoevolveAI/.claude/worktrees/openai-math-discovery`), `PY` for `~/AutoevolveAI/.venv/bin/python -I`, and `T` for the Rust build directory `~/rusty-target-ar`.
- The commands are reproduced exactly as they were run, so their placeholders and comments are in French: `SORTIE` is an output file or directory, `FICHIER(S)` a file (files), `DOSSIER` a directory, `TRAVAIL` a work directory, `NOUVEAU_DOSSIER` a new directory, `AFFIRMATION` a claim file, `ÉTIQUETTE` a label and `Module.Nom` a module name.
- For each computation, the run time and the output are marked either *rerun on 2026-10-26* (executed for this appendix) or *record* (a value read from the cited file, not rerun here).
- Several repositories are local branches, and some commits had not been pushed at the time of writing; we say so where it applies. Reproducing outside this machine assumes that these branches have been published.
- Reproducing a Tier X number does not change its tier.

E.1 The environment

E.1.1 Python

- **Main interpreter:** the virtual environment `~/AutoevolveAI/.venv`, Python 3.11.16. Versions recorded on 2026-10-26: numpy 2.4.6, scipy 1.17.1, sympy 1.14.0, matplotlib 3.11.2, pyarrow 25.0.1, mpmath 1.3.0, hypothesis 6.168.0. It is created by `uv sync -all-extras` from the `pyproject.toml` of the AutoevolveAI repository. *It contains neither* `h5py` *nor*

python-flint.

- **python-flint 0.6.0** (Arb): installed by `pip install -target` into `W/.scratchpad/b2_pylib`, built for CPython 3.11 (file `pyflint.cpython-311-x86_64-linux-gnu.so`). The B2 scripts and `code/appD_certifie.py` add it to `sys.path` themselves; they must therefore run with the Python 3.11 of the `.venv`.
- **System Python:** `python3` 3.10.12, with python-flint 0.9.0 in `~/.local` (invisible under `-I`). It is used to run the Sandier–Serfaty certificate (P1, §E.4), which uses only the standard library and `g++`. *Do not use it for pytest*: there is a numpy/zarr conflict in this environment.
- **Always `-I`.** The option isolates the interpreter (no `PYTHONPATH`, no current directory in `sys.path`): a planted `json.py` file in a folder of downloaded data will not be imported. Scripts that need a module outside the environment (LeanFlow, python-flint) add it explicitly.

E.1.2 Rust and rusty-SUNDIALS

- **Toolchain:** `rustc 1.96.1` and `cargo 1.96.1`, edition 2024 (declared in the repository's `Cargo.toml`).
- **Repository:** `https://github.com/xaviercallens/rusty-SUNDIALS`. The main checkout `~/rusty-SUNDIALS` is on a local branch (`merge-mcp-temp`); *do not build anything there for this book.* Each study has its own working copy (*worktree*), at the commit given in Table E.1.
- **Shared build directory:** all worktrees use `T = ~/rusty-target-ar`. The binaries in `T/release` are therefore those of the *latest* build, whichever worktree it came from. Rebuild from the right worktree before each chapter.

Table E.1: rusty-SUNDIALS worktrees (commits read by `git rev-parse HEAD` on 2026-10-26).

worktree `~/rusty-SUNDIALS-wt-...`	branch	commit (HEAD) and use
`tb2ch5`	`textbook2/ch5-cas-usage-2026-10-26`	`cda9a9057ec16820f2e16b560cc0e63cf7eb05b9`; Chapter 5 (U1–U5), Appendix D. On top of `origin/main` `7630f7e6436f6791a8abc9fea05b76868305bc35`; contains `qf-yukawa-crystal`, `qf-shell-cascade`, `qf-gpe2d`, `qf-popgen`
`p7`	`physics/p7-fixation-2026-10-25`	same commit `cda9a905...`; P7, `qf-popgen`
`ar-h1`	`physics/ar-dipolar-exciton-2026-10-25`	`272a09e62c1b9b36eabc167cdafaa2845fbe9198`; dipolar excitons
`ar-h2`	`physics/ar-yukawa-2026-10-25`	`89a1fb79daac676c41db3fed57940c071161d855` (later merge; paper code `cc5a8aed81b3228bd9e9eb69f2c326f926903a34`, results `7f2a45c32a9eb9722ca36c334329cb7a5a541951`, not pushed); Yukawa
`ar-h3`	`physics/ar-keldysh-2026-10-25`	`859de5bcd7c4b0d4319ab5dc5b950db22c9e2448` (paper commit `2e41505cc9df8c46d2b2d667c89ea53e977218ca`, an ancestor); Keldysh
`p2`	`physics/p2-zero-point-optimality-2026-10-25`	`9e85e15b23be0c74a665b6a5b74ec02b0bbffc5f`; P2, B1 (`lattice_phonon`, `scha.rs`)

Typical build (a single package set, two jobs, no network):

```
cargo build --manifest-path ~/rusty-SUNDIALS-wt-tb2ch5/Cargo.toml --target-dir ~/rusty-target-
    ar \
    --release -j 2 --offline -p qf-yukawa-crystal -p qf-shell-cascade -p qf-popgen
```

The option `-target-dir` is equivalent to the variable `CARGO_TARGET_DIR=~/rusty-target-ar` used in the records. Chapter 5 (§5.1) gives the same command in that form.

E.1.3 Lean

The book uses *two* Lean toolchains; do not mix them.

- **The pinned stack (Lean 4.34.1).** This is a copy of the `lean/` directory of openai/math, with its complete `.lake` (about 8 GB), in `/mnt/disks/disk-socrateai-local-1/callensxavier_home_data/openai_math_pinned/lean`. Lean 4.34.1, commit `5045d0056413266e57c625dcd7c365b10e377c52`; Mathlib `d13f23b723b8a846827a245b89c10fc7d3f11612` (read from `lake-manifest.json`, together with batteries `f2effa3d803fda822b1f97b806c47cf2adfbcbc2` and some thirty other packages). The copy is not a `git` repository; the openai/math clone it comes from is at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. The toolchain itself is in `.../callensxavier_home_data/elan/toolchains/leanprover--lean4---v4.34.1/bin` (with `lake` and `leanchecker`); a second copy of the same version is in `~/.elan/toolchains`, and LeanFlow uses it (core only). All the `formal_cert/` modules cited in Chapters 1 to 5 and the examples of Appendix C were checked with this stack.

- **Lean v4.34.0-rc2.** This is elan's default toolchain on the machine. It was used for the minimal file `T.lean` of Chapter 1 (no library), for the QuantumFluids environment of case U3 (`sBog_le_sqrt`) and for the project of the $\sigma = 3$ shell model (case U4), each of which has its own Mathlib, outside this repository. The last two are **[Tier A without replay]**: no `leanchecker` replay is recorded for them.

To compile a file with the pinned stack, from `W`:

```
taskset -c 6,7 PY scripts/certified_numerics/lean_pinned.py compile FICHIER.lean --json-out
    SORTIE.json \
    [--olean Module.Nom]
```

Without `-olean`, no `.olean` is written, and subsequent files cannot import this one. With `-olean`, the `.olean` files go to `W/formal_cert/.olean_out`, a directory hard-coded in the script; compile the modules in the order of their `imports`. The replay is done with the `leanchecker` binary of the same toolchain and the same `LEAN_PATH`; `code/appC_leanchecker.py` shows how (Appendix C).

E.1.4 LeanFlow

Repository `https://github.com/xaviercallens/SocrateAI-Scientific-Agora-LeanFlow`, worktree `~/LeanFlow-wt-p1`, branch `feat/lean-data-certificates-2026-10-25`, commit `06084ef5c40057c66c476feba18ea3da8fb150ba` (the commit that adds the `decide_kernel` tactic). On 2026-10-26 this commit was *local*: the remote branch stopped at `84e0c2899830fba671022eefcd408af9ecd565c8`, which does not know `decide_kernel`. Command-line interface:

```
# depuis ~/LeanFlow-wt-p1/src, sans -I (qui retirerait le répertoire courant de sys.path) :
~/AutoevolveAI/.venv/bin/python -m leanflow_mcp.lean AFFIRMATION.json --lean ~/.elan/toolchains
    /leanprover--lean4---v4.34.1/bin/lean
# code de sortie 0 = vérifié (niveau A), 1 = rejeté, 2 = erreur
```

(The first comment: run from `~/LeanFlow-wt-p1/src`, without `-I`, which would remove the current directory from `sys.path`. Exit code 0 = verified (Tier A), 1 = rejected, 2 = error.) Under `-I`, `~/LeanFlow-wt-p1/src` must be in `sys.path`: the book's scripts (`ch5_U5_fixation.py`, `appD_certifie.py`) add it with their `-leanflow-src` option.

E.1.5 The AutoevolveAI repository

The scripts, the records (`results/`), the Lean sources (`formal_cert/`) and this book (`papers/textbook_formalisation_2026/`) are on the branch `worktree-openai-math-discovery` of the AutoevolveAI repository. A few record directories were not yet tracked by `git` at the time of writing, notably `results/kernel_replay_2026-10-26/` (the `leanchecker` replay in progress) and `results/proposal2/2026-10-26/A1-continuation/`.

E.2 The datasets

Table E.2: Datasets. The revisions and hashes are those checked by the scripts; the licences are those shown on the dataset card, not verified upstream.

case	Hugging Face identifier, revision	file, sha256	licence; local copy
U1	`colabfit/JARVIS_DFT_3D_12_12_2022`, `f4de0085edcf92885870c67459363d2c7080c90b`	`co/co_0.parquet` (54 027 813 bytes), `9f6c21a2a3eda4316eee1f2be7dddc276eeb5f905de87dcce70f20e36132510b`	"unknown" [61]; `~/hf-data/dipolar_exciton/...`
U2	`colabfit/OC20_IS2RES_val_id`, `6a1dfef12aec878b4cda99edd54b16a6a254e135`	`co/co_10.parquet` (142 986 687 bytes), `9e4249ba20cf4cdc46f69e7fe6deb19be286dd19c9f1cf85d350816d2b1e748b`	CC-BY-4.0 [63]; disk 2, `hf_datasets/oc20_is2res_val_id/`
U3	no dataset on the Hub; Table 14.1 of Donnelly–Barenghi [41]	`W/papers/prog_superfluid_structure_factor/data/he4_Sk_donnelly_table14_1.csv`	secondary source **[Tier L]**
U4	`polymathic-ai/turbulent_radiative_layer_2D`, `2ee7756575ff6f90981d0308cbcb6a2ab5995bdc`	`data/test/turbulent_radiative_layer_tcool_0.32.hdf5`, `06cbd36c1dbea84e70906a363d20143b228765ee37850ec65351e247cf7df747`	CC-BY-4.0 [64]; `~/hf-data/sigma3_enstrophy/...`
U5	`katarinagresova/Genomic_Benchmarks_demo_coding_vs_intergenomic_seqs`, `85458172cab6cffad46cb172e64ca94780506ed6`	`data/test-00000-of-00001-6fccc31405110e30.parquet` (2 456 511 bytes), `dfe13715a0227204cbae1ff6a167b2138337b749ca4f748da7c5a838a0049afe`	no licence on the card [65]; `~/hf-data/textbook2/...`

The U1 and U5 scripts download the file themselves if it is missing, from the address `https://huggingface.co/datasets/<id>/resolve/<rev>/<file>` (`<rev>`: the pinned revision of Table E.2), and then *stop* if the hash differs. The U2 and U4 scripts check the hash if the file is present, and otherwise fall back on the derived data stored in the crate (`data/initial/*.csv` for U2, `results/sigma3/well_amps.txt` for U4). Extracting the spectra from The Well requires `h5py`, which is absent from the `.venv`: it was done separately by the crate's scripts `results/sigma3/scripts/well_spectrum.py` and `make_amps.py`.

E.3 Chapter 1

Poisson summation and controls. `PY papers/textbook_formalisation_2026/code/ch1_theta.py`, with no argument; it writes no file. *Rerun on 2026-10-26*: 3.4 s; output in agreement with the table of §1.1.5 (excerpt in §E.9).

Lean. The file `T.lean` of §1.4 compiles with `lean T.lean` (v4.34.0-rc2, no library). The examples of Appendix C are an extended version of it, using the pinned stack.

E.4 Chapter 2

Dipolar excitons (CVODE relaxations). Worktree `ar-h1`. Source: `W/results/autoresearch/2026-10-25/H1/contribution/NUMERICS.md`, section *Reproduce*.

```
cargo build --manifest-path ~/rusty-SUNDIALS-wt-ar-h1/Cargo.toml --target-dir ~/rusty-target-ar
    \
    --release -j 2 --offline -p qf-yukawa-crystal
D=~/rusty-SUNDIALS-wt-ar-h1/crates/qf-yukawa-crystal
T/release/dipolar_crystal refs $D ;  T/release/dipolar_crystal gaps $D ;  T/release/
    dipolar_crystal controls $D
T/release/dipolar_crystal main $D 0.066 8 2          # 8 uniformes, 2 recuits, 2
    incommensurables (défaut)
```

Actual subcommands (read in `src/bin/dipolar_crystal.rs`): `refs`, `gaps`, `controls`, `main`; arguments of `main`: directory, $\tilde{d}$, number of uniform starts, number of annealed starts, number of incommensurate starts (the listing's comment: 8 uniform, 2 annealed, 2 incommensurate by default). *Record:* 57 min 48 s for `main` at $\tilde{d} = 0.066$, 8 min 13 s for `controls`. The main relaxations were run with binaries from commit `17ea6be` of this branch. Chapter 5 reruns two of these relaxations (`main <dossier> 0.066 1 0 1`, about 8 min on two cores) and recovers the record digit for digit.

Yukawa (erratum, exact potential). Worktree `ar-h2`; source `W/papers/ar_yukawa_crystal/main.tex`, reproducibility section.

```
cargo build --manifest-path ~/rusty-SUNDIALS-wt-ar-h2/Cargo.toml --target-dir ~/rusty-target-ar
    \
    --release -j 2 --offline -p qf-yukawa-crystal
D=~/rusty-SUNDIALS-wt-ar-h2/crates/qf-yukawa-crystal
T/release/yukawa_crystal exact-refs $D                # énergies exactes, borne de queue,
    Ewald
T/release/yukawa_crystal exact-relax $D 2 1 0 0       # kappa nrun recuit premier (voir
    scripts/run_exact_relax.sh)
T/release/yukawa_crystal exact-saddle $D              # témoin négatif : réseau carré, col
PY $D/scripts/mpmath_lattice_control.py $D SORTIE.json # témoin positif, 40 chiffres
bash $D/scripts/check_crate.sh                        # fmt, clippy, tests
```

(The comments read: exact energies, tail bound, Ewald; `kappa`, number of runs, annealing, first start; negative control: square lattice, saddle; positive control, 40 digits.) *Record:* 60 to 158 s per relaxation at $\kappa = 2$, 208 to 1 486 s at $\kappa = 1$; `check_crate.sh`: 24 tests passed, 0 failed, 1 ignored.

Keldysh (jellium). Worktree `ar-h3`. First build from this worktree:

```
cargo build --manifest-path ~/rusty-SUNDIALS-wt-ar-h3/Cargo.toml --target-dir ~/rusty-target-ar
    \
    --release -j 2 --offline -p qf-yukawa-crystal
```

then run `T/release/keldysh_jellium scan results/keldysh/scan.json` (subcommands `scan` and `points`), then `PY scripts/keldysh/analyze.py final results/keldysh ~/hf-data/keldysh_tmd/tmd_rows.json`, from the crate. *Record:* the scan takes 0.33 s. The 30-digit check of route E (with two planted errors) is `PY W/results/autoresearch/2026-10-25/H3/contribution/route_e_check.py SORTIE.json` (*record:* relative error of the positive control 2.1×10^{-17}; negative controls 0.37 and 0.82).

Sandier–Serfaty, P1: the interval certificate (478 s). In the `verification` directory of the preprint, under the openai/math clone (`.../SocrateAI-Scientific-Agora-LeanMaster/`

lean4basesource/openai-math/preprints/Triangular-minimality-for-planar-Coulomb-renormalized-energy-September-23-2026/verification):

```
python3 -I -B run-certificate.py --output-dir NOUVEAU_DOSSIER
```

with the system python3 and g++ 11.4.0 (the script compiles with -std=c++17 -O3 -fopenmp and sets OMP_NUM_THREADS=32 itself). *Record* (W/results/programmes/2026-10-25/P1/): last line OK, TIMING full certificate 478.001 seconds (490 s in total), 22 pairs out of 22 identical. Negative controls: P1/s1_neg/launch_negctl.sh and launch_negctl_AB.sh.

A1 and B3.

- A1 (Lean port): compilation by lean_pinned.py and W/results/proposal2/2026-10-26/A1/scripts/compile_seq.py; cost measurements by A1/scripts/bench_run.py SORTIE FICHIERS... and the instrumented copy A1/cpp/patch_counters.py. *No compilation command is recorded for cpp/dump_driver.cpp*: this step is not reproducible as it stands.
- B3 (bcc in dimension 3), from W/papers/p2_3d_feasibility/: PY code/n1_energies.py, code/n3b_cubic.py, code/n2_voronoi.py, code/n3_green_obstruction.py (about 45 min), then make_all.py. Only n1 and n3b were rerun by the producer (byte-for-byte identical outputs); n2 and n3 never were.

E.5 Chapter 3

P2: zero-point energy. Worktree p2. Build: bash W/results/programmes/2026-10-25/P2/build.sh (it builds only the lattice_phonon binary, on cores 4–7). Actual subcommands: point, scan, frozen, ewald_static, duffy, curv, scha. Examples:

```
T/release/lattice_phonon point gauss:1 0.5 0.966025403784 96,192,384 400   # le concurrent
    rhombique
PY ~/rusty-SUNDIALS-wt-p2/crates/qf-yukawa-crystal/scripts/phonon_mechanism.py SORTIE.json
PY W/papers/prog_zero_point_optimality/human_check.py                      # 1.490278318 et
    1.503352352
```

(The first command evaluates the rhombic competitor.) The full chain of gate G1 (625 shapes, 12 potentials) is described step by step in W/results/programmes/2026-10-25/P2/GATE.md (the crate's scripts phonon_gen_taus.py, phonon_run_gate.py, phonon_edge_check.py). *Record:* phonon_mechanism.py gives the endpoints of the interval $a \in (0.37668, 1.58148)$ in 121.6 s.

B2: the certified curvature (448 s). From the directory W/results/proposal2/2026-10-26/B2/, with the Python of the .venv (mandatory: python-flint 0.6.0 for CPython 3.11, loaded from .scratchpad/b2_pylib by the script):

```
B2_REFINE_TOL=0.01 B2_MAX_DEPTH=3 PY code/b2_certify.py 1 shear 64 1/100000 21
    run_a1_shear_main.json
B2_REFINE_TOL=0.01 B2_MAX_DEPTH=3 PY code/b2_certify.py 1 diag 64 1/100000 21 run_a1_diag_ctl.
    json
B2_REFINE_TOL=0.01 B2_MAX_DEPTH=3 PY code/b2_certify.py 2 shear 64 1/100000 21
    run_a2_shear_negctl.json
B2_REFINE_TOL=0.01 B2_MAX_DEPTH=3 PY code/b2_certify.py 1/2 shear 64 1/100000 21
    run_a05_shear_limit.json
PY code/b2_leanflow_claims.py run_a1_shear_main.json leanflow_claim_a1.json b2_a1_shear
```

Record: 448.1 s, 206 468 cells, no failure, $U''(0) \in [-1.94771, -1.90017]$; controls: 677.9 s (diag), 1 342.4 s ($a = 2$), 1 238.6 s ($a = 1/2$). The file b2_certify.py was corrected

after these runs (tail factor, no effect according to the report) and has not been rerun: its current hash differs from the one in the record. Then, on the Lean side (from `W`): `lean_pinned.py compile formal_cert/Programmes/ZeroPointCert/Basic.lean -olean Programmes.ZeroPointCert.Basic`, then `Controls.lean`, and the two files that must fail, `RejectTampered.lean` and `RejectA2.lean`.

B1: SCHA. From `W`, with `B=results/proposal2/2026-10-26/B1`, on cores 6 and 7 (`B/TRANSITION.md`, section *Commands*):

```
PY B/code/b1a_curvature.py B/data/b1a_curvature.json        # ~1 min
PY B/code/b1a_scan.py B/data 1.0 0.3 0.4 0.45 0.5 0.6 0.7 0.8 0.9 1.1 1.2 1.3 1.4 1.5 2.0   #
    ~12 min
bash B/code/run_stage2.sh                                    # témoins + SCHA, ~70 min
PY B/code/b1_summary.py B
bash B/code/rust_check.sh                                    # 63 tests
T/release/lattice_phonon scha 1 0.5 0.8660254037844386 0.1 32
```

(`run_stage2.sh` runs the controls and the SCHA.) The Python–Rust agreement to 8.9×10^{-16} comes from `W/papers/p2_quantum_transition/ledger/scha_crosscheck.py`, which calls `T/release/lattice_phonon`: build from the `p2` worktree first. The planar Fourier transform of the shoulder potential (relative minimum -0.0448 at $kR_c = 4.82$) is recomputed by `PY W/papers/p2_quantum_transition/checks.py`.

Helium: the roton-region test. `PY W/results/programmes/2026-10-25/P6/scripts/analysis.py` (writes `results.json`); the Chapter 5 version is `code/ch5_U3_helium.py`, rerun below.

E.6 Chapter 4

```
cargo build --manifest-path ~/rusty-SUNDIALS-wt-p7/Cargo.toml --target-dir ~/rusty-target-ar \
    --release -j 2 --offline -p qf-popgen
T/release/fixation_ode DOSSIER            # 115 lignes : 106 acceptées, 9 témoins refusés
PY W/results/programmes/2026-10-25/P7/numerics_fixation.py     # fractions exactes
PY W/results/programmes/2026-10-25/P7/price_numerics.py
PY W/results/proposal3/2026-10-26/N4/numerics_evograph.py      # étoile n = 3 et 4, 4-cycle
```

(The comments read: 115 rows, 106 accepted, 9 controls refused; exact fractions; star $n = 3$ and 4, 4-cycle.) The three Python scripts take no argument and write their JSON file next to themselves (they therefore overwrite the record: compare with `git diff`). *Record:* star $n = 4$: 392/1265, 168/253, mean 728/1265; 4-cycle: up/down $(1/3, 1/6)$ and $(2/3, 1/3)$. The unit tests of `qf-popgen` were rerun for this appendix (§E.9). The `leanchecker` replay of the chapter's ten modules is described in §4.6. Three numbers of the chapter have no reproduction script: the `sympy` discovery of Fisher's increment, the rational recomputation of 7/19 and 277/361 (which the kernel checks anyway), and the table of Moran's ρ_{100}.

E.7 Chapter 5

First build from the `tb2ch5` worktree (command of §E.1). The scripts write their figure into `figures/` *next to the script*: run from the book, they overwrite the printed figures. For a mere check, copy the book's directory elsewhere and run the copy, as we did.

case	command (from the book's directory)	run time
U1	`PY code/ch5_U1_dipolar.py SORTIE -bin T/release`	about 11 s (record)
U2	`PY code/ch5_U2_yukawa.py SORTIE [-skip-relax]`	3 relaxations of 1 to 2 min (record)
U3	`PY code/ch5_U3_helium.py SORTIE [-gpe-json FICHIER]`	4.3 s (rerun)
U4	`PY code/ch5_U4_shell.py SORTIE`	2.5 to 6 min (record)
U5	`PY code/ch5_U5_fixation.py SORTIE`	20.4 s (rerun)

The Gross–Pitaevskii linear response (U3, panel b) is computed by the `qf-gpe2d` example:

```
cargo run --manifest-path ~/rusty-SUNDIALS-wt-tb2ch5/Cargo.toml --target-dir ~/rusty-target-ar
    \
    --release -j 2 --offline -p qf-gpe2d --example bogoliubov_dispersion -- GPE.json
PY code/ch5_U3_helium.py SORTIE --gpe-json GPE.json
```

(*Record:* 10 min 24 s on one core.) The subcommands called by the scripts do exist: `dipolar_crystal refs|gaps`, `yukawa_crystal exact-refs|exact-relax` (arguments: directory, κ, number of starts, annealing, first start), `sigma3_study full SORTIE.json well_amps.txt ÉTIQUETTE`, `fixation_ode DOSSIER`.

E.8 Appendices C and D

```
taskset -c 6,7 PY scripts/certified_numerics/lean_pinned.py compile \
    papers/textbook_formalisation_2026/code/lean/appC_1_logique.lean --json-out SORTIE.json
    # idem 2 à 6
taskset -c 6,7 PY papers/textbook_formalisation_2026/code/appC_leanchecker.py TRAVAIL SORTIE.
    json \
    appC_1_logique appC_2_calcul appC_3_recurrence appC_4_lire_mathlib appC_5_temoins
    appC_6_contrefacon
PY papers/textbook_formalisation_2026/code/appD_certifie.py SORTIE
cargo run --manifest-path ~/rusty-SUNDIALS-wt-tb2ch5/Cargo.toml --target-dir ~/rusty-target-ar
    \
    --release -j 2 --offline -p examples --example robertson
```

(The first command is repeated likewise for files 2 to 6.) Expected exit codes: 0 for compilations 1 to 6 (even example 5, which contains `sorry`); 0 for `appC_leanchecker.py`, which itself checks that only example 6 and the nonexistent module are refused; 0 for `appD_certifie.py`, which checks its seven controls.

E.9 What was rerun for this appendix

On 2026-10-26, the agent that wrote this appendix reran the following computations, from a *copy* of the book's directory (so as not to overwrite the figures), on two cores. The outputs are copied from the terminals.

`ch1_theta.py` (3.4 s).

```
A   a=0.5: Theta=2.008477918598  a^-1 Theta(1/a)=2.008477918598  diff=4.44e-16  sans facteur
    diff=1.00e+00
A   a=2.0: Theta=1.004238959299  a^-1 Theta(1/a)=1.004238959299  diff=2.22e-16  sans facteur
    diff=1.00e+00
A   a=3.0: Theta=1.000112671200  a^-1 Theta(1/a)=1.000112671200  diff=8.88e-16  sans facteur
    diff=2.00e+00
a=1.0: Theta_A=1.159595267  Theta_Z2=1.180340599  A<=Z2: True
all ok: True
exp(-sqrt t) [True, True, True, True, True, True, True, True, True]
cos t (controle negatif) [False, False, False, False, False, False, False, False, False]
```

(excerpt; "sans facteur" means "without the factor", and "controle negatif" "negative control"). The values are those of Chapter 1.

ch5_U3_helium.py (4.3 s, with an explicit -csv).

```
"kstar": 3.003789625995706,  "n_points": 99,  "delta_ds": 0.08200000000000007,
"counts": {"CONSISTENT": 29, "UNDECIDED": 7, "REFUTED": 63},
"roton_n": 21,  "roton_all_refuted": true,  "roton_ratio_range": [1.6166788059571657,
    2.491062323005841],
"pos_control_SB_fires": 0,  "neg_control_free_fires": "95/95",  "all_controls_pass": true
```

These are the numbers of case U3 (29, 7, 63; factor 1.62 to 2.49; controls 0 out of 99 and 95 out of 95).

ch5_U5_fixation.py (20.4 s).

```
"rows": 115,  "accepted": 106,  "refused": 9,  "max_err_vs_exact": 3.130636638815076e-10,
"dataset_rows": 25000,  "gc_fraction": 0.5381068 (label 0), 0.4003632 (label 1),
"leanflow": {"pos_cvode_within_1e-11_of_4_7": "ok", "neg_neutral_1_3": "rejected",
             "neg_too_tight_1e-14": "rejected", "pos_harmonic_moran_N3_r2": "ok",
             "neg_harmonic_neutral_i_over_N": "rejected"},
"leanflow_all_as_expected": true,  "literal": "0.5714285714284759"
```

(compacted layout). These are the numbers of case U5.

Tests of qf-popgen (15.9 s, compilation included).

```
cargo test --manifest-path ~/rusty-SUNDIALS-wt-tb2ch5/Cargo.toml --target-dir ~/rusty-target-ar
     \
    --release -j 2 --offline -p qf-popgen
running 8 tests
test tests::negative_corrupted_kernel_is_refused ... ok
test tests::negative_identity_kernel_never_absorbs ... ok
test tests::negative_horizon_too_short_is_refused ... ok
test tests::nonnegativity_check_refuses_cancelling_entries_that_the_signed_sums_accept ... ok
test tests::accepted_runs_record_a_nonnegative_minimum_component ... ok
test tests::positive_moran_n3_r2_matches_the_lean_values ... ok
test tests::positive_neutral_wright_fisher_gives_i_over_n ... ok
test tests::negative_flipped_and_neutral_targets_are_refused_under_selection ... ok
test result: ok. 8 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.04s
```

Appendices C and D. The six Lean compilations, the `leanchecker` replay, `appD_certifie.py` (1.7 s) and the `robertson` example (43 s, compilation included): outputs quoted in Appendices C and D.

Not rerun. U1, U2 and U4, the Gross–Pitaevskii linear response, and all the computations of Chapters 2 to 4 other than the `qf-popgen` tests: their run times and outputs above are those of the records.

E.10 Known failures and remedies

Cold disk cache. The first load of Mathlib, or of `import Lean`, reads gigabytes of `.olean` files from disk. In Appendix C, the same file compiled in 51.6 s the first time and in 4.4 s the second. Do not draw conclusions from a slow first measurement; compare run times only with a warm cache. The `leanchecker` replay takes about a minute per module, even for a tiny module; we have not measured the cause (see Appendix C).

CPU. Limit parallelism: `cargo -j 2`, `taskset -c 6,7` for Python and Lean. A process already running is restricted with `taskset -a -cp 0,1 PID`. The P1 certificate sets `OMP_NUM_THREADS=32` itself; on a small machine it will take much longer than 478 s.

Memory. We did not measure the peak memory of the Lean compilations; importing all of Mathlib needs markedly more than our targeted imports. LeanFlow accepts a limit (`max_memory_mb` of `LeanConfig`); exceeding it yields an `error`, never an `ok`.

Stale binaries. `T/release` is shared: if a number differs from the record, rebuild from the chapter's worktree (Table E.1) before looking further.

`cargo -offline` **fails.** The dependency cache (`~/.cargo/registry`) must be populated: run `cargo fetch -manifest-path .../Cargo.toml` once with network access.

A Lean module cannot be found. `unknown module prefix` or `Could not find any oleans`: the imported module was not compiled with `-olean`, or not in the right order. The `.olean` files of `lean_pinned.py` always go to `W/formal_cert/.olean_out`.

`ModuleNotFoundError: flint` **or an ABI error.** The script is running with the wrong Python: B2 and `appD_certifie.py` require the Python 3.11 of the `.venv`, and python-flint 0.6.0 from `b2_pylib`.

`ModuleNotFoundError: leanflow_mcp`**.** Pass `-leanflow-src ~/LeanFlow-wt-p1/src`; under `-I`, `PYTHONPATH` is ignored.

`sha256 mismatch`**.** The downloaded file is not the one of the pinned revision (or the download was truncated). Delete it and rerun; never bypass the check.

Overwritten figures. The `code/ch5_U*.py` scripts write into `figures/` next to themselves. Work on a copy of the book, or check with `git diff` before committing.

Exit code 0. Lean accepts `sorry` with exit code 0; so does `lean_pinned.py`. Always read the `#print axioms` lines, and replay with `leanchecker` before writing **[Tier A]**.

Appendix F

Code examples: Python, Rust (rusty-SUNDIALS) and LeanFlow

Chapters 1 to 5 state theorems and tell how they were verified. This appendix gives the reader the tools to *redo* the computations that accompany them. It contains twelve examples, spread over the themes of the book. Each one follows the same plan:

1. **the question**, asked before any computation;
2. **the code**, a short excerpt read directly from the executed file (the line numbers are those of the file);
3. **the actual output** of the program, copied by LaTeX from the log of the run;
4. **a figure**, when it helps to read the numbers;
5. **an exercise**.

Warning : what these examples are, and what they are not

- A floating-point computation is Tier X: it illustrates, it tests, it can *refute*, but it never proves. Only Examples F.10 and F.11 involve the Lean kernel.
- Every program runs at least one positive control (a case whose answer is known) and one negative control (a deliberately false case, which must be refused). It exits with code 1 if a control does not behave as expected. All the outputs printed here end with `temoins conformes: True` ("controls as expected") or `controls as expected: true`, except that of Example F.11. That one is the raw output of Lean, which compiles with exit code 0 even when the planted witness is present; the witness is then read off from `#print axioms`.
- The code, outputs and figures of this appendix were produced by an AI agent (Claude) under the author's direction, and checked only by that agent. No human has audited them.

Where the files are. Everything is stored in `code/examples/`:

- `python/`: the Python examples, and `appF_style.py`, the common style of the figures;
- `rust/`: a byte-for-byte copy of the `qf-textbook-examples` crate of rusty-SUNDIALS:
  - branch `textbook2/code-examples-2026-10-26`, branched from `main` at commit `7630f7e`;

  - commit `f01b82b730c86ccf05c1f773b11fe0a9b650cc7b`;

- `lean/`: the Lean file of Example F.11 and the two certificates generated by LeanFlow in Example F.10;

- `logs/`: the outputs printed here; the CSV files written by the Rust binaries are in `logs/rust_out/`.

Reproducing. The Python examples use numpy, scipy, mpmath and matplotlib. They are run with the isolated interpreter (option `-I`):

```
python -I code/examples/python/f01_poisson.py > code/examples/logs/f01_poisson.txt
```

The Rust crate is built in the rusty-SUNDIALS worktree:

```
CARGO_TARGET_DIR=~/rusty-target-ar cargo build --release -j 2 --offline -p qf-textbook-examples
~/rusty-target-ar/release/tb_moran_forward code/examples/logs/rust_out
```

Before committing, we ran three of the repository's continuous-integration checks, and all three passed:

- `cargo fmt -all - -check`;

- `cargo clippy -workspace -features experimental - -D warnings`;

- `cargo test -p qf-textbook-examples`, with 5 tests out of 5 passing.

The full test suite of the workspace was not rerun.

Table F.1: The twelve examples. Tier: that of the statement *checked by the program*, not that of the theorem it illustrates.

	language	question	tier
1	Python	Poisson summation for Gaussians	X
2	Python	energy of the triangular and square lattices (Gaussian, Yukawa, dipolar)	X
3	Python	signs of $(-1)^n g^{(n)}$ with mpmath, and two counterexamples	X
4	Python	exact Wright–Fisher powers, $P^n(i,N) \to i/N$	X (exact computation)
5	Python	fixation on an isothermal graph and on the star	X (exact computation)
6	Python	phonons and zero-point energy, local maximum	X
7	Rust	CVODE on a stiff ODE, then Moran's Kolmogorov equation	X
8	Rust	relaxation of a small Yukawa crystal	X
9	Rust	the invariants of the $\sigma = 3$ shell model	X
10	LeanFlow	certify fixation fractions, reject a planted claim	A (arithmetic)
11	Lean	read and check `ILENE`	A
12	Python	cross-check of CVODE against SciPy	X

F.1 Example 1 – Poisson summation for Gaussians

The question. The Poisson summation formula is the engine of Chapters 1 and 3. For a Gaussian in dimension 1 it reads

$$\sum_{n\in\mathbb{Z}} e^{-\pi a(n+x)^2} = a^{-1/2}\sum_{k\in\mathbb{Z}} e^{-\pi k^2/a}\cos(2\pi kx), \qquad a>0.$$

For the triangular lattice A of covolume 1, whose dual is a rotation of itself, it gives $\Theta_A(a) = a^{-1}\Theta_A(1/a)$. Can we see it to within 10^{-15}? And what is it good for in practice?

The code. The two sides of the formula, and the theta function of the triangular lattice ($|v|^2 = \frac{2}{\sqrt{3}}(m^2+mn+n^2)$):

```
def theta1(a: float, x: float, nmax: int) -> float:
    """Somme directe sum_{|n|<=nmax} exp(-pi a (n+x)^2)."""
    n = np.arange(-nmax, nmax + 1)
    return float(np.exp(-math.pi * a * (n + x) ** 2).sum())


def theta1_dual(a: float, x: float, kmax: int) -> float:
    """Cote dual de Poisson : a^(-1/2) sum_{|k|<=kmax} exp(-pi k^2/a) cos(2 pi k x)."""
    k = np.arange(-kmax, kmax + 1)
    return float((np.exp(-math.pi * k**2 / a) * np.cos(2 * math.pi * k * x)).sum() / math.sqrt(a))


def theta_tri(a: float, nmax: int = 40) -> float:
    """Theta_A(a) = sum_{v in A} exp(-pi a |v|^2), |v|^2 = A2 (m^2 + m n + n^2)."""
    m, n = np.meshgrid(np.arange(-nmax, nmax + 1), np.arange(-nmax, nmax + 1))
    return float(np.exp(-math.pi * a * A2 * (m * m + m * n + n * n)).sum())
```

Actual output (`f01_poisson.txt`):

```
== dimension 1 : sum_n exp(-pi a (n+x)^2) ==
a=0.1  x=0.0  direct=3.162277660168523 dual=3.162277660168523 ecart=4.4e-16 | sans facteur: ecart=2.16e
    +00
a=0.1  x=0.3  direct=3.162277660168336 dual=3.162277660168335 ecart=4.4e-16 | sans facteur: ecart=2.16e
    +00
a=3.0  x=0.3  direct=0.438043367661845 dual=0.438043367661845 ecart=1.1e-16 | sans facteur: ecart=3.21e
    -01
a=7.0  x=0.5  direct=0.008191649778702 dual=0.008191649778702 ecart=9.2e-17 | sans facteur: ecart=1.35e
    -02
== dimension 2 : Theta_A(a) = a^-1 Theta_A(1/a) ==
a=0.25 Theta_A(a)=4.0000119791720  Theta_A(1/a)/a=4.0000119791720  ecart=0.0e+00 | sans facteur: ecart
    =3.00e+00
a=0.5  Theta_A(a)=2.0084779185979  Theta_A(1/a)/a=2.0084779185979  ecart=0.0e+00 | sans facteur: ecart
    =1.00e+00
a=2.0  Theta_A(a)=1.0042389592990  Theta_A(1/a)/a=1.0042389592990  ecart=0.0e+00 | sans facteur: ecart
    =1.00e+00
== nombre de termes pour une erreur < 1e-12 a a=0.01, x=0.3 ==
cote direct: nmax=28   cote dual: kmax=0   valeur=10.000000000000
figure: appF_poisson.pdf
temoins conformes: True
```

Reading the output. The two sides agree to machine precision (10^{-16}). Without the factor $a^{-1/2}$ (or a^{-1} in dimension 2), the discrepancy is of order 1: this is the negative control. The last line shows the practical interest of the formula. For $a = 0.01$ the Gaussian is very wide: the direct sum needs $n_{\max} = 28$ to reach 10^{-12}, whereas the dual side is exact from its first term on. This principle, summing on the side where the function decays fast, is that of Ewald summation.

Exercise F.1 (Example 1)**.** (a) Prove the formula in dimension 1 from the Fourier transform of $e^{-\pi a x^2}$, which equals $a^{-1/2}e^{-\pi\xi^2/a}$. (b) Modify `theta1` for another even function, $e^{-|x|}$, whose transform is $2/(1+4\pi^2\xi^2)$, and check the corresponding formula numerically. Why are many more terms needed on the dual side?

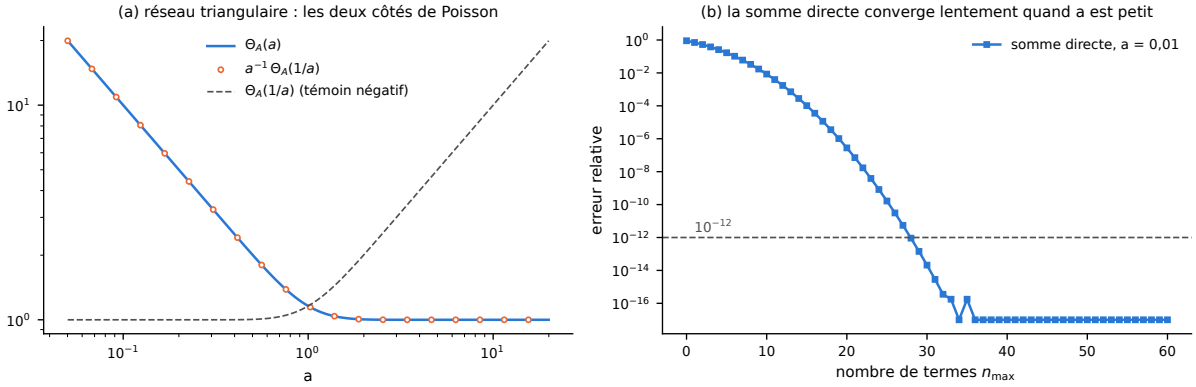


Figure F.1: Example 1 **[Tier X]**. (a) $\Theta_A(a)$ and $a^{-1}\Theta_A(1/a)$ coincide; without the factor a^{-1} (dashed), the curve is wrong. (b) Relative error of the direct sum as a function of the number of terms, for $a = 0.01$, $x = 0.3$.

F.2 Example 2 – Triangular versus square lattice

The question. At density 1, is the energy per particle $E(L) = \frac{1}{2}\sum_{v \in L\setminus 0} V(|v|)$ lower for the triangular lattice than for the square lattice? We take three potentials that are completely monotone in r^2:

- the Gaussian $e^{-\pi r^2}$, for which $E = (\Theta_L(1) - 1)/2$;
- the Yukawa potential $e^{-\kappa r}/r$, for $\kappa = 1$ and $\kappa = 2$;
- the dipolar potential $V = 2(1/r - 1/\sqrt{r^2 + d^2})$, for $d = 3$, in the units $U = V/d^2$ of the `qf-yukawa-crystal` crate.

For these three potentials, the optimality of the triangular lattice is **[Tier A]** (Chapter 2), conditional on the upstream theorem `universal_energy_minimum`.

The code. The dipolar potential decays like $1/r^3$. A sharp cutoff at $|v| < R$ leaves an error of order 10^{-5}, even after adding the continuous tail (the output shows it). We therefore cut off smoothly, with a weight $w = \frac{1}{2}\operatorname{erfc}((r - R)/\sigma)$, and integrate the remainder $(1 - w)U$ as a continuum of density 1:

```
def energy(pot: str, b: np.ndarray, rmax: float = 60.0) -> float:
    r = distances(b, rmax)
    if pot == "gauss":  # a = 1
        return 0.5 * float(np.exp(-math.pi * r * r).sum())
    if pot.startswith("yukawa"):
        kappa = float(pot.split(":")[1])
        return 0.5 * float((np.exp(-kappa * r) / r).sum())
    return dipolar(float(pot.split(":")[1]), b)


def dipolar(d: float, b: np.ndarray, big_r: float = 30.0, sig: float = 6.0) -> float:
    """U = V/d^2, queue en 1/r^3 : coupure douce w = erfc((r-R)/sig)/2 sur le reseau, et le reste
    (1-w) U integre comme un continuum de densite 1. Une coupure brutale laisserait une erreur ~1e-5.
    """
    u = lambda x: 2.0 / (x * math.sqrt(x * x + d * d) * (x + math.sqrt(x * x + d * d)))  # noqa: E731
    r = distances(b, big_r + 12 * sig)
    lattice = 0.5 * float(sum(u(x) * 0.5 * erfc((x - big_r) / sig) for x in r))
    rest = quad(lambda x: u(x) * (1 - 0.5 * erfc((x - big_r) / sig)) * math.pi * x, 1e-9, np.inf,
                limit=200, epsabs=1e-15, epsrel=1e-13)[0]
    return lattice + rest
```

Actual output (`f02_lattice_energies.txt`):

```
potentiel                E_tri             E_carre   carre-tri  ecart a la valeur enregistree
gauss        0.0797976334819642  0.0901702995080481  1.0373e-02  -
yukawa:1     1.6253213396062747  1.6348554953171981  9.5342e-03  2.2e-16
yukawa:2     0.3935428489323911  0.4009742487650575  7.4314e-03  5.6e-17
dipolar:3    1.6957619630355221  1.6980693676463541  2.3074e-03  4.7e-13
dipolaire, coupure brutale R=60 + queue continue: ecart 3.1e-05
== temoin negatif : pas egal a 1 (densites 2/sqrt(3) et 1) ==
yukawa:1   E_tri(pas 1)=1.9689324150 > E_carre(pas 1)=1.6348554953 : comparaison inversee
figure: appF_lattice_energies.pdf
temoins conformes: True
```

Reading the output. In all three cases, the square costs more. The last column compares with the values recorded by rusty-SUNDIALS:

- for Yukawa, with the exact sums of `exact_refs.jsonl`, with discrepancies of 10^{-16};
- for the dipolar potential, with the Ewald energies of `lattice_gaps.jsonl`, with a discrepancy of 5×10^{-13}.

This is the positive control. The negative control recalls a rule about the statements: at equal *spacing*, the triangular lattice is denser, its energy is higher and the comparison reverses. It is meaningful only at fixed density. In Figure F.2a, the three normalised curves almost coincide and increase without interruption from 60° to 90°. This is only one path among others in the space of lattices: the theorem concerns all configurations.

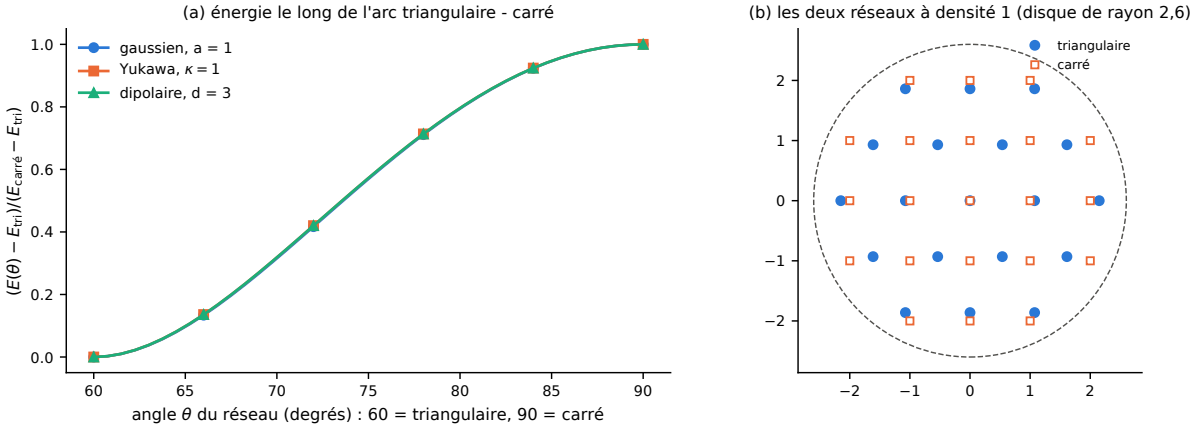


Figure F.2: Example 2 **[Tier X]**. (a) Energy along the lattices of shape $\tau = e^{i\theta}$, normalised between the triangular ($\theta = 60°$) and the square ($\theta = 90°$) lattice, for the three potentials. (b) The two lattices of covolume 1.

Exercise F.2 (Example 2)**.** (a) Replace `"gauss"` by a Gaussian $e^{-\pi a r^2}$ with arbitrary parameter a, and check on a grid of values of a that $\Theta_A(a) \leq \Theta_{\mathbb{Z}^2}(a)$ (Montgomery's theorem [75], **[Tier L]**). (b) For a potential that is *not* completely monotone, for instance $V = e^{-r^2} - \frac{1}{2}e^{-r^2/4}$, plot the energy along the same arc. Is the minimum still at 60°? What can one conclude from this, and what can one not conclude?

F.3 Example 3 – Complete monotonicity: what a computer can say

The question. A function g is completely monotone (CM) if $(-1)^n g^{(n)}(t) \geq 0$ for all n and all $t > 0$. A program tests only finitely many pairs (n, t). It can therefore *refute* complete

monotonicity, since a single wrong sign suffices, but never prove it: the proof belongs to the CM/Bernstein library and to the theorem `ALEXIS` of Chapter 5. What do we see on three CM potentials and on two functions that are not CM?

The code. `mpmath.diff` differentiates numerically with 40 significant digits, enough for order 8:

```
FUNCS = {
    "gauss": lambda t: mp.exp(-mp.pi * t),
    "yukawa": lambda t: mp.exp(-mp.sqrt(t)) / mp.sqrt(t),
    "dipolaire": lambda t: 2 * (1 / mp.sqrt(t) - 1 / mp.sqrt(t + 1)),
    "bernstein_signe_variable": lambda t: mp.exp(-t) - mp.exp(-2 * t) / 2,
    "lennard_jones": lambda t: t**-6 - t**-3,
}
TS = [mp.mpf(x) for x in ("0.05", "0.2", "0.5", "1", "2", "5", "10")]
NMAX = 8


def first_failure(g: object) -> tuple[int, str, str] | None:
    """Premier couple (n, t) ou (-1)^n g^(n)(t) < 0, ou None si tous les signes sont bons."""
    for n in range(NMAX + 1):
        for t in TS:
            val = (-1) ** n * mp.diff(g, t, n)
            if val < 0:
                return n, mp.nstr(t, 3), mp.nstr(val, 6)
    return None
```

Actual output (`f03_cm_signs.txt`):

```
gauss                     n<=8, 7 valeurs de t : tous les signes >= 0
yukawa                    n<=8, 7 valeurs de t : tous les signes >= 0
dipolaire                 n<=8, 7 valeurs de t : tous les signes >= 0
bernstein_signe_variable  n<=8, 7 valeurs de t : ECHEC n=2 t=0.05 valeur=-0.858445
lennard_jones             n<=8, 7 valeurs de t : ECHEC n=0 t=2.0 valeur=-0.109375
figure: appF_cm_signs.pdf
temoins conformes: True
```

Reading the output. The three physical potentials pass the 9×7 tests. The two counterexamples fail, and the output says where:

- $e^{-t} - \frac{1}{2}e^{-2t}$ is a mixture of exponentials whose Bernstein measure $\delta_1 - \frac{1}{2}\delta_2$ changes sign. It fails from $n = 2$ on, since $(-1)^n g^{(n)} = e^{-t} - 2^{n-1}e^{-2t} < 0$ for small t.
- Lennard-Jones fails already at $n = 0$: it becomes negative.

The first failure is a certificate of non-complete-monotonicity, which can be checked by hand. The successes, for their part, certify nothing.

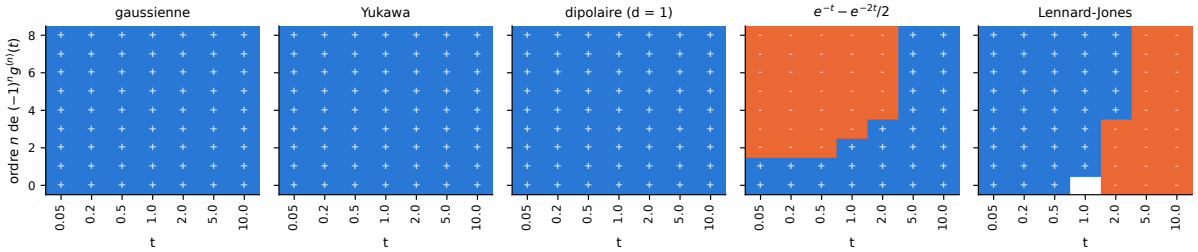


Figure F.3: Example 3 **[Tier X]**. Sign of $(-1)^n g^{(n)}(t)$ (blue: $+$, orange: $-$, white: exact zero) for $n \leq 8$ and seven values of t. The first three potentials are CM (**[Tier A]** theorems of Chapter 2); the last two are not.

Exercise F.3 (Example 3). (a) Prove by hand that $e^{-t} - \frac{1}{2}e^{-2t}$ is not CM, starting from the value of $g''(0^+)$. (b) The program says that $g(t) = e^{-\sqrt{t}}$ passes all the tests. Prove that it is CM by writing it as $f \circ \phi$, with $f = e^{-s}$ CM and $\phi = \sqrt{\cdot}$ a Bernstein function (Chapter 1). (c) Add the function $1/(1+t^2)$ to `FUNCS` and predict, before running the program, the order n of the first failure.

F.4 Example 4 – Wright–Fisher in exact fractions

The question. In the neutral Wright–Fisher model with N copies of a gene, the next generation is a binomial draw: $P(i,j) = \binom{N}{j}(i/N)^j(1-i/N)^{N-j}$. The theorem `KAL` (Kolmogorov absorption, **[Tier A]**) states that $P^n(i,N) \to h(i)$ for every harmonic function h with $h(0)=0$ and $h(N)=1$ (and a variance condition). Here $h(i) = i/N$. Can we see this *exactly*, without rounding? How fast does the convergence take place?

The code. Python's `fractions` module computes with exact rationals:

```
def wright_fisher(n_pop: int, r: Fraction = Fraction(1)) -> Matrix:
    """Noyau de Wright-Fisher ; r != 1 : tirage avec probabilite r i / (r i + N - i) (selection)."""
    rows = []
    for i in range(n_pop + 1):
        p = r * i / (r * i + n_pop - i)
        rows.append([math.comb(n_pop, j) * p**j * (1 - p) ** (n_pop - j) for j in range(n_pop + 1)])
    return rows


def matmul(a: Matrix, b: Matrix) -> Matrix:
    return [[sum(x * y for x, y in zip(row, col)) for col in zip(*b)] for row in a]
```

Actual output (`f04_wright_fisher.txt`):

```
N=6: stochastique=True  P h = h exactement=True  | temoin r=2: P h = h ? False
n=1   P^n(2,N) = 0.001371742112483   max_i |P^n(i,N) - i/N| = 5.789e-01   (taille du denominateur: 3
     chiffres)
n=2   P^n(2,N) = 0.017996706125421   max_i |P^n(i,N) - i/N| = 4.582e-01   (taille du denominateur: 6
     chiffres)
n=5   P^n(2,N) = 0.119879945240526   max_i |P^n(i,N) - i/N| = 2.539e-01   (taille du denominateur: 14
     chiffres)
n=10  P^n(2,N) = 0.243212026885335   max_i |P^n(i,N) - i/N| = 1.021e-01   (taille du denominateur: 27
     chiffres)
n=30  P^n(2,N) = 0.330965560064446   max_i |P^n(i,N) - i/N| = 2.664e-03   (taille du denominateur: 77
     chiffres)
n=60  P^n(2,N) = 0.333323358545381   max_i |P^n(i,N) - i/N| = 1.122e-05   (taille du denominateur: 152
     chiffres)
rapport des ecarts n=60/n=59 : 0.833333333333   1 - 1/N = 0.833333333333
figure: appF_wright_fisher.pdf
temoins conformes: True
```

Reading the output. Two equalities are *exact*, not merely to within 10^{-16}: the matrix is stochastic and $Ph = h$. With selection ($r = 2$), $h(i) = i/N$ is no longer harmonic: this is the negative control. The denominators grow fast, with 152 digits at $n = 60$. The ratio of two successive discrepancies equals $0.833333333333 = 1 - 1/N$, that is, the second eigenvalue of the Wright–Fisher matrix, $\lambda_2 = 1 - 1/N$.

Exercise F.4 (Example 4). (a) Prove that $\lambda_2 = 1 - 1/N$ is an eigenvalue of P. Hint: apply P to the polynomial $i(N-i)$, after computing $\mathbb{E}[X(N-X)]$ for a binomial X. (b) Deduce that $|P^n(i,N) - i/N| \le C(1-1/N)^n$. (c) Modify the script for Moran instead of Wright–Fisher, and compare the rates for equal N.

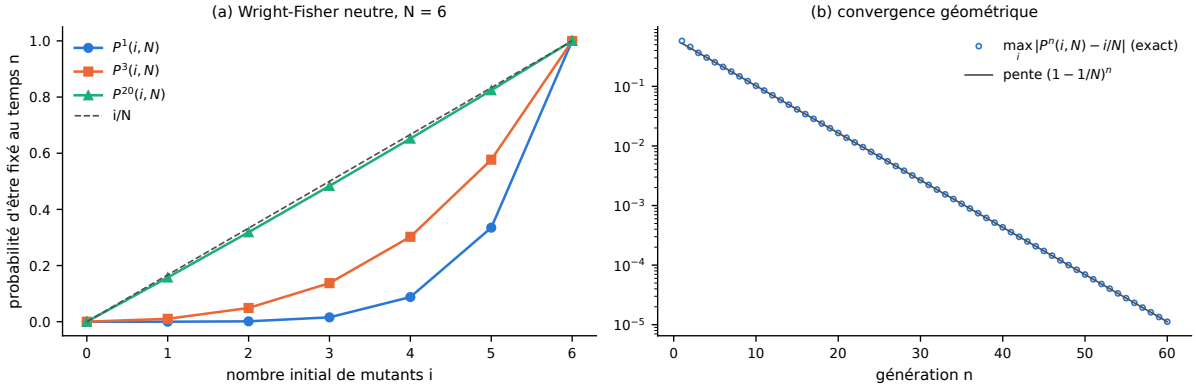


Figure F.4: Example 4 **[Tier X]**. (a) $P^n(i,N)$ for $n = 1$, 3 and 20, and the limit i/N, for $N = 6$. (b) The maximal discrepancy, computed exactly and then converted to floating point, decreases like $(1 - 1/N)^n$.

F.5 Example 5 – Isothermal graph versus star

The question. A mutant of fitness r is placed on a vertex of a graph, and the population evolves by birth–death (the model of N4, Chapter 4). The isothermal theorem `ILENE` (**[Tier A]**) states the following: if the replacement matrix is doubly stochastic, the fixation probability is Moran's, $(1 - r^{-1})/(1 - r^{-n})$, *whatever* the starting vertex. What happens on the star, which is not isothermal?

The code. The state is the set S of mutants, encoded by a bit mask, which gives 2^n states. The fixation probabilities solve a linear system, which we solve in fractions by Gaussian elimination:

```python
def transitions(w: Kernel, r: Fraction, s: int) -> dict[int, Fraction]:
    """Probabilites de transition depuis l'ensemble de mutants s (masque de bits)."""
    n = len(w)
    fit = [r if s >> i & 1 else Fraction(1) for i in range(n)]
    total = sum(fit)
    out: dict[int, Fraction] = {}
    for i in range(n):
        for j in range(n):
            if w[i][j] and (s >> i & 1) != (s >> j & 1):
                t = s | 1 << j if s >> i & 1 else s & ~(1 << j)
                out[t] = out.get(t, Fraction(0)) + fit[i] / total * w[i][j]
    return out
```

```python
def fixation(w: Kernel, r: Fraction) -> list[Fraction]:
    """x_S pour les 2^n ensembles S, par elimination de Gauss exacte."""
    n = len(w)
    full, inner = (1 << n) - 1, list(range(1, (1 << n) - 1))
    idx = {s: k for k, s in enumerate(inner)}
    a = [[Fraction(0)] * (len(inner) + 1) for _ in inner]
    for s in inner:
        row = a[idx[s]]
        for t, p in transitions(w, r, s).items():
            row[idx[s]] -= p
            if t == full:
                row[-1] -= p  # x_tout = 1 passe au second membre
            elif t in idx:
                row[idx[t]] += p
    for c in range(len(inner)):  # pivot de Gauss (la matrice est inversible)
        piv = next(k for k in range(c, len(inner)) if a[k][c] != 0)
        a[c], a[piv] = a[piv], a[c]
        for k in range(len(inner)):
            if k != c and a[k][c] != 0:
                f = a[k][c] / a[c][c]
                a[k] = [x - f * y for x, y in zip(a[k], a[c])]
    x = [Fraction(0)] * (full + 1)
```

```
    x[full] = Fraction(1)
    for s in inner:
        x[s] = a[idx[s]][-1] / a[idx[s]][idx[s]]
    return x
```

Actual output (`f05_graph_fixation.txt`):

```
cycle-4   r=2: mutant seul en v -> ['8/15', '8/15', '8/15', '8/15']  moyenne=8/15  Moran=8/15
etoile-3  r=2: mutant seul en v -> ['5/12', '2/3', '2/3']  moyenne=7/12  Moran=4/7
etoile-4  r=2: mutant seul en v -> ['392/1265', '168/253', '168/253', '168/253']  moyenne=728/1265  Moran
    =8/15
cycle-4, 2 mutants voisins: P+ = 1/3, P- = 1/6, P+/P- = 2 (biais par configuration)
cycle-4, 2 mutants opposes: P+ = 2/3, P- = 1/3, P+/P- = 2 (biais par configuration)
figure: appF_graph_fixation.pdf
temoins conformes: True
```

Reading the output. On the cycle with 4 vertices, which is isothermal, the value is 8/15 everywhere: it is Moran's. On the star with 3 vertices we recover the values of N4, all three **[Tier A]**:

- 5/12 for a mutant at the centre;
- 2/3 for a mutant on a leaf;
- 7/12 for a uniform start, against 4/7 for Moran.

The values of the star with 4 vertices, 392/1265, 168/253 and 728/1265, are only **[Tier X]** in N4: they come from an unverified script, and are recovered here by a second script. The last two lines show why the number of mutants is not a Markov chain on the cycle. With two mutants, the probability of moving up one step is 1/3 if they are neighbours and 2/3 if they are opposite. But the ratio P^+/P^- equals $r = 2$ in both cases: this is the per-configuration bias that N4 proves.

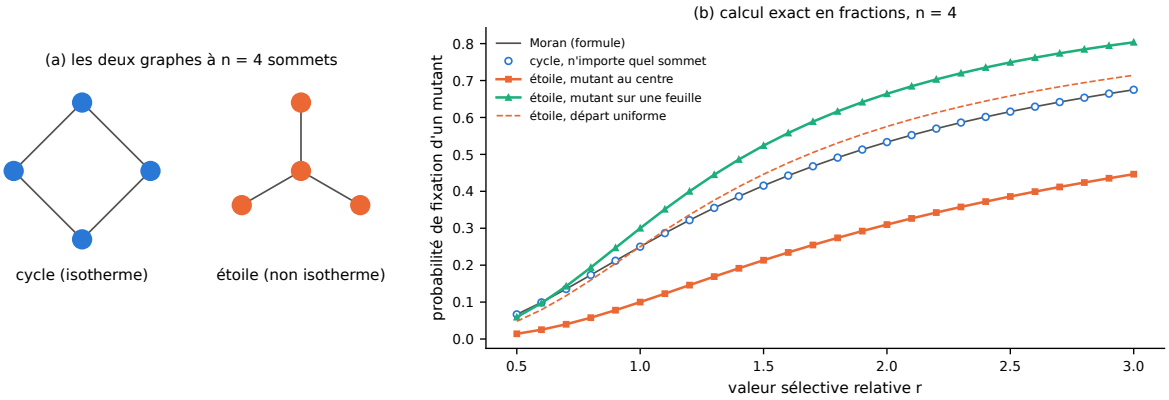


Figure F.5: Example 5 **[Tier X]**, exact computation. (a) The cycle and the star with 4 vertices. (b) Fixation probability of a mutant as a function of r. The cycle follows Moran whatever the vertex. On the star, a mutant at the centre is disadvantaged and a mutant on a leaf advantaged; on average, the star amplifies selection for $r > 1$ in this sample of values.

Exercise F.5 (Example 5)**.** (a) Check by hand that, on the star with 3 vertices and $r = 2$, the state `001` (mutant at the centre) moves to `011` and to `101` with probability 1/4 each, and to the empty state with probability 1/2. (b) Write the complete graph K_4 (with $W(i, j) = 1/3$ for $i \neq j$) and predict the result before running the program. (c) For the star with n vertices, conjecture the limit of the mean probability as $n \to \infty$, then compare with the literature (Lieberman, Hauert and Nowak, 2005 [74]). A conjecture drawn from a few exact values is Tier X.

F.6 Example 6 – Phonons and zero-point energy

The question. In Chapter 3, conjecture QU-1 asked whether the triangular lattice also minimises the harmonic zero-point energy $E_{zp} = \frac{1}{2}\langle \omega_1 + \omega_2 \rangle_{\mathrm{ZB}}$. For the Gaussian $e^{-\pi a r^2}$, the P2 programme found the opposite ([Tier X]). On $a \in (0.37668, 1.58148)$, the triangular lattice is a local *maximum* of E_{zp}. Can we recover this with thirty lines of numpy?

The code. The dynamical matrix $D(\mathbf{k}) = \sum_{R \neq 0} (1 - \cos \mathbf{k}\cdot R)\, \mathrm{Hess}\, V(R)$ is 2×2, so its eigenvalues have a closed form. The average over the Brillouin zone is taken on an $n \times n$ grid of the reciprocal cell. The fractional grid is the same for all shapes τ, which makes the grid error cancel in the differences:

```python
def dyn_eigs(k: np.ndarray, r: np.ndarray, a: float) -> np.ndarray:
    """Valeurs propres (nk, 2) de D(k) pour les vecteurs d'onde k (nk, 2)."""
    e = np.exp(-math.pi * a * (r * r).sum(axis=1))
    hxx = (-2 * math.pi * a + 4 * math.pi**2 * a * a * r[:, 0] ** 2) * e
    hyy = (-2 * math.pi * a + 4 * math.pi**2 * a * a * r[:, 1] ** 2) * e
    hxy = 4 * math.pi**2 * a * a * r[:, 0] * r[:, 1] * e
    c = 1.0 - np.cos(k @ r.T)
    dxx, dyy, dxy = c @ hxx, c @ hyy, c @ hxy
    mid, rad = 0.5 * (dxx + dyy), np.sqrt(0.25 * (dxx - dyy) ** 2 + dxy**2)
    return np.stack([mid - rad, mid + rad], axis=1)


def ezp(tau: complex, a: float, n: int) -> float:
    r, recip = lattice(tau, a)
    u = (np.arange(n) + 0.5) / n
    uu, vv = np.meshgrid(u, u)
    k = np.stack([uu.ravel(), vv.ravel()], axis=1) @ recip
    lam = dyn_eigs(k, r, a)
    if lam.min() < -1e-9 * lam.max():
        raise ValueError(f"reseau dynamiquement instable en tau = {tau}")
    return 0.5 * float(np.sqrt(np.clip(lam, 0.0, None)).sum(axis=1).mean())


def curvature(a: float, n: int, h: float = 0.01) -> float:
    e0 = ezp(RHO, a, n)
    lap = sum(ezp(RHO + d, a, n) for d in (h, -h, 1j * h, -1j * h)) - 4 * e0
    return 0.25 * RHO.imag**2 * lap / h**2
```

Actual output (`f06_phonons_zero_point.txt`):

```
== a = 1 : triangulaire rho contre le losange tau = 0.5 + 0.966025 i (convergence en grille) ==
n=48   E_zp(rho) = 1.5033540889 (P2: 1.5033523522)  E_zp(tau) = 1.4902819928 (P2: 1.4902783183)
     difference = -1.3072e-02
n=96   E_zp(rho) = 1.5033525690 (P2: 1.5033523522)  E_zp(tau) = 1.4902788709 (P2: 1.4902783183)
     difference = -1.3074e-02
n=192  E_zp(rho) = 1.5033523793 (P2: 1.5033523522)  E_zp(tau) = 1.4902784814 (P2: 1.4902783183)
     difference = -1.3074e-02
== courbure c(a) = Delta E_zp(rho)/4 (h = 0.01) ==
a=0.25  c(n=48)=+0.1879  c(n=96)=+0.1879  P2: +0.188  -> rho est un minimum local
a=0.5   c(n=48)=-0.4605  c(n=96)=-0.4605  P2: -0.461  -> rho est un MAXIMUM local
a=1.0   c(n=48)=-0.9621  c(n=96)=-0.9621  P2: -0.962  -> rho est un MAXIMUM local
a=2.0   c(n=48)=+0.4941  c(n=96)=+0.4941  P2: +0.495  -> rho est un minimum local
figure: appF_phonons.pdf
temoins conformes: True
```

Reading the output. The positive controls are the P2 values. For $a = 1$, the script recovers $E_{zp}(\rho) = 1.5033523522$ to within 3×10^{-8} on the grid $n = 192$. It also recovers P2's counterexample, the rhombus $\tau = 0.5 + 0.966025\,i$, lower by 1.3074×10^{-2}. The curvature $c(a)$ changes sign as announced:

- it is negative for $a = 0.5$ and $a = 1$, where ρ is a local maximum;
- it is positive for $a = 0.25$ and $a = 2$, where ρ is a local minimum.

The four P2 values are recovered to within one thousandth, and the curvature does not move between grids 48 and 96. All of this remains **[Tier X]**: it is a second floating-point computation, not a proof. The B2 certificate of Chapter 3 treats the case $a = 1$ with ball arithmetic.

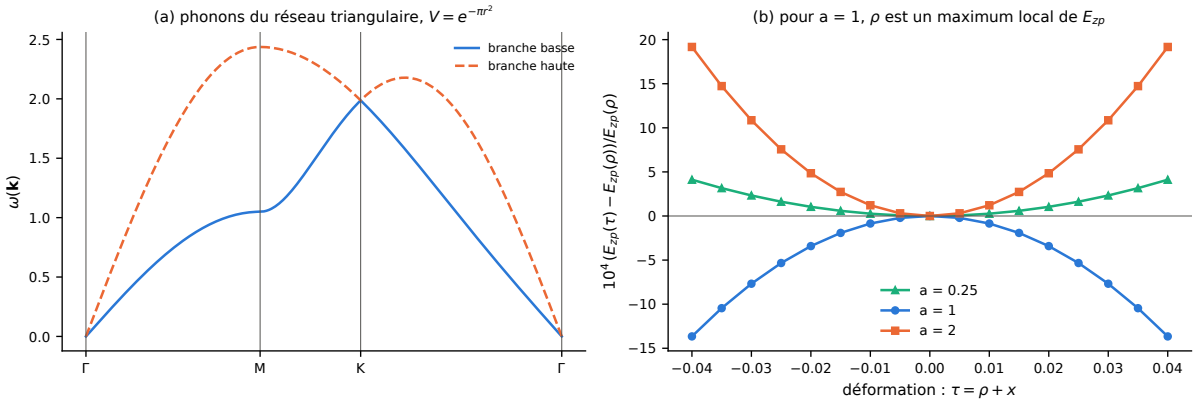


Figure F.6: Example 6 **[Tier X]**. (a) The two phonon branches of the triangular lattice for $V = e^{-\pi r^2}$, along Γ–M–K–Γ; they vanish at Γ, as for every crystal (acoustic modes). (b) Relative variation of E_{zp} when ρ is deformed into $\rho + x$: a downward parabola for $a = 1$ (local maximum), upward for $a = 0.25$ and $a = 2$.

Exercise F.6 (Example 6)**.** (a) Prove that $D(\mathbf{k})$ is symmetric and that $D(\mathbf{0}) = 0$. Why is $\omega(\mathbf{k})$ linear in $|\mathbf{k}|$ near Γ? (b) Modify `curvature` to plot $c(a)$ on $a \in [0.3, 1.7]$ and locate numerically the two sign changes (P2: 0.37668 and 1.58148). (c) Why can the statement "the triangular lattice is a local maximum of E_{zp} for $a = 1$" not become Tier A with this code alone?

F.7 Example 7 – CVODE in Rust: a stiff ODE, then the Kolmogorov equation

The question. rusty-SUNDIALS is a Rust rewrite of SUNDIALS. Its CVODE solver integrates stiff differential equations with the BDF formulas. How is it called from an independent crate? And why does an explicit scheme fail where BDF succeeds?

A stiff ODE. The Prothero–Robinson problem, $y' = \lambda(y - \cos t) - \sin t$ with $y(0) = 2$ and $\lambda = -10^4$, has the exact solution $y = \cos t + e^{\lambda t}$. Here is the call to CVODE (a function of the crate's library):

```rust
pub fn prothero_robinson(
    lambda: f64,
    y0: f64,
    t_out: &[f64],
    rtol: f64,
) -> Result<StiffRun, String> {
    let rhs = move |t: f64, y: &[f64], ydot: &mut [f64]| -> Result<(), String> {
        ydot[0] = lambda * (y[0] - t.cos()) - t.sin();
        Ok(())
    };
    let mut solver = Cvode::builder(Method::Bdf)
        .rtol(rtol)
        .atol(rtol * 1e-2)
        .max_steps(1_000_000)
        .jacobian(move |_t, _y, j: &mut DenseMat| {
            j.cols[0][0] = lambda;
            Ok(())
        })
        .build(rhs, 0.0, SerialVector::from_slice(&[y0]))
        .map_err(|e| format!("CVODE setup failed: {e}"))?;
```

```rust
    let mut samples = Vec::with_capacity(t_out.len());
    for &tout in t_out {
        let (t, y) = solver
            .solve(tout, Task::Normal)
            .map_err(|e| format!("CVODE failed at t = {tout}: {e}"))?;
        samples.push((t, y[0], prothero_exact(lambda, y0, t)));
    }
    Ok(StiffRun {
        samples,
        steps: solver.num_steps(),
        rhs_evals: solver.num_rhs_evals(),
    })
}
```

Actual output (`f07a_tb_stiff.txt`):

```
CVODE BDF, lambda = -1e4, rtol = 1e-8
       t                y_cvode                y_exact          |err|
  0.0001   1.3678794406493111e0   1.3678794361714424e0    4.48e-9
  0.0010   1.0000449057345198e0   1.0000448999298042e0    5.80e-9
  0.5000  8.7758256189050377e-1  8.7758256189037276e-1   1.31e-13
  1.0000  5.4030230589460859e-1  5.4030230586813977e-1   2.65e-11
  2.0000 -4.1614683654930645e-1 -4.1614683654714241e-1   2.16e-12
  5.0000  2.8366218546364519e-1  2.8366218546322625e-1   4.19e-13
 10.0000 -8.3907152908166083e-1 -8.3907152907645244e-1   5.21e-12
steps = 506, rhs evals = 1403
explicit Euler h = 1.0e-4 (10000 steps): y(1) = 0.5403023086
explicit Euler h = 2.5e-4 (100 steps):   y(0.025) = 4.066e17
explicit Euler h = 2.5e-4 (4000 steps):  y(1) = NaN
controls as expected: true
```

Reading the output. CVODE follows the exact solution to within 6×10^{-9} during the transient phase, where $e^{\lambda t}$ dies out within 10^{-4}, then to within 3×10^{-11} up to $t = 10$. It gets there in 506 steps. The explicit Euler scheme is stable only if $|1 + h\lambda| \le 1$, that is, $h \le 2/|\lambda| = 2 \times 10^{-4}$:

- with $h = 10^{-4}$ it succeeds, but in 10 000 steps;
- with $h = 2.5 \times 10^{-4}$, the error is multiplied by $|1 + h\lambda| = 1.5$ at each step, reaches 4×10^{17} in 100 steps and ends in `NaN`.

This is the negative control, and the very definition of stiffness.

University remark. **[Tier L]** The BDF formulas of orders 1 and 2 are A-stable: their stability region contains the whole left half-plane, so they accept any step $h > 0$ on $y' = \lambda y$ with $\mathrm{Re}\,\lambda < 0$. By Dahlquist's second barrier, no A-stable linear multistep method has order greater than 2; the higher-order BDF formulas used by CVODE (up to order 5) are only $A(\alpha)$-stable, with a stability sector that narrows as the order grows. CVODE varies both the step size and the order (1 to 5) during an integration [71] (standard).

Moran's Kolmogorov equation. The Moran chain (Chapter 4), uniformised in continuous time, obeys $\dot{\pi} = \pi(P - I)$. The mass absorbed at N tends to the fixation probability $\varphi_i = (1 - r^{-i})/(1 - r^{-N})$. The Jacobian is the constant matrix $(P - I)^{\mathsf{T}}$. We double the output time until the interior mass falls below 10^{-12}:

```rust
pub fn absorb(kernel: &[Vec<f64>], i: usize, tol: f64, tmax: f64) -> Result<Absorption, String> {
    let m = kernel.len();
    let k1 = kernel.to_vec();
    let k2 = kernel.to_vec();
    let rhs = move |_t: f64, y: &[f64], ydot: &mut [f64]| -> Result<(), String> {
        for (j, d) in ydot.iter_mut().enumerate() {
            *d = y.iter().zip(&k1).map(|(yi, row)| yi * row[j]).sum::<f64>() - y[j];
        }
        Ok(())
```

```
    };
    let jac = move |_t: f64, _y: &[f64], j: &mut DenseMat| -> Result<(), String> {
        for (c, col) in j.cols.iter_mut().enumerate() {
            for (r, v) in col.iter_mut().enumerate() {
                *v = k2[c][r] - if r == c { 1.0 } else { 0.0 };
            }
        }
        Ok(())
    };
    let mut y0 = vec![0.0; m];
    y0[i] = 1.0;
    let mut solver = Cvode::builder(Method::Bdf)
        .rtol(1e-11)
        .atol(1e-15)
        .max_steps(5_000_000)
        .jacobian(jac)
        .build(rhs, 0.0, SerialVector::from_slice(&y0))
        .map_err(|e| format!("CVODE setup failed: {e}"))?;
    let mut trace = Vec::new();
    let mut tout = 0.25;
    loop {
        let (t, y) = solver
            .solve(tout, Task::Normal)
            .map_err(|e| format!("CVODE failed at t = {tout}: {e}"))?;
        let interior: f64 = y[1..m - 1].iter().sum();
        trace.push((t, y[m - 1], interior));
        if interior < tol || t >= tmax {
            return Ok(Absorption {
                p_fix: y[m - 1],
                interior,
                mass_defect: (y.iter().sum::<f64>() - 1.0).abs(),
                t,
                steps: solver.num_steps(),
                trace,
            });
        }
        tout *= 2.0;
    }
}
```

Actual output (`f07b_tb_moran_forward.txt`):

```
N=3 r=2 i=1: p_fix = 0.5714285714284764 |p_fix - 4/7| = 9.50e-14 interior = 1.2e-13 t = 128 steps = 1287
N=10 r=0.8: max_i |p_cvode - p_exact| = 1.12e-14
N=10 r=1: max_i |p_cvode - p_exact| = 1.32e-14
N=10 r=1.5: max_i |p_cvode - p_exact| = 8.54e-13
negative control, identity kernel: interior mass at t = 1024 is 1.000 -> refused (NotAbsorbed): true
negative control, inverted exponent (1-r^i)/(1-r^N) = 0.142857 vs CVODE 0.571429 -> refused: true
controls as expected: true
```

Reading the output. For $N = 3$, $r = 2$ and $i = 1$, CVODE gives $4/7$ to within 9.5×10^{-14}. For $N = 10$, the 27 values ($r = 0.8$, 1 and 1.5, $i = 1, \ldots, 9$) follow the formula to within 10^{-12}. The two negative controls are refused:

- the identity kernel, which never absorbs;
- the formula with the inverted exponent, which gives $1/7$.

The `qf-popgen` crate of Chapter 5 does the same computation with a more complete absorption certificate. It is not yet merged into `main`, which is why this crate rewrites the equation directly.

Exercise F.7 (Example 7)**.** (a) For $y' = \lambda y$, show that the implicit Euler scheme (BDF of order 1) gives $y_{n+1} = y_n/(1 - h\lambda)$, which is stable for every $h > 0$ if $\lambda < 0$. (b) Why does the interior mass decay exponentially, and what rate should be expected for $N = 3$, $r = 2$? Compute the eigenvalues of the restriction of $P - I$ to the interior states. (c) Add a control: a kernel one of whose rows sums to 1.001. Which quantity computed by `absorb` detects it?

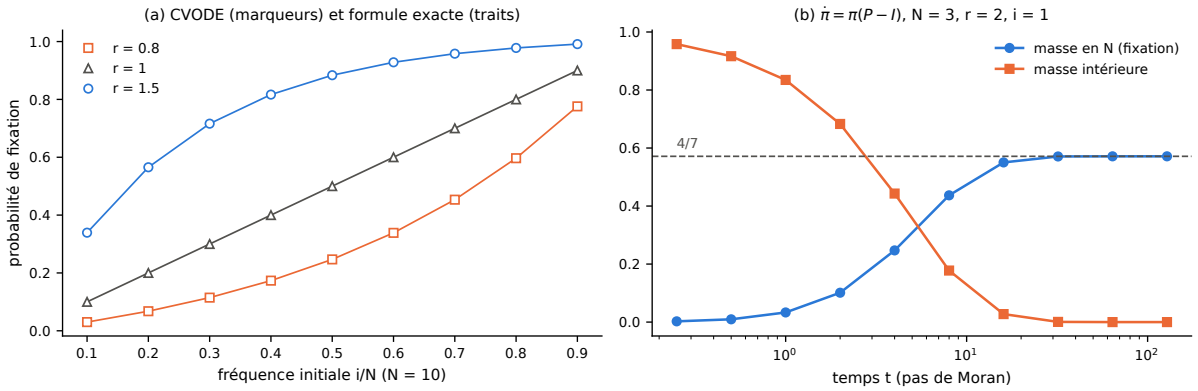


Figure F.7: Example 7 **[Tier X]**, plotted from the CSV files written by `tb_moran_forward`. (a) Fixation probability for $N = 10$: CVODE (markers) and exact formula (lines). (b) $N = 3$, $r = 2$, $i = 1$: the mass at N rises towards $4/7$ and the interior mass dies out, at the output times $0.25 \cdot 2^k$.

F.8 Example 8 – Relaxing a small Yukawa crystal

The question. 64 particles interact via e^{-2r}/r in a periodic box. If they are released from an arbitrary position and descend the energy gradient ($\dot{x} = F(x)$), do they recover the triangular lattice? The theorem says only that the triangular lattice is the *global minimum* (**[Tier A]**, conditional). It does not say that the descent reaches it.

The code. The crate calls the API of `qf-yukawa-crystal`:

- `ExactBox` for the energy, the forces and the Hessian summed over the images up to $R = 12$;
- `lattice_energy_exact` for the exact energy of the lattice.

The relaxation itself is a CVODE integration whose Jacobian is minus the Hessian (function `relax_traced` of the library). Here is the main program:

```rust
fn main() {
    let out = PathBuf::from(std::env::args().nth(1).unwrap_or_else(|| ".".into()));
    std::fs::create_dir_all(&out).expect("create OUT_DIR");
    let a = spacing_density_one();
    let e_tri = lattice_energy_exact(KAPPA, tri_basis(), RCUT).partial;
    let e_sq = lattice_energy_exact(KAPPA, square_basis(), RCUT).partial;
    println!(
        "exact lattice energies (kappa = 2, R = 12): E_tri = {e_tri:.15}, E_square = {e_sq:.15}"
    );
    let (lx, ly) = (8.0 * a, 4.0 * 3f64.sqrt() * a);
    let tri_box = ExactBox::new(KAPPA, RCUT, lx, ly);
    let sq_box = ExactBox::new(KAPPA, RCUT, 8.0, 8.0);
    let starts = [
        (
            "tri_perturbed",
            &tri_box,
            perturb(&triangular(8, 8, a), 0.15 * a, 7),
        ),
        ("random", &tri_box, random_start(lx, ly, 64, 0.5 * a, 11)),
        (
            "square_perturbed",
            &sq_box,
            perturb(&square(8, 1.0), 0.02, 13),
        ),
    ];
    let mut ok = true;
    for (name, sys, x0) in starts {
        let (x, trace) =
            qf_textbook_examples::relax_traced(sys, &x0, 1e-9, 1e6).expect("relaxation");
```

```
        let (t, e, f) = *trace.last().expect("non-empty trace");
        println!(
            "{name:<17} E/N: start {:.12} -> end {e:.15}  E/N - E_tri = {:+.3e}  max|F| = {f:.1e}  t =
    {t}",
            trace[0].1,
            e - e_tri
        );
        ok &= f < 1e-9;
        ok &= match name {
            "tri_perturbed" => (e - e_tri).abs() < 1e-10,
            "random" => e >= e_tri - 1e-12,
            _ => e < e_sq - 1e-6 && e > e_tri,
        };
        save(&out, name, &x, &trace);
    }
    println!("controls as expected: {ok}");
    std::process::exit(if ok { 0 } else { 1 });
}
```

Actual output (`f08_tb_yukawa_relax.txt`):

```
exact lattice energies (kappa = 2, R = 12): E_tri = 0.393542848871227, E_square = 0.400974248679901
tri_perturbed     E/N: start 0.419497990027 -> end 0.393542848871227  E/N - E_tri = +5.551e-17  max|F| =
    4.9e-10  t = 163.84
random            E/N: start 0.558760679165 -> end 0.395641230453870  E/N - E_tri = +2.098e-3  max|F| =
    2.4e-11  t = 5242.88
square_perturbed  E/N: start 0.401349760275 -> end 0.395723326453643  E/N - E_tri = +2.180e-3  max|F| =
    1.5e-11  t = 655.36
controls as expected: true
```

Reading the output. The three starts behave as expected.

- *Perturbed triangular lattice* (positive control): it returns to the exact lattice energy, to within 10^{-16}. This energy, 0.393542848871227 at $R = 12$, differs by 6×10^{-11} from the value at $R = 20$ recorded in `exact_refs.jsonl`, which is 0.39354284893239; this is the omitted tail.
- *Random start*: it stops 2.1×10^{-3} *above*, at a stationary point (forces below 10^{-9}) that has defects (Figure F.8b). We did not check the Hessian: nothing here says that it is a local minimum. A gradient descent does not always find the global minimum.
- *Square lattice in a square box* (negative control): it descends below the energy of the square lattice, which indicates that the square lattice is not a minimum, but it remains above the triangular lattice, which the 8×8 box cannot contain.

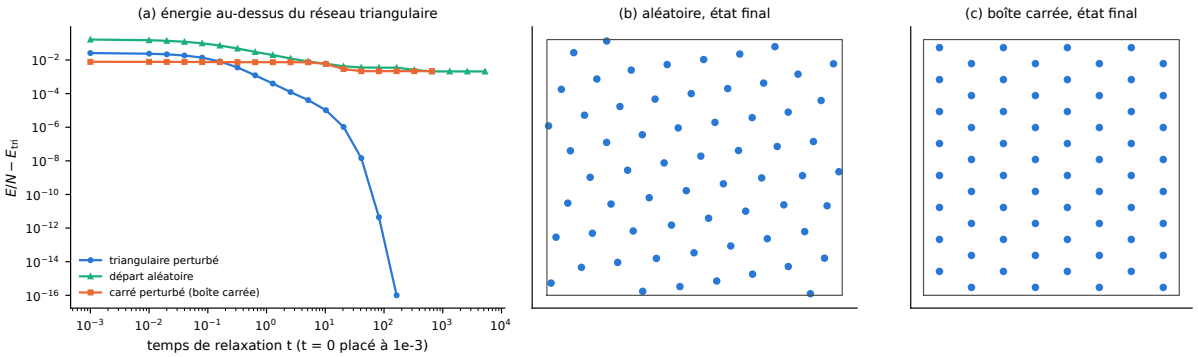


Figure F.8: Example 8 **[Tier X]**. (a) Energy per particle above the triangular lattice during the CVODE relaxation, at the output times $0.01 \cdot 2^k$. (b) Final state of the random start: triangular domains and defects. (c) Final state in the square box.

Exercise F.8 (Example 8). (a) Why can the box $8a \times 4\sqrt{3}\,a$ contain exactly the 8×8 triangular lattice of spacing a, whereas the 8×8 box cannot? (b) Change the seed of the random start (11) to five other values. How many starts reach the triangular lattice to within 10^{-10}? (c) Does this result contradict the theorem?

F.9 Example 9 – The invariants of the $\sigma = 3$ shell model

The question. The shell model of Section 5.5 conserves two quantities:

- $S = \sum |v_n|^2$;
- H_3, whose quadratic part equals $2D\Omega$, where Ω is the enstrophy.

Lean deduces from them the bound $2D\,\Omega(t) \le H_3(0) + S(0)^{3/2}$ along solutions (**[Tier A without replay]**, Chapter 5). Does a numerical integrator conserve these invariants? And does the bound hold along a trajectory?

The code. Six shells, $D = 0.05$, integration up to $T = 2$ with `integrate_cvode` of `qf-shell-cascade`, which evaluates the invariants at each internal CVODE step. The negative control multiplies one coupling by $1 + 10^{-2}$:

```
    let n_max = 5;
    // |v_n| = 2^{-1.5 n} with deterministic phases n * 0.7, normalised to S = 1.
    let v: Vec<Complex64> = (0..=n_max)
        .map(|n| Complex64::from_polar(2f64.powf(-1.5 * n as f64), 0.7 * n as f64))
        .collect();
    let s = mass(&v).sqrt();
    let v0: Vec<Complex64> = v.iter().map(|z| z / s).collect();
    let set = CvodeSettings {
        rtol: 1e-10,
        atol: 1e-13,
        max_steps: 20_000_000,
        analytic_jacobian: true,
        trace_every: 20,
    };
    let exact = SigmaModel {
        n_max,
        d: 0.05,
        sigma: 3.0,
        perturb: None,
    };
    let broken = SigmaModel {
        perturb: Some((1, 1e-2)),
        ..exact
    };
    let mut ok = true;
    for (name, model) in [("exact", exact), ("perturbed_coupling", broken)] {
        let run = integrate_cvode(model, &v0, 2.0, set).expect("CVODE run");
        println!(
            "{name:<19} T = 2: steps = {:>6}  max|dS|/S = {:.2e}  max|dH|/scale = {:.2e}  sup 2D Omega / (H(0)+S(0)^1.5) = {:.4}",
            run.steps, run.max_rel_ds, run.max_rel_dh, run.sup_ratio
        );
        if name == "exact" {
            ok &= run.max_rel_ds < 1e-7 && run.max_rel_dh < 1e-7 && run.sup_ratio <= 1.0;
            let mut csv = String::from("t,S,H,Omega\n");
            for (t, s, h, om) in &run.trace {
                let _ = writeln!(csv, "{t},{s:.17e},{h:.17e},{om:.17e}");
            }
            println!(
                "  H(0) = {:.12}, S(0) = {:.12}, bound H(0) + S(0)^1.5 = {:.12}",
                run.h0, run.s0, run.bound_rhs
            );
            std::fs::write(out.join("shell_sigma3_trace.csv"), csv).expect("write trace");
        } else {
            ok &= run.max_rel_ds > 1e-5;
```

```
        }
    }
```

Actual output (`f09_tb_shell_sigma3.txt`):

```
exact              T = 2: steps = 117613  max|dS|/S = 4.91e-10  max|dH|/scale = 9.22e-8  sup 2D Omega /
    (H(0)+S(0)^1.5) = 0.2905
  H(0) = 0.272167632338, S(0) = 1.000000000000, bound H(0) + S(0)^1.5 = 1.272167632338
perturbed_coupling  T = 2: steps = 117613  max|dS|/S = 3.07e-3  max|dH|/scale = 3.93e-3  sup 2D Omega / (
    H(0)+S(0)^1.5) = 0.2882
controls as expected: true
```

Reading the output. For the exact model, the invariants drift by 5×10^{-10} for S and by 9×10^{-8} for H_3, relative to the scale of its terms. These drifts come from the solver tolerance, not from the model. The Lean bound is far from attained, with a maximal ratio of 0.29. With the perturbed coupling, the drifts rise to 3 and 4×10^{-3}: conservation rests on the exact cancellation of the nonlinear terms, which the negative control breaks. We restrict ourselves to six shells: the fastest rotation, $\omega_n = Dk_n^3$ with $k_n = 2^n$, is multiplied by 8 with each added shell, and CVODE must resolve it step by step.

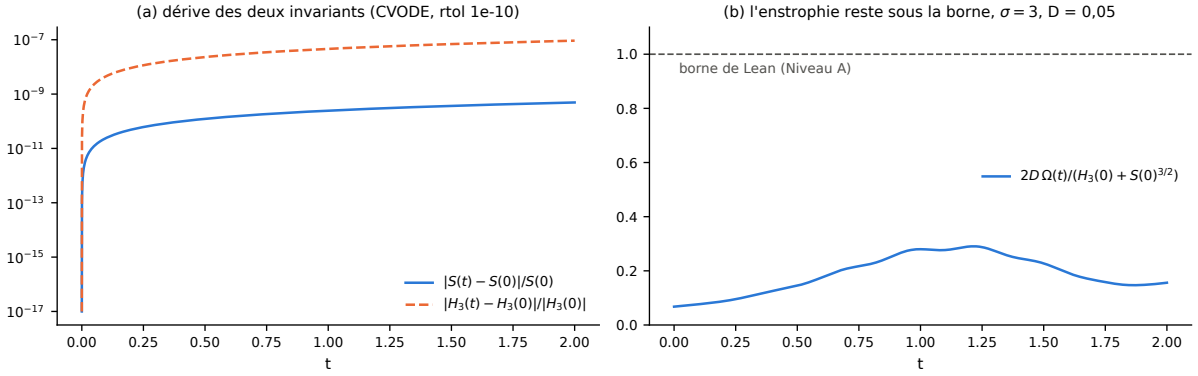


Figure F.9: Example 9 **[Tier X]**, plotted from the CSV written by `tb_shell_sigma3` (one internal step in twenty). (a) Relative drift of the two invariants. (b) The ratio $2D\Omega(t)/(H_3(0) + S(0)^{3/2})$ stays below 1, as the Lean theorem requires.

Exercise F.9 (Example 9)**.** (a) Prove that $\frac{d}{dt}\sum |v_n|^2 = 0$ for the model without dissipation. Group the coupling terms in pairs. (b) Replace `sigma:  3.0` by `2.0`. Are the invariants S and H_2 still conserved? Why does the enstrophy bound no longer follow in the same way?

F.10 Example 10 – LeanFlow: certifying fractions, rejecting a lie

The question. Example F.5 computed in Python the exact table of fixation probabilities of the star with 3 vertices. Python can make mistakes, or be badly programmed. How can these fractions be checked by the Lean kernel, without writing a single line of Lean?

The code. LeanFlow's `lean.certify` receives claims about rationals. It translates them into statements about integers, has them decided by `decide +kernel` (tactic `decide_kernel`), then audits `#print axioms`. An exact Python oracle must return the same verdict as the kernel; otherwise the result is an error, never a success. Here is the construction of the claim "the table x_S is harmonic":

```
def harmonic_claim(name: str, x: list[Fraction], r: Fraction) -> dict:
    """Affirmation LeanFlow : x_vide = 0, x_tout = 1 et sum_S' P(S,S') (x_S' - x_S) = 0 pour S
     interieur."""
```

```
    w, full = g.star(3), 7
    enc = {f"x{s}": [str(v), str(v)] for s, v in enumerate(x)}
    asr = [{"label": "x0 == 0", "endpoint": "x0.lo", "op": "==", "rhs": "0"},
           {"label": "x7 == 1", "endpoint": "x7.lo", "op": "==", "rhs": "1"}]
    for s in range(1, full):
        tr = g.transitions(w, r, s)
        terms = [[str(p), f"x{t}.lo"] for t, p in sorted(tr.items())] + [[str(-sum(tr.values())), f"x{s
     }.lo"]]
        asr.append({"label": f"harmonique en S={s:03b}", "terms": terms, "op": "==", "rhs": "0"})
    return {"name": name, "tactic": "decide_kernel", "enclosures": enc, "assertions": asr}


def compare_claim(name: str, value: Fraction, op: str, rhs: Fraction) -> dict:
    return {"name": name, "tactic": "decide_kernel", "enclosures": {"m": [str(value), str(value)]},
            "assertions": [{"label": f"m {op} {rhs}", "endpoint": "m.lo", "op": op, "rhs": str(rhs)}]}
```

Actual output (`f10_leanflow_certify.txt`):

```
tableau de l'etoile (S en binaire): 000:0, 001:5/12, 010:2/3, 011:5/6, 100:2/3, 101:5/6, 110:14/15, 111:1
pos_star_harmonic     statut=ok        niveau=A theoremes refuses=[] axiomes=[] rejeu={'compile_rc': 0, '
    leanchecker_rc': 0, 'forbidden_tokens': []}
pos_mean_above_moran  statut=ok        niveau=A theoremes refuses=[] axiomes=[] rejeu={'compile_rc': 0, '
    leanchecker_rc': 0, 'forbidden_tokens': []}
neg_moran_vector      statut=rejected  niveau=None theoremes refuses=['cert_2', 'cert_3', 'cert_4', '
    cert_5', 'cert_6', 'cert_7'] axiomes=['sorryAx']
neg_reversed          statut=rejected  niveau=None theoremes refuses=['cert_0'] axiomes=['sorryAx']
temoin du rejeu (module inexistant): leanchecker rc=1
temoins conformes: True
```

Here is the shorter of the two generated certificates, as the kernel decided it:

```
-- LeanFlow data certificate 'meanStar' (generated; tier A for the arithmetic statement only)
set_option maxRecDepth 100000
-- m > 4/7
theorem cert_0 : (0 : Int) < ((1 : Int) * (12 : Int)) ∧ ((1 : Int) * (7 : Int)) * (7 : Int) > (4 : Int)
      * ((1 : Int) * (12 : Int)) := by decide +kernel
```

Reading the output. The two true claims are accepted:

- the six first-step equations, satisfied exactly by the table $(0, \frac{5}{12}, \frac{2}{3}, \frac{5}{6}, \frac{2}{3}, \frac{5}{6}, \frac{14}{15}, 1)$;
- the inequality $7/12 > 4/7$.

The two planted claims are rejected, with `sorryAx` in the list of axioms:

- the Moran vector $(0, \frac{4}{7}, \frac{6}{7}, 1)$ indexed by $|S|$ violates the six interior equations, but it satisfies the two boundary conditions (`cert_0` and `cert_1` are not refused);
- the reversed inequality $7/12 < 4/7$ is refused.

LeanFlow checks only `#print axioms`, which can be fooled (Chapter 2). The script therefore completes the Tier A gate. Each accepted source is recompiled to `.olean` with Lean 4.34.1 and then replayed by `leanchecker`, with exit code 0 for both. None of the words `skipKernelTC`, `addDecl`, `native_decide` or `implemented_by` occurs in them. The replay control, a nonexistent module, exits with code 1. The two certificates are therefore **[Tier A]**, but for the *arithmetic statement only*. That this table is the fixation probability of the star requires in addition the uniqueness of the harmonic solution (Chapter 4) and the identification of the Python transitions with those of `bdK`. This link is a human argument, **[Tier C]**.

Exercise F.10 (Example 10, LeanFlow)**.** (a) Certify in the same way the table of the cycle with 4 vertices (2^4 states, value 8/15 for every single mutant). (b) Plant a one-ulp error: replace 5/12 by the fraction of the float `float(5/12)`. Does LeanFlow accept it? Why is a certificate on rationals more demanding than a floating-point comparison to within 10^{-15}?

F.11 Example 11 – Reading and checking the theorem ILENE

The question. Chapter 5 cites `ILENE`, the single-mutant version of the isothermal theorem. How can the reader see, on their own machine, the exact statement that Lean checked and the axioms its proof depends on? And how can a real proof be told apart from a fake one?

The code. A 21-line file, of which we reproduce lines 2 to 21, imports the compiled module and asks Lean four questions:

```
2
3  /-! Annexe F, exemple 11 : lire et verifier le theoreme ILENE (theoreme isotherme, un mutant).
4  Compile avec Lean 4.34.1 + Mathlib d13f23b7 (scripts/certified_numerics/lean_pinned.py). -/
5
6  open Acronymes
7
8  -- 1. L'enonce exact, tel que Lean le connait.
9  #check @ILENE
10
11 -- 2. Les axiomes dont depend la preuve (liste blanche : propext, Classical.choice, Quot.sound).
12 #print axioms ILENE
13
14 -- 3. La valeur limite pour n = 4 et r = 2, celle du cycle de l'exemple 5 : 8/15.
15 example : (1 - 1 / (2 : Real)) / (1 - (1 / (2 : Real)) ^ 4) = 8 / 15 := by norm_num
16
17 -- 4. Temoin negatif plante : une egalite fausse "prouvee" par sorry. Le fichier compile quand meme
18 --    (avertissement seulement) ; c'est #print axioms qui trahit la fraude.
19 theorem planted_wrong_value : (1 - 1 / (2 : Real)) / (1 - (1 / (2 : Real)) ^ 4) = 4 / 7 := by sorry
20
21 #print axioms planted_wrong_value
```

(The first line of the file, the `import` of the acronym module, is omitted here.) It is compiled with Lean 4.34.1 and Mathlib `d13f23b7`, through the script `lean_pinned.py` of the directory `scripts/certified_numerics/`. It gives:

Actual output (`f11_ilene.txt`):

```
{"module": null, "rc": 0, "seconds": 24.7}
@ILENE : ∀ {n : ℕ} {W : Fin n → Fin n → ℝ} {r : ℝ},
  0 < n →
    0 < r →
      r ≠ 1 →
        EvoGraph.Isothermal W →
          EvoGraph.StronglyConnected W →
            ∀ (v : Fin n),
              Filter.Tendsto (fun t => EvoGraph.sPow (EvoGraph.bdK (fun x => 1) W r) t {v} Finset.univ)
    Filter.atTop
                (nhds ((1 - r⁻¹) / (1 - r⁻¹ ^ n)))
'Acronymes.ILENE' depends on axioms: [propext, Classical.choice, Quot.sound]
/home/callensxavier_gmail_com/AutoevolveAI/.claude/worktrees/openai-math-discovery/papers/
    textbook_formalisation_2026/code/examples/lean/f11_ilene.lean:19:8: warning: declaration uses `sorry`
'planted_wrong_value' depends on axioms: [propext, sorryAx, Classical.choice, Quot.sound]
```

Reading the output. The output is read in four steps.

- `#check` displays the full statement. The hypotheses are: $0 < n$, $0 < r$, $r \neq 1$, W isothermal and strongly connected. The conclusion is a limit, that of $P^t(\{v\}, V)$ to $(1-r^{-1})/(1-r^{-n})$, for every vertex v.
- The axioms are exactly those of the whitelist. The acronyms module and its sources were replayed by `leanchecker`, with exit code 0 (record `acronymes_leanchecker.json`): `ILENE` is therefore **[Tier A]**.
- The `example` checks, by `norm_num`, that the limit value equals $8/15$ for $n = 4$ and $r = 2$. This is the value of the cycle of Example F.5.

- The planted witness asserts a false value, $4/7$, "proved" by `sorry`. The file compiles all the same, with exit code 0 and a mere warning. Only the `#print axioms` line betrays the fraud: it displays `sorryAx`.

This is the lesson of Chapter 2: a zero exit code proves nothing; one must read the axioms and replay the kernel.

Exercise F.11 (Example 11, Lean)**.** (a) Add `#check @KAL` and `#print axioms AURELIA`, and reread the statements with the help of the table of acronyms (Section 5.8). (b) Replace `by sorry` by `by norm_num` in the planted witness. What does Lean answer, and with which exit code? (c) Why is the equality $1/2 = 2^{-1}$, used implicitly when comparing the `example` with the statement of `ILENE`, a link to be checked, and how can this be done in Lean?

F.12 Example 12 – Cross-check: CVODE versus SciPy

The question. Two codes written by the same team can share the same error. Are the numbers of Example F.7 confirmed by *independent* methods?

The code. The script rereads the outputs of the Rust binary and recomputes the same quantities in two ways:

- for the stiff problem, with SciPy's Radau method, an implicit Runge–Kutta method and not BDF;
- for Moran, with the matrix exponential $\pi(t) = e_i\, e^{t(P-I)}$, without any time step.

```
    pts = cvode_stiff(d / "f07a_tb_stiff.txt")
    ts = [t for t, _ in pts]
    sol = solve_ivp(lambda t, y: LAM * (y - math.cos(t)) - math.sin(t), (0, ts[-1]), [2.0], method="
     Radau",
                    t_eval=ts, rtol=1e-12, atol=1e-14, jac=lambda t, y: [[LAM]])
    print(f"Radau: {sol.nfev} evaluations, statut {sol.status}")
    for (t, yc), ys in zip(pts, sol.y[0]):
        print(f"t={t:<8} CVODE={yc:+.15f}  SciPy Radau={ys:+.15f}  ecart={abs(yc - ys):.1e}")
        ok = ok and abs(yc - ys) < 1e-7
    y_shift = solve_ivp(lambda t, y: LAM * (y - math.cos(t)) - math.sin(t), (0, 1.001), [2.0], method="
     Radau",
                        rtol=1e-12, atol=1e-14).y[0][-1]
    yc1 = dict(pts)[1.0]
    ok = ok and abs(yc1 - y_shift) > 1e-4
    print(f"temoin negatif (t decale de 1e-3): ecart={abs(yc1 - y_shift):.2e}")
    with (d / "rust_out" / "moran_trace_N3.csv").open() as f:
        trace = [(float(r["t"]), float(r["p_absorbed_N"])) for r in csv.DictReader(f)]
    q = moran_kernel(3, 2.0) - np.eye(4)
    worst = 0.0
    for t, pc in trace:
        pe = expm(t * q)[1, 3]
        worst = max(worst, abs(pc - pe))
    print(f"Moran N=3 r=2 i=1: {len(trace)} temps de sortie, max |CVODE - expm| = {worst:.1e},"
          f" a t={trace[-1][0]:g}: CVODE={trace[-1][1]:.15f} expm={expm(trace[-1][0] * q)[1, 3]:.15f}")
    ok = ok and worst < 1e-8
```

Actual output (`f12_scipy_crosscheck.txt`):

```
Radau: 19213 evaluations, statut 0
t=0.0001    CVODE=+1.367879440649311  SciPy Radau=+1.367879436171424  ecart=4.5e-09
t=0.001     CVODE=+1.000044905734520  SciPy Radau=+1.000044899929879  ecart=5.8e-09
t=0.5       CVODE=+0.877582561890504  SciPy Radau=+0.877582561892962  ecart=2.5e-12
t=1.0       CVODE=+0.540302305894609  SciPy Radau=+0.540302305869891  ecart=2.5e-11
t=2.0       CVODE=-0.416146836549306  SciPy Radau=-0.416146836548008  ecart=1.3e-12
t=5.0       CVODE=+0.283662185463645  SciPy Radau=+0.283662185460866  ecart=2.8e-12
t=10.0      CVODE=-0.839071529081661  SciPy Radau=-0.839071529076606  ecart=5.1e-12
temoin negatif (t decale de 1e-3): ecart=8.42e-04
```

```
Moran N=3 r=2 i=1: 10 temps de sortie, max |CVODE - expm| = 1.3e-11, a t=128: CVODE=0.571428571428476
     expm=0.571428571428477
temoins conformes: True
```

Reading the output. After the transient phase, CVODE and Radau agree to within 3×10^{-11}. At the first two instants, the discrepancy of 5×10^{-9} is that of CVODE from the exact solution: the Radau value coincides with $\cos t + e^{\lambda t}$, and the CVODE tolerance was 10^{-8}. For Moran, CVODE and the matrix exponential agree to within 1.3×10^{-11} at the ten output times. The negative control shifts one time by 10^{-3}, and the discrepancy, of 8×10^{-4}, becomes immediately visible: the comparison therefore has the power to detect an error.

Exercise F.12 (Example 12)**.** (a) Replace `method="Radau"` by `"RK45"`, an explicit method. How many evaluations are needed, and why? (b) For Moran, the matrix exponential is exact up to rounding. Why does it not replace CVODE for $N = 1000$?

F.13 Summary

The twelve examples were run on 2026-10-26, and all their controls behaved as expected. They illustrate the three ways of handling a number in this book.

- *Floating-point computation* (Examples 1, 2, 3, 6 to 9 and 12) explores, checks and refutes. It recovers the published values to the expected precision, but it remains Tier X.
- *Exact rational computation* (Examples 4 and 5) removes rounding, but not the risk of a programming error.
- *The Lean kernel* (Examples 10 and 11) removes that risk for the checked statement, and for it alone. The link between that statement and the question asked remains a link to be checked separately.

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